

Theory of Mathematics
Volume VIII: Algebraic Topology

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Part I

Homotopy

Chapter 1

Homotopies of Maps

Algebraic topology begins by weakening the notion of equality of maps. Two continuous maps need not agree pointwise to carry the same large-scale topological information. The basic relation is *homotopy*: one map may be continuously deformed into the other while the domain and target remain fixed. This chapter develops the definition, the elementary calculus of homotopies, relative and based variants, and the first explicit deformation formulas.

1.1 The unit interval as a deformation parameter

Write

$$I = [0, 1].$$

A point of $X \times I$ is written (x, t) , with t interpreted as time. At $t = 0$ we see the initial map and at $t = 1$ the final map.

Definition 1.1 (Homotopy). Let $f, g : X \rightarrow Y$ be continuous maps. A *homotopy from f to g* is a continuous map

$$H : X \times I \longrightarrow Y$$

such that

$$H(x, 0) = f(x), \quad H(x, 1) = g(x)$$

for every $x \in X$. We write

$$f \simeq g.$$

For each fixed t , the map

$$H_t : X \rightarrow Y, \quad H_t(x) = H(x, t),$$

is a time slice of the deformation.

Warning 1.2 (Pointwise paths are not enough). For every x , the curve $t \mapsto H(x, t)$ is a path in Y , but choosing an arbitrary path from $f(x)$ to $g(x)$ for each x does not automatically produce a homotopy. The dependence on x and t must be jointly continuous.

1.2 First explicit homotopies

Example 1.3 (Straight-line homotopy). If Y is a convex subset of a real topological vector space and $f, g : X \rightarrow Y$ are continuous, then

$$H(x, t) = (1 - t)f(x) + tg(x)$$

is a homotopy from f to g .

Example 1.4 (Maps into an interval). Any two continuous maps $f, g : X \rightarrow [a, b]$ are homotopic by the straight-line formula. Thus the interval has no obstruction to deforming one incoming map into another.

Example 1.5 (Rotation of the circle). For a fixed angle α , the identity map on $S^1 \subset \mathbb{C}$ is homotopic to the rotation $R_\alpha(z) = e^{i\alpha}z$ via

$$H(z, t) = e^{it\alpha}z.$$

1.3 Homotopy is an equivalence relation

Theorem 1.6 (Equivalence relation). *Homotopy is reflexive, symmetric, and transitive on the set of continuous maps $X \rightarrow Y$.*

Proof. Reflexivity is given by the constant homotopy $H(x, t) = f(x)$. If H runs from f to g , then

$$\overline{H}(x, t) = H(x, 1 - t)$$

runs from g to f . For transitivity, suppose $H : f \simeq g$ and $K : g \simeq h$. Define

$$L(x, t) = \begin{cases} H(x, 2t), & 0 \leq t \leq \frac{1}{2}, \\ K(x, 2t - 1), & \frac{1}{2} \leq t \leq 1. \end{cases}$$

The two formulas agree at $t = \frac{1}{2}$, where both equal $g(x)$. The gluing lemma gives continuity. $\square$

Remark 1.7 (Homotopy classes). The equivalence class of $f : X \rightarrow Y$ is denoted $[f]$. The set of homotopy classes is often written

$$[X, Y].$$

At this stage it is only a set; later chapters supply algebraic structure in important cases.

1.4 Compatibility with composition

Proposition 1.8 (Postcomposition). *If $f \simeq g : X \rightarrow Y$ and $u : Y \rightarrow Z$ is continuous, then*

$$u \circ f \simeq u \circ g.$$

Proof. If H is a homotopy from f to g , use

$$(x, t) \mapsto u(H(x, t)).$$

$\square$

Proposition 1.9 (Precomposition). *If $f \simeq g : X \rightarrow Y$ and $v : W \rightarrow X$ is continuous, then*

$$f \circ v \simeq g \circ v.$$

Proof. Use

$$(w, t) \mapsto H(v(w), t).$$

$\square$

Corollary 1.10 (Two-sided compatibility). *If $f_0 \simeq f_1 : X \rightarrow Y$ and $g_0 \simeq g_1 : Y \rightarrow Z$, then*

$$g_0 \circ f_0 \simeq g_1 \circ f_1.$$

1.5 Null-homotopic maps

Definition 1.11 (Null-homotopic map). A map $f : X \rightarrow Y$ is *null-homotopic* if it is homotopic to a constant map.

Example 1.12 (Maps into a convex set). Every map into a convex subset $C \subset \mathbb{R}^n$ is null-homotopic: choose $c_0 \in C$ and use

$$H(x, t) = (1 - t)f(x) + tc_0.$$

Proposition 1.13 (Factorization through a point). A map is null-homotopic precisely when it is homotopic to a map factoring as

$$X \longrightarrow \{*\} \longrightarrow Y.$$

1.6 Homotopies relative to a subspace

Often a deformation must leave part of the domain fixed.

Definition 1.14 (Homotopy relative to A). Let $A \subset X$. A homotopy $H : f \simeq g$ is a *homotopy relative to A* , written

$$f \simeq g \pmod{A},$$

if

$$H(a, t) = f(a) = g(a)$$

for all $a \in A$ and $t \in I$.

Remark 1.15. A relative homotopy can exist only if $f|_A = g|_A$. The condition says more: the common restriction is fixed throughout the deformation.

Example 1.16 (Fixing endpoints). For paths $\alpha, \beta : I \rightarrow Y$ with common endpoints, a homotopy relative to $\{0, 1\}$ is exactly an endpoint-fixing deformation of one path into the other. This is the relation used later in the fundamental group.

1.7 Based homotopy

Let (X, x_0) and (Y, y_0) be pointed spaces.

Definition 1.17 (Based map and based homotopy). A map $f : (X, x_0) \rightarrow (Y, y_0)$ is *based* if $f(x_0) = y_0$. A homotopy through based maps is a homotopy relative to $\{x_0\}$:

$$H(x_0, t) = y_0 \quad \text{for all } t.$$

Warning 1.18 (Free versus based homotopy). Two based maps can be freely homotopic without being based-homotopic. The distinction becomes essential when loops and fundamental groups enter.

1.8 Reparameterizing a homotopy

Lemma 1.19 (Monotone reparameterization). If $H : f \simeq g$ and $\rho : I \rightarrow I$ is continuous with $\rho(0) = 0$ and $\rho(1) = 1$, then

$$K(x, t) = H(x, \rho(t))$$

is another homotopy from f to g .

This permits slowing, speeding, or pausing a deformation without changing its homotopy class.

1.9 Homotopy through a subspace

Proposition 1.20 (Image control). *Suppose $A \subset Y$, and a homotopy $H : X \times I \rightarrow Y$ satisfies $H(X \times I) \subset A$. Then H , regarded as a map into A , is a homotopy in the subspace A .*

The proposition is elementary, but important: whether a deformation is allowed depends on the target in which the deformation is required to live.

1.10 A punctured-plane obstruction preview

The straight-line formula can fail when the target is not convex.

Example 1.21 (Why the ambient target matters). In $\mathbb{R}^2 \setminus \{0\}$, the formula

$$H(x, t) = (1 - t)x + t(-x)$$

passes through 0 at $t = \frac{1}{2}$. Therefore it is not a homotopy in the punctured plane. Later invariants will explain why certain maps cannot be deformed while avoiding the missing point.

1.11 Cylinder viewpoint

A homotopy $H : X \times I \rightarrow Y$ is simply a map out of the cylinder on X , with prescribed values on the two ends

$$X \times \{0\}, \quad X \times \{1\}.$$

This point of view anticipates mapping cylinders, cofibrations, and relative homotopy constructions.

Proposition 1.22 (Restriction to time intervals). *If $H : f \simeq g$ and $0 \leq a < b \leq 1$, then after affine reparameterization the restriction of H to $X \times [a, b]$ is a homotopy from H_a to H_b .*

1.12 Products of homotopies

Proposition 1.23 (Product homotopy). *If $f_0 \simeq f_1 : X \rightarrow Y$ and $g_0 \simeq g_1 : W \rightarrow Z$, then*

$$f_0 \times g_0 \simeq f_1 \times g_1 : X \times W \rightarrow Y \times Z.$$

Proof. For homotopies H and K , set

$$L((x, w), t) = (H(x, t), K(w, t)).$$

□

1.13 What a homotopy does not claim

A homotopy does not assert that any H_t is injective, surjective, an embedding, a homeomorphism, or a diffeomorphism. It is a deformation in the space of continuous maps. If every time slice is required to be an embedding or homeomorphism, one obtains stronger notions such as isotopy.

Warning 1.24 (Homotopy is weaker than isotopy). Do not infer geometric rigidity from a homotopy. Intermediate maps may fold, collapse, or identify points unless additional hypotheses prohibit this.

1.14 Exercises

Exercise 1.25. Write the defining endpoint equations for a homotopy $H : f \simeq g$.

Hint. Evaluate at $t = 0$ and $t = 1$. □

Solution. They are $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$ for every $x \in X$. □

Exercise 1.26. Show that every map is homotopic to itself.

Hint. Do not move the map in time. □

Solution. Use $H(x, t) = f(x)$. □

Exercise 1.27. Reverse a given homotopy $H : f \simeq g$.

Hint. Reverse the interval parameter. □

Solution. Set $\bar{H}(x, t) = H(x, 1 - t)$. □

Exercise 1.28. Concatenate homotopies $H : f \simeq g$ and $K : g \simeq h$.

Hint. Give each homotopy half the time interval. □

Solution. Use $H(x, 2t)$ for $t \leq 1/2$ and $K(x, 2t - 1)$ for $t \geq 1/2$. □

Exercise 1.29. Why is the concatenated homotopy continuous at $t = 1/2$?

Hint. Compare the two values there. □

Solution. Both pieces equal $g(x)$ at the joining time, so the gluing lemma applies. □

Exercise 1.30. Give a homotopy from $f(x) = x$ to $g(x) = 0$ on $\mathbb{R}^n$.

Hint. Use scalar multiplication. □

Solution. $H(x, t) = (1 - t)x$. □

Exercise 1.31. Show that two maps $X \rightarrow [a, b]$ are homotopic.

Hint. The interval is convex. □

Solution. Use $H(x, t) = (1 - t)f(x) + tg(x)$. □

Exercise 1.32. Construct a homotopy from the identity of S^1 to rotation by angle $\pi/3$.

Hint. Rotate gradually. □

Solution. $H(z, t) = e^{it\pi/3}z$. □

Exercise 1.33. If $f \simeq g$, prove $u \circ f \simeq u \circ g$.

Hint. Postcompose the homotopy. □

Solution. For $H : f \simeq g$, use $K(x, t) = u(H(x, t))$. □

Exercise 1.34. If $f \simeq g$, prove $f \circ v \simeq g \circ v$.

Hint. Precompose the homotopy. □

Solution. Use $K(w, t) = H(v(w), t)$. □

Exercise 1.35. Define a null-homotopic map.

Hint. Compare with a constant map. □

Solution. It is a map homotopic to a constant map. □

Exercise 1.36. Show every map into a convex subset of $\mathbb{R}^n$ is null-homotopic.

Hint. Choose a point in the convex set. □

Solution. For c_0 in the target, $H(x, t) = (1 - t)f(x) + tc_0$ remains in the convex set. □

Exercise 1.37. State what $f \simeq g \pmod{A}$ means.

Hint. The subspace must not move. □

Solution. It means there is a homotopy $H : f \simeq g$ with $H(a, t) = f(a) = g(a)$ for all $a \in A$. □

Exercise 1.38. What preliminary condition on f and g is necessary for a homotopy rel A ?

Hint. Set $t = 0, 1$ on A . □

Solution. They must agree on A : $f|_A = g|_A$. □

Exercise 1.39. Interpret homotopy rel $\{0, 1\}$ for paths.

Hint. The endpoints are the chosen subspace. □

Solution. It is a deformation of paths that keeps both endpoints fixed at all times. □

Exercise 1.40. What is a based homotopy?

Hint. Fix the basepoint throughout time. □

Solution. It is a homotopy H satisfying $H(x_0, t) = y_0$ for every t . □

Exercise 1.41. Why can an arbitrary collection of paths from $f(x)$ to $g(x)$ fail to be a homotopy?

Hint. Continuity has two variables. □

Solution. The chosen paths may fail to depend continuously on x , so the resulting map $X \times I \rightarrow Y$ need not be continuous. □

Exercise 1.42. Reparameterize a homotopy so that it pauses at $t = 1/2$.

Hint. Choose a continuous $\rho : I \rightarrow I$ that is constant on a small interval. □

Solution. Choose any continuous endpoint-preserving ρ that is constant near $1/2$, and use $K(x, t) = H(x, \rho(t))$. □

Exercise 1.43. Show that the product of two null-homotopic maps is null-homotopic.

Hint. Take the product of their homotopies. □

Solution. If H and K contract the two maps to constants, then (H, K) contracts their product to the product constant. □

Exercise 1.44. Why does the straight-line deformation from x to $-x$ fail in $\mathbb{R}^2 \setminus \{0\}$?

Hint. Evaluate at $t = 1/2$. □

Solution. It equals 0 halfway through, which is not in the target. □

Exercise 1.45. If $H : X \times I \rightarrow A \subset Y$, in which two targets can the same formula be viewed?

Hint. Use the inclusion $A \hookrightarrow Y$. □

Solution. It is a homotopy into A , and after composing with the inclusion it is also a homotopy into Y . □

Exercise 1.46. Does a homotopy require every time slice to be injective?

Hint. Return to the definition. □

Solution. No. Only continuity and the endpoint conditions are required. □

Exercise 1.47. Does a homotopy between homeomorphisms have to pass through homeomorphisms?

Hint. Compare homotopy with isotopy. □

Solution. No. That stronger requirement belongs to isotopy. □

Exercise 1.48. Explain why composition descends to homotopy classes.

Hint. Use pre- and postcomposition compatibility. □

Solution. Replacing either factor by a homotopic map changes the composite only by a homotopy, so the homotopy class of a composite depends only on the classes of its factors. □

1.15 Solved problem dossiers

Problem 1.49 (Piecewise concatenation). Verify directly that concatenating two relative homotopies rel A again gives a homotopy rel A .

Solution. Use the usual two-piece concatenation. On A , both original homotopies are constant at the common restriction $f|_A = g|_A = h|_A$, so both pieces of the concatenation are constant there as well. □

Problem 1.50 (Radial contraction). Let $D^n = \{x : \|x\| \leq 1\}$. Construct a homotopy from the identity of D^n to the constant map at 0.

Solution. Set $H(x, t) = (1 - t)x$. Since $\|H(x, t)\| = (1 - t)\|x\| \leq 1$, the image remains in D^n . □

Problem 1.51 (Affine homotopy in a convex target). Prove that the straight-line formula is jointly continuous.

Solution. The maps $(x, t) \mapsto (1 - t)f(x)$ and $(x, t) \mapsto tg(x)$ are continuous by continuity of scalar multiplication and of f, g ; their sum is continuous. Convexity ensures the value remains in the target. □

Problem 1.52 (Homotopy of constant maps). If Y is path connected, prove that any two constant maps $X \rightarrow Y$ are homotopic.

Solution. If the constants are y_0, y_1 , choose a path $\gamma : I \rightarrow Y$ from y_0 to y_1 and define $H(x, t) = \gamma(t)$. $\square$

Problem 1.53 (Path components detected by maps from a point). Describe $[\{*\}, Y]$.

Solution. A map from a point chooses a point of Y . Two such maps are homotopic exactly when the chosen points are joined by a path. Hence $[\{*\}, Y]$ is the set of path components of Y . $\square$

Problem 1.54 (Restriction of a relative homotopy). If $H : f \simeq g \pmod{A}$ and $B \subset A$, show H is a homotopy rel B .

Solution. The defining fixed-point equation on all of A in particular holds on its subset B . $\square$

Problem 1.55 (Composition of based homotopies). Show that composing based homotopies produces a based homotopy.

Solution. If $H(x_0, t) = y_0$ and $K(y_0, t) = z_0$, the standard two-sided compatibility construction keeps the distinguished point at z_0 throughout. $\square$

Problem 1.56 (A stationary first half). Given $H : f \simeq g$, construct a homotopy that stays equal to f until time $1/2$ and then performs H .

Solution. Use

$$K(x, t) = \begin{cases} f(x), & 0 \leq t \leq 1/2, \\ H(x, 2t - 1), & 1/2 \leq t \leq 1. \end{cases}$$

The formulas agree at $t = 1/2$. $\square$

Problem 1.57 (Product deformation). If X and W each admit a contraction to a chosen point, write the contraction of $X \times W$.

Solution. For contractions H and K , use

$$L((x, w), t) = (H(x, t), K(w, t)).$$

$\square$

Problem 1.58 (Target inclusion). Let $A \subset Y$ and suppose $f, g : X \rightarrow A$ are homotopic as maps into A . Prove they are homotopic as maps into Y .

Solution. Compose the homotopy $X \times I \rightarrow A$ with the inclusion $A \hookrightarrow Y$. $\square$

Problem 1.59 (Two-stage deformation). Suppose $f \simeq c$ and $c \simeq g$, where c is constant. Prove $f \simeq g$.

Solution. This is transitivity: concatenate the two homotopies. $\square$

Problem 1.60 (Bridge to VIII/02). Explain why a homotopy $f \simeq g$ alone does not say that the spaces X and Y have the same homotopy type.

Solution. The relation compares two maps with the same domain and target. To compare spaces, one needs maps in both directions whose composites are homotopic to the appropriate identity maps. That is the homotopy-equivalence notion developed in VIII/02. $\square$

Chapter 2

Homotopy Equivalence and Contractibility

Homotopy of maps becomes a relation between spaces when maps are available in both directions. The resulting notion, homotopy equivalence, is much weaker than homeomorphism but strong enough to preserve the invariants that algebraic topology is designed to compute. Contractible spaces, deformation retracts, and homotopy type are the central themes of this chapter.

2.1 Homotopy equivalence

Definition 2.1 (Homotopy equivalence). Spaces X and Y are *homotopy equivalent*, written

$$X \simeq Y,$$

if there exist continuous maps

$$f : X \rightarrow Y, \quad g : Y \rightarrow X$$

such that

$$g \circ f \simeq \text{id}_X, \quad f \circ g \simeq \text{id}_Y.$$

The maps f and g are called homotopy inverses.

Warning 2.2 (Not an actual inverse). A homotopy inverse need not satisfy $g \circ f = \text{id}_X$ or $f \circ g = \text{id}_Y$ pointwise. Equality is replaced by homotopy.

2.2 Homotopy type

Definition 2.3 (Homotopy type). Two spaces have the same *homotopy type* if they are homotopy equivalent.

Proposition 2.4 (Homotopy equivalence is an equivalence relation). *Homotopy equivalence is reflexive, symmetric, and transitive.*

Proof. Reflexivity uses the identity. Symmetry is built into the definition. For transitivity, if $X \simeq Y$ via f, g and $Y \simeq Z$ via u, v , then $u \circ f : X \rightarrow Z$ and $g \circ v : Z \rightarrow X$ are homotopy inverses. The verification uses compatibility of homotopy with composition from VIII/01. $\square$

2.3 Contractible spaces

Definition 2.5 (Contractible space). A space X is *contractible* if

$$\text{id}_X$$

is homotopic to a constant map.

Theorem 2.6 (Contractible means homotopy equivalent to a point). *A nonempty space X is contractible if and only if*

$$X \simeq \{*\}.$$

Proof. If X is contractible to x_0 , let $p : X \rightarrow \{*\}$ be the unique map and $i : \{*\} \rightarrow X$ send the point to x_0 . Then $p \circ i$ is the identity on the point, while $i \circ p$ is the constant map at x_0 , homotopic to id_X . The converse is the same argument read backwards. $\square$

2.4 Convex and star-shaped examples

Proposition 2.7 (Convex sets are contractible). *Every nonempty convex subset C of a real topological vector space is contractible.*

Proof. Fix $c_0 \in C$ and use

$$H(x, t) = (1 - t)x + tc_0.$$

$\square$

Definition 2.8 (Star-shaped set). A subset $A \subset \mathbb{R}^n$ is star-shaped with center $a \in A$ if the line segment from a to every $x \in A$ lies in A .

Proposition 2.9 (Star-shaped sets are contractible). *Every nonempty star-shaped subset of $\mathbb{R}^n$ is contractible.*

Example 2.10. Intervals, disks, balls, cubes, and $\mathbb{R}^n$ are contractible.

2.5 Retractions

Definition 2.11 (Retraction). If $A \subset X$, a *retraction* is a map

$$r : X \rightarrow A$$

such that

$$r|_A = \text{id}_A.$$

Then A is called a retract of X .

Let $i : A \hookrightarrow X$ denote inclusion. The retraction condition is

$$r \circ i = \text{id}_A.$$

Warning 2.12 (Retraction alone is not a deformation retract). A retraction provides an exact one-sided inverse to inclusion, but does not require $i \circ r$ to be homotopic to id_X .

2.6 Deformation retracts

Definition 2.13 (Deformation retract). A subspace $A \subset X$ is a *deformation retract* of X if there is a retraction $r : X \rightarrow A$ such that

$$i \circ r \simeq \text{id}_X.$$

Equivalently, there is a homotopy from id_X to a map with image in A , ending at the retraction followed by inclusion.

Theorem 2.14 (Deformation retracts preserve homotopy type). *If A is a deformation retract of X , then*

$$A \simeq X.$$

Proof. The inclusion $i : A \rightarrow X$ and retraction $r : X \rightarrow A$ satisfy $r \circ i = \text{id}_A$ and $i \circ r \simeq \text{id}_X$. $\square$

2.7 Strong deformation retracts

Definition 2.15 (Strong deformation retract). A deformation retract $A \subset X$ is *strong* if the deformation

$$H : X \times I \rightarrow X$$

from id_X to $i \circ r$ fixes A pointwise:

$$H(a, t) = a \quad (a \in A, t \in I).$$

Thus a strong deformation retract is a relative homotopy from VIII/01.

2.8 Radial examples

Example 2.16 (Punctured Euclidean space). The sphere S^{n-1} is a strong deformation retract of $\mathbb{R}^n \setminus \{0\}$. Put

$$r(x) = \frac{x}{\|x\|}$$

and

$$H(x, t) = \left((1 - t) + \frac{t}{\|x\|} \right) x.$$

The scalar coefficient is positive, so the homotopy never reaches 0. For $x \in S^{n-1}$, $H(x, t) = x$.

Corollary 2.17.

$$\mathbb{R}^n \setminus \{0\} \simeq S^{n-1}.$$

Example 2.18 (Annulus). For

$$A = \{x \in \mathbb{R}^2 : 1 \leq \|x\| \leq 2\},$$

the unit circle is a strong deformation retract by radial normalization.

2.9 Cylinder and circle

Example 2.19 (Cylinder). The cylinder $S^1 \times I$ strongly deformation retracts onto $S^1 \times \{0\}$ by

$$H((z, s), t) = (z, (1 - t)s).$$

Hence

$$S^1 \times I \simeq S^1.$$

2.10 Graphs with dangling trees

A contractible branch attached to a graph often contributes no new homotopy type if it can be collapsed continuously while the remaining graph is fixed.

Example 2.20 (Circle with a whisker). Attach an interval to S^1 at one endpoint. Collapsing the interval linearly to its attachment point gives a strong deformation retract onto S^1 .

This elementary picture foreshadows cell collapses and CW complexes.

2.11 Contractibility versus path connectedness

Proposition 2.21 (Contractible implies path connected). *Every nonempty contractible space is path connected.*

Proof. Let H contract X to x_0 . For each x , the path $t \mapsto H(x, t)$ joins x to x_0 . Concatenating such paths joins any two points. $\square$

Warning 2.22. The converse fails. The circle S^1 is path connected but is not contractible. Proving noncontractibility requires an invariant; degree in VIII/03 gives one route, and later fundamental-group or homology calculations give others.

2.12 Homotopy invariants

Definition 2.23 (Homotopy invariant). A construction F on spaces is a *homotopy invariant* if

$$X \simeq Y \implies F(X) \cong F(Y)$$

in the appropriate category.

The central strategy of algebraic topology is:

$$\text{space} \longmapsto \text{algebraic invariant},$$

then use an invariant mismatch to prove that two spaces are not homotopy equivalent.

2.13 Homotopy equivalent need not mean homeomorphic

Example 2.24 (Interval and point). The interval I is contractible, so $I \simeq \{*\}$, but I is not homeomorphic to a point.

Example 2.25 (Disk and point). The closed disk D^n has the homotopy type of a point while having many more points and nontrivial local geometry.

Warning 2.26 (Do not transfer non-homotopy properties). Homotopy equivalence does not preserve dimension, cardinality, smooth structure, metric data, or local geometric shape in general.

2.14 A useful criterion

Proposition 2.27 (One-sided strict inverse plus homotopy). *If maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ satisfy*

$$g \circ f = \text{id}_X \quad \text{and} \quad f \circ g \simeq \text{id}_Y,$$

then f and g are homotopy inverses.

Deformation retracts are the main source of this pattern.

2.15 Exercises

Exercise 2.28. State the two equations defining homotopy inverse maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$.

Hint. Compose in both orders. □

Solution. $g \circ f \simeq \text{id}_X$ and $f \circ g \simeq \text{id}_Y$. □

Exercise 2.29. Why is every space homotopy equivalent to itself?

Hint. Use the identity map twice. □

Solution. The identity is its own strict inverse, hence also its own homotopy inverse. □

Exercise 2.30. Prove symmetry of homotopy equivalence.

Hint. The same pair of maps works in the opposite order. □

Solution. If f and g are homotopy inverses, then g and f are homotopy inverses after swapping the roles of X and Y . □

Exercise 2.31. Define a contractible space.

Hint. Contract the identity map. □

Solution. X is contractible when id_X is homotopic to a constant map. □

Exercise 2.32. Show $\mathbb{R}^n$ is contractible.

Hint. Contract to the origin. □

Solution. Use $H(x, t) = (1 - t)x$. □

Exercise 2.33. Show a convex set is contractible.

Hint. Choose any point c_0 in it. □

Solution. The straight-line homotopy $(1 - t)x + tc_0$ remains in the convex set. □

Exercise 2.34. Show a star-shaped set is contractible.

Hint. Use its star center. □

Solution. The segment from each x to the center lies in the set, giving the usual linear contraction. □

Exercise 2.35. Define a retraction $r : X \rightarrow A$.

Hint. What must happen on A ? □

Solution. It must satisfy $r(a) = a$ for all $a \in A$. □

Exercise 2.36. Translate the retraction condition using the inclusion $i : A \hookrightarrow X$.

Hint. Compose r after i . □

Solution. $r \circ i = \text{id}_A$. □

Exercise 2.37. What additional condition turns a retract into a deformation retract?

Hint. Compare $i \circ r$ with the identity of X . □

Solution. One requires $i \circ r \simeq \text{id}_X$. □

Exercise 2.38. What makes a deformation retract strong?

Hint. The subspace should remain fixed during the homotopy. □

Solution. The deformation from id_X to $i \circ r$ must be relative to A . □

Exercise 2.39. Prove a deformation retract has the same homotopy type as the ambient space.

Hint. Use inclusion and retraction as the two maps. □

Solution. They satisfy $ri = \text{id}_A$ and $ir \simeq \text{id}_X$, so they are homotopy inverses. □

Exercise 2.40. Write the radial retraction $\mathbb{R}^n \setminus \{0\} \rightarrow S^{n-1}$.

Hint. Normalize the vector. □

Solution. $r(x) = x/\|x\|$. □

Exercise 2.41. Why does the radial deformation avoid the origin?

Hint. Inspect the scalar multiplying x . □

Solution. The coefficient $(1-t) + t/\|x\|$ is positive, so a nonzero x remains nonzero. □

Exercise 2.42. Show the radial deformation fixes S^{n-1} .

Hint. Set $\|x\| = 1$. □

Solution. The scalar coefficient becomes $(1-t) + t = 1$, so $H(x, t) = x$. □

Exercise 2.43. Show $S^1 \times I \simeq S^1$.

Hint. Collapse the interval coordinate. □

Solution. The homotopy $H((z, s), t) = (z, (1-t)s)$ strongly deformation retracts onto $S^1 \times \{0\} \cong S^1$. □

Exercise 2.44. Why is a contractible space path connected?

Hint. Follow each point along its contraction path. □

Solution. Every point is joined by the contraction to the same terminal point, so any two points can be joined through that point. □

Exercise 2.45. Give a path-connected space that is not contractible.

Hint. The basic example is one-dimensional and closed. □

Solution. S^1 . □

Exercise 2.46. Give homotopy-equivalent spaces with different cardinalities.

Hint. Use a contractible continuum and a point. □

Solution. The interval I and a one-point space are homotopy equivalent but have different cardinalities. $\square$

Exercise 2.47. Does homotopy equivalence preserve dimension?

Hint. Compare D^n with a point. $\square$

Solution. No. Every disk D^n is homotopy equivalent to a point. $\square$

Exercise 2.48. If $X \simeq Y$ and $Y \simeq Z$, what maps exhibit $X \simeq Z$?

Hint. Compose the forward maps and compose the backward maps. $\square$

Solution. If $f : X \rightarrow Y$, $g : Y \rightarrow X$, $u : Y \rightarrow Z$, $v : Z \rightarrow Y$, use $uf : X \rightarrow Z$ and $gv : Z \rightarrow X$. $\square$

Exercise 2.49. Why is a circle with a contractible whisker homotopy equivalent to the circle?

Hint. Collapse the whisker to its attachment point. $\square$

Solution. That collapse is a strong deformation retraction onto the circle. $\square$

Exercise 2.50. What is a homotopy invariant?

Hint. It cannot distinguish homotopy-equivalent spaces. $\square$

Solution. It is a construction whose value is isomorphic for homotopy-equivalent spaces. $\square$

Exercise 2.51. How can a homotopy invariant prove $X \not\simeq Y$?

Hint. Compare the invariant values. $\square$

Solution. If the invariant values are not isomorphic, homotopy invariance implies the spaces cannot be homotopy equivalent. $\square$

2.16 Solved problem dossiers

Problem 2.52 (Interval versus point). Write explicit homotopy-inverse maps between I and a point.

Solution. Let $p : I \rightarrow \{*\}$ be unique and $i(*) = 0$. Then pi is the identity of the point and ip is the constant map at 0, homotopic to id_I via $H(s, t) = (1 - t)s$. $\square$

Problem 2.53 (Annulus collapse). Construct a strong deformation retraction of $\{x : 1 \leq \|x\| \leq 2\}$ onto S^1 .

Solution. Use

$$H(x, t) = \left((1 - t) + \frac{t}{\|x\|} \right) x.$$

Its radius is $(1 - t)\|x\| + t$, which remains in $[1, 2]$, and unit vectors are fixed. $\square$

Problem 2.54 (Punctured plane). Determine the homotopy type of $\mathbb{R}^2 \setminus \{0\}$.

Solution. Radial normalization gives a strong deformation retraction onto S^1 , so $\mathbb{R}^2 \setminus \{0\} \simeq S^1$. $\square$

Problem 2.55 (Punctured three-space). Determine the homotopy type of $\mathbb{R}^3 \setminus \{0\}$.

Solution. The same radial construction retracts it strongly onto S^2 . □

Problem 2.56 (Cylinder core). Show $S^1 \times [-1, 1]$ has the homotopy type of S^1 .

Solution. Collapse the second coordinate linearly: $H((z, s), t) = (z, (1 - t)s)$. □

Problem 2.57 (Whisker graph). Let $X = S^1 \cup_p I$ attach $0 \in I$ to $p \in S^1$. Give a strong deformation retraction $X \rightarrow S^1$.

Solution. Fix every point of S^1 , and send a whisker point $s \in I$ to $(1 - t)s$ along the whisker. At $t = 1$ the entire whisker is at p . □

Problem 2.58 (Product of contractible spaces). Prove $X \times Y$ is contractible if X and Y are contractible.

Solution. Take contractions H of X and K of Y , and define $L((x, y), t) = (H(x, t), K(y, t))$. □

Problem 2.59 (Homotopy type of a strip). Determine the homotopy type of $\mathbb{R} \times [0, 1]$.

Solution. The strip is convex, hence contractible and therefore homotopy equivalent to a point. □

Problem 2.60 (Retract versus deformation retract). Explain logically why the equation $ri = \text{id}_A$ alone does not establish $A \simeq X$.

Solution. Homotopy equivalence requires information about both composites. The retraction equation controls ri , but without $ir \simeq \text{id}_X$ there is no homotopy inverse condition in the other direction. □

Problem 2.61 (Transporting an invariant). Suppose F is a homotopy invariant and A deformation retracts from X . What relation holds between $F(A)$ and $F(X)$?

Solution. Since $A \simeq X$, homotopy invariance gives $F(A) \cong F(X)$. □

Problem 2.62 (Noncontractibility strategy). What would be enough to prove that S^1 is not contractible?

Solution. It is enough to find any homotopy invariant that takes different values on S^1 and on a point. VIII/03 introduces degree as an early obstruction, while later chapters supply fundamental group and homology. □

Problem 2.63 (Bridge to VIII/03). Why does the notion of homotopy invariant naturally lead to degree?

Solution. To distinguish maps and spaces up to homotopy, one seeks quantities unchanged by deformation. Degree assigns an integer to maps between oriented spheres (or, more generally, oriented closed manifolds) and is invariant under homotopy, so it supplies the first quantitative obstruction in this volume. □

Chapter 3

Degree of Maps

Homotopy theory becomes effective when a deformation-invariant quantity can be computed. Degree is the first such integer-valued invariant in this volume. For a map between oriented spaces of the same dimension, degree records the signed number of times the domain covers the target. It detects maps that cannot be deformed into one another and provides immediate applications to spheres and antipodal maps.

3.1 Orientation and the signed-counting idea

For a smooth map

$$f : M^n \rightarrow N^n$$

between oriented n -manifolds, a point $x \in M$ is regular if the derivative

$$df_x : T_x M \rightarrow T_{f(x)} N$$

is an isomorphism. At such a point, df_x either preserves or reverses orientation.

Definition 3.1 (Local sign). At a regular point x , define

$$\operatorname{sgn}_x(f) = \begin{cases} +1, & df_x \text{ preserves orientation,} \\ -1, & df_x \text{ reverses orientation.} \end{cases}$$

3.2 Degree by a regular value

Definition 3.2 (Degree for a smooth map). Let M, N be connected compact oriented n -manifolds without boundary, and let $f : M \rightarrow N$ be smooth. If $y \in N$ is a regular value, define

$$\deg(f) = \sum_{x \in f^{-1}(y)} \operatorname{sgn}_x(f).$$

Compactness makes the regular fiber finite.

Theorem 3.3 (Independence of regular value). *The signed sum above is independent of the chosen regular value y .*

This theorem is the key reason degree is global rather than merely local. A full proof can be given by oriented cobordism of inverse images of a path between regular values. Later homology theory will provide a shorter algebraic formulation.

3.3 Continuous maps

Theorem 3.4 (Continuous degree). *Every continuous map $f : M \rightarrow N$ between connected closed oriented n -manifolds has an integer degree. It agrees with the signed regular-value count for smooth maps and is invariant under homotopy.*

One construction approximates a continuous map by a smooth map and proves that different sufficiently close approximations are homotopic. Later, after homology is available, degree will be characterized by the induced map on the top-dimensional fundamental class.

3.4 Normalization

Proposition 3.5 (Identity and constant map). *For a connected oriented closed n -manifold,*

$$\deg(\text{id}) = 1.$$

If $n > 0$, every constant map has degree 0.

Proof. The identity has one preimage of a regular value and preserves orientation. A constant map misses every point other than its image, so choosing a regular value outside that image gives an empty signed sum. $\square$

3.5 Homotopy invariance

Theorem 3.6 (Homotopy invariance of degree). *If*

$$f \simeq g : M \rightarrow N,$$

then

$$\deg(f) = \deg(g).$$

Corollary 3.7 (Nonzero degree obstructs null-homotopy). *If $n > 0$ and $\deg(f) \neq 0$, then f is not null-homotopic.*

Proof. A null-homotopic map is homotopic to a constant map, whose degree is 0. $\square$

3.6 Multiplicativity

Theorem 3.8 (Degree of a composite). *For maps of connected closed oriented n -manifolds*

$$M \xrightarrow{f} N \xrightarrow{g} P,$$

one has

$$\deg(g \circ f) = \deg(g) \deg(f).$$

Corollary 3.9 (Degree of a homotopy equivalence). *If $f : M \rightarrow N$ is a homotopy equivalence between connected closed oriented n -manifolds, then*

$$\deg(f) = \pm 1.$$

Proof. For a homotopy inverse g ,

$$1 = \deg(\text{id}_M) = \deg(g \circ f) = \deg(g) \deg(f).$$

The only units in $\mathbb{Z}$ are ± 1 . $\square$

3.7 Degree on the circle

Identify

$$S^1 = \{z \in \mathbb{C} : |z| = 1\}.$$

Theorem 3.10 (Power map). *For*

$$p_k : S^1 \rightarrow S^1, \quad p_k(z) = z^k,$$

with $k \in \mathbb{Z}$,

$$\deg(p_k) = k.$$

For $k > 0$, a regular value has k preimages, each with positive local sign. For $k < 0$, complex conjugation reverses orientation and contributes the negative sign.

Example 3.11.

$$\deg(z \mapsto z^5) = 5, \quad \deg(z \mapsto z^{-3}) = -3.$$

3.8 Degree classifies circle maps up to homotopy

Theorem 3.12 (Classification of maps $S^1 \rightarrow S^1$). *Two continuous maps*

$$f, g : S^1 \rightarrow S^1$$

are homotopic if and only if

$$\deg(f) = \deg(g).$$

This is an early instance of a complete homotopy classification by an algebraic invariant. Later the fundamental group identifies the same integer as a winding number.

3.9 Orientation-reversing maps

Proposition 3.13 (Reflection). *A reflection of S^n across a hyperplane has degree -1 .*

Proposition 3.14 (Orientation-preserving homeomorphism). *An orientation-preserving homeomorphism of an oriented closed manifold has degree $+1$; an orientation-reversing one has degree -1 .*

3.10 The antipodal map

Let

$$a : S^n \rightarrow S^n, \quad a(x) = -x.$$

Theorem 3.15 (Degree of the antipodal map). *The antipodal map on S^n has degree*

$$\deg(a) = (-1)^{n+1}.$$

Proof. The antipodal map is the restriction of $-I : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^{n+1}$. Its determinant is

$$\det(-I) = (-1)^{n+1}.$$

The induced orientation sign on the sphere agrees with this value. $\square$

Thus the antipodal map has degree $+1$ for odd-dimensional spheres and degree -1 for even-dimensional spheres.

3.11 No-retraction preview

Degree can obstruct retractions.

Theorem 3.16 (No retraction of a ball onto its boundary). *For $n \geq 1$, there is no retraction*

$$r : D^{n+1} \rightarrow S^n.$$

Proof. Suppose r existed and let $i : S^n \hookrightarrow D^{n+1}$ be inclusion. Then

$$r \circ i = \text{id}_{S^n}.$$

Because D^{n+1} is contractible, i is null-homotopic as a map into the disk, so $r \circ i$ is null-homotopic by postcomposition. But the identity of S^n has degree 1, while a null-homotopic map has degree 0, a contradiction. $\square$

3.12 Surjectivity consequence

Proposition 3.17 (Nonzero degree implies surjective). *If $f : M \rightarrow N$ is a continuous map between connected closed oriented n -manifolds and*

$$\deg(f) \neq 0,$$

then f is surjective.

Proof. If f missed a point $y \in N$, then in the smooth regular-value picture a nearby regular value outside the image would have empty preimage and hence degree 0. Equivalently, a map missing a point factors through a punctured target, which has no top-dimensional degree contribution. $\square$

3.13 Degree and homotopy equivalence of spheres

Corollary 3.18. *A self-map $f : S^n \rightarrow S^n$ of degree other than ± 1 cannot be a homotopy equivalence.*

For example, $z \mapsto z^k$ on S^1 is a homotopy equivalence exactly when $k = \pm 1$.

3.14 Regular-value computation protocol

For a smooth map $f : M^n \rightarrow N^n$, a practical degree computation is:

1. choose a regular value $y \in N$;
2. solve $f(x) = y$;
3. determine whether each df_x preserves or reverses orientation;
4. sum the local signs.

Warning 3.19 (Unsigned counting is not degree). The number $|f^{-1}(y)|$ is generally not homotopy invariant. Oppositely oriented sheets can be created or cancelled in pairs. Degree is the *signed* count.

3.15 Exercises

Exercise 3.20. Define the local sign of a smooth map at a regular point.

Hint. Inspect whether the derivative preserves orientation. □

Solution. It is $+1$ if df_x preserves orientation and -1 if it reverses orientation. □

Exercise 3.21. Write the regular-value formula for degree.

Hint. Sum local signs over the inverse image. □

Solution.

$$\deg(f) = \sum_{x \in f^{-1}(y)} \operatorname{sgn}_x(f).$$

□

Exercise 3.22. What is the degree of the identity map?

Hint. A regular value has one preimage. □

Solution. 1. □

Exercise 3.23. What is the degree of a constant map $S^n \rightarrow S^n$ for $n > 0$?

Hint. Choose a point not equal to the constant value. □

Solution. 0. □

Exercise 3.24. If $f \simeq g$, how are their degrees related?

Hint. Degree is a homotopy invariant. □

Solution. $\deg(f) = \deg(g)$. □

Exercise 3.25. Why can a nonzero-degree map not be null-homotopic?

Hint. Compare with a constant map. □

Solution. A null-homotopic map has the degree of a constant map, namely 0. □

Exercise 3.26. State the multiplicativity formula.

Hint. Take the degree of a composite. □

Solution. $\deg(g \circ f) = \deg(g) \deg(f)$. □

Exercise 3.27. What degrees are possible for a homotopy equivalence of oriented closed manifolds?

Hint. Its degree must be a unit in $\mathbb{Z}$. □

Solution. Only $+1$ and -1 . □

Exercise 3.28. Compute $\deg(z \mapsto z^7)$ on S^1 .

Hint. Use the power-map theorem. □

Solution. 7. □

Exercise 3.29. Compute $\deg(z \mapsto z^{-4})$ on S^1 .

Hint. Negative exponent reverses winding orientation. ☐

Solution. -4 . ☐

Exercise 3.30. When are two maps $S^1 \rightarrow S^1$ homotopic?

Hint. Degree is complete in this case. ☐

Solution. Exactly when they have the same degree. ☐

Exercise 3.31. What is the degree of a reflection of S^n ?

Hint. A reflection reverses orientation. ☐

Solution. -1 . ☐

Exercise 3.32. Compute the degree of the antipodal map on S^1 .

Hint. Use $(-1)^{n+1}$ with $n = 1$. ☐

Solution. $+1$. ☐

Exercise 3.33. Compute the degree of the antipodal map on S^2 .

Hint. Use $(-1)^{n+1}$ with $n = 2$. ☐

Solution. -1 . ☐

Exercise 3.34. Compute the degree of the antipodal map on S^3 .

Hint. Use parity. ☐

Solution. $+1$. ☐

Exercise 3.35. If $\deg f = 2$ and $\deg g = -3$, compute $\deg(g \circ f)$.

Hint. Multiply. ☐

Solution. -6 . ☐

Exercise 3.36. Can a degree-2 self-map of S^n be a homotopy equivalence?

Hint. A homotopy equivalence has degree ± 1 . ☐

Solution. No. ☐

Exercise 3.37. Why does nonzero degree force surjectivity?

Hint. A missed regular value has no preimages. ☐

Solution. If the map missed a regular value, the signed inverse-image sum would be empty, forcing degree 0. ☐

Exercise 3.38. Why is ordinary preimage cardinality not a homotopy invariant?

Hint. Pairs with opposite local signs can appear or disappear. ☐

Solution. Homotopies can create or cancel pairs of preimages whose local signs sum to zero, changing cardinality while preserving degree. ☐

Exercise 3.39. What degree would a retraction argument force on $r \circ i : S^n \rightarrow S^n$?

Hint. A retraction satisfies $ri = \text{id}$. □

Solution. It would have degree 1. □

Exercise 3.40. Why is $r \circ i$ null-homotopic if $i : S^n \hookrightarrow D^{n+1}$?

Hint. The disk is contractible. □

Solution. The inclusion i is homotopic in the disk to a constant map; postcomposing that homotopy by r makes ri null-homotopic. □

Exercise 3.41. What contradiction proves the no-retraction theorem?

Hint. Compare degrees 1 and 0. □

Solution. A retraction would make $ri = \text{id}_{S^n}$ have degree 1, while the contraction through the disk would make ri null-homotopic and hence degree 0. □

Exercise 3.42. If $\deg f = 1$, must f be a homeomorphism?

Hint. Degree measures global signed covering, not injectivity. □

Solution. No. Degree 1 does not imply injectivity or local homeomorphism behavior. □

Exercise 3.43. What topic does the antipodal degree formula prepare for next?

Hint. Look at VIII/04. □

Solution. It prepares the study of spheres and antipodal maps, including parity-sensitive obstructions and Borsuk–Ulam-type phenomena. □

3.16 Solved problem dossiers

Problem 3.44 (Power map). Compute the degree of $p_6(z) = z^6$.

Solution. A regular value has six preimages, each orientation preserving, so $\deg(p_6) = 6$. □

Problem 3.45 (Negative power). Compute the degree of $q(z) = z^{-5}$.

Solution. The map winds five times with reversed orientation, so $\deg(q) = -5$. □

Problem 3.46 (Composition). Let $f(z) = z^3$ and $g(z) = z^{-2}$. Compute the degree of $g \circ f$.

Solution.

$$\deg(g \circ f) = (-2)(3) = -6.$$

Indeed $g(f(z)) = z^{-6}$. □

Problem 3.47 (Homotopy obstruction). Can $z \mapsto z^2$ be homotopic to $z \mapsto z^3$?

Solution. No. Their degrees are 2 and 3, and homotopic maps have equal degree. □

Problem 3.48 (Null-homotopy test). Can $z \mapsto z^{-1}$ be null-homotopic as a map $S^1 \rightarrow S^1$?

Solution. No. Its degree is -1 , whereas a null-homotopic circle map has degree 0. □

Problem 3.49 (Antipodal parity). For which n does the antipodal map $S^n \rightarrow S^n$ have degree $+1$?

Solution. When n is odd, because $(-1)^{n+1} = +1$. □

Problem 3.50 (Antipodal parity II). For which n does the antipodal map have degree -1 ?

Solution. When n is even. □

Problem 3.51 (No retraction). Give the degree contradiction for a hypothetical retraction $D^2 \rightarrow S^1$.

Solution. For inclusion $i : S^1 \hookrightarrow D^2$, a retraction r gives $ri = \text{id}_{S^1}$, degree 1. But i contracts through D^2 , so ri is null-homotopic, degree 0. Contradiction. □

Problem 3.52 (Surjectivity). Suppose $f : S^2 \rightarrow S^2$ has degree 4. What can be said about its image?

Solution. It must be all of S^2 , since nonzero degree implies surjectivity. □

Problem 3.53 (Homotopy equivalence test). Suppose $f : S^3 \rightarrow S^3$ has degree -1 . Does degree alone prove that f is a homotopy equivalence?

Solution. Degree -1 is necessary and, for sphere self-maps, it is sufficient because sphere maps of the same degree are homotopic and a degree -1 reflection is a homeomorphism. In general manifolds, degree ± 1 alone need not imply homotopy equivalence. □

Problem 3.54 (Classification on the circle). Classify the homotopy class of $f(z) = \bar{z}^4$.

Solution. Since $\bar{z} = z^{-1}$ on S^1 , $f(z) = z^{-4}$, so $\deg f = -4$. Its homotopy class is the unique circle-map class of degree -4 . □

Problem 3.55 (Bridge to VIII/04). Explain what the formula $\deg(a) = (-1)^{n+1}$ says qualitatively about antipodal maps.

Solution. Their homotopy behavior depends on the parity of the sphere dimension: odd-dimensional spheres give degree $+1$, even-dimensional spheres give degree -1 . VIII/04 turns this parity into concrete sphere and antipodal-map phenomena. □

Chapter 4

Spheres and Antipodal Maps

Spheres are the first family on which homotopy and degree interact systematically. The involution $x \mapsto -x$ exposes a parity phenomenon, while the same geometry leads to Borsuk–Ulam obstructions, projective quotients, and the cell structures used in the next chapters.

4.1 Spheres, disks, and hemispheres

Write

$$S^n = \{x \in \mathbb{R}^{n+1} : \|x\| = 1\}, \quad D^{n+1} = \{x \in \mathbb{R}^{n+1} : \|x\| \leq 1\}.$$

Then $\partial D^{n+1} = S^n$. The upper and lower closed hemispheres of S^n are copies of D^n meeting in the equatorial S^{n-1} .

Proposition 4.1 (Two-disk description). *For $n \geq 1$, S^n is obtained by gluing two copies of D^n along their boundary S^{n-1} by the identity.*

4.2 The antipodal map

Definition 4.2 (Antipodal map). The antipodal map is

$$a : S^n \rightarrow S^n, \quad a(x) = -x.$$

It is a fixed-point-free involution: $a^2 = \text{id}$.

Theorem 4.3 (Antipodal degree). *With the standard orientation,*

$$\deg(a) = (-1)^{n+1}.$$

Corollary 4.4 (Parity obstruction). *If n is even, a is not homotopic to the identity map of S^n .*

4.3 Odd and even maps

Definition 4.5 (Odd and even). A map $f : S^n \rightarrow \mathbb{R}^m$ is odd if $f(-x) = -f(x)$, and even if $f(-x) = f(x)$.

Theorem 4.6 (Odd self-map theorem). *Every continuous odd map $f : S^n \rightarrow S^n$ has odd degree.*

Corollary 4.7. *An odd self-map of S^n is not null-homotopic.*

4.4 Borsuk–Ulam

Theorem 4.8 (Borsuk–Ulam). *For every continuous $f : S^n \rightarrow \mathbb{R}^n$, there exists $x \in S^n$ such that*

$$f(x) = f(-x).$$

Theorem 4.9 (Equivalent no-odd-map form). *There is no continuous odd map $S^n \rightarrow S^{n-1}$.*

If a map $f : S^n \rightarrow \mathbb{R}^n$ separated every antipodal pair, then

$$g(x) = \frac{f(x) - f(-x)}{\|f(x) - f(-x)\|}$$

would be an odd map $S^n \rightarrow S^{n-1}$. This proves the equivalence of the two formulations.

4.5 The one-dimensional case

For $f : S^1 \rightarrow \mathbb{R}$, put $h(x) = f(x) - f(-x)$. Then $h(-x) = -h(x)$, so the intermediate value theorem on a semicircle forces a zero of h . Thus some antipodal pair has the same value.

4.6 No retraction

Theorem 4.10 (No boundary retraction). *There is no retraction $D^{n+1} \rightarrow S^n$.*

For $n \geq 1$, degree proves this: if r were a retraction and $i : S^n \hookrightarrow D^{n+1}$, then $ri = \text{id}$ has degree 1, while i is null-homotopic in the disk and hence ri would have degree 0.

4.7 Real projective space

Definition 4.11 (Real projective space).

$$\mathbb{R}P^n = S^n / (x \sim -x).$$

Equivalently, $\mathbb{R}P^n$ is the set of one-dimensional linear subspaces of $\mathbb{R}^{n+1}$.

Let $q : S^n \rightarrow \mathbb{R}P^n$ be the quotient map.

Proposition 4.12 (Even-map factorization). *If $f : S^n \rightarrow Y$ is continuous and even, then it factors uniquely as*

$$S^n \xrightarrow{q} \mathbb{R}P^n \xrightarrow{\bar{f}} Y.$$

4.8 Low-dimensional projective spaces

$$\mathbb{R}P^0 \cong \{*\}, \quad \mathbb{R}P^1 \cong S^1.$$

Under a standard identification $\mathbb{R}P^1 \cong S^1$, the quotient map is the degree-two circle map $z \mapsto z^2$.

4.9 Projective filtration

There are natural inclusions

$$\mathbb{R}P^0 \subset \mathbb{R}P^1 \subset \cdots \subset \mathbb{R}P^n.$$

Moreover,

$$\mathbb{R}P^k \setminus \mathbb{R}P^{k-1} \cong \mathbb{R}^k,$$

so $\mathbb{R}P^n$ has one open cell in each dimension $0, \dots, n$.

4.10 One-point compactification

Stereographic projection gives

$$S^n \setminus \{N\} \cong \mathbb{R}^n,$$

so S^n is the one-point compactification of $\mathbb{R}^n$.

4.11 Suspension

The sphere also satisfies

$$S^{n+1} \cong \Sigma S^n.$$

This viewpoint connects sphere geometry to cones and mapping cones.

4.12 Bridge to cell attachment

The projective filtration can be written schematically as

$$\mathbb{R}P^n = \mathbb{R}P^{n-1} \cup e^n.$$

VIII/05 formalizes this operation as attachment of an n -disk along its boundary.

Warning 4.13 (Quotient versus subspace). The antipodal quotient identifies points; it is not an inclusion. Retraction, subspace, and quotient arguments must therefore be kept distinct.

4.13 Exercises

Exercise 4.14. Compute a^2 for $a(x) = -x$.

Hint. Apply the map twice.

□

Solution. $a(a(x)) = x$.

□

Exercise 4.15. Does a have a fixed point on S^n ?

Hint. Solve $x = -x$.

□

Solution. No; that would force $x = 0$.

□

Exercise 4.16. Find $\deg a$ on S^4 .

Hint. Use $(-1)^{n+1}$.

□

Solution. -1 .

□

Exercise 4.17. Find $\deg a$ on S^5 .

Hint. Use parity.

□

Solution. 1 .

□

Exercise 4.18. When does degree rule out $a \simeq \text{id}$?

Hint. Compare their degrees.

□

Solution. For even n .

□

Exercise 4.19. State Borsuk–Ulam on S^2 .

Hint. The target is $\mathbb{R}^2$. □

Solution. Every continuous $S^2 \rightarrow \mathbb{R}^2$ identifies an antipodal pair. □

Exercise 4.20. Why would an odd $S^n \rightarrow S^{n-1}$ violate Borsuk–Ulam?

Hint. Embed S^{n-1} in $\mathbb{R}^n$. □

Solution. Oddness prevents equal antipodal images. □

Exercise 4.21. Construct an odd map from an antipode-separating $f : S^n \rightarrow \mathbb{R}^n$.

Hint. Normalize $f(x) - f(-x)$. □

Solution. $g(x) = \frac{f(x) - f(-x)}{\|f(x) - f(-x)\|}$. □

Exercise 4.22. Show that this g is odd.

Hint. Replace x by $-x$. □

Solution. Its numerator changes sign and denominator does not. □

Exercise 4.23. State Borsuk–Ulam for $S^1 \rightarrow \mathbb{R}$.

Hint. Set $n = 1$. □

Solution. Some antipodal pair has equal value. □

Exercise 4.24. Define $\mathbb{R}P^n$ as a quotient.

Hint. Identify antipodes. □

Solution. $\mathbb{R}P^n = S^n / (x \sim -x)$. □

Exercise 4.25. Give the linear-algebra description of $\mathbb{R}P^n$.

Hint. Use lines through the origin. □

Solution. One-dimensional linear subspaces of $\mathbb{R}^{n+1}$. □

Exercise 4.26. When does $f : S^n \rightarrow Y$ factor through q ?

Hint. Be constant on quotient fibers. □

Solution. When $f(x) = f(-x)$. □

Exercise 4.27. Identify $\mathbb{R}P^0$.

Hint. There is one line in $\mathbb{R}$. □

Solution. A point. □

Exercise 4.28. Identify $\mathbb{R}P^1$.

Hint. Antipodal circle quotient. □

Solution. It is homeomorphic to S^1 . □

Exercise 4.29. What circle map models $q : S^1 \rightarrow \mathbb{R}P^1 \cong S^1$?

Hint. Two-to-one covering.

□

Solution. $z \mapsto z^2$.

□

Exercise 4.30. What is the degree of that map?

Hint. Use the power-map theorem.

□

Solution. 2.

□

Exercise 4.31. Why is S^n a one-point compactification of $\mathbb{R}^n$?

Hint. Use stereographic projection.

□

Solution. Removing one point gives $\mathbb{R}^n$.

□

Exercise 4.32. Express S^{n+1} as a suspension.

Hint. Suspend S^n .

□

Solution. $S^{n+1} \cong \Sigma S^n$.

□

Exercise 4.33. How many standard cells does $\mathbb{R}P^n$ have?

Hint. One in each dimension $0, \dots, n$.

□

Solution. $n + 1$.

□

Exercise 4.34. What is the equator of S^n ?

Hint. Set the last coordinate to zero.

□

Solution. A copy of S^{n-1} .

□

Exercise 4.35. Why is a closed hemisphere a disk?

Hint. Project to the first n coordinates.

□

Solution. It is homeomorphic to D^n .

□

Exercise 4.36. Why is S^n noncontractible for $n \geq 1$?

Hint. Use degree of the identity.

□

Solution. A contraction would force degree $1 = 0$, impossible.

□

Exercise 4.37. What is the bridge to VIII/05?

Hint. Use projective filtration.

□

Solution. $\mathbb{R}P^n$ is built from $\mathbb{R}P^{n-1}$ by adding one n -cell.

□

4.14 Solved problem dossiers

Problem 4.38 (Antipodal parity). Use degree to compare the antipodal map with the identity.

Solution. For even n , the degrees are -1 and 1 , so they are not homotopic. For odd n , both degrees are 1 , so degree gives no obstruction. $\square$

Problem 4.39 (Circle antipodes). Prove directly that every continuous $f : S^1 \rightarrow \mathbb{R}$ has equal values at some antipodal pair.

Solution. Set $h(x) = f(x) - f(-x)$. On a semicircle from x to $-x$, the endpoint values have opposite signs, so the intermediate value theorem gives $h = 0$. $\square$

Problem 4.40 (No odd map to S^0). Prove there is no odd continuous map $S^1 \rightarrow S^0$.

Solution. A continuous image of connected S^1 in discrete S^0 is a point, but a constant map cannot be odd. $\square$

Problem 4.41 (Equivalent Borsuk–Ulam form). Derive the no-odd-map formulation and its converse.

Solution. An odd $S^n \rightarrow S^{n-1} \subset \mathbb{R}^n$ contradicts equal antipodal images. Conversely, an antipode-separating f yields the normalized odd difference map. $\square$

Problem 4.42 (Odd self-map). Explain why an odd self-map of S^n cannot be null-homotopic.

Solution. Its degree is odd and therefore nonzero, while a null-homotopic sphere map has degree zero. $\square$

Problem 4.43 (Antipodal S^2). Compare the antipodal map on S^2 with the identity and a constant map.

Solution. Its degree is -1 , versus 1 for the identity and 0 for a constant; hence it is homotopic to neither. $\square$

Problem 4.44 (Antipodal S^3). What does degree say about the antipodal map on S^3 ?

Solution. Its degree is 1 , equal to the identity degree. Thus degree gives no obstruction to a homotopy with the identity; this chapter does not infer the homotopy from degree alone. $\square$

Problem 4.45 (No retraction). Recover the no-retraction theorem from degree.

Solution. If $ri = \text{id}_{S^n}$, then degree is 1 ; but i contracts through the disk, making ri null-homotopic of degree 0 . $\square$

Problem 4.46 (Even-map quotient). Prove the factorization of an even map through $\mathbb{R}P^n$.

Solution. Define $\bar{f}([x]) = f(x)$. Evenness gives well-definedness, and the quotient property of q gives continuity and uniqueness. $\square$

Problem 4.47 (Projective line). Construct $\mathbb{R}P^1 \cong S^1$.

Solution. Represent a line by angle θ modulo π and send it to $e^{2i\theta}$. This descends to a continuous bijection from compact $\mathbb{R}P^1$ to Hausdorff S^1 . $\square$

Problem 4.48 (Quotient degree). Compute the degree of $S^1 \rightarrow \mathbb{R}P^1 \cong S^1$.

Solution. It is $z \mapsto z^2$, hence degree 2. $\square$

Problem 4.49 (Sphere from hemispheres). Describe S^n as two disks glued along the boundary.

Solution. Glue D_+^n and D_-^n by the identity on S^{n-1} ; the quotient is S^n . $\square$

Problem 4.50 (One-point compactification). Explain the stereographic model of S^n .

Solution. Stereographic projection identifies $S^n \setminus \{N\}$ with $\mathbb{R}^n$; the north pole becomes the point at infinity. $\square$

Problem 4.51 (Suspension sphere). Explain geometrically why $\Sigma S^n \cong S^{n+1}$.

Solution. Use height in $[-1, 1]$; at the two extreme heights all directions collapse to the suspension points. $\square$

Problem 4.52 (Antipodal orbit size). Why is $q : S^n \rightarrow \mathbb{R}P^n$ two-to-one?

Solution. Each orbit is exactly $\{x, -x\}$, and $x \neq -x$ on a unit sphere. $\square$

Problem 4.53 (Projective inclusion). Explain $\mathbb{R}P^{n-1} \subset \mathbb{R}P^n$.

Solution. Lines lying in the coordinate hyperplane $\mathbb{R}^n \subset \mathbb{R}^{n+1}$ form the embedded lower projective space. $\square$

Problem 4.54 (Projective open cell). Why is $\mathbb{R}P^n \setminus \mathbb{R}P^{n-1} \cong \mathbb{R}^n$?

Solution. Normalize a line not in the coordinate hyperplane so its last coordinate is 1; the remaining coordinates are arbitrary in $\mathbb{R}^n$. $\square$

Problem 4.55 (Borsuk–Ulam application). Apply Borsuk–Ulam to two continuous measurements on S^2 .

Solution. For $f = (f_1, f_2) : S^2 \rightarrow \mathbb{R}^2$, some antipodal pair has both measurements equal. $\square$

Problem 4.56 (Odd-vector obstruction). Why can there be no continuous rule $v : S^n \rightarrow S^{n-1}$ with $v(-x) = -v(x)$?

Solution. That is exactly an odd map $S^n \rightarrow S^{n-1}$, forbidden by Borsuk–Ulam. $\square$

Problem 4.57 (Bridge to cells). Explain how projective geometry motivates cell attachment.

Solution. The filtration $\mathbb{R}P^{n-1} \subset \mathbb{R}P^n$ adds an open n -cell; VIII/05 abstracts the gluing of its closed disk along the boundary. $\square$

Part II

CW Complexes

Chapter 5

Cell Attachments

Cell attachment turns local disks into global spaces. Starting from X , an n -disk is glued along its boundary by a map

$$\varphi : S^{n-1} \rightarrow X.$$

The quotient remembers the old space, the new cell dimension, and the attaching map. This chapter develops the full adjunction-space language and the major problem families mapped to cell attachments, while reserving the theorem about homotopic attaching maps for VIII/08.

5.1 Adjunction spaces

Definition 5.1 (Attaching an n -cell).

$$X \cup_{\varphi} D^n = (X \sqcup D^n) / \sim,$$

where $u \sim \varphi(u)$ for $u \in S^{n-1} = \partial D^n$.

5.2 Characteristic map and open cell

Let $q : X \sqcup D^n \rightarrow X \cup_{\varphi} D^n$ be the quotient map.

Definition 5.2 (Characteristic map). The characteristic map is

$$\chi = q|_{D^n} : D^n \rightarrow X \cup_{\varphi} D^n.$$

Its restriction to $\mathring{D}^n$ is a homeomorphism onto the open n -cell e^n .

Warning 5.3 (Boundary identifications). The characteristic map need not embed the closed disk. Distinct boundary points can be identified by the attaching map.

5.3 Universal property

Proposition 5.4 (Gluing criterion). *To define $F : X \cup_{\varphi} D^n \rightarrow Y$ is equivalent to giving continuous maps $f : X \rightarrow Y$ and $g : D^n \rightarrow Y$ satisfying*

$$g|_{S^{n-1}} = f \circ \varphi.$$

This is the quotient, equivalently pushout, universal property.

5.4 Zero- and one-cells

A 0-cell is a point and has empty boundary, so attaching it adds a new disjoint point. A 1-cell has boundary

$$S^0 = \{-1, +1\},$$

so its attaching map chooses two points of the old space, possibly equal.

Example 5.5 (Loop). Attaching both endpoints of D^1 to one vertex produces S^1 .

Example 5.6 (Graphs). Finite graphs arise from finitely many zero-cells followed by finitely many one-cell attachments.

5.5 Constant attachments

Proposition 5.7 (Constant attachment). *If $S^{n-1} \rightarrow X$ is constant at a basepoint x_0 , the whole boundary of the new disk collapses at x_0 . In the standard pointed setting the result is*

$$X \vee S^n.$$

In particular,

$$D^n / S^{n-1} \cong S^n.$$

5.6 Degree- m two-cell attachments

For

$$\varphi_m : S^1 \rightarrow S^1, \quad \varphi_m(z) = z^m,$$

define

$$X_m = S^1 \cup_{\varphi_m} D^2.$$

Example 5.8 (Degree one). $X_1 \cong D^2$.

Example 5.9 (Degree zero). For a constant attaching map,

$$X_0 \cong S^1 \vee S^2.$$

Example 5.10 (Projective plane).

$$\mathbb{R}P^2 \cong S^1 \cup_{\varphi_2} D^2.$$

The identical cell counts can therefore carry very different topology because the attaching maps differ.

5.7 Sphere with two cells

Proposition 5.11 (Minimal sphere structure). *For $n \geq 1$, S^n is obtained from one zero-cell by attaching one n -cell with constant attaching map.*

5.8 Torus polygon

The standard torus has one 0-cell, two 1-cells a, b , and one 2-cell. Its one-skeleton is

$$S^1 \vee S^1,$$

and the two-cell attaches by the commutator word

$$aba^{-1}b^{-1}.$$

5.9 Closed orientable surfaces

A genus- g orientable surface has one 0-cell, $2g$ one-cells, and one two-cell, with attaching word

$$[a_1, b_1] \cdots [a_g, b_g].$$

5.10 Projective filtration

Proposition 5.12 (Projective cells). $\mathbb{R}P^n$ is obtained from $\mathbb{R}P^{n-1}$ by attaching one n -cell. Hence it has one cell in every dimension $0, \dots, n$.

5.11 Attaching several cells

For a family $\varphi_\alpha : S^{n-1} \rightarrow X$, attach all cells simultaneously:

$$\left(X \sqcup \coprod_{\alpha} D_{\alpha}^n \right) / \sim, \quad u \sim \varphi_{\alpha}(u).$$

Distinct disk interiors remain disjoint.

5.12 Successive attachment

Dimension-by-dimension construction yields

$$X^0 \subset X^1 \subset X^2 \subset \cdots,$$

where X^n is obtained from X^{n-1} by attaching n -cells. VIII/06 formalizes these X^n as skeleta.

5.13 Attaching versus characteristic maps

The attaching map

$$\varphi : S^{n-1} \rightarrow X$$

describes how the boundary meets the old space. The characteristic map

$$\chi : D^n \rightarrow X \cup_{\varphi} D^n$$

parametrizes the closure of the new cell.

5.14 Pushout formulation

Cell attachment is the pushout

$$\begin{array}{ccc} S^{n-1} & \xrightarrow{\varphi} & X \\ \downarrow & & \downarrow \\ D^n & \longrightarrow & X \cup_{\varphi} D^n. \end{array}$$

5.15 Dependence on the attaching map

Warning 5.13 (The map matters). The number and dimensions of cells do not generally determine the resulting space. The attaching maps are essential data.

Homotopic attaching maps often lead to homotopy-equivalent adjunction spaces, but the rigorous theorem and its homotopy-extension argument belong to VIII/08. Blocks explicitly mapped there are therefore not absorbed into this chapter.

5.16 Mapping-cone preview

For a map $f : A \rightarrow X$, the mapping cone is

$$C_f = X \cup_f CA.$$

Since $CS^{n-1} \cong D^n$, ordinary cell attachment is the sphere-domain special case. VIII/07 develops mapping cones in full.

5.17 Why attachments compute topology

Cellular homology later converts n -cell attaching maps into boundary homomorphisms. The degree- m family already predicts the integer m that will appear in a cellular boundary map.

Warning 5.14 (Use the quotient topology). The construction is not merely a set-theoretic union. Continuity is controlled by the quotient topology induced by the boundary identifications.

5.18 Exercises

Exercise 5.15. Write the quotient defining $X \cup_{\varphi} D^n$.

Hint. Begin with the disjoint union. □

Solution. $(X \sqcup D^n)/\sim$, with $u \sim \varphi(u)$ on S^{n-1} . □

Exercise 5.16. What is the domain of an n -cell attaching map?

Hint. Use the disk boundary. □

Solution. S^{n-1} . □

Exercise 5.17. Define the characteristic map.

Hint. Restrict the quotient map. □

Solution. $\chi = q|_{D^n}$. □

Exercise 5.18. Where is χ a homeomorphism?

Hint. Use the disk interior. □

Solution. On $\mathring{D}^n$, onto the open cell. □

Exercise 5.19. State the compatibility for maps $f : X \rightarrow Y$, $g : D^n \rightarrow Y$.

Hint. Compare on the boundary. □

Solution. $g|_{S^{n-1}} = f \circ \varphi$. □

Exercise 5.20. What does a zero-cell attachment do?

Hint. Its boundary is empty. □

Solution. It adds a new disjoint point. □

Exercise 5.21. What data specifies a one-cell attachment?

Hint. S^0 has two points. □

Solution. The images of the two endpoints. □

Exercise 5.22. Build S^1 from one zero-cell and one one-cell.

Hint. Attach both endpoints to one vertex. □

Solution. The quotient $D^1/\partial D^1$ is S^1 . □

Exercise 5.23. How are finite graphs built cellularly?

Hint. Use dimensions 0 and 1. □

Solution. Attach finitely many one-cells to finitely many vertices. □

Exercise 5.24. What is D^n/S^{n-1} ?

Hint. Collapse the boundary. □

Solution. S^n . □

Exercise 5.25. What does a constant n -attachment give in a pointed space?

Hint. The new sphere meets at the basepoint. □

Solution. $X \vee S^n$. □

Exercise 5.26. Define the degree- m attaching map on S^1 .

Hint. Use a power map. □

Solution. $\varphi_m(z) = z^m$. □

Exercise 5.27. Identify $S^1 \cup_{\varphi_1} D^2$.

Hint. Boundary goes once around. □

Solution. D^2 . □

Exercise 5.28. Identify a constant two-cell attachment to S^1 .

Hint. Use a wedge. □

Solution. $S^1 \vee S^2$. □

Exercise 5.29. Give the cell attachment for $\mathbb{R}P^2$.

Hint. Use degree two. □

Solution. $S^1 \cup_{\varphi_2} D^2$. □

Exercise 5.30. Give minimal cell counts for S^n .

Hint. One old point, one new cell. □

Solution. One 0-cell and one n -cell. □

Exercise 5.31. Give the torus cell counts.

Hint. Use the square model. □

Solution. One 0-cell, two 1-cells, one 2-cell. □

Exercise 5.32. What word attaches the torus two-cell?

Hint. Use the commutator. □

Solution. $aba^{-1}b^{-1}$. □

Exercise 5.33. Give genus- g surface cell counts.

Hint. Generalize the torus. □

Solution. One 0-cell, $2g$ one-cells, one two-cell. □

Exercise 5.34. Give its attaching word.

Hint. Multiply commutators. □

Solution. $[a_1, b_1] \cdots [a_g, b_g]$. □

Exercise 5.35. How is $\mathbb{R}P^n$ built from $\mathbb{R}P^{n-1}$?

Hint. Use projective filtration. □

Solution. Attach one n -cell. □

Exercise 5.36. What is an n -skeleton informally?

Hint. Stop at dimension n . □

Solution. The union of all cells of dimension at most n . □

Exercise 5.37. Why is attachment a mapping-cone construction?

Hint. A cone on S^{n-1} is D^n . □

Solution. $X \cup_{\varphi} D^n = X \cup_{\varphi} CS^{n-1}$. □

Exercise 5.38. Why do cell dimensions alone not determine a space?

Hint. Compare degree- m attachments. □

Solution. Different attaching maps with identical cell counts can yield different topology. □

5.19 Solved problem dossiers

Problem 5.39 (Adjunction universal property). Prove the gluing criterion for maps out of $X \cup_{\varphi} D^n$.

Solution. A map out restricts to compatible maps on X and D^n . Conversely, compatible maps are constant on the quotient equivalence classes and therefore descend uniquely by the quotient property. □

Problem 5.40 (One-cell loop). Attach one interval to one vertex at both endpoints and identify the result.

Solution. The quotient $D^1/\partial D^1$ is a circle. □

Problem 5.41 (One-cell edge). Attach an interval between two distinct vertices.

Solution. The result is a graph with two vertices joined by one edge, homeomorphic to a closed interval. $\square$

Problem 5.42 (Parallel edges). Attach two one-cells between the same two vertices. Determine the homotopy type.

Solution. The two edges form a circle: go out along one and return along the other. The space is homeomorphic to S^1 . $\square$

Problem 5.43 (Bouquet of circles). Attach r one-cells with both endpoints at one vertex.

Solution. Each interval becomes a circle and all circles meet only at the common vertex, giving $\bigvee_{j=1}^r S^1$. $\square$

Problem 5.44 (Constant attachment). Prove $D^n/S^{n-1} \cong S^n$.

Solution. The quotient is the one-point compactification of the open disk $\mathring{D}^n \cong \mathbb{R}^n$, hence S^n . $\square$

Problem 5.45 (Minimal sphere). Give the two-cell structure of S^n .

Solution. Start from one vertex and attach one n -disk by the constant boundary map. The quotient is $D^n/S^{n-1} \cong S^n$. $\square$

Problem 5.46 (Degree-one two-cell). Identify $S^1 \cup_{\text{id}} D^2$.

Solution. The existing circle becomes exactly the boundary of the new disk, so the result is D^2 . $\square$

Problem 5.47 (Constant two-cell). Identify $S^1 \cup_c D^2$.

Solution. The new disk boundary collapses at the chosen point, turning the disk into S^2 wedged with the old S^1 . $\square$

Problem 5.48 (Degree-two two-cell). Explain the $\mathbb{R}P^2$ disk model.

Solution. Identify antipodal points of ∂D^2 . The quotient boundary is $\mathbb{R}P^1 \cong S^1$, and its attaching map is the two-to-one map of degree 2. $\square$

Problem 5.49 (Degree- m family). Why can $X_m = S^1 \cup_{\varphi_m} D^2$ vary with m despite fixed cell counts?

Solution. The boundary winds m times around the one-skeleton. Cellular homology later records this as multiplication by m , so the attaching data are topologically significant. $\square$

Problem 5.50 (Torus polygon). Derive the torus attaching word.

Solution. Label the square boundary in order a, b, a^{-1}, b^{-1} ; after opposite edge identifications this becomes the loop $aba^{-1}b^{-1}$. $\square$

Problem 5.51 (Genus-two surface). Give a cell attachment for the closed genus-two surface.

Solution. Use one vertex, four one-cells a_1, b_1, a_2, b_2 , and attach one two-cell along $[a_1, b_1][a_2, b_2]$. $\square$

Problem 5.52 (Projective filtration). Why does $\mathbb{R}P^n$ have one cell in every dimension?

Solution. Each difference $\mathbb{R}P^k \setminus \mathbb{R}P^{k-1}$ is $\mathbb{R}^k$, so stage k adds exactly one open k -cell. $\square$

Problem 5.53 (Simultaneous cells). Describe the quotient for a family of n -cell attachments.

Solution. Use $X \sqcup \coprod_{\alpha} D_{\alpha}^n$ and identify each boundary point u with $\varphi_{\alpha}(u)$; disk interiors remain mutually disjoint. $\square$

Problem 5.54 (Characteristic boundary). Why can a characteristic map fail to be injective on the boundary?

Solution. If distinct $u, v \in S^{n-1}$ satisfy $\varphi(u) = \varphi(v)$, the quotient identifies them. $\square$

Problem 5.55 (Open-cell embedding). Why is the interior embedded?

Solution. The attachment relation only touches boundary points. No two disk interior points are identified, and none is identified with the old space. $\square$

Problem 5.56 (Pushout uniqueness). State the uniqueness clause for the cell-attachment pushout.

Solution. Compatible maps on X and D^n determine exactly one map on the quotient because their images cover the adjunction space. $\square$

Problem 5.57 (Successive attachments). What data determine a two-dimensional complex built on a graph?

Solution. The one-skeleton graph plus one attaching loop $S^1 \rightarrow X^1$ for every two-cell. $\square$

Problem 5.58 (Attaching word). Why can a word in graph edges specify a two-cell attachment?

Solution. Reading the oriented edges in sequence defines a loop in the graph, hence a map $S^1 \rightarrow X^1$ used as the attaching map. $\square$

Problem 5.59 (Cell as cone). Show an n -cell attachment is a mapping cone of a sphere map.

Solution. $CS^{n-1} \cong D^n$, with the cone base equal to the disk boundary, so $X \cup_{\varphi} CS^{n-1}$ is the same adjunction space. $\square$

Problem 5.60 (Homotopic attaching maps boundary). What conclusion is expected for homotopic attaching maps, and why is it reserved for VIII/08?

Solution. Under standard hypotheses their adjunction spaces are homotopy equivalent, but the proof requires homotopy extension/pushout control; VIII/08 supplies that theorem rather than treating the quotients as automatically homeomorphic. $\square$

Problem 5.61 (Quotient topology). Why is the quotient topology essential?

Solution. It is what makes the gluing universal property valid and determines continuity of maps into and out of the adjunction space. $\square$

Problem 5.62 (Bridge to CW complexes). How does repeated cell attachment lead to VIII/06?

Solution. Build $X^0, X^1, X^2, \dots$ dimension by dimension; a CW complex formalizes this filtration together with closure-finiteness and weak-topology conditions. $\square$

Chapter 6

CW Complexes

Cell attachments become a general language when organized into skeleta. CW complexes are built from disks in increasing dimension, together with closure finiteness and the weak topology. They include graphs, spheres, surfaces, projective spaces, and most spaces used in classical algebraic topology.

6.1 Cells and characteristic maps

An n -cell is homeomorphic to $\mathring{D}^n \cong \mathbb{R}^n$. A characteristic map

$$\chi_\alpha : D^n \rightarrow X$$

restricts to a homeomorphism from $\mathring{D}^n$ onto its open cell and sends the boundary into lower-dimensional cells.

6.2 Skeleta

Definition 6.1 (Skeleton). The n -skeleton X^n is the union of all cells of dimension at most n , with $X^{-1} = \emptyset$.

Inductively, X^n is obtained from X^{n-1} by attaching n -cells.

6.3 CW complex

Definition 6.2 (CW complex). A CW complex is a cell decomposition satisfying:

1. every cell has a characteristic map with boundary in the lower skeleton;
2. closure finiteness: the closure of each cell meets only finitely many cells;
3. weak topology: $A \subset X$ is closed exactly when $A \cap \bar{e}$ is closed in $\bar{e}$ for every cell e .

The letters C and W refer to closure finiteness and weak topology.

6.4 Skeletal pushouts

At stage n ,

$$\begin{array}{ccc} \coprod_\alpha S_\alpha^{n-1} & \longrightarrow & X^{n-1} \\ \downarrow & & \downarrow \\ \coprod_\alpha D_\alpha^n & \longrightarrow & X^n \end{array}$$

is a pushout. This is VIII/05 repeated dimension by dimension.

6.5 Dimension

Definition 6.3 (Dimension).

$$\dim X = \sup\{n : X \text{ has an } n\text{-cell}\}.$$

A zero-dimensional CW complex is discrete; a one-dimensional CW complex is a topological graph.

6.6 Finite and locally finite complexes

Definition 6.4 (Finite CW complex). A CW complex is finite if it has only finitely many cells.

Proposition 6.5 (Finite implies compact). *Every finite CW complex is compact.*

Definition 6.6 (Locally finite). A CW complex is locally finite if every point has a neighborhood meeting only finitely many cells.

Warning 6.7 (Different finiteness notions). Closure finiteness and local finiteness are distinct conditions.

6.7 Subcomplexes

Definition 6.8 (Subcomplex). A subspace $A \subset X$ is a CW subcomplex if it is a union of cells and, with every cell it contains, it contains the closure of that cell.

Every skeleton is a subcomplex.

Proposition 6.9 (Subcomplex closedness). *Every subcomplex of a CW complex is closed.*

6.8 Cellular maps

Definition 6.10 (Cellular map). A map $f : X \rightarrow Y$ of CW complexes is cellular if

$$f(X^n) \subset Y^n$$

for every n .

6.9 Standard examples

Example 6.11 (Sphere). S^n has one 0-cell and one n -cell.

Example 6.12 (Graph). Graphs are one-dimensional CW complexes with vertices as zero-cells and edge interiors as one-cells.

Example 6.13 (Torus). The torus has one 0-cell, two 1-cells, and one 2-cell.

Example 6.14 (Projective space). $\mathbb{R}P^n$ has one cell in each dimension $0, \dots, n$.

6.10 CW structure is extra data

The same underlying space may carry different CW structures. For example, S^1 can have one zero-cell and one one-cell, or two zero-cells and two one-cells.

Warning 6.15 (Raw cell counts). Individual cell counts depend on the CW structure. Invariant combinations, such as Euler characteristic for finite CW complexes, require proof of independence from the decomposition.

6.11 Infinite examples

The ray $[0, \infty)$ has a locally finite CW structure with zero-cells at nonnegative integers and one-cells $(k, k+1)$. Also,

$$\mathbb{R}P^\infty = \bigcup_{n \geq 0} \mathbb{R}P^n$$

has one cell in every nonnegative dimension.

6.12 Weak topology and cellwise continuity

Proposition 6.16 (Cellwise criterion). *A function $f : X \rightarrow Y$ from a CW complex is continuous if each composite*

$$f \circ \chi_\alpha : D^{n_\alpha} \rightarrow Y$$

is continuous.

6.13 Compact subsets

Theorem 6.17 (Compact-cell finiteness). *Every compact subset of a CW complex meets only finitely many open cells.*

Consequently a compact image lies in a finite subcomplex after adjoining the finitely many cell closures that it meets.

6.14 Connectedness and the one-skeleton

Proposition 6.18 (One-skeleton detects components). *For a nonempty CW complex, path components are already detected by the one-skeleton. In particular, X is path connected exactly when X^1 is path connected.*

6.15 CW pairs

Definition 6.19 (CW pair). A CW pair (X, A) consists of a CW complex X and a subcomplex $A \subset X$.

6.16 Quotients by subcomplexes

Proposition 6.20 (CW quotient). *If $A \subset X$ is a subcomplex, then X/A has a natural CW structure: A becomes a zero-cell and cells not in A survive with their dimensions.*

6.17 Euler characteristic preview

For a finite CW complex with c_n n -cells, define

$$\chi(X) = \sum_n (-1)^n c_n.$$

For the standard torus structure,

$$\chi(T^2) = 1 - 2 + 1 = 0,$$

while

$$\chi(S^n) = 1 + (-1)^n.$$

Later homology proves this alternating sum is independent of the finite CW structure.

6.18 Why CW complexes are effective

CW complexes combine constructive flexibility with strong control: compact data live in finite subcomplexes, maps can often be made cellular, subcomplex inclusions support homotopy extension, and homology can be computed from the skeletal filtration and attaching maps. VIII/07 and VIII/08 exploit exactly these features.

6.19 Exercises

Exercise 6.21. What do C and W stand for?

Hint. Recall the defining global conditions.

☐

Solution. Closure finite and weak topology.

☐

Exercise 6.22. Define X^n .

Hint. Collect cells up to dimension n .

☐

Solution. The union of all cells of dimension at most n .

☐

Exercise 6.23. What is X^{-1} ?

Hint. It is the initial skeleton.

☐

Solution. $\emptyset$.

☐

Exercise 6.24. How is X^n built from X^{n-1} ?

Hint. Use VIII/05.

☐

Solution. Attach the n -cells along maps $S^{n-1} \rightarrow X^{n-1}$.

☐

Exercise 6.25. Define CW dimension.

Hint. Take the largest cell dimension or supremum.

☐

Solution. $\dim X = \sup\{n : X \text{ has an } n\text{-cell}\}$.

☐

Exercise 6.26. Describe a zero-dimensional CW complex.

Hint. It has only points.

☐

Solution. It is discrete.

☐

Exercise 6.27. Describe a one-dimensional CW complex.

Hint. Use vertices and edges.

☐

Solution. It is a topological graph.

☐

Exercise 6.28. Define a finite CW complex.

Hint. Count all cells. ☐

Solution. It has finitely many cells. ☐

Exercise 6.29. Why is a finite CW complex compact?

Hint. Use characteristic disks. ☐

Solution. It is a continuous image of a finite union of compact closed disks. ☐

Exercise 6.30. Define a subcomplex.

Hint. Require closure under cell closures. ☐

Solution. A union of cells containing the closure of every cell it contains. ☐

Exercise 6.31. Are skeleta subcomplexes?

Hint. Apply the definition. ☐

Solution. Yes. ☐

Exercise 6.32. Define a cellular map.

Hint. Respect dimensions. ☐

Solution. $f(X^n) \subset Y^n$ for all n . ☐

Exercise 6.33. Give minimal cell counts for S^n .

Hint. Use VIII/05. ☐

Solution. One 0-cell and one n -cell. ☐

Exercise 6.34. Give torus cell counts.

Hint. Use the polygon model. ☐

Solution. One 0-cell, two 1-cells, one 2-cell. ☐

Exercise 6.35. Give $\mathbb{R}P^n$ cell counts.

Hint. Use projective filtration. ☐

Solution. One cell in every dimension $0, \dots, n$. ☐

Exercise 6.36. Can one space have different CW structures?

Hint. Think of S^1 . ☐

Solution. Yes. ☐

Exercise 6.37. Define local finiteness.

Hint. Look at neighborhoods. ☐

Solution. Every point has a neighborhood meeting finitely many cells. ☐

Exercise 6.38. Is closure finiteness the same as local finiteness?

Hint. They quantify different things. ☐

Solution. No. □

Exercise 6.39. State the cellwise continuity test.

Hint. Use characteristic maps. □

Solution. Continuity follows if the composite with every characteristic map is continuous. □

Exercise 6.40. What is true of compact subsets and cells?

Hint. Use compact-cell finiteness. □

Solution. A compact subset meets only finitely many open cells. □

Exercise 6.41. Define a CW pair.

Hint. Choose a subcomplex. □

Solution. A pair (X, A) with A a subcomplex of CW complex X . □

Exercise 6.42. Describe X/A cellularly.

Hint. Collapse the subcomplex. □

Solution. A becomes a zero-cell; other cells survive with their dimensions. □

Exercise 6.43. Compute $\chi(T^2)$.

Hint. Use $1 - 2 + 1$. □

Solution. 0. □

Exercise 6.44. Compute $\chi(S^n)$.

Hint. Use one 0-cell and one n -cell. □

Solution. $1 + (-1)^n$. □

6.20 Solved problem dossiers

Problem 6.45 (Skeletal pushout). Write the construction of X^n from X^{n-1} .

Solution. Take the pushout of $\coprod S_\alpha^{n-1} \rightarrow X^{n-1}$ and the boundary inclusions $\coprod S_\alpha^{n-1} \hookrightarrow \coprod D_\alpha^n$. □

Problem 6.46 (Two circle structures). Compare the one-vertex/one-edge and two-vertex/two-edge structures on S^1 .

Solution. They have the same underlying space and different raw cell counts. Both have Euler alternating sum zero. □

Problem 6.47 (Sphere skeletons). Describe the minimal CW filtration of S^n .

Solution. There is one zero-cell, no cells in dimensions $1, \dots, n-1$, and one n -cell attached constantly. □

Problem 6.48 (Torus skeleta). Describe the standard torus skeleta.

Solution. X^0 is a point, $X^1 = S^1 \vee S^1$, and X^2 attaches one disk along $aba^{-1}b^{-1}$. □

Problem 6.49 (Projective skeleta). Identify the k -skeleton of standard $\mathbb{R}P^n$.

Solution. It is $\mathbb{R}P^k$, because the projective filtration adds exactly one cell at each dimension. $\square$

Problem 6.50 (Finite implies compact). Give the compactness proof for a finite CW complex.

Solution. Take the finite disjoint union of all compact characteristic disks and map it continuously and surjectively to the complex. $\square$

Problem 6.51 (Subcomplex closure). Why does a subcomplex contain cell boundaries?

Solution. By definition it contains the closure of each cell it contains, so all cells appearing in those boundaries are included. $\square$

Problem 6.52 (Skeleton subcomplex). Prove X^n is a subcomplex.

Solution. The closure of a k -cell with $k \leq n$ lies in cells of dimension at most k , hence remains in X^n . $\square$

Problem 6.53 (Cellular map test). For minimal sphere CW structures, when is $f : S^n \rightarrow S^n$ cellular?

Solution. It suffices that the unique zero-cell maps to the unique zero-cell; the full n -skeleton condition is automatic. $\square$

Problem 6.54 (Graph realization). Explain how a multigraph becomes a CW complex.

Solution. Use vertices as zero-cells and attach one interval per edge to its endpoint vertices, allowing loops and parallel edges. $\square$

Problem 6.55 (Infinite ray). Verify the integer subdivision of $[0, \infty)$ is locally finite.

Solution. A small neighborhood of a nonvertex meets one edge; near an integer it meets at most that vertex and its adjacent edges. $\square$

Problem 6.56 (Finiteness distinction). Contrast closure finiteness and local finiteness.

Solution. Closure finiteness starts from one cell and studies its closure; local finiteness starts from one point and studies a neighborhood. $\square$

Problem 6.57 (Weak topology). Why do compatible cellwise continuous maps assemble globally?

Solution. Preimages of closed sets intersect every closed cell in closed sets; the weak topology then makes the full preimage closed. $\square$

Problem 6.58 (Compact image). What does a continuous map from compact K into a CW complex reduce to?

Solution. Its image meets finitely many cells and is contained in a finite subcomplex after adjoining their closures. $\square$

Problem 6.59 (CW quotient). Describe cells of X/A for subcomplex A .

Solution. Collapse A to one zero-cell and retain each open cell outside A , with the attaching map followed by the quotient. $\square$

Problem 6.60 (CW pair). Give a natural example of a CW pair.

Solution. With compatible cell structures, (D^n, S^{n-1}) is a CW pair because the boundary is a subcomplex. $\square$

Problem 6.61 (Euler sphere). Compute Euler characteristics of even and odd spheres.

Solution. $\chi(S^{2k}) = 2$ and $\chi(S^{2k+1}) = 0$. $\square$

Problem 6.62 (Euler projective). Compute the cellular Euler sum for $\mathbb{R}P^n$.

Solution. $\sum_{k=0}^n (-1)^k$, equal to 1 for even n and 0 for odd n . $\square$

Problem 6.63 (One-skeleton connectedness). Why can higher-dimensional cells not join two distinct components of the lower skeleton?

Solution. For $n \geq 2$, the connected boundary sphere of an attached n -cell maps into a single component, so the disk is added to that component only. $\square$

Problem 6.64 (Bridge to mapping cones). Why are CW attachments naturally related to mapping cones?

Solution. Each disk is CS^{n-1} ; attaching it to X^{n-1} is the mapping cone of its sphere attaching map, the subject of VIII/07. $\square$

Chapter 7

Mapping Cones

Cell attachment is a special case of a more flexible construction. Given a map $f : A \rightarrow X$, instead of attaching only a disk along a sphere, one can attach the whole cone on A . The resulting mapping cone records the map itself as part of a new space. Mapping cones organize quotient spaces, cofibers, suspensions, and many exact-sequence constructions that appear later in homology.

7.1 The cone on a space

Definition 7.1 (Cone). For a space A , the cone on A is

$$CA = (A \times I)/(A \times \{1\}),$$

where all points of $A \times \{1\}$ are identified to one cone vertex.

The copy $A \times \{0\}$ is the base of the cone.

Proposition 7.2 (Cones are contractible). *For every nonempty space A , the cone CA is contractible.*

Proof. Push every point linearly toward the cone vertex in the I -coordinate. The quotient description makes the terminal slice a single point. $\square$

7.2 The suspension

Definition 7.3 (Suspension). The suspension of A is

$$\Sigma A = (A \times I)/(A \times \{0\}, A \times \{1\}),$$

where each end is collapsed to a point.

Equivalently, ΣA is obtained by gluing two cones on A along their common base. In particular,

$$\Sigma S^n \cong S^{n+1}.$$

7.3 Definition of the mapping cone

Definition 7.4 (Mapping cone). For a continuous map $f : A \rightarrow X$, the mapping cone is

$$C_f = X \cup_f CA,$$

where the base $A \subset CA$ is identified with X using f .

Thus C_f is the pushout

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ \downarrow & & \downarrow \\ CA & \longrightarrow & C_f. \end{array}$$

7.4 Cell attachment as a mapping cone

If $A = S^{n-1}$, then

$$CA \cong D^n.$$

Therefore

$$X \cup_{\varphi} D^n \cong C_{\varphi}$$

for every attaching map $\varphi : S^{n-1} \rightarrow X$.

Proposition 7.5 (Cell-attachment specialization). *Ordinary n -cell attachment is the mapping-cone construction for a map whose domain is S^{n-1} .*

7.5 Mapping cone of the identity

Proposition 7.6 (Cone of the identity). *For every space A ,*

$$C_{\text{id}_A} \cong CA.$$

Hence it is contractible.

The copy of A already present as $X = A$ is exactly the base to which the cone is attached.

7.6 Mapping cone of a constant map

Proposition 7.7 (Constant map cone). *If $f : A \rightarrow X$ is constant at a chosen basepoint x_0 , then under standard pointed hypotheses,*

$$C_f \simeq X \vee \Sigma A.$$

In the literal quotient model, the cone base is attached at one point of X .

For $X = \{*\}$,

$$C_f \cong \Sigma A.$$

7.7 Mapping cone of an inclusion

Let $i : A \hookrightarrow X$ be an inclusion.

Proposition 7.8 (Cone of a cofibration-like inclusion). *When $A \subset X$ is a CW subcomplex, the mapping cone C_i is naturally homotopy equivalent to the quotient*

$$X/A.$$

For CW pairs this is one of the main reasons mapping cones model relative topology.

7.8 The cofiber viewpoint

The sequence

$$A \xrightarrow{f} X \longrightarrow C_f$$

is called a cofiber sequence at its first stage. After suspending, one obtains the extended pattern

$$A \xrightarrow{f} X \longrightarrow C_f \longrightarrow \Sigma A \longrightarrow \Sigma X \longrightarrow \cdots.$$

Later homology turns this geometric sequence into exact algebraic sequences.

7.9 Examples from spheres

For the degree- m map

$$f_m : S^1 \rightarrow S^1, \quad f_m(z) = z^m,$$

the mapping cone is

$$C_{f_m} = S^1 \cup_{f_m} D^2.$$

Thus:

$$C_{f_0} \simeq S^1 \vee S^2, \quad C_{f_1} \cong D^2, \quad C_{f_2} \cong \mathbb{R}P^2.$$

7.10 Moore-space preview

A basic Moore space can be obtained by attaching an $(n+1)$ -cell to S^n along a degree- m map:

$$M(\mathbb{Z}/m, n) = S^n \cup_m D^{n+1}.$$

Its cellular homology later realizes a single $\mathbb{Z}/m$ group in the desired degree.

7.11 Functorial behavior

Suppose there is a commutative square

$$\begin{array}{ccc} A & \xrightarrow{f} & X \\ \downarrow u & & \downarrow v \\ B & \xrightarrow{g} & Y. \end{array}$$

The maps u and v induce a natural map

$$C_f \rightarrow C_g$$

by applying v on the old space and the cone on u on the cone part.

Proposition 7.9 (Map of mapping cones). *A strictly commutative square of maps induces a continuous map between the corresponding mapping cones.*

7.12 Quotient description

One can write

$$C_f = (X \sqcup A \times I) / \sim,$$

where

$$(a, 0) \sim f(a)$$

and all $(a, 1)$ are identified to the cone vertex.

This description is useful for checking maps explicitly.

7.13 Mapping cylinders versus mapping cones

Definition 7.10 (Mapping cylinder). The mapping cylinder of $f : A \rightarrow X$ is

$$M_f = (A \times I) \sqcup X / ((a, 0) \sim f(a)).$$

The mapping cone is obtained from M_f by collapsing the free end $A \times \{1\}$ to a point.

Proposition 7.11 (Cylinder retract). *The mapping cylinder M_f deformation retracts onto the copy of X .*

7.14 Null-homotopic maps

If $f : A \rightarrow X$ is null-homotopic, the mapping cone is closely related to $X \vee \Sigma A$. For CW-type spaces this relationship is a homotopy equivalence.

Theorem 7.12 (Null-map splitting). *For a based null-homotopic map between suitable CW spaces,*

$$C_f \simeq X \vee \Sigma A.$$

7.15 Homotopy invariance preview

If $f_0 \simeq f_1 : A \rightarrow X$, one expects

$$C_{f_0} \simeq C_{f_1}.$$

This is the mapping-cone version of the homotopic-attaching-map theorem. VIII/08 proves the relevant principle in the cell-attachment setting.

Warning 7.13 (Not automatically homeomorphic). Homotopic maps need not produce literally identical quotient spaces. The natural invariant conclusion is homotopy equivalence, not homeomorphism.

7.16 Exercises

Exercise 7.14. Define the cone CA .

Hint. Collapse one end of $A \times I$. □

Solution. $CA = (A \times I) / (A \times \{1\})$. □

Exercise 7.15. Why is CA contractible?

Hint. Move every point toward the cone vertex. □

Solution. The homotopy that increases the I -coordinate to 1 contracts the cone to its vertex. □

Exercise 7.16. Define ΣA .

Hint. Collapse both ends. □

Solution. $\Sigma A = (A \times I) / (A \times \{0\}, A \times \{1\})$. □

Exercise 7.17. Express S^{n+1} as a suspension.

Hint. Suspend S^n . □

Solution. $S^{n+1} \cong \Sigma S^n$. □

Exercise 7.18. Define C_f for $f : A \rightarrow X$.

Hint. Attach the cone base using f . □

Solution. $C_f = X \cup_f CA$. □

Exercise 7.19. How is an n -cell attachment a mapping cone?

Hint. Use $CS^{n-1} \cong D^n$. □

Solution. $X \cup_\varphi D^n \cong C_\varphi$. □

Exercise 7.20. What is C_{id_A} ?

Hint. The cone is attached to its own base copy. □

Solution. It is homeomorphic to CA , hence contractible. □

Exercise 7.21. What is the mapping cone of a constant map to a point?

Hint. The target is one point. □

Solution. It is ΣA . □

Exercise 7.22. For a CW subcomplex inclusion $A \hookrightarrow X$, what quotient models the cone?

Hint. Collapse A . □

Solution. $C_i \simeq X/A$. □

Exercise 7.23. Write the first three terms of a cofiber sequence.

Hint. Start with f . □

Solution. $A \xrightarrow{f} X \rightarrow C_f$. □

Exercise 7.24. What follows C_f in the extended cofiber sequence?

Hint. Suspension of the domain. □

Solution. ΣA . □

Exercise 7.25. Identify $C_{z \mapsto z}$ for $S^1 \rightarrow S^1$.

Hint. Degree one attachment. □

Solution. It is D^2 . □

Exercise 7.26. Identify $C_{z \mapsto z^2}$.

Hint. Recall the projective-plane attachment. □

Solution. It is $\mathbb{R}P^2$. □

Exercise 7.27. Identify the constant-map cone $S^1 \rightarrow S^1$ up to homotopy.

Hint. Use the null-map splitting. □

Solution. $S^1 \vee S^2$. □

Exercise 7.28. What is a Moore-space attachment?

Hint. Attach by a degree- m map. □

Solution. $S^n \cup_m D^{n+1}$. □

Exercise 7.29. Define the mapping cylinder M_f .

Hint. Do not collapse the free end. □

Solution. $(A \times I) \sqcup X / ((a, 0) \sim f(a))$. □

Exercise 7.30. How do you get C_f from M_f ?

Hint. Collapse the free end. □

Solution. Collapse $A \times \{1\}$ to one point. □

Exercise 7.31. To what does M_f deformation retract?

Hint. Push the cylinder into the glued end. □

Solution. To the copy of X . □

Exercise 7.32. What does a commutative square induce on cones?

Hint. Map the target part and cone part separately. □

Solution. A natural map $C_f \rightarrow C_g$. □

Exercise 7.33. What splitting holds for a suitable null-homotopic map?

Hint. Use suspension. □

Solution. $C_f \simeq X \vee \Sigma A$. □

Exercise 7.34. Are mapping cones of homotopic maps necessarily homeomorphic?

Hint. The expected invariant is weaker. □

Solution. No; the natural conclusion is homotopy equivalence. □

Exercise 7.35. Write the quotient model of C_f .

Hint. Use $X \sqcup A \times I$. □

Solution. Identify $(a, 0)$ with $f(a)$ and collapse all $(a, 1)$ to the cone vertex. □

Exercise 7.36. Why are mapping cones useful for relative topology?

Hint. Consider a subcomplex inclusion. □

Solution. For $A \subset X$, $C_i \simeq X/A$, so relative information becomes an ordinary quotient space. □

Exercise 7.37. What is the bridge from VIII/07 to VIII/08?

Hint. Vary the attaching map through a homotopy. □

Solution. VIII/08 proves that homotopic attaching maps give the same homotopy type under the standard CW hypotheses. □

7.17 Solved problem dossiers

Problem 7.38 (Cone contraction). Construct an explicit contraction of CA to its vertex.

Solution. Represent a cone point by $[a, s]$. Define $H([a, s], t) = [a, (1 - t)s + t]$. At $t = 1$ every point has coordinate 1, hence is the cone vertex. $\square$

Problem 7.39 (Suspension of a point). Identify $\Sigma\{*\}$.

Solution. The interval has each endpoint separately collapsed, which changes nothing; the result is an interval, hence contractible. $\square$

Problem 7.40 (Suspension of S^0). Show $\Sigma S^0 \cong S^1$.

Solution. Two intervals, one for each point of S^0 , share both suspension endpoints. Their union is a circle. $\square$

Problem 7.41 (Identity cone). Prove $C_{\text{id}_A} \cong CA$.

Solution. The target copy A is glued pointwise to the cone base. It contributes no additional points beyond that base, leaving precisely the cone. $\square$

Problem 7.42 (Constant cone). Compute the mapping cone of the constant map $S^{n-1} \rightarrow \{*\}$.

Solution. It is the suspension $\Sigma S^{n-1} \cong S^n$. $\square$

Problem 7.43 (Degree-one circle cone). Compute C_f for $f : S^1 \rightarrow S^1$ of degree 1 represented by the identity.

Solution. The cone is a disk attached along its full boundary by the identity, so the result is D^2 , hence contractible. $\square$

Problem 7.44 (Degree-two circle cone). Explain why the cone of $z \mapsto z^2$ is $\mathbb{R}P^2$.

Solution. The cone on S^1 is D^2 ; attaching its boundary to S^1 by the degree-two quotient map is the standard cell model of $\mathbb{R}P^2$. $\square$

Problem 7.45 (Null circle cone). Compute the cone of a constant $S^1 \rightarrow S^1$.

Solution. The disk boundary collapses at one point of the old circle, producing $S^1 \vee S^2$. $\square$

Problem 7.46 (Inclusion cone). For the inclusion $S^{n-1} \hookrightarrow D^n$, identify the quotient model D^n/S^{n-1} .

Solution. It is S^n . Thus the mapping cone of the boundary inclusion has homotopy type S^n . $\square$

Problem 7.47 (Pair quotient). Let A be a CW subcomplex of X . Explain why C_i models X/A .

Solution. The cone on A contracts the entire attached copy of A to the cone vertex while the base is identified with the original $A \subset X$. For a cofibration such as a CW-subcomplex inclusion, this yields the quotient homotopy type. $\square$

Problem 7.48 (Mapping-cylinder retraction). Write an explicit deformation retraction $M_f \rightarrow X$.

Solution. On the cylinder send $[a, s]$ to $[a, (1 - t)s]$ during the homotopy, keeping X fixed. At $t = 1$ every cylinder point lies at $s = 0$, identified with $f(a) \in X$. $\square$

Problem 7.49 (Cone from cylinder). Describe C_f as a quotient of M_f .

Solution. Collapse the free copy $A \times \{1\} \subset M_f$ to a single point. The result is exactly the mapping cone. $\square$

Problem 7.50 (Map of cones). Given $vf = gu$, construct $C_f \rightarrow C_g$.

Solution. Use v on X and send $[a, t] \in CA$ to $[u(a), t] \in CB$. Commutativity ensures agreement on the glued base. $\square$

Problem 7.51 (Moore-space preview). For $M = S^n \cup_m D^{n+1}$, predict the nontrivial reduced homology group.

Solution. Cellular homology later gives a boundary map multiplication by m , so the reduced homology in degree n is $\mathbb{Z}/m$. $\square$

Problem 7.52 (Cofiber continuation). Explain geometrically why a map $C_f \rightarrow \Sigma A$ exists.

Solution. Collapse the copy of $X \subset C_f$ to a point. The cone part then has its base collapsed as well as its vertex, producing the suspension ΣA . $\square$

Problem 7.53 (Null-map splitting). Why should a null-homotopy of $f : A \rightarrow X$ lead to $C_f \simeq X \vee \Sigma A$?

Solution. The null-homotopy lets the cone base attachment deform to a constant attachment. A constant attachment wedges the suspended domain onto X . $\square$

Problem 7.54 (Two homotopic maps). Suppose $f_0 \simeq f_1 : A \rightarrow X$. What mapping-cone relation should be expected?

Solution. Under the standard CW/cofibration hypotheses, $C_{f_0} \simeq C_{f_1}$. VIII/08 develops the corresponding attaching-map theorem. $\square$

Problem 7.55 (Quotient map check). Verify that the natural $X \rightarrow C_f$ is continuous.

Solution. It is the composite of the inclusion $X \hookrightarrow X \sqcup CA$ with the quotient map defining C_f , hence continuous. $\square$

Problem 7.56 (Suspension as double cone). Show that ΣA is obtained from two copies of CA glued along A .

Solution. Split $A \times I$ at $t = 1/2$. Each half becomes a cone after collapsing its outer end, and their common middle slice is a copy of A . $\square$

Problem 7.57 (Bridge to exact sequences). Why does the cofiber sequence foreshadow homology exact sequences?

Solution. Homology converts the geometric passage $A \rightarrow X \rightarrow C_f \rightarrow \Sigma A$ into groups linked by induced maps; suspension shifts degree, yielding the connecting morphisms of the long exact sequence. $\square$

Chapter 8

Homotopic Attaching Maps

The topology of an adjunction space depends on its attaching map, but it does not depend on that map pointwise. Under the standard CW/cofibration hypotheses, deforming the attaching map through a homotopy preserves the homotopy type of the resulting space. This chapter makes that principle precise and develops the homotopy-extension mechanism behind it.

8.1 The basic problem

Let

$$\varphi_0, \varphi_1 : S^{n-1} \rightarrow X$$

be homotopic maps. Form

$$Y_i = X \cup_{\varphi_i} D^n.$$

The two quotient spaces need not be literally identical, but one expects

$$Y_0 \simeq Y_1.$$

8.2 Homotopy extension property

Definition 8.1 (Homotopy extension property). A pair (X, A) has the homotopy extension property if, whenever

$$f : X \rightarrow Y$$

and

$$H : A \times I \rightarrow Y$$

satisfy $H(a, 0) = f(a)$, there exists

$$\tilde{H} : X \times I \rightarrow Y$$

with

$$\tilde{H}(x, 0) = f(x), \quad \tilde{H}|_{A \times I} = H.$$

Definition 8.2 (Cofibration). An inclusion $i : A \hookrightarrow X$ is a cofibration if the pair (X, A) has the homotopy extension property.

8.3 CW pairs are cofibrations

Theorem 8.3 (CW subcomplex inclusion). *If A is a CW subcomplex of a CW complex X , then*

$$A \hookrightarrow X$$

is a cofibration.

This theorem is one of the main reasons CW complexes form a convenient category for homotopy theory.

8.4 Homotopic attachment theorem

Theorem 8.4 (Homotopic attaching maps). *Let X be a CW complex and let*

$$\varphi_0, \varphi_1 : S^{n-1} \rightarrow X$$

be homotopic. Then

$$X \cup_{\varphi_0} D^n \simeq X \cup_{\varphi_1} D^n.$$

The conclusion is homotopy equivalence, not generally homeomorphism.

8.5 Idea of the proof

Let

$$H : S^{n-1} \times I \rightarrow X$$

be a homotopy from φ_0 to φ_1 . One can insert a collar between the old space and the disk boundary and use H along this collar. The resulting intermediate adjunction space contains both attachments as deformation retracts.

Equivalently, one can use a mapping-cylinder construction to replace the attaching map by a homotopic representative without changing homotopy type.

8.6 A cylinder model

Attach

$$S^{n-1} \times I$$

to X using H , and place D^n at the far end. At one end the boundary is attached by φ_0 ; at the other by φ_1 . Sliding along the cylinder provides the comparison homotopies.

Proposition 8.5 (Interpolation space). *There is a space Z_H constructed from the homotopy H that deformation retracts onto each of*

$$X \cup_{\varphi_0} D^n \quad \text{and} \quad X \cup_{\varphi_1} D^n.$$

8.7 Constant versus null-homotopic attachment

A null-homotopic attaching map need not be literally constant, but it produces the same homotopy type as a constant attachment.

Corollary 8.6 (Null attachment splitting). *If*

$$\varphi : S^{n-1} \rightarrow X$$

is null-homotopic, then

$$X \cup_{\varphi} D^n \simeq X \vee S^n$$

under the usual pointed CW hypotheses.

8.8 Degree classifies circle attachments up to homotopy

For maps

$$S^1 \rightarrow S^1,$$

homotopy classes are classified by degree.

Corollary 8.7 (Degree determines two-cell attachment type). *If*

$$\varphi, \psi : S^1 \rightarrow S^1$$

have the same degree, then

$$S^1 \cup_{\varphi} D^2 \simeq S^1 \cup_{\psi} D^2.$$

This does not say the two quotient spaces are homeomorphic for arbitrary representatives; it says the homotopy type is determined by the homotopy class of the attachment.

8.9 Several cells

Suppose a family of n -cells is attached by maps

$$\varphi_{\alpha} : S^{n-1} \rightarrow X.$$

If each φ_{α} is replaced by a homotopic map ψ_{α} , then under the standard CW hypotheses the resulting spaces are homotopy equivalent.

Theorem 8.8 (Simultaneous homotopic attachments). *Homotopic replacement of a family of attaching maps preserves the homotopy type of the corresponding CW attachment stage.*

8.10 Relative version

If the attaching-map homotopy is fixed on a specified subspace of the boundary, the comparison can be performed relative to the corresponding attached subspace. Relative homotopies therefore fit naturally into the attachment theory developed in VIII/01.

8.11 Mapping-cone formulation

For maps

$$f_0, f_1 : A \rightarrow X$$

with $f_0 \simeq f_1$, mapping cones satisfy

$$C_{f_0} \simeq C_{f_1}$$

under the standard cofibration/CW assumptions.

Theorem 8.9 (Homotopy invariance of mapping cones). *Homotopic maps between suitable CW spaces have homotopy-equivalent mapping cones.*

This is the general form of the cell-attachment theorem.

8.12 Why cofibrations matter

A naive quotient argument can fail if the inclusion being pushed out behaves badly. Cofibrations guarantee that a homotopy on the attaching subspace can be extended and controlled inside the ambient construction.

Warning 8.10 (Do not replace cofibration hypotheses silently). The theorem is robust for CW attachments because sphere-boundary inclusions and CW-subcomplex inclusions are cofibrations. For arbitrary quotient constructions, one should not assume homotopy invariance without checking the relevant hypotheses.

8.13 Application: different formulas, same cell complex homotopy type

Two polygonal attaching loops can look quite different as parametrized maps but represent the same element of

$$[S^1, X].$$

When they are homotopic, attaching a two-cell along either loop yields the same homotopy type.

8.14 Application to null loops

If a loop

$$\varphi : S^1 \rightarrow X$$

is null-homotopic, adding a two-cell along φ has homotopy type

$$X \vee S^2.$$

This is often easier to recognize than the original quotient.

8.15 Dependence only on homotopy class

The correct conceptual object attached to a cell is not the raw parametrized formula

$$\varphi : S^{n-1} \rightarrow X,$$

but its homotopy class

$$[\varphi] \in [S^{n-1}, X],$$

as far as the resulting homotopy type is concerned.

8.16 Limits of the conclusion

Homotopy-equivalent attachment spaces may differ in geometric or quotient-level details. The theorem does not preserve a chosen cell parametrization, metric, smooth structure, or literal point-set presentation.

Warning 8.11 (Homotopy equivalence is the invariant). Never strengthen the conclusion

$$X \cup_{\varphi_0} D^n \simeq X \cup_{\varphi_1} D^n$$

to homeomorphism without additional proof.

8.17 Bridge to the fundamental group

For two-cell attachments to a graph, the homotopy class of the attaching loop is encoded by an element of the fundamental group. Thus the next chapter's group law on loops gives an algebraic language for the attachment data of two-dimensional CW complexes.

8.18 Exercises

Exercise 8.12. Define the homotopy extension property.

Hint. Extend a homotopy from the subspace while preserving the initial map. □

Solution. A pair (X, A) has HEP when every compatible $f : X \rightarrow Y$ and $H : A \times I \rightarrow Y$ extend to $\tilde{H} : X \times I \rightarrow Y$. □

Exercise 8.13. Define a cofibration.

Hint. Use HEP. □

Solution. An inclusion $A \hookrightarrow X$ is a cofibration when (X, A) has the homotopy extension property. □

Exercise 8.14. What important inclusions in CW complexes are cofibrations?

Hint. Think subcomplexes. □

Solution. Every CW-subcomplex inclusion $A \hookrightarrow X$. □

Exercise 8.15. State the homotopic-attaching-map theorem.

Hint. Compare two maps $S^{n-1} \rightarrow X$. □

Solution. If $\varphi_0 \simeq \varphi_1$, then $X \cup_{\varphi_0} D^n \simeq X \cup_{\varphi_1} D^n$ under the standard CW hypotheses. □

Exercise 8.16. What is the conclusion: homeomorphism or homotopy equivalence?

Hint. Use the weaker invariant. □

Solution. Homotopy equivalence. □

Exercise 8.17. What geometric device interpolates between two attaching maps?

Hint. Use the homotopy parameter as a collar. □

Solution. A cylinder $S^{n-1} \times I$ carrying the homotopy. □

Exercise 8.18. What happens if the attaching map is null-homotopic?

Hint. Replace it by a constant attachment. □

Solution. $X \cup_{\varphi} D^n \simeq X \vee S^n$. □

Exercise 8.19. When are two maps $S^1 \rightarrow S^1$ homotopic?

Hint. Use degree. □

Solution. Exactly when they have the same degree. □

Exercise 8.20. What does equal degree imply for two two-cell attachments to S^1 ?

Hint. Apply the attachment theorem. □

Solution. The resulting adjunction spaces are homotopy equivalent. □

Exercise 8.21. Can several attaching maps be changed simultaneously by homotopy?

Hint. Use the family version. □

Solution. Yes, under the standard CW hypotheses the resulting stage has the same homotopy type. □

Exercise 8.22. What relative condition can be imposed on an attaching homotopy?

Hint. Fix a subspace during the deformation. □

Solution. The homotopy may be taken relative to a specified boundary subspace, leading to a relative comparison. □

Exercise 8.23. State the mapping-cone version of the theorem.

Hint. Replace S^{n-1} by a general domain. □

Solution. If $f_0 \simeq f_1 : A \rightarrow X$, then $C_{f_0} \simeq C_{f_1}$ under suitable CW/cofibration hypotheses. □

Exercise 8.24. Why are cofibrations used?

Hint. They control extension of homotopies. □

Solution. They ensure a homotopy on the attaching subspace extends compatibly to the ambient construction. □

Exercise 8.25. What point-set warning accompanies the theorem?

Hint. Do not confuse equivalence notions. □

Solution. Homotopy-equivalent quotients need not be homeomorphic. □

Exercise 8.26. On what datum does the homotopy type of a cell attachment depend?

Hint. Pass from map to class. □

Solution. On the homotopy class $[\varphi] \in [S^{n-1}, X]$, under the standard hypotheses. □

Exercise 8.27. If $\varphi : S^1 \rightarrow X$ is null-homotopic, what is the two-cell attachment type?

Hint. Set $n = 2$. □

Solution. $X \vee S^2$ up to homotopy. □

Exercise 8.28. What is an interpolation space Z_H ?

Hint. Use a cylinder carrying H . □

Solution. A space built from the homotopy between attachments that deformation retracts onto both endpoint adjunction spaces. □

Exercise 8.29. Why can different formulas represent the same attachment homotopy type?

Hint. Parametrized maps may be homotopic. □

Solution. The theorem depends on the homotopy class, not the literal formula. □

Exercise 8.30. Does HEP say every homotopy on every subset extends?

Hint. The pair matters. □

Solution. No. HEP is a property of a specified inclusion $A \hookrightarrow X$. □

Exercise 8.31. Why is $S^{n-1} \hookrightarrow D^n$ a good attaching inclusion?

Hint. It is a CW-subcomplex inclusion. □

Solution. It is a cofibration. □

Exercise 8.32. How does VIII/08 refine VIII/05?

Hint. VIII/05 said the map matters. □

Solution. VIII/08 says the homotopy class matters for homotopy type, so pointwise changes within the same class do not matter. □

Exercise 8.33. How does VIII/08 refine VIII/07?

Hint. Apply the same principle to cones. □

Solution. It proves homotopy invariance of mapping cones under suitable hypotheses. □

Exercise 8.34. What algebraic object will encode attaching loops in dimension two?

Hint. Go to VIII/09. □

Solution. The fundamental group. □

Exercise 8.35. Why is this theorem especially useful for CW complexes?

Hint. Attaching maps define skeleta. □

Solution. It permits replacement of complicated attaching maps by simpler homotopic representatives without changing the homotopy type. □

8.19 Solved problem dossiers

Problem 8.36 (Null attachment). Let $\varphi : S^{n-1} \rightarrow X$ be null-homotopic. Derive the splitting of the attachment.

Solution. Choose a homotopy from φ to a constant map c . The homotopic-attaching-map theorem gives $X \cup_{\varphi} D^n \simeq X \cup_c D^n$, and the constant attachment has homotopy type $X \vee S^n$. □

Problem 8.37 (Degree-three representatives). Let $\varphi, \psi : S^1 \rightarrow S^1$ both have degree 3. Compare the two attached spaces.

Solution. Circle maps of equal degree are homotopic. Therefore $S^1 \cup_{\varphi} D^2 \simeq S^1 \cup_{\psi} D^2$. □

Problem 8.38 (Degree mismatch). Suppose $\deg \varphi = 2$ and $\deg \psi = 3$. Can VIII/08 identify their attachments?

Solution. No. The maps are not homotopic, so the theorem does not apply. Later cellular homology in fact distinguishes the corresponding degree attachments. □

Problem 8.39 (Constant versus wiggly null loop). A loop in X is visibly complicated but null-homotopic. What two-cell attachment homotopy type results?

Solution. Replace the loop by the constant loop through a null-homotopy. The result is homotopy equivalent to $X \vee S^2$. $\square$

Problem 8.40 (Cylinder interpolation). Explain how $S^{n-1} \times I$ records a homotopy between two attachments.

Solution. At time 0 its base is glued to X by φ_0 , at time 1 by φ_1 , and intermediate slices follow $H(-, t)$. Sliding along this collar produces comparison deformation retracts. $\square$

Problem 8.41 (Why not homeomorphic?). Why does the theorem not automatically produce a homeomorphism between the endpoint quotient spaces?

Solution. The construction controls deformations and deformation retracts, not pointwise quotient identifications. Homotopy equivalence is invariant under the interpolation, while homeomorphism would require stronger point-set control. $\square$

Problem 8.42 (Mapping-cone corollary). Deduce the homotopy invariance of C_f from the attachment principle.

Solution. A mapping cone is obtained by attaching the cone CA along its base A using f . Replacing f by a homotopic map changes the attachment through a homotopy, hence preserves the cone homotopy type under the cofibration hypotheses. $\square$

Problem 8.43 (CW-pair role). Why does the theorem apply cleanly to a cell attachment in a CW complex?

Solution. The boundary inclusion $S^{n-1} \hookrightarrow D^n$ and CW-subcomplex inclusions are cofibrations, so the required homotopies can be extended in the pushout construction. $\square$

Problem 8.44 (Two-cell graph attachment). Let X be a graph and let two loops $\alpha, \beta : S^1 \rightarrow X$ be homotopic rel basepoint. Compare the attached two-complexes.

Solution. Attaching one two-cell along α or β produces homotopy-equivalent spaces. The based homotopy is stronger than needed for the unbased conclusion. $\square$

Problem 8.45 (Relative attachment). Suppose a homotopy of attaching maps is fixed on a subset $B \subset S^{n-1}$. What extra control should the comparison preserve?

Solution. The interpolation can be performed relative to the image of B , so the resulting homotopy equivalence can preserve the corresponding attached subspace. $\square$

Problem 8.46 (Family of null attachments). Attach r n -cells to X by null-homotopic maps. Predict the homotopy type.

Solution. Replace all attaching maps simultaneously by constants. Under the standard pointed hypotheses the result is homotopy equivalent to $X \vee \bigvee_{j=1}^r S^n$. $\square$

Problem 8.47 (HEP extension). Given $f : X \rightarrow Y$ and a homotopy $H : A \times I \rightarrow Y$ starting at $f|_A$, what does HEP produce?

Solution. It produces $\tilde{H} : X \times I \rightarrow Y$ extending H and satisfying $\tilde{H}(-, 0) = f$. $\square$

Problem 8.48 (Subcomplex example). Give a concrete cofibration from the previous chapters.

Solution. The inclusion $S^{n-1} \hookrightarrow D^n$ is an inclusion of a CW subcomplex and therefore a cofibration. $\square$

Problem 8.49 (Attachment simplification). Why is it legitimate to replace an attaching loop by a simpler loop in the same homotopy class before computing invariants?

Solution. VIII/08 ensures the replacement preserves the homotopy type of the resulting CW complex, and homotopy invariants therefore have the same values. $\square$

Problem 8.50 (Same word, different parametrization). Two polygon boundary maps traverse the same loop word but with different speeds and pauses. Compare their attachments.

Solution. Reparameterization gives a homotopy of the loops, so their two-cell attachments are homotopy equivalent. $\square$

Problem 8.51 (Reverse parametrization warning). Is a loop necessarily homotopic to the same geometric loop traversed backward?

Solution. No. Reversal represents the inverse fundamental-group element and need not be homotopic to the original loop. VIII/09 makes this distinction algebraic. $\square$

Problem 8.52 (Degree-zero circle attachment). Let $\varphi : S^1 \rightarrow S^1$ have degree 0. Identify the attachment homotopy type.

Solution. Every degree-zero circle map is null-homotopic, so $S^1 \cup_{\varphi} D^2 \simeq S^1 \vee S^2$. $\square$

Problem 8.53 (Degree-one circle attachment). Let $\varphi : S^1 \rightarrow S^1$ have degree 1. Compare its attachment with the identity attachment.

Solution. φ is homotopic to the identity, so the attachment is homotopy equivalent to $S^1 \cup_{\text{id}} D^2 \cong D^2$, hence contractible. $\square$

Problem 8.54 (Homotopy-class principle). State precisely what attachment data VIII/08 says can be simplified.

Solution. For homotopy-type questions under the standard hypotheses, the attaching map may be replaced by any representative of its homotopy class in $[S^{n-1}, X]$. $\square$

Problem 8.55 (Bridge to π_1). Why is the fundamental group the natural next tool for two-cell attachments?

Solution. An attaching map for a two-cell is a loop $S^1 \rightarrow X^1$. Based homotopy classes of loops form $\pi_1(X^1)$, converting the attaching data into group elements and words. $\square$

Part III

Fundamental Groups and Coverings

Chapter 9

Paths and Fundamental Groups

The first genuinely algebraic invariant of a space is built from loops. Paths record how points are connected, while loops based at a chosen point can be concatenated. Passing to homotopy classes converts this concatenation into a group law. The resulting fundamental group detects holes, organizes covering spaces, and encodes two-cell attaching loops in low-dimensional CW complexes.

9.1 Paths

Definition 9.1 (Path). A path in X from x_0 to x_1 is a continuous map

$$\alpha : I \rightarrow X$$

with

$$\alpha(0) = x_0, \quad \alpha(1) = x_1.$$

A loop based at x_0 is a path satisfying

$$\alpha(0) = \alpha(1) = x_0.$$

9.2 Path reversal

For a path α , define

$$\bar{\alpha}(t) = \alpha(1 - t).$$

Then $\bar{\alpha}$ runs from the endpoint of α back to its starting point.

9.3 Concatenation

If

$$\alpha(1) = \beta(0),$$

define

$$(\alpha * \beta)(t) = \begin{cases} \alpha(2t), & 0 \leq t \leq \frac{1}{2}, \\ \beta(2t - 1), & \frac{1}{2} \leq t \leq 1. \end{cases}$$

Warning 9.2 (Convention). This chapter reads $\alpha * \beta$ as first traverse α , then β . Some texts adopt the opposite multiplication convention.

9.4 Path homotopy

Definition 9.3 (Homotopy rel endpoints). Two paths $\alpha, \beta : I \rightarrow X$ with the same endpoints are path-homotopic if there exists

$$H : I \times I \rightarrow X$$

such that

$$H(s, 0) = \alpha(s), \quad H(s, 1) = \beta(s),$$

and

$$H(0, t) = x_0, \quad H(1, t) = x_1.$$

This is exactly homotopy relative to $\{0, 1\}$ from VIII/01.

9.5 The fundamental group

Definition 9.4 (Fundamental group). The fundamental group of X at x_0 is

$$\pi_1(X, x_0) = \{\text{loops at } x_0\} / \text{path homotopy},$$

with multiplication

$$[\alpha][\beta] = [\alpha * \beta].$$

Theorem 9.5 (Group structure). *Path concatenation induces a well-defined group structure on $\pi_1(X, x_0)$.*

9.6 Identity and inverse

The identity element is represented by the constant loop

$$c_{x_0}(t) = x_0.$$

The inverse of $[\alpha]$ is

$$[\bar{\alpha}].$$

Associativity of literal path concatenation holds only up to reparameterization, but that is enough because reparameterized paths are homotopic rel endpoints.

9.7 Well-defined multiplication

Proposition 9.6 (Compatibility with path homotopy). *If*

$$\alpha \simeq \alpha' \quad \text{and} \quad \beta \simeq \beta'$$

rel endpoints, then

$$\alpha * \beta \simeq \alpha' * \beta'$$

rel endpoints.

Hence multiplication depends only on homotopy classes.

9.8 Simply connected spaces

Definition 9.7 (Simply connected). A space X is simply connected if it is path connected and

$$\pi_1(X, x_0) = 0$$

for one, equivalently every, basepoint.

Contractible spaces are simply connected.

Example 9.8. For $n \geq 2$,

$$\pi_1(S^n) = 0.$$

9.9 The circle

Theorem 9.9 (Fundamental group of the circle).

$$\pi_1(S^1, 1) \cong \mathbb{Z}.$$

The integer records winding number. The loop

$$\omega_n(t) = e^{2\pi i n t}$$

represents the integer n .

9.10 Winding and degree

A loop in S^1 is equivalently a based map

$$S^1 \rightarrow S^1.$$

Its fundamental-group class is measured by the same integer that appeared as degree in VIII/03.

Proposition 9.10 (Degree and $\pi_1(S^1)$). *Under the identification $\pi_1(S^1) \cong \mathbb{Z}$,*

$$[\omega_n] \longleftrightarrow n.$$

9.11 Basepoint change

Let γ be a path from x_0 to x_1 .

Theorem 9.11 (Change of basepoint). *There is an isomorphism*

$$\gamma_{\#} : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$$

given by

$$[\alpha] \longmapsto [\bar{\gamma} * \alpha * \gamma]$$

with the concatenation convention used in this chapter, after interpreting the order consistently.

Different choices of connecting path may change the isomorphism by an inner automorphism.

9.12 Induced homomorphisms

For a based map

$$f : (X, x_0) \rightarrow (Y, y_0),$$

define

$$f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$$

by

$$f_*([\alpha]) = [f \circ \alpha].$$

Theorem 9.12 (Functoriality).

$$(\text{id}_X)_* = \text{id}_{\pi_1(X)}, \quad (g \circ f)_* = g_* \circ f_*.$$

9.13 Homotopy invariance

Theorem 9.13 (Based homotopy invariance). *Based-homotopic maps induce the same homomorphism on fundamental groups.*

Corollary 9.14 (Homotopy equivalence). *A homotopy equivalence of path-connected spaces induces an isomorphism of fundamental groups, after the usual basepoint identifications.*

9.14 Products

Theorem 9.15 (Product formula). *For path-connected based spaces,*

$$\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0).$$

Hence

$$\pi_1(T^2) \cong \mathbb{Z}^2.$$

9.15 Wedges and free groups: preview

For a wedge of r circles,

$$\bigvee_{j=1}^r S^1,$$

the fundamental group is the free group

$$F_r.$$

A complete computation uses van Kampen's theorem, developed later in the fundamental-group arc.

9.16 Attaching loops and relations

A two-cell attaching map

$$\varphi : S^1 \rightarrow X^1$$

represents a loop class in

$$\pi_1(X^1).$$

Attaching the two-cell makes that loop null-homotopic in the enlarged space. Thus two-cell attachments impose relations on the fundamental group.

Example 9.16 (Torus relation). The torus one-skeleton has two loop generators a, b . Its two-cell attaches along

$$aba^{-1}b^{-1},$$

forcing the commutator to become trivial. This is the geometric source of

$$\pi_1(T^2) \cong \mathbb{Z}^2.$$

9.17 Fundamental group as a homotopy invariant

If two path-connected spaces have nonisomorphic fundamental groups, then they cannot be homotopy equivalent. In particular,

$$S^1 \not\cong \{*\}$$

because

$$\pi_1(S^1) \cong \mathbb{Z}$$

whereas the point has trivial fundamental group.

9.18 Bridge to covering spaces

The circle computation is most naturally understood using the covering map

$$p: \mathbb{R} \rightarrow S^1, \quad p(t) = e^{2\pi it}.$$

Loops lift to paths in $\mathbb{R}$, and the endpoint difference is an integer. VIII/10 develops covering spaces systematically and turns this observation into a general lifting theory.

9.19 Exercises

Exercise 9.17. Define a path from x_0 to x_1 .

Hint. Use a continuous map on I . □

Solution. A continuous $\alpha: I \rightarrow X$ with $\alpha(0) = x_0$ and $\alpha(1) = x_1$. □

Exercise 9.18. Define a loop at x_0 .

Hint. Make the endpoints equal. □

Solution. A path α with $\alpha(0) = \alpha(1) = x_0$. □

Exercise 9.19. Define the reverse path $\bar{\alpha}$.

Hint. Reverse the parameter. □

Solution. $\bar{\alpha}(t) = \alpha(1 - t)$. □

Exercise 9.20. When is $\alpha * \beta$ defined?

Hint. Match the endpoint of the first to the start of the second. □

Solution. When $\alpha(1) = \beta(0)$. □

Exercise 9.21. Write the concatenation formula.

Hint. Give each path half the interval. ☐

Solution. $(\alpha * \beta)(t) = \alpha(2t)$ for $t \leq 1/2$, and $\beta(2t - 1)$ for $t \geq 1/2$. ☐

Exercise 9.22. What must a path homotopy keep fixed?

Hint. Use homotopy rel endpoints. ☐

Solution. The initial and terminal endpoints. ☐

Exercise 9.23. Define $\pi_1(X, x_0)$.

Hint. Take based loops modulo endpoint-fixing homotopy. ☐

Solution. The set of based loop homotopy classes with concatenation. ☐

Exercise 9.24. What is the identity in $\pi_1(X, x_0)$?

Hint. Use the constant loop. ☐

Solution. $[c_{x_0}]$. ☐

Exercise 9.25. What is the inverse of $[\alpha]$?

Hint. Reverse the loop. ☐

Solution. $[\bar{\alpha}]$. ☐

Exercise 9.26. Why is multiplication well-defined?

Hint. Concatenate representatives of homotopies. ☐

Solution. Path-homotopic representatives have path-homotopic concatenations. ☐

Exercise 9.27. Is literal path concatenation strictly associative?

Hint. Compare parametrizations. ☐

Solution. No; it is associative up to endpoint-fixing reparameterization homotopy. ☐

Exercise 9.28. Define simply connected.

Hint. Require connectivity and trivial π_1 . ☐

Solution. Path connected with trivial fundamental group. ☐

Exercise 9.29. What is $\pi_1(S^n)$ for $n \geq 2$?

Hint. Higher spheres are simply connected. ☐

Solution. Trivial. ☐

Exercise 9.30. What is $\pi_1(S^1)$?

Hint. Use winding number. ☐

Solution. $\mathbb{Z}$. ☐

Exercise 9.31. What integer does $\omega_n(t) = e^{2\pi i n t}$ represent?

Hint. Count windings. ☐

Solution. n . □

Exercise 9.32. How is degree related to $\pi_1(S^1)$?

Hint. Both classify circle maps. □

Solution. The degree equals the winding-number integer under $\pi_1(S^1) \cong \mathbb{Z}$. □

Exercise 9.33. What does a based map $f : X \rightarrow Y$ induce?

Hint. Compose loops with f . □

Solution. A homomorphism $f_* : \pi_1(X) \rightarrow \pi_1(Y)$. □

Exercise 9.34. State functoriality for $g \circ f$.

Hint. Induced maps compose. □

Solution. $(g \circ f)_* = g_* \circ f_*$. □

Exercise 9.35. What do based-homotopic maps induce?

Hint. Use homotopy invariance. □

Solution. The same homomorphism on π_1 . □

Exercise 9.36. What does a homotopy equivalence induce on π_1 ?

Hint. Use a homotopy inverse. □

Solution. An isomorphism, after appropriate basepoint identification. □

Exercise 9.37. State the product formula.

Hint. Project loops to both factors. □

Solution. $\pi_1(X \times Y) \cong \pi_1(X) \times \pi_1(Y)$. □

Exercise 9.38. Compute $\pi_1(T^2)$.

Hint. Use $T^2 = S^1 \times S^1$. □

Solution. $\mathbb{Z}^2$. □

Exercise 9.39. What does attaching a two-cell along a loop do to that loop class?

Hint. The disk fills the loop. □

Solution. It makes the attaching loop null-homotopic in the enlarged space. □

Exercise 9.40. What is the bridge to VIII/10?

Hint. Use the universal cover of the circle. □

Solution. Covering-space lifting turns loop classes into endpoint data of lifted paths. □

9.20 Solved problem dossiers

Problem 9.41 (Reverse twice). Show $\overline{\overline{\alpha}} = \alpha$.

Solution. By definition, $\overline{\overline{\alpha}}(t) = \overline{\alpha}(1 - t) = \alpha(1 - (1 - t)) = \alpha(t)$. $\square$

Problem 9.42 (Concatenation endpoints). Verify the endpoints of $\alpha * \beta$.

Solution. At $t = 0$ the value is $\alpha(0)$; at $t = 1$ it is $\beta(1)$. At $t = 1/2$, both formulas equal the common point $\alpha(1) = \beta(0)$. $\square$

Problem 9.43 (Identity loop). Why is $\alpha * c_{x_1}$ homotopic rel endpoints to α ?

Solution. Reparameterize the traversal of α so that it reaches the endpoint later and removes the terminal pause. Endpoint-preserving reparameterization gives the required homotopy. $\square$

Problem 9.44 (Inverse loop). Why is $\alpha * \overline{\alpha}$ null-homotopic rel basepoint?

Solution. Shrink the out-and-back traversal symmetrically toward the basepoint; at each stage traverse only an initial segment of α and immediately return along it. $\square$

Problem 9.45 (Associativity). Explain why $(\alpha * \beta) * \gamma$ and $\alpha * (\beta * \gamma)$ represent the same fundamental-group element.

Solution. They traverse the same three paths in the same order with different time allocations. A continuous reparameterization homotopy rel endpoints changes one allocation into the other. $\square$

Problem 9.46 (Circle generator). Show that $\omega_1(t) = e^{2\pi it}$ generates $\pi_1(S^1) \cong \mathbb{Z}$.

Solution. Every loop class has an integer winding number n , and ω_n is homotopic to the n -fold concatenation of ω_1 for $n > 0$, with reversed copies for $n < 0$. $\square$

Problem 9.47 (Degree and winding). For $f(z) = z^m$, compute the induced map $f_* : \pi_1(S^1) \rightarrow \pi_1(S^1)$.

Solution. The generator loop ω_1 is sent to $t \mapsto e^{2\pi imt} = \omega_m$. Thus f_* is multiplication by m . $\square$

Problem 9.48 (Contractible target). Show that a contractible path-connected space has trivial fundamental group.

Solution. The identity is homotopic to a constant map. Homotopy invariance makes the identity homomorphism on π_1 equal to the zero homomorphism induced by the constant map, so the group is trivial. $\square$

Problem 9.49 (Product loop). Describe a loop in $X \times Y$ in terms of loops in the factors.

Solution. A loop $\alpha(t) = (\alpha_X(t), \alpha_Y(t))$ is exactly a pair of loops. Homotopies and concatenation occur componentwise, yielding the direct-product formula. $\square$

Problem 9.50 (Torus group). Compute $\pi_1(T^2)$ from the product formula.

Solution. $\pi_1(S^1 \times S^1) \cong \mathbb{Z} \times \mathbb{Z}$. $\square$

Problem 9.51 (Basepoint path). Explain why path-connected spaces have isomorphic fundamental groups at different basepoints.

Solution. Choose a path joining the basepoints and conjugate loops by that path and its reverse. This gives an isomorphism, with different path choices differing by inner automorphisms. $\square$

Problem 9.52 (Homotopy-equivalence obstruction). Why can S^1 not be homotopy equivalent to a point?

Solution. Homotopy equivalence would induce an isomorphism of fundamental groups, but $\pi_1(S^1) \cong \mathbb{Z}$ and the point has trivial fundamental group. $\square$

Problem 9.53 (Higher sphere comparison). Use π_1 to distinguish S^1 from S^2 .

Solution. $\pi_1(S^1) \cong \mathbb{Z}$ while $\pi_1(S^2) = 0$, so the spaces are not homotopy equivalent. $\square$

Problem 9.54 (Wedge preview). What group should be expected for $S^1 \vee S^1$?

Solution. The expected group is the free group F_2 on the two circle loops. Van Kampen's theorem gives the formal computation. $\square$

Problem 9.55 (Torus attaching relation). Explain geometrically why the torus two-cell kills the commutator.

Solution. The boundary of the two-cell is the loop $aba^{-1}b^{-1}$. Once the disk is attached, that boundary loop bounds the disk and is therefore null-homotopic, imposing $[a, b] = 1$. $\square$

Problem 9.56 (Null two-cell attachment). If a two-cell attaches along a null-homotopic loop in X , what does VIII/08 predict?

Solution. The result is homotopy equivalent to $X \vee S^2$, because the attaching loop can be replaced by the constant loop. $\square$

Problem 9.57 (Functoriality check). Prove $(g \circ f)_* = g_* \circ f_*$ on a loop class.

Solution. For $[\alpha]$, $(g \circ f)_*[\alpha] = [g \circ f \circ \alpha] = g_*([f \circ \alpha]) = (g_* \circ f_*)[\alpha]$. $\square$

Problem 9.58 (Constant map). What homomorphism on fundamental groups is induced by a constant based map?

Solution. Every loop is sent to the constant loop, so the induced homomorphism is trivial. $\square$

Problem 9.59 (Circle covering preview). Lift $\omega_n(t) = e^{2\pi i n t}$ to a path in $\mathbb{R}$ starting at 0.

Solution. The lift is $\tilde{\omega}_n(t) = nt$, because $e^{2\pi i (nt)} = \omega_n(t)$. Its endpoint is the integer n . $\square$

Problem 9.60 (Why coverings next?). Explain how the circle cover suggests a general method for studying π_1 .

Solution. Loop classes can be detected by how their lifts move between points in a fiber. VIII/10 generalizes path lifting and endpoint behavior to arbitrary covering spaces. $\square$

Problem 9.61 (Loop reparameterization). Show that changing the speed of a loop by an endpoint-preserving map $\rho : I \rightarrow I$ does not change its fundamental-group class.

Solution. The homotopy $H(s, t) = \alpha((1-t)s + t\rho(s))$ works when the interpolation remains endpoint preserving; more generally VIII/01's reparameterization principle provides the endpoint-fixing homotopy. $\square$

Problem 9.62 (Path-connected basepoint independence). Why is the statement ' $\pi_1(X)$ ' usually acceptable without naming a basepoint for a path-connected space?

Solution. All basepoint fundamental groups are isomorphic. The isomorphism is not completely canonical in nonabelian groups, but the group isomorphism type is independent of basepoint. $\square$

Problem 9.63 (Two-cell relation principle). State the conceptual relation between VIII/08 and VIII/09 for two-dimensional CW complexes.

Solution. VIII/08 says only the homotopy class of the attaching loop matters. VIII/09 packages based loop homotopy classes into π_1 , so attaching data become group elements and relations. $\square$

Problem 9.64 (Bridge dossier). Describe the role of $p : \mathbb{R} \rightarrow S^1$, $p(t) = e^{2\pi it}$, in the next chapter.

Solution. It is the model covering map. Paths and loops lift uniquely after choosing a starting point, and a loop's winding number is recovered from the endpoint of its lift. $\square$

Chapter 10

Covering Spaces

Covering spaces turn global topology into locally trivial sheets. Around every point of the base, a covering map looks like a disjoint union of copies of one open neighborhood. This simple local model gives strong global information: fibers are discrete, connected covers have a constant number of sheets, the circle admits the exponential cover by the real line, and coverings interact directly with fundamental groups.

10.1 Covering maps

Definition 10.1 (Covering map). A continuous surjection

$$p : \tilde{X} \rightarrow X$$

is a covering map if every $x \in X$ has an open neighborhood U such that

$$p^{-1}(U) = \coprod_{\alpha \in A} V_{\alpha}$$

with each V_{α} open in $\tilde{X}$ and

$$p|_{V_{\alpha}} : V_{\alpha} \rightarrow U$$

a homeomorphism.

Such a neighborhood U is called evenly covered.

10.2 Sheets and fibers

For $x \in X$, the fiber is

$$p^{-1}(x).$$

Proposition 10.2 (Fibers are discrete). *Every fiber of a covering map is a discrete subspace of $\tilde{X}$.*

If U is evenly covered, each sheet above U contains exactly one point of the fiber over each $x \in U$.

10.3 Number of sheets

Proposition 10.3 (Constant sheet number on connected bases). *If X is connected, then the cardinality*

$$|p^{-1}(x)|$$

is independent of $x \in X$.

A cover with n points in each fiber is called an n -sheeted cover.

10.4 The universal example

Example 10.4 (The exponential cover). The map

$$p : \mathbb{R} \rightarrow S^1, \quad p(t) = e^{2\pi it},$$

is a covering map. Every sufficiently short open arc in S^1 is evenly covered by countably many translated intervals.

The fiber over $1 \in S^1$ is

$$p^{-1}(1) = \mathbb{Z}.$$

10.5 Finite circle covers

For $n \geq 1$,

$$p_n : S^1 \rightarrow S^1, \quad p_n(z) = z^n$$

is an n -sheeted covering map.

Example 10.5. The map $z \mapsto z^2$ is a two-sheeted covering of the circle by itself.

10.6 Trivial covers

Definition 10.6 (Trivial covering). For a discrete set F , the projection

$$X \times F \rightarrow X$$

is a trivial covering.

Every covering is locally of this form over an evenly covered neighborhood, but need not be globally a product.

10.7 Local homeomorphisms

Proposition 10.7 (Coverings are local homeomorphisms). *Every covering map is a local homeomorphism.*

Warning 10.8 (The converse fails). A surjective local homeomorphism need not be a covering map without additional global control on the sheets.

10.8 Restriction of a covering

Proposition 10.9 (Restriction). *If $p : \tilde{X} \rightarrow X$ is a covering and $A \subset X$, then the restricted map*

$$p^{-1}(A) \rightarrow A$$

is a covering with the subspace topologies.

10.9 Pullback coverings

Given $p : \tilde{X} \rightarrow X$ and a map $f : Y \rightarrow X$, define

$$f^* \tilde{X} = \{(y, \tilde{x}) \in Y \times \tilde{X} : f(y) = p(\tilde{x})\}.$$

Theorem 10.10 (Pullback covering). *The projection*

$$f^* \tilde{X} \rightarrow Y$$

is a covering map.

This gives a systematic way to transport coverings along maps.

10.10 Connected coverings

A covering space $\tilde{X}$ may be disconnected even when X is connected. Its connected components themselves cover the base when the base is connected and locally path connected.

Proposition 10.11 (Components of a covering). *For a connected locally path-connected base, every connected component of a covering space maps onto the base and is itself a covering space.*

10.11 Universal covers

Definition 10.12 (Universal covering space). A covering

$$p : \tilde{X} \rightarrow X$$

is universal if $\tilde{X}$ is simply connected.

The real line is the universal cover of S^1 .

Example 10.13. For $n \geq 2$,

$$S^n \rightarrow \mathbb{R}P^n$$

is a two-sheeted cover, and S^n is simply connected. Thus it is the universal cover of $\mathbb{R}P^n$.

10.12 Evenly covered neighborhoods

If $U \subset X$ is evenly covered, choosing a point

$$\tilde{x} \in p^{-1}(x), \quad x \in U,$$

selects a unique sheet V above U . The inverse

$$(p|_V)^{-1} : U \rightarrow V$$

is a local section.

10.13 Sections

Definition 10.14 (Section). A section of $p : \tilde{X} \rightarrow X$ is a map

$$s : X \rightarrow \tilde{X}$$

such that

$$p \circ s = \text{id}_X.$$

Every covering has local sections over evenly covered neighborhoods. A global section is much stronger and can force a connected cover to be trivial in many situations.

10.14 Covering isomorphisms

Definition 10.15 (Isomorphism of coverings). Two coverings $p_i : \tilde{X}_i \rightarrow X$ are isomorphic if there is a homeomorphism

$$h : \tilde{X}_1 \rightarrow \tilde{X}_2$$

such that

$$p_2 \circ h = p_1.$$

The base is held fixed.

10.15 Coverings and fundamental groups

A connected covering determines a subgroup of the base fundamental group. If

$$p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0),$$

then

$$p_* : \pi_1(\tilde{X}, \tilde{x}_0) \rightarrow \pi_1(X, x_0)$$

is injective.

Theorem 10.16 (Fundamental-group injection). *For a covering map, the induced homomorphism on fundamental groups is injective.*

The proof uses lifting uniqueness, developed systematically in VIII/11.

10.16 Index and number of sheets

Under the standard hypotheses used in covering-space classification, a connected covering corresponds to a subgroup

$$H \leq \pi_1(X, x_0),$$

and the number of sheets is the index

$$[\pi_1(X, x_0) : H].$$

10.17 Universal cover and the trivial subgroup

For a simply connected universal cover,

$$\pi_1(\tilde{X}) = 0,$$

so the corresponding subgroup of $\pi_1(X)$ is trivial.

10.18 Classification preview

For path-connected, locally path-connected, semilocally simply connected spaces, connected coverings are classified by conjugacy classes of subgroups of $\pi_1(X)$. Pointed connected coverings correspond to actual subgroups.

This theorem relies on the lifting machinery of VIII/11 and the deck-group structure of VIII/12.

10.19 Why coverings matter

Covering spaces convert loops into sheet changes. A loop in the base lifts after a starting point is chosen; whether its lift closes or ends at another point of the fiber detects its fundamental-group behavior. The next chapter develops this lifting principle precisely.

10.20 Exercises

Exercise 10.17. Define an evenly covered neighborhood.

Hint. Write the preimage as disjoint sheets. □

Solution. An open $U \subset X$ is evenly covered if $p^{-1}(U) = \coprod V_\alpha$ and every $p|_{V_\alpha} : V_\alpha \rightarrow U$ is a homeomorphism. □

Exercise 10.18. Define a covering map.

Hint. Every base point needs an evenly covered neighborhood. □

Solution. A continuous surjection locally decomposing into disjoint homeomorphic sheets. □

Exercise 10.19. What is a fiber?

Hint. Take the inverse image of one point. □

Solution. $p^{-1}(x)$. □

Exercise 10.20. Why are covering fibers discrete?

Hint. Use one evenly covered neighborhood. □

Solution. Each sheet contains one fiber point, isolated inside that sheet. □

Exercise 10.21. What is an n -sheeted cover?

Hint. Count fiber points. □

Solution. A covering whose fibers have exactly n points. □

Exercise 10.22. When is sheet number constant?

Hint. Use connectedness of the base. □

Solution. For a connected base, fiber cardinality is constant. □

Exercise 10.23. Give the universal cover of S^1 .

Hint. Use the exponential map. □

Solution. $p : \mathbb{R} \rightarrow S^1$, $p(t) = e^{2\pi it}$. □

Exercise 10.24. What is the fiber over 1 in the exponential cover?

Hint. Solve $e^{2\pi it} = 1$. □

Solution. $\mathbb{Z}$. □

Exercise 10.25. Give an n -sheeted self-cover of S^1 .

Hint. Use the power map. ☐

Solution. $z \mapsto z^n$. ☐

Exercise 10.26. Define a trivial covering.

Hint. Use a discrete fiber. ☐

Solution. The projection $X \times F \rightarrow X$ for discrete F . ☐

Exercise 10.27. What local property does every covering have?

Hint. Restrict to one sheet. ☐

Solution. It is a local homeomorphism. ☐

Exercise 10.28. Does every surjective local homeomorphism automatically give a covering?

Hint. Recall the warning. ☐

Solution. Not in complete generality; global sheet control can fail. ☐

Exercise 10.29. What happens when a covering is restricted to a subspace $A \subset X$?

Hint. Restrict domain to the inverse image. ☐

Solution. $p^{-1}(A) \rightarrow A$ is again a covering. ☐

Exercise 10.30. Define the pullback covering set.

Hint. Impose $f(y) = p(\tilde{x})$. ☐

Solution. $f^*\tilde{X} = \{(y, \tilde{x}) : f(y) = p(\tilde{x})\}$. ☐

Exercise 10.31. What map is the pullback covering projection?

Hint. Project to Y . ☐

Solution. $(y, \tilde{x}) \mapsto y$. ☐

Exercise 10.32. Define a universal cover.

Hint. Require the total space to be simply connected. ☐

Solution. A covering $p : \tilde{X} \rightarrow X$ with simply connected $\tilde{X}$. ☐

Exercise 10.33. What universal cover does $\mathbb{R}P^n$ have for $n \geq 2$?

Hint. Use the antipodal quotient. ☐

Solution. $S^n \rightarrow \mathbb{R}P^n$. ☐

Exercise 10.34. Define a section of a covering.

Hint. It is a right inverse. ☐

Solution. A map $s : X \rightarrow \tilde{X}$ with $p \circ s = \text{id}_X$. ☐

Exercise 10.35. Do coverings always have local sections?

Hint. Use an evenly covered neighborhood. ☐

Solution. Yes. □

Exercise 10.36. Define an isomorphism of coverings over X .

Hint. Use a homeomorphism commuting with projections. □

Solution. A homeomorphism $h : \tilde{X}_1 \rightarrow \tilde{X}_2$ with $p_2 h = p_1$. □

Exercise 10.37. What does a covering induce on π_1 ?

Hint. Use VIII/11 lifting uniqueness. □

Solution. An injective homomorphism $p_* : \pi_1(\tilde{X}) \hookrightarrow \pi_1(X)$. □

Exercise 10.38. What subgroup corresponds to a universal cover?

Hint. The total fundamental group is trivial. □

Solution. The trivial subgroup. □

Exercise 10.39. How is sheet number related to subgroup index in the standard classification setting?

Hint. Count cosets. □

Solution. It equals $[\pi_1(X) : H]$. □

Exercise 10.40. What is the bridge to VIII/11?

Hint. Loops and paths must be lifted. □

Solution. VIII/11 proves path lifting, homotopy lifting, uniqueness, and the general lifting criterion. □

10.21 Solved problem dossiers

Problem 10.41 (Exponential evenly covered arc). Show a sufficiently short open arc $U \subset S^1$ is evenly covered by $p(t) = e^{2\pi it}$.

Solution. Choose U avoiding one point so that it admits a continuous argument interval (a, b) of length less than 1. Then $p^{-1}(U) = \coprod_{k \in \mathbb{Z}} (a + k, b + k)$, and p is a homeomorphism from each interval to U . □

Problem 10.42 (Two-sheeted circle cover). Show $p(z) = z^2$ is a two-sheeted cover of S^1 .

Solution. Every point $w \in S^1$ has two square roots. On a sufficiently short arc choose a continuous square-root branch; the two branches give disjoint sheets each mapped homeomorphically onto the arc. □

Problem 10.43 (Degree- n circle cover). Show $z \mapsto z^n$ is n -sheeted.

Solution. Each $w = e^{i\theta}$ has the n roots $e^{i(\theta+2\pi k)/n}$, and short arcs admit n disjoint continuous root branches. □

Problem 10.44 (Trivial finite cover). Describe the n -sheeted trivial covering of X .

Solution. Take $X \times \{1, \dots, n\} \rightarrow X$. Its n components are sheets globally homeomorphic to X . □

Problem 10.45 (Discrete fiber proof). Prove directly that $p^{-1}(x)$ is discrete.

Solution. Choose evenly covered $U \ni x$. Each fiber point lies in a different sheet V_α , and $V_\alpha \cap p^{-1}(x)$ is that single point. $\square$

Problem 10.46 (Sheet-number constancy). Why is fiber cardinality locally constant on the base?

Solution. Over an evenly covered neighborhood, every fiber has one point in each sheet. Hence all fibers over that neighborhood have the same cardinality. On a connected base a locally constant cardinal-valued function is constant. $\square$

Problem 10.47 (Restriction cover). Prove $p^{-1}(A) \rightarrow A$ is a covering for arbitrary $A \subset X$.

Solution. If U evenly covers $x \in A$, then $U \cap A$ is open in A , and the sheets $V_\alpha \cap p^{-1}(A)$ map homeomorphically onto $U \cap A$. $\square$

Problem 10.48 (Pullback cover). Prove the pullback of a covering is a covering.

Solution. Choose an evenly covered U around $f(y)$. Over $f^{-1}(U)$, the pullback splits into copies indexed by the sheets above U , with each copy projected homeomorphically to $f^{-1}(U)$. $\square$

Problem 10.49 (Projective cover). Why is $S^n \rightarrow \mathbb{R}P^n$ two-sheeted?

Solution. Each projective point is a line with exactly two unit representatives $x, -x$. Small neighborhoods of a line lift to disjoint neighborhoods around these two representatives. $\square$

Problem 10.50 (Universal projective cover). For $n \geq 2$, why is $S^n \rightarrow \mathbb{R}P^n$ universal?

Solution. The antipodal quotient is a covering, and S^n is simply connected for $n \geq 2$. $\square$

Problem 10.51 (Local sections). Construct a local section over an evenly covered neighborhood.

Solution. Choose one sheet V over U and set $s = (p|_V)^{-1} : U \rightarrow V \subset \tilde{X}$. $\square$

Problem 10.52 (Global section warning). Why can the exponential cover $\mathbb{R} \rightarrow S^1$ have no global section?

Solution. A global section would induce a homomorphism splitting $p_* : 0 = \pi_1(\mathbb{R}) \rightarrow \pi_1(S^1) \cong \mathbb{Z}$, impossible because $p \circ s = \text{id}$ would imply $p_* s_* = \text{id}_{\mathbb{Z}}$. $\square$

Problem 10.53 (Covering isomorphism). Show an isomorphism of coverings maps each fiber bijectively to the corresponding fiber.

Solution. If $p_2 h = p_1$, then $p_1(\tilde{x}) = x$ implies $p_2(h(\tilde{x})) = x$. Since h is a homeomorphism, the induced fiber map is bijective. $\square$

Problem 10.54 (Connected component cover). Why does a component of a cover over a connected locally path-connected base map onto the base?

Solution. The image of a component is open by local sheets and closed under local continuation; equivalently path lifting from one image point reaches points along base paths. Connectedness forces the image to be all of the base. $\square$

Problem 10.55 (Universal cover subgroup). What subgroup of $\pi_1(S^1)$ corresponds to $\mathbb{R} \rightarrow S^1$?

Solution. The total space is simply connected, so the subgroup $p_*\pi_1(\mathbb{R})$ is $0 \leq \mathbb{Z}$. $\square$

Problem 10.56 (Finite circle subgroup). What subgroup of $\pi_1(S^1) \cong \mathbb{Z}$ corresponds to $z \mapsto z^n$?

Solution. The induced map on π_1 is multiplication by n , so the subgroup is $n\mathbb{Z}$, of index n . $\square$

Problem 10.57 (Product pullback). Pull back the trivial cover $X \times F \rightarrow X$ along $f : Y \rightarrow X$. Identify the result.

Solution. The pullback consists of triples effectively determined by $(y, a) \in Y \times F$, so it is naturally the trivial cover $Y \times F \rightarrow Y$. $\square$

Problem 10.58 (Cover of a simply connected base). What can be said about a connected covering of a simply connected, suitably nice base?

Solution. The corresponding subgroup of the trivial fundamental group is trivial, so the connected covering is isomorphic to the identity cover under the standard classification hypotheses. $\square$

Problem 10.59 (Cover and local topology). Why do base and total space have the same local topological type under a covering?

Solution. Near every total point, choose the sheet over an evenly covered neighborhood; the covering restriction is a homeomorphism onto the base neighborhood. $\square$

Problem 10.60 (Fiber cardinality of disconnected cover). Can a covering over a connected base be disconnected while having constant fiber size?

Solution. Yes. A trivial n -sheeted covering $X \times \{1, \dots, n\} \rightarrow X$ has n disconnected components when X is connected. $\square$

Problem 10.61 (Identity cover). Show $\text{id}_X : X \rightarrow X$ is a one-sheeted covering.

Solution. Every open neighborhood is evenly covered by itself, with one sheet. $\square$

Problem 10.62 (Composition preview). Under suitable hypotheses, why is a composition of covering maps again locally a covering?

Solution. Choose evenly covered neighborhoods successively so the upper sheets map homeomorphically through the intermediate cover to the base neighborhood. The local sheet decomposition composes. $\square$

Problem 10.63 (Classification preview). State the subgroup object associated with a pointed connected cover.

Solution. Choose $\tilde{x}_0$ above x_0 ; the associated subgroup is $p_*\pi_1(\tilde{X}, \tilde{x}_0) \leq \pi_1(X, x_0)$. $\square$

Problem 10.64 (Bridge to lifting). Why is lifting the mechanism behind covering classification?

Solution. Loops lift uniquely from a chosen sheet. Whether a lifted loop closes detects membership in the associated subgroup; maps lift precisely when their fundamental-group images land in that subgroup. $\square$

Chapter 11

Lifting Properties

Covering maps are powerful because maps into the base can often be lifted to the covering space once an initial point is chosen. Paths always lift, homotopies of paths lift compatibly, and more general maps lift exactly when their fundamental-group image lands inside the subgroup represented by the cover. This chapter develops those principles carefully.

11.1 Lifts

Definition 11.1 (Lift). Let

$$p : \tilde{X} \rightarrow X$$

be a covering and $f : Y \rightarrow X$ a map. A lift of f is a map

$$\tilde{f} : Y \rightarrow \tilde{X}$$

such that

$$p \circ \tilde{f} = f.$$

A lift therefore makes the triangle commute.

11.2 Path lifting

Theorem 11.2 (Path lifting). *Let*

$$\alpha : I \rightarrow X$$

be a path and choose

$$\tilde{x}_0 \in p^{-1}(\alpha(0)).$$

Then there exists a unique path

$$\tilde{\alpha} : I \rightarrow \tilde{X}$$

such that

$$\tilde{\alpha}(0) = \tilde{x}_0 \quad \text{and} \quad p \circ \tilde{\alpha} = \alpha.$$

The proof subdivides I into finitely many intervals whose images lie in evenly covered neighborhoods and continues sheet by sheet.

11.3 Uniqueness of lifts

Theorem 11.3 (Uniqueness on connected domains). *Let Y be connected. If*

$$\tilde{f}_0, \tilde{f}_1 : Y \rightarrow \tilde{X}$$

are lifts of the same map $f : Y \rightarrow X$ and agree at one point of Y , then

$$\tilde{f}_0 = \tilde{f}_1.$$

For locally path-connected applications, this is often proved by showing the agreement set is both open and closed.

11.4 Homotopy lifting

Theorem 11.4 (Homotopy lifting). *Let*

$$H : Y \times I \rightarrow X$$

be a homotopy and suppose

$$\tilde{H}_0 : Y \rightarrow \tilde{X}$$

lifts the initial map $H(-, 0)$. Then there is a unique lift

$$\tilde{H} : Y \times I \rightarrow \tilde{X}$$

with

$$\tilde{H}(-, 0) = \tilde{H}_0.$$

Path lifting is the case $Y = \{*\}$.

11.5 Endpoint behavior of lifted homotopies

Corollary 11.5 (Homotopic paths have matching lifted endpoints). *If paths $\alpha, \beta : I \rightarrow X$ are homotopic rel endpoints and their lifts begin at the same point, then their lifts end at the same point.*

This makes the endpoint of a lifted loop depend only on its fundamental-group class.

11.6 Loop lifting criterion

Let

$$p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$$

be a pointed covering.

Proposition 11.6 (Closed-lift criterion). *A loop α at x_0 lifts to a loop at $\tilde{x}_0$ if and only if*

$$[\alpha] \in p_*\pi_1(\tilde{X}, \tilde{x}_0).$$

11.7 Injection on fundamental groups

Theorem 11.7 (Coverings inject on π_1). *The homomorphism*

$$p_* : \pi_1(\tilde{X}, \tilde{x}_0) \rightarrow \pi_1(X, x_0)$$

induced by a covering map is injective.

Proof. If a loop upstairs projects to a null-homotopic loop downstairs, lift the null-homotopy beginning with the original loop. The lifted final path is the constant loop at $\tilde{x}_0$, so the original loop is null-homotopic. $\square$

11.8 General lifting criterion

Theorem 11.8 (Lifting criterion). *Let Y be path connected and locally path connected, choose $y_0 \in Y$, and let*

$$f : (Y, y_0) \rightarrow (X, x_0)$$

be a based map. For a covering

$$p : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0),$$

a based lift

$$\tilde{f} : (Y, y_0) \rightarrow (\tilde{X}, \tilde{x}_0)$$

exists if and only if

$$f_*\pi_1(Y, y_0) \subset p_*\pi_1(\tilde{X}, \tilde{x}_0).$$

This is one of the central structural results of covering-space theory.

11.9 Construction of the lifted map

Assume the subgroup inclusion in the lifting criterion. For each $y \in Y$, choose a path γ from y_0 to y . Lift

$$f \circ \gamma$$

starting at $\tilde{x}_0$, and define

$$\tilde{f}(y)$$

to be its endpoint. The subgroup condition guarantees independence of the chosen path.

11.10 Why path independence works

If γ_0, γ_1 both run from y_0 to y , then

$$\gamma_0 * \overline{\gamma_1}$$

is a loop. Its image under f lies in the covering subgroup, so its lift from $\tilde{x}_0$ is closed. Therefore the lifts of $f\gamma_0$ and $f\gamma_1$ end at the same point.

11.11 Lifting maps to the universal cover

If $p : \tilde{X} \rightarrow X$ is universal, then

$$p_*\pi_1(\tilde{X}) = 0.$$

Hence a based map $f : Y \rightarrow X$ lifts to $\tilde{X}$ exactly when

$$f_*\pi_1(Y) = 0.$$

Corollary 11.9 (Simply connected domains lift). *If Y is simply connected and satisfies the standard local hypotheses, then every based map $Y \rightarrow X$ lifts to every covering of X after a lift of the basepoint is chosen.*

11.12 The logarithm as a lift

For the exponential cover

$$p : \mathbb{R} \rightarrow S^1, \quad p(t) = e^{2\pi it},$$

a lift of a map

$$f : Y \rightarrow S^1$$

is a continuous real-valued argument function

$$\tilde{f} : Y \rightarrow \mathbb{R}.$$

Thus the lifting criterion becomes a precise obstruction to the existence of a global continuous argument.

11.13 Roots as lifting problems

The map

$$p_n : S^1 \rightarrow S^1, \quad p_n(z) = z^n,$$

is a covering. A continuous n th root of

$$f : Y \rightarrow S^1$$

is exactly a lift of f through p_n .

11.14 Winding obstruction

For a loop

$$f : S^1 \rightarrow S^1$$

of degree m , a lift through

$$z \mapsto z^n$$

exists if and only if

$$m \in n\mathbb{Z}.$$

Equivalently,

$$n \mid m.$$

11.15 Monodromy on a fiber

Fix $x_0 \in X$. A loop α at x_0 lifts uniquely from every

$$\tilde{x} \in p^{-1}(x_0).$$

Taking the endpoint defines a permutation of the fiber. Homotopic loops rel basepoint give the same permutation.

This produces an action

$$\pi_1(X, x_0) \curvearrowright p^{-1}(x_0),$$

called monodromy.

11.16 Connectedness and transitivity

For a connected covering of a path-connected locally path-connected base, the monodromy action on the fiber is transitive. Any two points in the fiber can be joined upstairs by a path; its projection is a loop downstairs whose lift moves the first fiber point to the second.

11.17 Lifting and covering classification

The subgroup

$$H = p_*\pi_1(\tilde{X}, \tilde{x}_0)$$

controls which loops close upstairs and which maps lift. This subgroup is the algebraic fingerprint of the pointed connected covering.

11.18 Bridge to deck transformations

Deck transformations are self-homeomorphisms of the covering space that lie over the identity of the base. Lifting uniqueness makes them rigid: on a connected cover, a deck transformation is completely determined by the image of one point. VIII/12 develops the resulting group action.

11.19 Exercises

Exercise 11.10. Define a lift of $f : Y \rightarrow X$.

Hint. Make the triangle commute. □

Solution. A map $\tilde{f} : Y \rightarrow \tilde{X}$ with $p\tilde{f} = f$. □

Exercise 11.11. What extra datum is needed to lift a path uniquely?

Hint. Choose a starting point upstairs. □

Solution. A point $\tilde{x}_0 \in p^{-1}(\alpha(0))$. □

Exercise 11.12. State path-lift uniqueness.

Hint. Same path and same initial lift point. □

Solution. There is exactly one lifted path with the chosen starting point. □

Exercise 11.13. How is path lifting proved locally?

Hint. Use evenly covered neighborhoods. ☐

Solution. Subdivide the parameter interval and continue the lift sheet by sheet. ☐

Exercise 11.14. When are two lifts of one map equal on a connected domain?

Hint. They need agree once. ☐

Solution. If they agree at one point, uniqueness implies they agree everywhere. ☐

Exercise 11.15. State homotopy lifting.

Hint. Lift the initial time slice first. ☐

Solution. A homotopy $H : Y \times I \rightarrow X$ has a unique lift once a lift of $H(-, 0)$ is fixed. ☐

Exercise 11.16. What happens to lifted endpoints of endpoint-homotopic paths?

Hint. Lift the path homotopy. ☐

Solution. Starting at the same point, the lifted paths end at the same point. ☐

Exercise 11.17. When does a loop lift to a loop at $\tilde{x}_0$?

Hint. Use the covering subgroup. ☐

Solution. When its class lies in $p_*\pi_1(\tilde{X}, \tilde{x}_0)$. ☐

Exercise 11.18. What does a covering induce on fundamental groups?

Hint. Use lifted null-homotopies. ☐

Solution. An injective homomorphism. ☐

Exercise 11.19. State the general lifting criterion.

Hint. Compare subgroup images. ☐

Solution. A based lift exists iff $f_*\pi_1(Y) \subset p_*\pi_1(\tilde{X})$, under the standard path/local hypotheses. ☐

Exercise 11.20. How is a lift constructed from the criterion?

Hint. Choose a path from the basepoint. ☐

Solution. Lift $f \circ \gamma$ from $\tilde{x}_0$ and define $\tilde{f}(y)$ as its endpoint. ☐

Exercise 11.21. Why is the constructed value independent of path?

Hint. Compare two paths via a loop. ☐

Solution. Their difference loop maps into the covering subgroup, so its lift closes. ☐

Exercise 11.22. What is the covering subgroup of a universal cover?

Hint. The total space is simply connected. ☐

Solution. The trivial subgroup. ☐

Exercise 11.23. When does a map lift to a universal cover?

Hint. Use the trivial subgroup criterion. □

Solution. Exactly when $f_*\pi_1(Y) = 0$. □

Exercise 11.24. Why does a simply connected domain lift?

Hint. Its π_1 is trivial. □

Solution. The subgroup criterion is automatically satisfied. □

Exercise 11.25. What is a lift to $\mathbb{R}$ of a map into S^1 ?

Hint. Use the exponential cover. □

Solution. A continuous real-valued argument. □

Exercise 11.26. How is a continuous n th root a lifting problem?

Hint. Use $z \mapsto z^n$. □

Solution. It is a lift through the n -sheeted power covering. □

Exercise 11.27. When does a degree- m circle map have an n th-root lift?

Hint. Compare $m\mathbb{Z}$ with $n\mathbb{Z}$. □

Solution. Exactly when $n \mid m$. □

Exercise 11.28. What is monodromy?

Hint. Lift loops from points of the fiber. □

Solution. The permutation action of $\pi_1(X, x_0)$ on $p^{-1}(x_0)$ by lifted endpoints. □

Exercise 11.29. Why does monodromy depend only on loop homotopy class?

Hint. Use homotopy lifting rel endpoints. □

Solution. Homotopic loops have lifts with the same endpoints. □

Exercise 11.30. When is monodromy transitive?

Hint. Use connectedness upstairs. □

Solution. For a connected cover over a path-connected locally path-connected base. □

Exercise 11.31. What subgroup records a pointed connected covering?

Hint. Image of upstairs π_1 . □

Solution. $p_*\pi_1(\tilde{X}, \tilde{x}_0) \leq \pi_1(X, x_0)$. □

Exercise 11.32. What does this subgroup tell about loops?

Hint. Use the closed-lift criterion. □

Solution. It is exactly the set of loop classes whose lifts from $\tilde{x}_0$ close. □

Exercise 11.33. What is the bridge to VIII/12?

Hint. Use lifting uniqueness for self-lifts. □

Solution. Deck transformations are lifts of the identity and are determined by one point on a connected cover. □

11.20 Solved problem dossiers

Problem 11.34 (Lift a circle loop). Lift $\alpha(t) = e^{2\pi i n t}$ through $\mathbb{R} \rightarrow S^1$ starting at 0.

Solution. The unique lift is $\tilde{\alpha}(t) = nt$. □

Problem 11.35 (Lift from another sheet). Lift the same loop starting at $k \in \mathbb{Z}$.

Solution. The unique lift is $\tilde{\alpha}(t) = k + nt$, ending at $k + n$. □

Problem 11.36 (Closed lift test on the universal circle cover). Which loops in S^1 lift to loops in $\mathbb{R}$?

Solution. Only degree-zero loops. The covering subgroup is trivial, so the lift closes exactly for the zero class in $\mathbb{Z}$. □

Problem 11.37 (Finite circle cover criterion). For $p_n(z) = z^n$, which loop classes in $\pi_1(S^1) = \mathbb{Z}$ lift to loops?

Solution. The subgroup is $n\mathbb{Z}$, so exactly multiples of n lift closed. □

Problem 11.38 (Injection proof). Prove p_* is injective.

Solution. If an upstairs loop projects to a null-homotopic loop, lift a null-homotopy starting with the upstairs loop. The opposite edge is the constant lift, showing the original loop is null-homotopic. □

Problem 11.39 (Uniqueness of map lifts). Why do two lifts agreeing at one point agree everywhere on a connected domain?

Solution. Locally, both lifts must lie in the same sheet and equal the inverse sheet map applied to f , so the agreement set is open; its complement is also open. Connectedness forces global agreement. □

Problem 11.40 (Homotopy endpoint invariant). Why do homotopic based loops induce the same endpoint permutation of a fiber?

Solution. Lift the based homotopy from a chosen fiber point. The boundary of the lifted square forces the two lifted loops to have the same endpoint. □

Problem 11.41 (Continuous argument on an interval). Why does every map $f : I \rightarrow S^1$ admit a continuous real argument after choosing the initial value?

Solution. The interval is simply connected, so the lifting criterion to $\mathbb{R} \rightarrow S^1$ is automatic. □

Problem 11.42 (No global argument on S^1). Why does the identity map $S^1 \rightarrow S^1$ not lift to $\mathbb{R}$?

Solution. Its induced map on π_1 is the identity $\mathbb{Z} \rightarrow \mathbb{Z}$, not contained in the trivial covering subgroup. □

Problem 11.43 (Square root of a circle map). When does $f : S^1 \rightarrow S^1$ admit a continuous square root?

Solution. A lift through $z \mapsto z^2$ exists iff $\deg f \in 2\mathbb{Z}$, so exactly when the degree is even. □

Problem 11.44 (Cube root). When does a degree-8 circle map admit a continuous cube root?

Solution. Never, because $3 \nmid 8$. □

Problem 11.45 (Fourth root). When does a degree-12 circle map admit a continuous fourth root?

Solution. Yes, because $4 \mid 12$. □

Problem 11.46 (Lift from simply connected domain). Let $Y = D^2$ and $f : D^2 \rightarrow X$. Why does f lift to any cover after choosing the initial lift point?

Solution. $\pi_1(D^2) = 0$, so the lifting subgroup criterion is automatically satisfied. □

Problem 11.47 (Universal-cover lift). Let $f : Y \rightarrow X$ induce the trivial map on π_1 . What can be said about a lift to a universal cover?

Solution. Under the standard path-connected/local hypotheses and a chosen basepoint lift, a unique based lift exists. □

Problem 11.48 (Path-independence calculation). Explain explicitly how two paths γ_0, γ_1 from y_0 to y are compared in the lifting criterion proof.

Solution. Form the loop $\gamma_0 * \overline{\gamma_1}$. Its image class lies in the covering subgroup, so its lift closes. Hence the lifted endpoints of $f\gamma_0$ and $f\gamma_1$ coincide. □

Problem 11.49 (Monodromy of $z \mapsto z^n$). Describe the action of the generator of $\pi_1(S^1)$ on the n -point fiber over 1.

Solution. Label the roots by $0, \dots, n-1$. Lifting one positive turn sends label k to $k+1 \pmod{n}$, an n -cycle. □

Problem 11.50 (Transitivity). Why is that monodromy action transitive?

Solution. The generator acts by one n -cycle, so any fiber point can be reached from any other. □

Problem 11.51 (Disconnected trivial cover monodromy). Describe monodromy for the trivial cover $X \times F \rightarrow X$.

Solution. Every lifted loop stays in the same global sheet, so the action on F is trivial and therefore not transitive when $|F| > 1$. □

Problem 11.52 (Connectedness test preview). What does transitivity of monodromy suggest about a cover?

Solution. For standard path-connected locally path-connected bases, transitive monodromy corresponds to connectedness of the covering space. □

Problem 11.53 (Lift of identity). What are lifts of $\text{id}_X : X \rightarrow X$ through $p : \tilde{X} \rightarrow X$?

Solution. They are sections $s : X \rightarrow \tilde{X}$ satisfying $ps = \text{id}$. □

Problem 11.54 (Deck transformation as a lift). Why is a deck transformation a lift of $p : \tilde{X} \rightarrow X$ through itself?

Solution. If h is a deck transformation, $ph = p$. Thus $h : \tilde{X} \rightarrow \tilde{X}$ is a lift of the map p to the covering p . □

Problem 11.55 (Bridge to deck rigidity). Why is a deck transformation on a connected cover determined by one point?

Solution. Two deck transformations are lifts of the same map p ; if they agree at one point, uniqueness of lifts on the connected domain forces them to agree everywhere. □

Chapter 12

Deck Transformations and Group Actions

A covering space has internal symmetries that preserve every point of the base. These symmetries are deck transformations. On connected covers they are extremely rigid: one point determines the whole transformation. Their group acts on fibers, regular coverings are characterized by transitivity of this action, and universal covers recover the base fundamental group as a deck group.

12.1 Deck transformations

Definition 12.1 (Deck transformation). Let

$$p : \tilde{X} \rightarrow X$$

be a covering. A deck transformation is a homeomorphism

$$h : \tilde{X} \rightarrow \tilde{X}$$

such that

$$p \circ h = p.$$

The deck transformations form a group under composition:

$$\text{Deck}(p).$$

12.2 Deck maps preserve fibers

Proposition 12.2 (Fiber preservation). *For every $x \in X$,*

$$h(p^{-1}(x)) = p^{-1}(x)$$

for each deck transformation h .

Thus the deck group acts by permutations on every fiber.

12.3 Rigidity on connected covers

Theorem 12.3 (One point determines a deck transformation). *If $\tilde{X}$ is connected, two deck transformations that agree at one point are equal everywhere.*

Proof. Both maps are lifts of the same covering map

$$p : \tilde{X} \rightarrow X$$

through the covering p . Apply uniqueness of lifts from VIII/11. $\square$

Corollary 12.4 (Free action). *If a deck transformation of a connected covering fixes one point, it is the identity. Hence the deck group acts freely on $\tilde{X}$.*

12.4 The universal circle cover

For

$$p : \mathbb{R} \rightarrow S^1, \quad p(t) = e^{2\pi it},$$

the translations

$$T_n(t) = t + n, \quad n \in \mathbb{Z},$$

are deck transformations.

Theorem 12.5 (Deck group of the circle cover).

$$\text{Deck}(\mathbb{R} \rightarrow S^1) \cong \mathbb{Z}.$$

12.5 Finite circle covers

For

$$p_n : S^1 \rightarrow S^1, \quad p_n(z) = z^n,$$

the rotations

$$R_k(z) = e^{2\pi ik/n} z$$

are deck transformations.

Theorem 12.6 (Deck group of z^n).

$$\text{Deck}(p_n) \cong \mathbb{Z}/n.$$

12.6 Antipodal covering

For $n \geq 2$,

$$q : S^n \rightarrow \mathbb{R}P^n$$

has the nontrivial deck transformation

$$a(x) = -x.$$

Proposition 12.7 (Projective deck group).

$$\text{Deck}(S^n \rightarrow \mathbb{R}P^n) \cong \mathbb{Z}/2.$$

12.7 Deck action on a fiber

Fix $x_0 \in X$. Restriction gives an action

$$\text{Deck}(p) \curvearrowright p^{-1}(x_0).$$

Because the action is free for connected covers, no nonidentity deck transformation fixes a fiber point.

12.8 Regular coverings

Definition 12.8 (Regular covering). A connected covering is regular, normal, or Galois if the deck group acts transitively on one, equivalently every, fiber.

Theorem 12.9 (Normal-subgroup criterion). *Under the standard covering-classification hypotheses, a connected pointed cover corresponding to*

$$H \leq \pi_1(X, x_0)$$

is regular if and only if

$$H \trianglelefteq \pi_1(X, x_0).$$

12.9 Deck group as a quotient

Theorem 12.10 (Deck group of a regular cover). *If a connected covering corresponds to a normal subgroup*

$$H \trianglelefteq G = \pi_1(X, x_0),$$

then

$$\text{Deck}(p) \cong G/H.$$

For a universal cover $H = 1$, so

$$\text{Deck}(\tilde{X} \rightarrow X) \cong \pi_1(X, x_0)$$

under the standard hypotheses.

12.10 Universal covers and group actions

If $\tilde{X}$ is a universal cover, the fundamental group acts on $\tilde{X}$ by deck transformations. The quotient by this action recovers the base:

$$X \cong \tilde{X}/\pi_1(X).$$

This is the geometric realization of the fundamental group as a symmetry group of the universal cover.

12.11 Properly discontinuous actions

A group G acting on a space $\tilde{X}$ is properly discontinuous in the covering-space sense when every point has a neighborhood whose distinct translates do not overlap in a way that destroys the local quotient model.

Theorem 12.11 (Quotient construction). *If G acts freely and properly discontinuously on a suitable space $\tilde{X}$, then*

$$\tilde{X} \rightarrow \tilde{X}/G$$

is a covering map, and G embeds into its deck group.

Under standard connectedness hypotheses and an effective action realizing all deck symmetries, the deck group is exactly G .

12.12 Translations on the line

The action

$$\mathbb{Z} \curvearrowright \mathbb{R}, \quad n \cdot t = t + n,$$

is free and properly discontinuous. Its quotient is

$$\mathbb{R}/\mathbb{Z} \cong S^1.$$

12.13 Lattice actions

For a lattice

$$\Lambda \cong \mathbb{Z}^n$$

acting by translations on $\mathbb{R}^n$,

$$\mathbb{R}^n/\Lambda$$

is an n -torus, and

$$\mathbb{R}^n \rightarrow \mathbb{R}^n/\Lambda$$

is a universal covering map with deck group Λ .

In particular,

$$\text{Deck}(\mathbb{R}^2 \rightarrow T^2) \cong \mathbb{Z}^2.$$

12.14 Monodromy versus deck action

Monodromy is the action

$$\pi_1(X, x_0) \curvearrowright p^{-1}(x_0)$$

obtained by lifting loops. The deck group also acts on the same fiber. These actions are related but not identical in arbitrary covers.

For a regular cover, the deck action is transitive and can be identified with a quotient of the fundamental group.

12.15 The normalizer formula

Let a connected pointed cover correspond to

$$H \leq G = \pi_1(X, x_0).$$

Theorem 12.12 (Normalizer formula). *Under the standard classification hypotheses,*

$$\text{Deck}(p) \cong N_G(H)/H,$$

where

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

If H is normal, then $N_G(H) = G$, recovering the regular-cover quotient formula.

12.16 Deck transformations and basepoint choices

A deck transformation can move the chosen point

$$\tilde{x}_0 \in p^{-1}(x_0)$$

to another point of the same fiber. Changing the pointed cover in this way conjugates the associated subgroup. This explains why unpointed connected coverings correspond to conjugacy classes of subgroups.

12.17 Automorphisms of coverings

Deck transformations are precisely automorphisms of the covering object over the fixed base X . Thus

$$\text{Deck}(p) = \text{Aut}_X(\tilde{X}).$$

12.18 Regularity examples

The universal circle cover is regular because the trivial subgroup of $\mathbb{Z}$ is normal. The n -fold circle cover corresponds to $n\mathbb{Z} \trianglelefteq \mathbb{Z}$, so it is regular with deck group

$$\mathbb{Z}/n.$$

The universal projective cover corresponds to the trivial subgroup of $\mathbb{Z}/2$, hence is regular.

12.19 Bridge to $SU(2) \rightarrow SO(3)$

The next chapter studies one of the most important geometric two-sheeted covers:

$$SU(2) \rightarrow SO(3).$$

Its kernel is

$$\{\pm I\},$$

and the nontrivial deck transformation is multiplication by $-I$. This provides a Lie-group realization of the same covering and quotient principles developed here.

12.20 Exercises

Exercise 12.13. Define a deck transformation.

Hint. Use a homeomorphism over the identity of the base. □

Solution. A homeomorphism $h : \tilde{X} \rightarrow \tilde{X}$ satisfying $ph = p$. □

Exercise 12.14. Why does a deck transformation preserve each fiber?

Hint. Apply p after h . □

Solution. If $p(\tilde{x}) = x$, then $p(h(\tilde{x})) = p(\tilde{x}) = x$. □

Exercise 12.15. What group do deck transformations form?

Hint. Use composition. □

Solution. $\text{Deck}(p)$. □

Exercise 12.16. On a connected cover, how much data determines a deck transformation?

Hint. Use lift uniqueness. □

Solution. Its value at one point. □

Exercise 12.17. What happens if a deck transformation fixes one point of a connected cover?

Hint. Compare with the identity deck map. ☐

Solution. It is the identity. ☐

Exercise 12.18. What deck transformations act on $\mathbb{R} \rightarrow S^1$?

Hint. Use integer translations. ☐

Solution. $T_n(t) = t + n$. ☐

Exercise 12.19. What is the deck group of $\mathbb{R} \rightarrow S^1$?

Hint. Translations are indexed by integers. ☐

Solution. $\mathbb{Z}$. ☐

Exercise 12.20. What deck transformations act on $z \mapsto z^n$?

Hint. Rotate by n th roots of unity. ☐

Solution. $R_k(z) = e^{2\pi i k/n} z$. ☐

Exercise 12.21. What is the deck group of $z \mapsto z^n$?

Hint. There are n rotations. ☐

Solution. $\mathbb{Z}/n$. ☐

Exercise 12.22. What is the nontrivial deck map of $S^n \rightarrow \mathbb{R}P^n$?

Hint. Use the antipodal involution. ☐

Solution. $x \mapsto -x$. ☐

Exercise 12.23. What is that deck group?

Hint. Two elements. ☐

Solution. $\mathbb{Z}/2$. ☐

Exercise 12.24. Define a regular covering.

Hint. Use the deck action on a fiber. ☐

Solution. A connected cover whose deck group acts transitively on a fiber. ☐

Exercise 12.25. What subgroup condition characterizes regular covers?

Hint. Use normality. ☐

Solution. The corresponding subgroup H is normal in $\pi_1(X)$. ☐

Exercise 12.26. For a regular cover corresponding to $H \trianglelefteq G$, what is the deck group?

Hint. Take the quotient. ☐

Solution. G/H . ☐

Exercise 12.27. What is the deck group of a universal cover?

Hint. Set $H = 1$. ☐

Solution. It is isomorphic to $\pi_1(X)$ under the standard hypotheses. $\square$

Exercise 12.28. How can a group action produce a covering?

Hint. Require free proper discontinuity. $\square$

Solution. A free properly discontinuous action gives $\tilde{X} \rightarrow \tilde{X}/G$ as a covering under standard hypotheses. $\square$

Exercise 12.29. What quotient does $\mathbb{Z} \curvearrowright \mathbb{R}$ produce?

Hint. Translate by integers. $\square$

Solution. $\mathbb{R}/\mathbb{Z} \cong S^1$. $\square$

Exercise 12.30. What quotient does $\mathbb{Z}^2 \curvearrowright \mathbb{R}^2$ produce?

Hint. Use a lattice. $\square$

Solution. A two-torus. $\square$

Exercise 12.31. What is the deck group of $\mathbb{R}^2 \rightarrow T^2$?

Hint. Use lattice translations. $\square$

Solution. $\mathbb{Z}^2$. $\square$

Exercise 12.32. State the normalizer formula.

Hint. Use $N_G(H)$. $\square$

Solution. $\text{Deck}(p) \cong N_G(H)/H$. $\square$

Exercise 12.33. What happens to the normalizer formula when H is normal?

Hint. Then $N_G(H) = G$. $\square$

Solution. It reduces to G/H . $\square$

Exercise 12.34. Why do unpointed connected covers correspond to conjugacy classes?

Hint. Change the chosen lift basepoint. $\square$

Solution. Moving the basepoint in a fiber conjugates the associated subgroup. $\square$

Exercise 12.35. How can deck transformations be described categorically?

Hint. Automorphisms over the fixed base. $\square$

Solution. $\text{Deck}(p) = \text{Aut}_X(\tilde{X})$. $\square$

Exercise 12.36. What is the bridge to VIII/13?

Hint. Use the two-sheeted Lie-group cover. $\square$

Solution. VIII/13 studies $SU(2) \rightarrow SO(3)$ with deck group $\{\pm I\} \cong \mathbb{Z}/2$. $\square$

12.21 Solved problem dossiers

Problem 12.37 (Circle translations). Prove $T_n(t) = t + n$ is a deck transformation of $\mathbb{R} \rightarrow S^1$.

Solution. $p(T_n(t)) = e^{2\pi i(t+n)} = e^{2\pi it} = p(t)$, and T_n is a homeomorphism. $\square$

Problem 12.38 (All circle deck maps). Show every deck transformation of $\mathbb{R} \rightarrow S^1$ is an integer translation.

Solution. A deck map sends 0 to a point in the fiber $\mathbb{Z}$, say n . The translation T_n is another deck map with the same value at 0; connected-cover rigidity gives equality. $\square$

Problem 12.39 (Finite circle rotations). Verify $R_k(z) = e^{2\pi i k/n} z$ is a deck map of $p_n(z) = z^n$.

Solution. $p_n(R_k(z)) = e^{2\pi i k} z^n = z^n = p_n(z)$. $\square$

Problem 12.40 (All finite circle deck maps). Why are there exactly n deck transformations of $z \mapsto z^n$?

Solution. A deck transformation is determined by the image of 1, which can be any of the n points in the fiber over 1. Each choice is realized by one root-of-unity rotation. $\square$

Problem 12.41 (Antipodal deck map). Verify $a(x) = -x$ is a deck transformation of $S^n \rightarrow \mathbb{R}P^n$.

Solution. The quotient map sends x and $-x$ to the same projective line, so $qa = q$. $\square$

Problem 12.42 (Projective deck group). Why are identity and antipodal map the only deck transformations of the connected cover $S^n \rightarrow \mathbb{R}P^n$?

Solution. A deck transformation is determined by the image of one point, and the fiber contains only x and $-x$. $\square$

Problem 12.43 (Free action proof). Prove the deck action on a connected covering is free.

Solution. If $h(\tilde{x}) = \tilde{x}$, then h and the identity deck map agree at one point. Lift uniqueness makes them equal. $\square$

Problem 12.44 (Universal-cover quotient). Explain why a universal cover can be viewed as a quotient by its deck group.

Solution. The deck group acts freely and properly discontinuously, and its orbits are fibers for a regular universal cover. The quotient therefore recovers the base. $\square$

Problem 12.45 (Torus lattice cover). Show $\mathbb{R}^2 \rightarrow \mathbb{R}^2/\mathbb{Z}^2$ has deck group $\mathbb{Z}^2$.

Solution. Each lattice translation is a deck map. Any deck map is determined by the image of 0, which must be another point of the fiber $\mathbb{Z}^2$, so it is the corresponding translation. $\square$

Problem 12.46 (Regularity of z^n). Why is the n -fold circle cover regular?

Solution. It corresponds to $n\mathbb{Z} \trianglelefteq \mathbb{Z}$. Equivalently, the root-of-unity deck rotations act transitively on every fiber. $\square$

Problem 12.47 (Quotient deck group). Use subgroup theory to compute the deck group of the cover corresponding to $n\mathbb{Z} \leq \mathbb{Z}$.

Solution. Since $\mathbb{Z}$ is abelian, $n\mathbb{Z}$ is normal and the deck group is $\mathbb{Z}/n\mathbb{Z}$. $\square$

Problem 12.48 (Universal circle quotient). Apply the regular-cover formula to the trivial subgroup of $\mathbb{Z}$.

Solution. The deck group is $\mathbb{Z}/0 \cong \mathbb{Z}$, matching integer translations of $\mathbb{R}$. $\square$

Problem 12.49 (Normalizer example). If $H \leq G$ is self-normalizing, what is the deck group of the corresponding connected cover?

Solution. The normalizer formula gives $N_G(H)/H = H/H$, so the deck group is trivial. $\square$

Problem 12.50 (Normal subgroup example). If $H \trianglelefteq G$ has index 5, what can be said about the regular cover?

Solution. It is five-sheeted and its deck group G/H has five elements. $\square$

Problem 12.51 (Deck versus monodromy). Explain one difference between deck action and monodromy action.

Solution. Monodromy is an action of the base fundamental group on a fiber obtained by lifting loops. Deck action is an action of automorphisms of the entire covering. They coincide only through additional regularity identifications. $\square$

Problem 12.52 (Connected-cover rigidity). Why can a deck transformation not permute one fiber arbitrarily?

Solution. The image of one point determines the whole map, but not every fiber permutation need extend continuously to a deck transformation. Only permutations compatible with the global covering structure do. $\square$

Problem 12.53 (Trivial cover automorphisms). Describe deck transformations of $X \times F \rightarrow X$ when X is connected and F is discrete.

Solution. Any fixed permutation $\sigma \in \text{Sym}(F)$ defines $h(x, a) = (x, \sigma(a))$. Conversely continuity over connected X forces the permutation to be constant, so the deck group is $\text{Sym}(F)$. $\square$

Problem 12.54 (Why connectedness mattered). Why does one-point rigidity fail for a disconnected covering space?

Solution. A deck map can agree with another on one component but differ on another component, so agreement at one point need not control all components. $\square$

Problem 12.55 (Conjugate subgroup viewpoint). How does moving $\tilde{x}_0$ in the fiber change the associated subgroup?

Solution. A path/monodromy element relating the two fiber points conjugates $H = p_*\pi_1(\tilde{X}, \tilde{x}_0)$ inside $G = \pi_1(X, x_0)$. $\square$

Problem 12.56 (Regular cover fiber action). Why does transitivity of the deck group on a fiber imply every fiber point is equivalent as a pointed cover?

Solution. For any two fiber points, a deck transformation carries one to the other while commuting with the projection, giving an isomorphism of the corresponding pointed covering structures. $\square$

Problem 12.57 (Deck group as quotient symmetry). Interpret G/H geometrically for a regular cover.

Solution. Cosets label sheets/fiber positions, and quotient elements act by translating those positions; this algebraic action is realized by deck transformations. $\square$

Problem 12.58 (Proper discontinuity of translations). Why is the integer-translation action on $\mathbb{R}$ properly discontinuous?

Solution. For any t , choose an interval of length less than 1. Its nonzero integer translates are disjoint from it, giving a local quotient sheet. $\square$

Problem 12.59 (Free action necessity). What goes wrong for covering quotients if a group action has a fixed point?

Solution. At a fixed point the quotient map cannot look locally like disjoint homeomorphic sheets indexed by group elements; distinct orbit representatives collapse at the fixed point. $\square$

Problem 12.60 (Bridge to $SU(2)$). What deck symmetry should appear in the cover $SU(2) \rightarrow SO(3)$?

Solution. The kernel is $\{\pm I\}$. Multiplication by $-I$ is the nontrivial deck transformation, so the deck group is $\mathbb{Z}/2$. $\square$

Chapter 13

$SU(2)$ to $SO(3)$

The double cover $SU(2) \rightarrow SO(3)$ is a central example where topology, group structure, and geometry reinforce one another. The group $SU(2)$ is a 3-sphere, while $SO(3)$ is obtained by identifying antipodal points. Unit quaternions provide the most transparent formula for the covering map.

13.1 The group $SU(2)$

Definition 13.1 (Special unitary group).

$$SU(2) = \{A \in M_2(\mathbb{C}) : A^* A = I, \det A = 1\}.$$

Every element has the form

$$A = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Proposition 13.2 (Three-sphere model). *As a topological space,*

$$SU(2) \cong S^3.$$

Indeed $(\alpha, \beta) \in \mathbb{C}^2 \cong \mathbb{R}^4$ lies on the unit 3-sphere.

13.2 Unit quaternions

Write

$$\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}.$$

The unit quaternions

$$S^3 = \{q \in \mathbb{H} : |q| = 1\}$$

form a group under quaternion multiplication and identify with $SU(2)$.

The purely imaginary quaternions

$$\text{Im } \mathbb{H} = \{bi + cj + dk\}$$

are naturally $\mathbb{R}^3$.

13.3 Quaternion conjugation

For a unit quaternion q , define

$$R_q(v) = qvq^{-1}, \quad v \in \operatorname{Im} \mathbb{H}.$$

Theorem 13.3 (Quaternion rotation). *R_q is an orientation-preserving orthogonal transformation of $\mathbb{R}^3$. Hence*

$$R_q \in SO(3).$$

Thus

$$\rho : S^3 \cong SU(2) \longrightarrow SO(3), \quad q \longmapsto R_q$$

is a group homomorphism.

13.4 Kernel

Theorem 13.4 (Kernel of the rotation map).

$$\ker \rho = \{\pm 1\}.$$

Proof. Both 1 and -1 commute with every imaginary quaternion and therefore act trivially. Conversely, a unit quaternion acting trivially by conjugation commutes with all imaginary quaternions, hence lies in the real center of $\mathbb{H}$, so it is ± 1 . $\square$

13.5 Axis-angle formula

Let $u \in \operatorname{Im} \mathbb{H}$ be a unit vector. Put

$$q = \cos \frac{\theta}{2} + u \sin \frac{\theta}{2}.$$

Theorem 13.5 (Half-angle formula). *Conjugation by q rotates $\mathbb{R}^3$ through angle θ about the axis u .*

The half-angle is the geometric source of the double cover.

13.6 Surjectivity

Every rotation in $SO(3)$ has an axis and angle. The half-angle formula therefore gives:

Theorem 13.6 (Surjectivity). *The map*

$$\rho : SU(2) \rightarrow SO(3)$$

is surjective.

Together with the kernel computation,

$$SO(3) \cong SU(2)/\{\pm I\}.$$

13.7 Double covering

Theorem 13.7 (Double cover).

$$\rho : SU(2) \rightarrow SO(3)$$

is a two-sheeted covering map with deck group

$$\{\pm I\} \cong \mathbb{Z}/2.$$

The nontrivial deck transformation is multiplication by $-I$.

13.8 $SO(3)$ as projective 3-space

Since

$$SU(2) \cong S^3$$

and the kernel identifies q with $-q$,

$$SO(3) \cong S^3/\{\pm 1\} \cong \mathbb{R}P^3.$$

Corollary 13.8.

$$SO(3) \cong \mathbb{R}P^3$$

as topological manifolds.

13.9 Fundamental group of $SO(3)$

Because S^3 is simply connected, the double cover is universal.

Theorem 13.9 (Fundamental group).

$$\pi_1(SO(3)) \cong \mathbb{Z}/2.$$

13.10 A 2π rotation loop

A path in $SU(2)$ from 1 to -1 projects to a loop in $SO(3)$. For example

$$q(t) = \cos(\pi t) + u \sin(\pi t)$$

projects to rotation through angle $2\pi t$ about u .

Its lift is not closed, so the projected loop represents the nontrivial element of $\pi_1(SO(3))$.

13.11 A 4π rotation

Traversing twice gives a lift from 1 to 1. Hence the 4π -rotation loop is null-homotopic in $SO(3)$.

This is the topological content of the familiar distinction between 2π and 4π rotation paths.

13.12 Deck symmetry and lifting

A loop in $SO(3)$ lifts from $I \in SU(2)$ either back to I or to $-I$. These two possible endpoints detect the two elements of

$$\pi_1(SO(3)) \cong \mathbb{Z}/2.$$

13.13 Matrix picture

The quaternion formula is equivalent to the adjoint action of $SU(2)$ on its three-dimensional Lie algebra. After identifying the Lie algebra with $\mathbb{R}^3$, the adjoint representation is precisely

$$\text{Ad} : SU(2) \rightarrow SO(3).$$

13.14 Central quotient principle

The covering map is a group quotient by a discrete central subgroup:

$$1 \longrightarrow \{\pm I\} \longrightarrow SU(2) \longrightarrow SO(3) \longrightarrow 1.$$

This is a model for many Lie-group covering constructions.

13.15 Bridge to covering graphs

The $SU(2) \rightarrow SO(3)$ example shows how a simply connected total space carries a deck action realizing the base fundamental group. VIII/14 develops the same principle combinatorially for graphs, where universal covers become trees and fundamental groups become free groups.

13.16 Exercises

Exercise 13.10. Define $SU(2)$.

Hint. Unitary matrices with determinant one. □

Solution. $SU(2) = \{A : A^*A = I, \det A = 1\}$. □

Exercise 13.11. Write the standard form of an $SU(2)$ matrix.

Hint. Use two complex parameters. □

Solution. $\begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}$ with $|\alpha|^2 + |\beta|^2 = 1$. □

Exercise 13.12. Why is $SU(2) \cong S^3$?

Hint. Count real coordinates. □

Solution. $(\alpha, \beta) \in \mathbb{C}^2 \cong \mathbb{R}^4$ lies on the unit sphere. □

Exercise 13.13. What group is another model of $SU(2)$?

Hint. Use quaternions. □

Solution. The unit quaternions. □

Exercise 13.14. Which quaternion subspace is $\mathbb{R}^3$?

Hint. Discard the real part. □

Solution. $\text{Im } \mathbb{H}$. □

Exercise 13.15. Define R_q .

Hint. Conjugate imaginary quaternions. ☐

Solution. $R_q(v) = qvq^{-1}$. ☐

Exercise 13.16. In which group does R_q lie?

Hint. It preserves orientation and norm. ☐

Solution. $SO(3)$. ☐

Exercise 13.17. What is $\ker \rho$?

Hint. Find unit quaternions commuting with all imaginary ones. ☐

Solution. $\{\pm 1\}$. ☐

Exercise 13.18. Why do q and $-q$ give the same rotation?

Hint. The signs cancel in conjugation. ☐

Solution. $(-q)v(-q)^{-1} = qvq^{-1}$. ☐

Exercise 13.19. Write the quaternion for rotation by θ about unit axis u .

Hint. Use half-angle. ☐

Solution. $q = \cos(\theta/2) + u \sin(\theta/2)$. ☐

Exercise 13.20. Why is ρ surjective?

Hint. Every 3D rotation has axis-angle form. ☐

Solution. Choose the half-angle quaternion for that axis and angle. ☐

Exercise 13.21. How many points are in each fiber of ρ ?

Hint. Use the kernel. ☐

Solution. Two. ☐

Exercise 13.22. What is the deck group?

Hint. Use multiplication by kernel elements. ☐

Solution. $\mathbb{Z}/2$. ☐

Exercise 13.23. Identify $SO(3)$ as a quotient of S^3 .

Hint. Identify antipodes. ☐

Solution. $S^3/\{\pm 1\}$. ☐

Exercise 13.24. Identify that quotient as a familiar space.

Hint. Use real projective space. ☐

Solution. $\mathbb{RP}^3$. ☐

Exercise 13.25. What is $\pi_1(SO(3))$?

Hint. The cover is universal. ☐

Solution. $\mathbb{Z}/2$.

□

Exercise 13.26. What does a 2π -rotation loop lift to?

Hint. Start at 1.

□

Solution. A path ending at -1 .

□

Exercise 13.27. Is the 2π -rotation loop null-homotopic?

Hint. Its lift is not closed.

□

Solution. No.

□

Exercise 13.28. What does a 4π -rotation loop lift to?

Hint. Traverse the nontrivial loop twice.

□

Solution. A loop from 1 back to 1.

□

Exercise 13.29. What is the homotopy class of the 4π -rotation loop?

Hint. Use the order-two fundamental group.

□

Solution. It is trivial.

□

Exercise 13.30. What representation gives the matrix version of ρ ?

Hint. Use conjugation on the Lie algebra.

□

Solution. The adjoint representation $\text{Ad} : SU(2) \rightarrow SO(3)$.

□

Exercise 13.31. Write the short exact sequence.

Hint. Kernel, source, quotient.

□

Solution. $1 \rightarrow \{\pm I\} \rightarrow SU(2) \rightarrow SO(3) \rightarrow 1$.

□

Exercise 13.32. Why is the kernel central?

Hint. $\pm I$ commute with every matrix.

□

Solution. Both scalar matrices lie in the center of $SU(2)$.

□

Exercise 13.33. What is the bridge to VIII/14?

Hint. Compare universal covers.

□

Solution. Graphs have simply connected universal covers that are trees, with free groups acting as deck groups.

□

13.17 Solved problem dossiers

Problem 13.34 (Three-sphere parametrization). Verify the usual two-complex-parameter form of $SU(2)$.

Solution. Unitarity forces the second column to be a unit orthogonal vector to the first, hence $(-\bar{\beta}, \bar{\alpha})^T$ up to phase; determinant one fixes the phase. The norm condition is $|\alpha|^2 + |\beta|^2 = 1$. $\square$

Problem 13.35 (Quaternion norm preservation). Why does $v \mapsto qvq^{-1}$ preserve Euclidean norm on imaginary quaternions?

Solution. Quaternion norm is multiplicative, so $|qvq^{-1}| = |q||v||q|^{-1} = |v|$. Conjugation also preserves the imaginary subspace. $\square$

Problem 13.36 (Orientation). Why does quaternion conjugation land in $SO(3)$, not merely $O(3)$?

Solution. The map depends continuously on $q \in S^3$, and at $q = 1$ it has determinant 1. Since determinant takes values ± 1 and S^3 is connected, the determinant is always 1. $\square$

Problem 13.37 (Kernel computation). Prove the kernel is exactly $\{\pm 1\}$.

Solution. An element in the kernel commutes with i, j, k , hence with all quaternions. The center of $\mathbb{H}$ is $\mathbb{R}$; a real unit quaternion is ± 1 . $\square$

Problem 13.38 (Half-angle check). Why does a 2π rotation correspond to the quaternion -1 ?

Solution. Set $\theta = 2\pi$: $q = \cos \pi + u \sin \pi = -1$. $\square$

Problem 13.39 (Double cover). Why does each rotation have exactly two quaternion lifts?

Solution. Surjectivity gives at least one lift q . The fiber is the coset $q \ker \rho = \{q, -q\}$. $\square$

Problem 13.40 (Projective identification). Derive $SO(3) \cong \mathbb{R}P^3$.

Solution. Use $SU(2) \cong S^3$ and quotient by the central kernel $\{\pm 1\}$, which is exactly the antipodal equivalence relation defining $\mathbb{R}P^3$. $\square$

Problem 13.41 (Universal cover). Why is $SU(2) \rightarrow SO(3)$ universal?

Solution. $SU(2) \cong S^3$ and S^3 is simply connected. A covering with simply connected total space is universal. $\square$

Problem 13.42 (Fundamental group). Recover $\pi_1(SO(3))$ from deck transformations.

Solution. For a universal cover, the deck group is isomorphic to the base fundamental group. The deck group here is $\{\pm I\} \cong \mathbb{Z}/2$. $\square$

Problem 13.43 (Two-pi loop). Construct a loop in $SO(3)$ representing the nontrivial element.

Solution. Fix an axis u and project $q(t) = \cos(\pi t) + u \sin(\pi t)$. The endpoints $1, -1$ project to the same rotation, so the projection is a loop whose lift is not closed. $\square$

Problem 13.44 (Four-pi loop). Explain why a 4π rotation path is null-homotopic.

Solution. It represents twice the generator of $\mathbb{Z}/2$, hence the identity. Equivalently its lift to S^3 is closed, and S^3 is simply connected. $\square$

Problem 13.45 (Fiber endpoint test). How can one decide which element of $\pi_1(SO(3))$ a loop represents?

Solution. Lift it from $I \in SU(2)$. If the lift ends at I , the class is trivial; if it ends at $-I$, it is the nontrivial class. $\square$

Problem 13.46 (Axis reversal). Compare axis-angle data (u, θ) and $(-u, -\theta)$.

Solution. They give the same quaternion $\cos(\theta/2) + u \sin(\theta/2)$, hence the same rotation. $\square$

Problem 13.47 (Angle $2\pi - \theta$). Explain why distinct quaternion signs can encode the same rotation.

Solution. Changing q to $-q$ alters the quaternion representative but not conjugation. Axis-angle parametrizations reflect this double-valuedness at the group level. $\square$

Problem 13.48 (Central quotient). Why is $SU(2)/\{\pm I\}$ naturally a group?

Solution. $\{\pm I\}$ is a normal central subgroup, so the quotient group is defined. The homomorphism theorem identifies it with $SO(3)$. $\square$

Problem 13.49 (Deck action). Describe the nontrivial deck transformation explicitly.

Solution. It is $q \mapsto -q$, or matrix multiplication $A \mapsto -A$. It preserves every $SO(3)$ image. $\square$

Problem 13.50 (Quaternion example). What rotation is represented by $q = \cos(\pi/4) + k \sin(\pi/4)$?

Solution. The half-angle is $\pi/4$, so the rotation angle is $\pi/2$ about the k -axis, i.e. the z -axis. $\square$

Problem 13.51 (Quaternion i). What rotation does $q = i$ represent?

Solution. Write $i = \cos(\pi/2) + i \sin(\pi/2)$; it represents rotation by π about the x -axis. $\square$

Problem 13.52 (Adjoint viewpoint). Why does the adjoint action have three real dimensions?

Solution. The Lie algebra $\mathfrak{su}(2)$ is three-dimensional over $\mathbb{R}$, and its invariant inner product identifies the adjoint action with orthogonal rotations of $\mathbb{R}^3$. $\square$

Problem 13.53 (Bridge to graph covers). State the common pattern between $SU(2) \rightarrow SO(3)$ and a graph universal cover.

Solution. A simply connected total space carries a free deck action by the base fundamental group, and the quotient by that action recovers the base. $\square$

Chapter 14

Free Groups and Covering Graphs

Graphs are the combinatorial laboratory of covering-space theory. Their fundamental groups are free, their universal covers are trees, and connected covering graphs correspond to subgroups of free groups. This chapter develops the algebra of reduced words together with the geometry of graph covers.

14.1 Free groups

Definition 14.1 (Free group). Let S be a set. The free group $F(S)$ is the group of reduced finite words in symbols

$$s^{\pm 1}, \quad s \in S,$$

with multiplication given by concatenation followed by cancellation of adjacent inverse pairs.

If $S = \{x_1, \dots, x_r\}$, write

$$F_r = F(x_1, \dots, x_r).$$

14.2 Reduced words

A word is reduced if it contains no adjacent subword

$$xx^{-1} \quad \text{or} \quad x^{-1}x.$$

Proposition 14.2 (Unique reduced form). *Every element of a free group has a unique reduced-word representative.*

This normal form is the basis for concrete computations.

14.3 Universal property

Theorem 14.3 (Universal property of a free group). *Given a set map*

$$f : S \rightarrow G$$

into a group G , there exists a unique homomorphism

$$\bar{f} : F(S) \rightarrow G$$

extending f .

Thus a homomorphism out of a free group is completely determined by the images of its free generators.

14.4 Graphs as one-dimensional CW complexes

A connected graph X is a one-dimensional CW complex. Choose a base vertex v_0 . Edge paths determine loops and homotopies can be simplified combinatorially by canceling immediate backtracking.

14.5 Spanning trees

Definition 14.4 (Spanning tree). A spanning tree $T \subset X$ of a connected graph is a connected subgraph containing every vertex and having no cycles.

A finite connected graph always has a spanning tree.

14.6 Collapsing a spanning tree

Theorem 14.5 (Graph collapse). *If T is a spanning tree of a connected graph X , then the quotient*

$$X/T$$

is a wedge of circles, one for each edge of $X \setminus T$, and

$$X \simeq X/T.$$

Hence graph fundamental groups are free.

14.7 Rank formula

For a finite connected graph with V vertices and E edges, a spanning tree has $V - 1$ edges. Therefore the number of edges outside the tree is

$$E - (V - 1) = E - V + 1.$$

Theorem 14.6 (Fundamental group of a finite connected graph).

$$\pi_1(X) \cong F_{E-V+1}.$$

14.8 Wedges of circles

For the bouquet

$$\bigvee_{j=1}^r S^1,$$

one obtains

$$\pi_1 \left(\bigvee_{j=1}^r S^1 \right) \cong F_r.$$

Each circle contributes one free generator.

14.9 Trees

Definition 14.7 (Tree). A tree is a connected graph containing no nontrivial cycles.

Theorem 14.8 (Trees are simply connected). *Every tree is contractible and therefore simply connected.*

A finite tree collapses edge by edge to a vertex. Infinite trees admit the same path-uniqueness intuition and are simply connected as graphs.

14.10 Universal covers of graphs

Theorem 14.9 (Universal covering graph). *The universal cover of a connected graph is a tree.*

Conversely, a tree with a free properly discontinuous graph automorphism action gives a graph quotient whose universal cover is that tree.

14.11 Cayley graphs

For a group G generated by a set S , the Cayley graph has vertex set G and directed edges

$$g \longrightarrow gs$$

for $s \in S$.

Example 14.10 (Cayley tree of a free group). The Cayley graph of F_r with respect to its free generators is a regular tree of valence $2r$.

14.12 Deck action of a free group

The universal cover of the wedge of r circles is the Cayley tree of F_r . The group acts by left multiplication on vertices and edges.

Theorem 14.11 (Free-group deck action).

$$F_r \cong \text{Deck}(\tilde{X} \rightarrow X)$$

for the universal cover of the r -circle wedge.

14.13 Covering graphs

A graph map

$$p : \tilde{X} \rightarrow X$$

is a covering graph when the incident edge pattern around every vertex maps bijectively to the incident edge pattern around its image. This is the combinatorial form of an evenly covered neighborhood.

14.14 Subgroups and connected graph covers

For a connected based graph X , pointed connected covers correspond to subgroups

$$H \leq \pi_1(X).$$

Since $\pi_1(X)$ is free, graph coverings give a geometric realization of subgroups of free groups.

14.15 Schreier graphs

Let F_r act on the right cosets of a subgroup H . The associated Schreier graph has vertices

$$Hg, \quad g \in F_r,$$

and edges labeled by free generators.

This graph realizes the covering corresponding to H .

14.16 Nielsen–Schreier theorem

Theorem 14.12 (Nielsen–Schreier). *Every subgroup of a free group is free.*

A covering-space proof is immediate in spirit: a subgroup determines a connected covering graph, and the fundamental group of that graph is free.

14.17 Finite-index rank formula

Theorem 14.13 (Schreier index formula). *If*

$$H \leq F_r$$

has finite index d , then

$$\text{rank}(H) = 1 + d(r - 1).$$

This can be seen by counting vertices and edges in the finite covering graph.

14.18 Regular graph covers

A covering graph corresponding to

$$H \leq F_r$$

is regular exactly when H is normal. Then the deck group is

$$F_r/H.$$

14.19 Core graphs

For finitely generated subgroups of free groups, one can prune hanging trees from a covering graph and retain a finite core containing every reduced loop. This gives a compact combinatorial model of the subgroup.

14.20 Edge paths and reduced words

In a rose with labeled oriented edges

$$x_1, \dots, x_r,$$

a based edge loop reads a word in the symbols $x_i^{\pm 1}$. Removing immediate backtracking corresponds exactly to free reduction.

Thus graph homotopy and free-group reduction are two versions of the same operation.

14.21 Bridge to simplicial complexes

Graphs are one-dimensional simplicial complexes after subdividing loops and multiple edges if necessary. VIII/15 generalizes the vertex-edge language to higher-dimensional simplices and prepares the chain complexes used in homology.

14.22 Exercises

Exercise 14.14. Define $F(S)$.

Hint. Use reduced words. ☐

Solution. The group of reduced finite words in $s^{\pm 1}$ for $s \in S$. ☐

Exercise 14.15. What is a reduced word?

Hint. No adjacent inverse pair. ☐

Solution. A word containing neither xx^{-1} nor $x^{-1}x$ adjacently. ☐

Exercise 14.16. What is F_r ?

Hint. Choose r free generators. ☐

Solution. The free group on r generators. ☐

Exercise 14.17. State the universal property of $F(S)$.

Hint. Extend any set map to a homomorphism. ☐

Solution. Every map $S \rightarrow G$ extends uniquely to $F(S) \rightarrow G$. ☐

Exercise 14.18. Define a spanning tree.

Hint. All vertices, connected, no cycles. ☐

Solution. A connected cycle-free subgraph containing every vertex. ☐

Exercise 14.19. How many edges does a finite spanning tree on V vertices have?

Hint. Use the tree formula. ☐

Solution. $V - 1$. ☐

Exercise 14.20. What is X/T for a spanning tree T ?

Hint. Collapse the tree. ☐

Solution. A wedge of circles indexed by edges outside T . ☐

Exercise 14.21. Give the rank of $\pi_1(X)$ for a finite connected graph.

Hint. Count non-tree edges. ☐

Solution. $E - V + 1$. ☐

Exercise 14.22. What is $\pi_1(\bigvee^r S^1)$?

Hint. Each circle gives one generator. ☐

Solution. F_r . □

Exercise 14.23. Define a tree.

Hint. Connected graph with no cycles. □

Solution. A connected cycle-free graph. □

Exercise 14.24. What is π_1 of a tree?

Hint. Trees are contractible. □

Solution. Trivial. □

Exercise 14.25. What is the universal cover of a connected graph?

Hint. Use simple connectivity. □

Solution. A tree. □

Exercise 14.26. What is the Cayley graph vertex set of G ?

Hint. One vertex per group element. □

Solution. G . □

Exercise 14.27. What valence has the standard Cayley graph of F_r ?

Hint. Each generator and inverse gives an edge direction. □

Solution. $2r$. □

Exercise 14.28. What group acts by deck transformations on the universal cover of the r -rose?

Hint. Use the fundamental group. □

Solution. F_r . □

Exercise 14.29. What local condition defines a covering graph?

Hint. Incident edges must match bijectively. □

Solution. At every vertex, the star maps bijectively to the star of its image. □

Exercise 14.30. What algebraic objects classify pointed connected graph covers?

Hint. Use covering classification. □

Solution. Subgroups of the graph fundamental group. □

Exercise 14.31. What is a Schreier graph?

Hint. Use cosets of a subgroup. □

Solution. A labeled graph whose vertices are cosets Hg and whose edges encode multiplication by free generators. □

Exercise 14.32. State Nielsen–Schreier.

Hint. Subgroups preserve freeness. □

Solution. Every subgroup of a free group is free. □

Exercise 14.33. State Schreier's finite-index rank formula.

Hint. Use index d . □

Solution. $\text{rank}(H) = 1 + d(r - 1)$. □

Exercise 14.34. When is a connected graph cover regular?

Hint. Use subgroup normality. □

Solution. When the corresponding subgroup is normal. □

Exercise 14.35. What is its deck group in the normal case?

Hint. Take quotient. □

Solution. F_r/H . □

Exercise 14.36. What does free reduction correspond to geometrically?

Hint. Cancel immediate edge backtracking. □

Solution. Removing an edge immediately followed by the same edge in reverse. □

Exercise 14.37. What is the bridge to VIII/15?

Hint. Graphs are 1-dimensional simplicial objects. □

Solution. VIII/15 generalizes vertices and edges to higher-dimensional simplices. □

14.23 Solved problem dossiers

Problem 14.38 (Reduce a word). Reduce $aba^{-1}ab^{-1}b$ in the free group $F(a, b)$.

Solution. The terminal $b^{-1}b$ cancels, leaving $aba^{-1}a$. No further adjacent inverse pair occurs. □

Problem 14.39 (Free universal property). Define a homomorphism $F(a, b) \rightarrow \mathbb{Z}$ by $a \mapsto 2$, $b \mapsto -1$. What is the image of $ab^{-1}a$?

Solution. By the universal property, extend additively: $2 - (-1) + 2 = 5$. □

Problem 14.40 (Triangle graph rank). Compute the fundamental-group rank of a triangle graph.

Solution. It has $V = 3$, $E = 3$, so the rank is $3 - 3 + 1 = 1$, hence $\pi_1 \cong \mathbb{Z}$. □

Problem 14.41 (Square with diagonal). Compute the rank for a square with one diagonal.

Solution. Here $V = 4$, $E = 5$, so $\text{rank} = 5 - 4 + 1 = 2$. □

Problem 14.42 (Complete graph K_4). Compute the rank of $\pi_1(K_4)$.

Solution. K_4 has $V = 4$, $E = 6$, so the rank is 3, hence $\pi_1(K_4) \cong F_3$. □

Problem 14.43 (Tree criterion). Show a finite connected graph is a tree iff $E = V - 1$.

Solution. A spanning tree already has $V - 1$ edges. If the connected graph has no extra edges it is the tree; any extra edge creates a cycle. Conversely every tree has $V - 1$ edges. $\square$

Problem 14.44 (Rose cover). Describe the universal cover of $S^1 \vee S^1$.

Solution. It is the 4-regular Cayley tree of F_2 , with edges labeled by the two generators and their inverse orientations. $\square$

Problem 14.45 (Deck action). How does F_2 act on that tree?

Solution. By left multiplication on vertex labels $g \mapsto hg$, carrying each labeled edge to the corresponding labeled edge. $\square$

Problem 14.46 (Index-two subgroup rank). If $H \leq F_3$ has index 2, find its rank.

Solution. Schreier gives $1 + 2(3 - 1) = 5$. $\square$

Problem 14.47 (Index-five subgroup of F_2). Find the rank.

Solution. $1 + 5(2 - 1) = 6$. $\square$

Problem 14.48 (Subgroup of $\mathbb{Z}$). Interpret $n\mathbb{Z} \leq F_1 \cong \mathbb{Z}$ as a graph cover.

Solution. It corresponds to the n -sheeted circle cover $S^1 \rightarrow S^1$, with covering graph an n -cycle. $\square$

Problem 14.49 (Nielsen–Schreier proof idea). Explain the covering-space proof that every subgroup $H \leq F_r$ is free.

Solution. Realize F_r as π_1 of the r -rose. The subgroup gives a connected covering graph. Fundamental groups of graphs are free, and that group identifies with H . $\square$

Problem 14.50 (Normal subgroup cover). What extra symmetry appears when $H \trianglelefteq F_r$?

Solution. The associated cover is regular, and deck transformations act transitively on fibers with group F_r/H . $\square$

Problem 14.51 (Non-normal subgroup). What does the deck group become for a general subgroup?

Solution. Under the covering classification, it is $N_{F_r}(H)/H$, which can be smaller than the full quotient of F_r . $\square$

Problem 14.52 (Schreier graph vertices). If $H \leq F_r$ has index d , how many vertices does its Schreier graph have?

Solution. Exactly d , one for each right coset. $\square$

Problem 14.53 (Schreier edge count). For a degree- d cover of the r -rose, how many unoriented edges are there?

Solution. Each of the r base loops lifts with one outgoing edge from each of d vertices, giving rd unoriented labeled edges. $\square$

Problem 14.54 (Derive Schreier rank). Use the previous count to recover the index formula.

Solution. The connected covering graph has $V = d$, $E = rd$, hence $\text{rank } E - V + 1 = rd - d + 1 = 1 + d(r - 1)$. $\square$

Problem 14.55 (Core graph). Why can hanging trees be removed without changing the subgroup represented by a covering graph?

Solution. Hanging trees contain no reduced closed loops. Collapsing or pruning them does not alter the fundamental group, leaving the loop-carrying core. $\square$

Problem 14.56 (Backtracking cancellation). Why does an edge followed immediately by its reverse represent a null segment in a graph path?

Solution. That out-and-back segment contracts through the edge, so deleting it is an endpoint-fixing path homotopy. $\square$

Problem 14.57 (Bridge to simplices). How does a graph become a simplicial complex?

Solution. Use vertices as 0-simplices and non-loop edges as 1-simplices; subdivide loops or multiple-edge configurations as needed to fit the abstract simplicial-complex rules. $\square$

Part IV

Homology

Chapter 15

Simplicial Complexes

Simplicial complexes replace a topological space by vertices, edges, triangles, tetrahedra, and higher-dimensional simplices glued face to face. They are the first combinatorial models used systematically in homology. This chapter develops abstract and geometric simplicial complexes, realizations, links and stars, subdivisions, orientations, and simplicial maps, while leaving the full chain-complex formalism to VIII/16.

15.1 Geometric simplices

Let

$$v_0, \dots, v_k \in \mathbb{R}^N$$

be affinely independent.

Definition 15.1 (Geometric simplex). The k -simplex spanned by these vertices is

$$[v_0, \dots, v_k] = \left\{ \sum_{i=0}^k t_i v_i : t_i \geq 0, \sum_{i=0}^k t_i = 1 \right\}.$$

The coefficients t_i are barycentric coordinates.

15.2 Faces

A face of

$$[v_0, \dots, v_k]$$

is obtained by retaining a subset of its vertices.

The i th codimension-one face is

$$[v_0, \dots, \widehat{v_i}, \dots, v_k].$$

Definition 15.2 (Dimension). A simplex with $k + 1$ vertices has dimension k .

Thus vertices are 0-simplices, edges are 1-simplices, triangles are 2-simplices, and tetrahedra are 3-simplices.

15.3 Geometric simplicial complexes

Definition 15.3 (Geometric simplicial complex). A geometric simplicial complex K is a collection of geometric simplices such that:

1. every face of a simplex in K also lies in K ;
2. the intersection of two simplices of K is either empty or a common face of both.

The second condition rules out uncontrolled crossings.

15.4 Abstract simplicial complexes

Definition 15.4 (Abstract simplicial complex). An abstract simplicial complex K on a vertex set V is a collection of finite nonempty subsets of V such that

$$\sigma \in K, \quad \emptyset \neq \tau \subset \sigma \implies \tau \in K.$$

Members $\sigma \in K$ are simplices, and

$$\dim \sigma = |\sigma| - 1.$$

15.5 Dimension of a complex

Definition 15.5 (Complex dimension).

$$\dim K = \sup_{\sigma \in K} \dim \sigma.$$

The dimension may be finite or infinite.

15.6 Subcomplexes

Definition 15.6 (Subcomplex). A subcollection $L \subset K$ is a subcomplex if it is itself a simplicial complex: whenever $\sigma \in L$, every nonempty face of σ also lies in L .

15.7 Geometric realization

An abstract complex becomes a topological space by assigning basis vectors to vertices and realizing each abstract simplex as their convex hull.

Definition 15.7 (Geometric realization). The geometric realization of K , denoted

$$|K|,$$

is the union of its realized simplices with the natural weak topology.

Different geometric embeddings of the same abstract complex yield homeomorphic realizations.

15.8 Basic examples

Example 15.8 (Simplex and boundary). The full simplex on $n + 1$ vertices realizes as D^n , while its proper-face subcomplex realizes as

$$S^{n-1}.$$

Example 15.9 (Cycle). A polygonal cycle with $m \geq 3$ vertices realizes as S^1 .

Example 15.10 (Triangulated disk). Two triangles sharing one edge can realize a quadrilateral disk.

15.9 Triangulations

Definition 15.11 (Triangulation). A triangulation of a topological space X is a simplicial complex K together with a homeomorphism

$$|K| \cong X.$$

Not every arbitrary decomposition is a triangulation; the simplicial intersection conditions must hold.

15.10 Stars

Definition 15.12 (Open star). For a vertex $v \in K$, the open star $\text{st}^\circ(v)$ is the union of interiors of all simplices containing v .

Definition 15.13 (Closed star). The closed star $\text{st}(v)$ is the subcomplex generated by all simplices containing v .

The closed star is contractible: it is a cone with vertex v .

15.11 Links

Definition 15.14 (Link). The link of a vertex v is

$$\text{lk}(v) = \{\sigma \in K : v \notin \sigma, \sigma \cup \{v\} \in K\}.$$

Informally, the link records the directions surrounding v .

Proposition 15.15 (Star as a cone).

$$\text{st}(v) = v * \text{lk}(v),$$

the simplicial cone on the link.

15.12 Manifold links

In a triangulated n -manifold, an interior vertex should have link homeomorphic to

$$S^{n-1},$$

while a boundary vertex has a link resembling

$$D^{n-1}.$$

This local criterion motivates later combinatorial manifold theory.

15.13 Barycentric subdivision

Definition 15.16 (Barycentric subdivision). The barycentric subdivision $\text{sd } K$ has one vertex for each nonempty simplex of K . A collection

$$\sigma_0, \dots, \sigma_r$$

forms a simplex in $\text{sd } K$ exactly when, after reordering,

$$\sigma_0 \subsetneq \sigma_1 \subsetneq \dots \subsetneq \sigma_r.$$

Theorem 15.17 (Subdivision preserves realization).

$$|\text{sd } K| \cong |K|.$$

Subdivision refines the combinatorics without changing the underlying topological space.

15.14 Orientation of simplices

An orientation of a k -simplex is an ordering

$$[v_0, \dots, v_k]$$

of its vertices, modulo even permutations.

Reversing orientation changes the sign:

$$[v_{\pi(0)}, \dots, v_{\pi(k)}] = \text{sgn}(\pi)[v_0, \dots, v_k].$$

This sign convention prepares the simplicial boundary operator in VIII/16.

15.15 Boundary preview

For an oriented simplex, the formal alternating boundary is

$$\partial[v_0, \dots, v_k] = \sum_{i=0}^k (-1)^i [v_0, \dots, \widehat{v_i}, \dots, v_k].$$

The central identity

$$\partial^2 = 0$$

will become the defining algebraic condition for chain complexes in VIII/16.

15.16 Simplicial maps

Definition 15.18 (Simplicial map). Let K, L be abstract simplicial complexes. A map on vertices

$$f : V(K) \rightarrow V(L)$$

is simplicial if for every simplex

$$\sigma = \{v_0, \dots, v_k\} \in K,$$

the image set

$$f(\sigma) = \{f(v_0), \dots, f(v_k)\}$$

is a simplex of L .

Distinct source vertices are allowed to have the same image.

15.17 Realization of a simplicial map

A simplicial map extends linearly over each simplex:

$$|f| \left(\sum_i t_i v_i \right) = \sum_i t_i f(v_i).$$

Theorem 15.19 (Continuity of realization). *A simplicial map induces a continuous map*

$$|f| : |K| \rightarrow |L|.$$

15.18 Composition

Proposition 15.20 (Functoriality). *The composite of simplicial maps is simplicial, and*

$$|g \circ f| = |g| \circ |f|.$$

15.19 Simplicial isomorphisms

Definition 15.21 (Simplicial isomorphism). A simplicial map is a simplicial isomorphism if it is bijective on vertices and its inverse vertex map is also simplicial.

A simplicial isomorphism induces a homeomorphism of realizations.

15.20 Why subdivision helps maps

A continuous map between polyhedra need not be simplicial with respect to a given triangulation. After sufficiently fine subdivisions, it can often be approximated by a simplicial map. This is the content of the simplicial approximation theorem, developed later as needed.

15.21 Bridge to chain complexes

Once each oriented k -simplex is treated as a generator of an abelian group, the alternating face formula becomes a homomorphism

$$\partial_k : C_k \rightarrow C_{k-1}.$$

The cancellation

$$\partial_{k-1}\partial_k = 0$$

creates the chain-complex structure of VIII/16.

15.22 Exercises

Exercise 15.22. Define a geometric k -simplex.

Hint. Use convex combinations of $k + 1$ affinely independent vertices. □

Solution. $[v_0, \dots, v_k] = \{\sum t_i v_i : t_i \geq 0, \sum t_i = 1\}$. □

Exercise 15.23. How many vertices does a k -simplex have?

Hint. Dimension is one less than vertex count. □

Solution. $k + 1$. □

Exercise 15.24. What is a face of a simplex?

Hint. Retain a subset of vertices. □

Solution. The simplex spanned by a nonempty subset of the original vertices. □

Exercise 15.25. State the intersection condition in a geometric simplicial complex.

Hint. Two simplices meet cleanly. □

Solution. Their intersection is empty or a common face of both. □

Exercise 15.26. Define an abstract simplicial complex.

Hint. Use finite subsets closed under nonempty subsets. □

Solution. A family K of finite nonempty vertex sets such that every nonempty subset of a simplex is again a simplex. □

Exercise 15.27. What is $\dim \sigma$ for an abstract simplex?

Hint. Count vertices. □

Solution. $|\sigma| - 1$. □

Exercise 15.28. Define a subcomplex.

Hint. Close under faces. □

Solution. A subcollection that contains every face of each of its simplices. □

Exercise 15.29. What is $|K|$?

Hint. Pass from combinatorics to topology. □

Solution. The geometric realization of the abstract simplicial complex. □

Exercise 15.30. What space does a full n -simplex realize?

Hint. It is convex. □

Solution. D^n . □

Exercise 15.31. What does its boundary subcomplex realize?

Hint. Use the simplex boundary. □

Solution. S^{n-1} . □

Exercise 15.32. Define a triangulation of X .

Hint. Use a simplicial complex plus homeomorphism. □

Solution. A complex K with a homeomorphism $|K| \cong X$. □

Exercise 15.33. Define the closed star of a vertex.

Hint. Collect simplices containing the vertex and all their faces. □

Solution. The subcomplex generated by all simplices containing the vertex. □

Exercise 15.34. Why is a closed star contractible?

Hint. It is a cone. □

Solution. $\text{st}(v) = v * \text{lk}(v)$. □

Exercise 15.35. Define the link of v .

Hint. Take faces opposite v . □

Solution. $\text{lk}(v) = \{\sigma : v \notin \sigma, \sigma \cup \{v\} \in K\}$. □

Exercise 15.36. What is the expected link of an interior vertex in an n -manifold triangulation?

Hint. One dimension lower. □

Solution. S^{n-1} . □

Exercise 15.37. What are vertices of $\text{sd } K$?

Hint. Use simplices of K . □

Solution. Nonempty simplices of K , represented by their barycenters. □

Exercise 15.38. When do vertices form a simplex in $\text{sd } K$?

Hint. Require a strict inclusion chain. □

Solution. When the corresponding simplices of K form a chain under inclusion. □

Exercise 15.39. What happens to the realization under barycentric subdivision?

Hint. Topology is unchanged. □

Solution. $|\text{sd } K| \cong |K|$. □

Exercise 15.40. Define an orientation of a simplex.

Hint. Order vertices modulo even permutations. □

Solution. An equivalence class of vertex orderings under even permutations. □

Exercise 15.41. Write the alternating boundary of $[v_0, \dots, v_k]$.

Hint. Delete one vertex at a time. □

Solution. $\partial[v_0, \dots, v_k] = \sum_i (-1)^i [v_0, \dots, \widehat{v_i}, \dots, v_k]$. □

Exercise 15.42. Define a simplicial map.

Hint. Images of vertex sets must remain simplices. □

Solution. A vertex map $f : V(K) \rightarrow V(L)$ sending every simplex of K to a simplex of L . □

Exercise 15.43. May a simplicial map identify two vertices?

Hint. Injectivity is not required. □

Solution. Yes, provided the resulting image set is a simplex. □

Exercise 15.44. How is $|f|$ defined on barycentric coordinates?

Hint. Extend linearly. □

Solution. $|f|(\sum t_i v_i) = \sum t_i f(v_i)$. □

Exercise 15.45. What is the bridge to VIII/16?

Hint. Turn oriented simplices into generators. □

Solution. The alternating face maps become boundary homomorphisms in a chain complex. □

15.23 Solved problem dossiers

Problem 15.46 (Triangle faces). List all nonempty faces of the abstract simplex $\{0, 1, 2\}$.

Solution. The three vertices, the three edges $\{0, 1\}$, $\{0, 2\}$, $\{1, 2\}$, and the triangle $\{0, 1, 2\}$. $\square$

Problem 15.47 (Tetrahedron count). Count the 0-, 1-, 2-, and 3-faces of a tetrahedron.

Solution. There are 4 vertices, $\binom{4}{2} = 6$ edges, $\binom{4}{3} = 4$ triangular faces, and one tetrahedron. $\square$

Problem 15.48 (Boundary of a triangle). Identify the realization of the proper-face subcomplex of a 2-simplex.

Solution. It is the three-edge cycle, homeomorphic to S^1 . $\square$

Problem 15.49 (Boundary of a tetrahedron). What space is realized by the boundary complex of a 3-simplex?

Solution. The four triangular faces glue to a 2-sphere S^2 . $\square$

Problem 15.50 (Invalid geometric complex). Why do two triangles crossing through their interiors fail to form a geometric simplicial complex?

Solution. Their intersection is not empty and is not a common face of the two triangles, violating the intersection axiom. $\square$

Problem 15.51 (Cycle complex). Build an abstract simplicial complex realizing a square circle.

Solution. Use vertices 1, 2, 3, 4, all singleton vertices, and edges $\{1, 2\}$, $\{2, 3\}$, $\{3, 4\}$, $\{4, 1\}$. Its realization is S^1 . $\square$

Problem 15.52 (Filled versus unfilled triangle). Compare the complexes with and without the 2-simplex $\{0, 1, 2\}$.

Solution. With the 2-simplex the realization is a disk; with only its proper faces the realization is a circle. $\square$

Problem 15.53 (Star and link in a triangle). For vertex 0 in a filled triangle, identify its closed star and link.

Solution. The star is the entire triangle and all faces incident/generated by it. The link consists of vertices 1, 2 and edge $\{1, 2\}$, hence realizes as an interval. $\square$

Problem 15.54 (Link in a cycle). What is the link of a vertex in a triangulated circle?

Solution. It consists of the two neighboring vertices as two isolated 0-simplices, realizing S^0 . $\square$

Problem 15.55 (Link in a triangulated surface). What local link should an interior vertex have?

Solution. A circle S^1 , matching the $n = 2$ manifold link rule. $\square$

Problem 15.56 (First barycentric subdivision of an edge). Describe sd of a single edge.

Solution. The original two vertices and the edge barycenter become subdivision vertices; the edge is replaced by two edges meeting at the barycenter. $\square$

Problem 15.57 (Subdivision of a triangle). How many small triangles appear in the first barycentric subdivision of one 2-simplex?

Solution. There are $3! = 6$ maximal chains vertex $\subset$ edge $\subset$ triangle, hence six small triangles. $\square$

Problem 15.58 (Orientation reversal). Compare $[v_0, v_1, v_2]$ and $[v_1, v_0, v_2]$.

Solution. The transposition is odd, so the orientations differ by a minus sign. $\square$

Problem 15.59 (Triangle boundary). Compute the formal oriented boundary of $[v_0, v_1, v_2]$.

Solution. $\partial[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$. $\square$

Problem 15.60 (Boundary-squared preview). Check formally that $\partial^2[v_0, v_1, v_2] = 0$.

Solution. Apply $\partial[v_i, v_j] = [v_j] - [v_i]$. Every vertex occurs twice with opposite signs, so the sum is zero. $\square$

Problem 15.61 (Constant simplicial map). When is a constant vertex map $K \rightarrow L$ simplicial?

Solution. If its common image vertex belongs to L , every simplex image is that singleton vertex, hence a simplex. Thus any constant map to a vertex is simplicial. $\square$

Problem 15.62 (Vertex identification). Map a triangle's vertices 0, 1, 2 to a, a, b . When is the map simplicial?

Solution. It is simplicial provided $\{a, b\}$ is a simplex of the target; the triangle image set is then the edge $\{a, b\}$. $\square$

Problem 15.63 (Composition). Prove a composite of simplicial maps is simplicial.

Solution. If $\sigma \in K$, then $f(\sigma)$ is a simplex of L , and $g(f(\sigma))$ is a simplex of M . Hence $g \circ f$ is simplicial. $\square$

Problem 15.64 (Realization continuity). Why is the realization of a simplicial map continuous?

Solution. On each simplex it is affine and hence continuous; the formulas agree on common faces, and the weak topology of the realization glues them into a global continuous map. $\square$

Problem 15.65 (Simplicial isomorphism). Why does a simplicial isomorphism induce a homeomorphism?

Solution. Both the map and its inverse are simplicial, so their realizations are continuous and inverse to one another. $\square$

Problem 15.66 (Graph as a simplicial complex). What restriction must be handled when viewing an arbitrary graph as an abstract simplicial complex?

Solution. Abstract simplicial complexes do not allow loop edges with repeated vertices or multiple distinct edges with the same endpoint set. Subdivision can remove these obstacles. $\square$

Problem 15.67 (Triangulated disk Euler preview). For a square cut by one diagonal, count vertices, edges, and triangles.

Solution. There are 4 vertices, 5 edges, and 2 triangles. The alternating count is $4 - 5 + 2 = 1$, foreshadowing disk Euler characteristic. $\square$

Problem 15.68 (Triangulated sphere Euler preview). For the tetrahedral sphere, compute $V - E + F$.

Solution. $4 - 6 + 4 = 2$, foreshadowing $\chi(S^2) = 2$. □

Problem 15.69 (Bridge to chains). Explain why orientations and alternating faces are exactly the data needed for simplicial homology.

Solution. Oriented k -simplices become generators of free abelian groups, and the signed face sum defines the boundary map. The identity $\partial^2 = 0$ makes these groups into the chain complex developed next. □

Chapter 16

Chain Complexes

Homology begins with a simple algebraic pattern: a sequence of abelian groups linked by homomorphisms whose consecutive composites vanish. The groups encode objects in each dimension, while the homomorphisms encode boundaries. This chapter develops the algebra of chain complexes before any particular topological chain model is chosen.

16.1 Chain complexes

Definition 16.1 (Chain complex). A chain complex $C_\bullet$ of abelian groups is a sequence

$$\cdots \xrightarrow{\partial_{n+2}} C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}} \cdots$$

such that

$$\partial_n \partial_{n+1} = 0$$

for every n .

The homomorphisms ∂_n are boundary operators.

16.2 Cycles and boundaries

Definition 16.2 (Cycles and boundaries). Define

$$Z_n(C) = \ker \partial_n, \quad B_n(C) = \operatorname{im} \partial_{n+1}.$$

Elements of $Z_n(C)$ are n -cycles and elements of $B_n(C)$ are n -boundaries.

The identity $\partial^2 = 0$ gives

$$B_n(C) \subset Z_n(C).$$

16.3 Homology of a chain complex

Definition 16.3 (Homology group). The n th homology group of $C_\bullet$ is

$$H_n(C) = Z_n(C)/B_n(C).$$

At this stage homology is purely algebraic. VIII/17 constructs chain complexes from spaces and simplicial complexes.

16.4 Concentrated complexes

A group A may be viewed as a chain complex concentrated in degree k :

$$C_k = A, \quad C_n = 0 \text{ for } n \neq k.$$

Example 16.4. For this complex,

$$H_k(C) \cong A$$

and every other homology group vanishes.

16.5 Two-term complexes

Consider

$$0 \longrightarrow A \xrightarrow{f} B \longrightarrow 0,$$

with A in degree 1 and B in degree 0.

Proposition 16.5 (Two-term homology).

$$H_1 \cong \ker f, \quad H_0 \cong B / \operatorname{im} f.$$

Thus kernels and cokernels are already homology groups.

16.6 Exact chain complexes

Definition 16.6 (Exactness). A chain complex is exact at C_n if

$$\operatorname{im} \partial_{n+1} = \ker \partial_n.$$

It is exact if this holds in every degree.

Proposition 16.7 (Exact iff acyclic). *A chain complex is exact if and only if*

$$H_n(C) = 0$$

for all n .

16.7 Chain maps

Definition 16.8 (Chain map). A chain map

$$f_\bullet : C_\bullet \rightarrow D_\bullet$$

is a family of homomorphisms

$$f_n : C_n \rightarrow D_n$$

such that

$$\partial_n^D f_n = f_{n-1} \partial_n^C$$

for every n .

Equivalently, every boundary square commutes.

16.8 Induced maps on homology

Theorem 16.9 (Homology is functorial on chain maps). *A chain map sends cycles to cycles and boundaries to boundaries, and therefore induces*

$$H_n(f) : H_n(C) \rightarrow H_n(D)$$

by

$$[z] \mapsto [f_n(z)].$$

16.9 Identity and composition

If $f : C \rightarrow D$ and $g : D \rightarrow E$ are chain maps, then $g \circ f$ is a chain map and

$$H_n(g \circ f) = H_n(g) \circ H_n(f).$$

Also

$$H_n(\text{id}_C) = \text{id}_{H_n(C)}.$$

16.10 Chain isomorphisms

Definition 16.10 (Chain isomorphism). A chain map $f : C \rightarrow D$ is a chain isomorphism if every

$$f_n : C_n \rightarrow D_n$$

is an isomorphism.

A chain isomorphism induces isomorphisms on every homology group.

16.11 Subcomplexes

Definition 16.11 (Chain subcomplex). A chain subcomplex $A_\bullet \subset C_\bullet$ consists of subgroups

$$A_n \subset C_n$$

with

$$\partial_C(A_n) \subset A_{n-1}.$$

The restricted maps form a chain complex.

16.12 Quotient complexes

If $A_\bullet \subset C_\bullet$ is a chain subcomplex, define

$$(C/A)_n = C_n/A_n.$$

Proposition 16.12 (Quotient boundary). *The formula*

$$\partial[c] = [\partial c]$$

is well defined and makes $C_\bullet/A_\bullet$ a chain complex.

16.13 Short exact sequences of complexes

Definition 16.13 (Short exact sequence of chain complexes). A sequence

$$0 \rightarrow A_{\bullet} \xrightarrow{i} B_{\bullet} \xrightarrow{p} C_{\bullet} \rightarrow 0$$

is short exact if

$$0 \rightarrow A_n \xrightarrow{i_n} B_n \xrightarrow{p_n} C_n \rightarrow 0$$

is exact for every n , and all maps are chain maps.

Such a sequence produces a long exact sequence in homology. The construction of the connecting homomorphism is developed in VIII/19.

16.14 Direct sums

Given chain complexes $C_{\bullet}^{\alpha}$, define

$$\left(\bigoplus_{\alpha} C^{\alpha} \right)_n = \bigoplus_{\alpha} C_n^{\alpha}$$

with boundary acting componentwise.

Proposition 16.14 (Homology and direct sums). *For ordinary direct sums of abelian-group chain complexes,*

$$H_n \left(\bigoplus_{\alpha} C^{\alpha} \right) \cong \bigoplus_{\alpha} H_n(C^{\alpha}).$$

16.15 Augmented complexes

An augmented chain complex continues one step beyond degree zero:

$$C_1 \xrightarrow{\partial_1} C_0 \xrightarrow{\epsilon} A \rightarrow 0,$$

with

$$\epsilon \partial_1 = 0.$$

For simplicial complexes, the augmentation

$$\epsilon : C_0 \rightarrow \mathbb{Z}$$

sends every vertex to 1. This leads to reduced homology in VIII/17.

16.16 Reduced chain complexes

The reduced degree-zero information is obtained by replacing the ordinary degree-zero homology with the kernel of the augmentation. For a connected space this removes the universal $\mathbb{Z}$ summand of H_0 .

16.17 Graded viewpoint

A chain complex may be written compactly as a graded group

$$C_* = \bigoplus_n C_n$$

equipped with a degree- -1 map

$$\partial : C_* \rightarrow C_*$$

satisfying

$$\partial^2 = 0.$$

This notation becomes especially convenient for chain homotopies in VIII/22.

16.18 Matrices for finite free complexes

If every C_n is finitely generated free abelian, choose bases and represent

$$\partial_n$$

by an integer matrix D_n . The chain condition becomes

$$D_{n-1}D_n = 0.$$

Homology can then be computed using kernels, images, and Smith normal form.

16.19 Smith normal form preview

For an integer matrix D , unimodular row and column operations can put D into diagonal form

$$\text{diag}(d_1, \dots, d_r, 0, \dots, 0), \quad d_i \mid d_{i+1}.$$

This is the standard computational tool for finitely generated homology groups.

16.20 Euler characteristic preview

For a finite chain complex of finitely generated free abelian groups, let

$$c_n = \text{rank } C_n.$$

The alternating chain rank

$$\sum_n (-1)^n c_n$$

will later agree with the alternating sum of homology ranks.

16.21 Bridge to topological chains

VIII/15 provided oriented simplices and the alternating face formula. In VIII/17 those simplices become generators of chain groups and the face formula becomes a boundary operator satisfying

$$\partial^2 = 0.$$

The present chapter supplies the algebra into which those constructions fit.

16.22 Exercises

Exercise 16.15. State the chain condition.

Hint. Compose consecutive boundaries. □

Solution. $\partial_n \partial_{n+1} = 0$. □

Exercise 16.16. Define $Z_n(C)$.

Hint. Take a kernel. □

Solution. $Z_n(C) = \ker \partial_n$. □

Exercise 16.17. Define $B_n(C)$.

Hint. Take the preceding image. □

Solution. $B_n(C) = \operatorname{im} \partial_{n+1}$. □

Exercise 16.18. Why is $B_n \subset Z_n$?

Hint. Use $\partial^2 = 0$. □

Solution. If $b = \partial_{n+1}c$, then $\partial_n b = \partial_n \partial_{n+1}c = 0$. □

Exercise 16.19. Define $H_n(C)$.

Hint. Cycles modulo boundaries. □

Solution. $H_n(C) = Z_n(C)/B_n(C)$. □

Exercise 16.20. What is the homology of a complex concentrated in degree k at A ?

Hint. All boundaries vanish. □

Solution. $H_k \cong A$, all other $H_n = 0$. □

Exercise 16.21. For $0 \rightarrow A \xrightarrow{f} B \rightarrow 0$, what is H_1 ?

Hint. Take the kernel of f . □

Solution. $\ker f$. □

Exercise 16.22. For the same complex, what is H_0 ?

Hint. Take the cokernel. □

Solution. $B/\operatorname{im} f$. □

Exercise 16.23. Define exactness at C_n .

Hint. Image equals kernel. □

Solution. $\operatorname{im} \partial_{n+1} = \ker \partial_n$. □

Exercise 16.24. When is a chain complex acyclic?

Hint. All homology vanishes. □

Solution. When $H_n(C) = 0$ for all n . □

Exercise 16.25. Define a chain map.

Hint. Boundary squares commute. ☐

Solution. $\partial^D f_n = f_{n-1} \partial^C$. ☐

Exercise 16.26. Why does a chain map send cycles to cycles?

Hint. Apply the commuting square. ☐

Solution. If $\partial z = 0$, then $\partial f(z) = f(\partial z) = 0$. ☐

Exercise 16.27. Why does a chain map send boundaries to boundaries?

Hint. Move the boundary through f . ☐

Solution. $f(\partial c) = \partial f(c)$. ☐

Exercise 16.28. What map does a chain map induce?

Hint. Pass to quotient classes. ☐

Solution. $H_n(f) : H_n(C) \rightarrow H_n(D)$. ☐

Exercise 16.29. What is a chain isomorphism?

Hint. Degreewise isomorphism. ☐

Solution. A chain map whose every component f_n is an isomorphism. ☐

Exercise 16.30. Define a chain subcomplex.

Hint. Boundary preserves the subgroups. ☐

Solution. Subgroups $A_n \subset C_n$ with $\partial(A_n) \subset A_{n-1}$. ☐

Exercise 16.31. How is the quotient boundary defined?

Hint. Apply boundary before taking the coset. ☐

Solution. $\partial[c] = [\partial c]$. ☐

Exercise 16.32. Define short exactness for chain complexes.

Hint. Require exactness degreewise. ☐

Solution. Every $0 \rightarrow A_n \rightarrow B_n \rightarrow C_n \rightarrow 0$ is exact. ☐

Exercise 16.33. What does a short exact sequence of complexes later produce?

Hint. Recall VIII/19. ☐

Solution. A long exact sequence in homology. ☐

Exercise 16.34. How is the boundary on a direct sum defined?

Hint. Componentwise. ☐

Solution. $\partial((c_\alpha)) = (\partial c_\alpha)$. ☐

Exercise 16.35. What is an augmentation?

Hint. Continue beyond C_0 . □

Solution. A map $\epsilon : C_0 \rightarrow A$ satisfying $\epsilon \partial_1 = 0$. □

Exercise 16.36. What does the standard simplicial augmentation send a vertex to?

Hint. Every vertex contributes one component. □

Solution. $1 \in \mathbb{Z}$. □

Exercise 16.37. What matrix identity expresses the chain condition?

Hint. Choose bases. □

Solution. $D_{n-1}D_n = 0$. □

Exercise 16.38. What is the bridge to VIII/17?

Hint. Use oriented simplices. □

Solution. VIII/17 turns simplices into chain generators and the alternating face sum into the boundary operator. □

16.23 Solved problem dossiers

Problem 16.39 (Zero differential). Let $C_n = A_n$ with every boundary zero. Compute $H_n(C)$.

Solution. Every element is a cycle and no nonzero element is a boundary, so $H_n(C) \cong A_n$ for every n . □

Problem 16.40 (Identity differential warning). Can a nonzero chain complex have every differential equal to the identity?

Solution. Not in consecutive nonzero degrees, because the chain condition would give $\text{id} \circ \text{id} = 0$, impossible on a nonzero group. □

Problem 16.41 (Multiplication-by- m complex). Compute the homology of $0 \rightarrow \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \rightarrow 0$ for $m \neq 0$.

Solution. The map is injective, so $H_1 = 0$. Its image is $m\mathbb{Z}$, so $H_0 \cong \mathbb{Z}/m$. □

Problem 16.42 (Zero multiplication). Repeat the previous computation for $m = 0$.

Solution. The boundary is zero, hence $H_1 \cong \mathbb{Z}$ and $H_0 \cong \mathbb{Z}$. □

Problem 16.43 (Exactness test). Show $0 \rightarrow \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \rightarrow 0$ is exact.

Solution. The identity has zero kernel and full image. Thus the degree-one and degree-zero homology groups both vanish. □

Problem 16.44 (Cycle-to-cycle proof). Prove a chain map preserves cycles.

Solution. For $z \in Z_n(C)$, $\partial^D f_n(z) = f_{n-1} \partial^C(z) = 0$, so $f_n(z) \in Z_n(D)$. □

Problem 16.45 (Boundary-to-boundary proof). Prove a chain map preserves boundaries.

Solution. If $b = \partial_{n+1}^C c$, then $f_n(b) = f_n \partial^C c = \partial^D f_{n+1}(c)$, so it is a boundary. □

Problem 16.46 (Well-defined induced map). Why does $[z] \mapsto [f(z)]$ not depend on the chosen cycle representative?

Solution. If $z - z' = \partial c$, then $f(z) - f(z') = f(\partial c) = \partial f(c)$, so the images differ by a boundary. $\square$

Problem 16.47 (Composition on homology). Verify $H_n(gf) = H_n(g)H_n(f)$.

Solution. On a class $[z]$, both sides equal $[g_n(f_n(z))]$. $\square$

Problem 16.48 (Quotient differential). Prove $\partial[c] = [\partial c]$ is well defined on C/A .

Solution. If $c - c' \in A_n$, then $\partial(c - c') \in A_{n-1}$, so $[\partial c] = [\partial c']$. $\square$

Problem 16.49 (Quotient chain condition). Verify the quotient differential squares to zero.

Solution. $\partial^2[c] = [\partial^2 c] = [0]$. $\square$

Problem 16.50 (Direct sum homology). Why do cycles and boundaries of a direct sum split componentwise?

Solution. The direct-sum differential acts componentwise, so kernels and images are direct sums of the component kernels and images, giving the homology direct-sum formula. $\square$

Problem 16.51 (Augmentation of a connected graph). Why does sending each vertex generator to 1 annihilate every edge boundary?

Solution. An oriented edge has boundary $v_1 - v_0$, and the augmentation sends this to $1 - 1 = 0$. $\square$

Problem 16.52 (Matrix chain condition). Let $D_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $D_1 = \begin{pmatrix} 1 & -1 \end{pmatrix}$. Verify the chain condition.

Solution. $D_1 D_2 = 1 - 1 = 0$. $\square$

Problem 16.53 (Matrix homology). For the preceding complex $0 \rightarrow \mathbb{Z} \xrightarrow{D_2} \mathbb{Z}^2 \xrightarrow{D_1} \mathbb{Z} \rightarrow 0$, compute the homology.

Solution. D_2 is injective, so $H_2 = 0$. The kernel of D_1 is generated by $(1, 1)$, exactly the image of D_2 , so $H_1 = 0$. D_1 is surjective, so $H_0 = 0$. $\square$

Problem 16.54 (Nonexact matrix example). Replace D_2 above by multiplication of the generator by $(2, 2)$. Compute H_1 .

Solution. $\ker D_1 = \mathbb{Z}(1, 1)$ and $\operatorname{im} D_2 = 2\mathbb{Z}(1, 1)$, so $H_1 \cong \mathbb{Z}/2$. $\square$

Problem 16.55 (Chain isomorphism). Why must a chain isomorphism induce homology isomorphisms?

Solution. Its inverse is also a chain map, and functoriality gives mutually inverse induced maps on homology. $\square$

Problem 16.56 (Exact iff zero homology). Prove the equivalence.

Solution. Exactness says $Z_n = B_n$ in every degree, which is equivalent to the quotient Z_n/B_n being zero. $\square$

Problem 16.57 (Short exact quotient example). Given a subcomplex $A \subset C$, write the canonical short exact sequence.

Solution. $0 \rightarrow A_{\bullet} \rightarrow C_{\bullet} \rightarrow (C/A)_{\bullet} \rightarrow 0$, degree-wise the standard subgroup-quotient sequence. $\square$

Problem 16.58 (Smith-form preview). What does the complex $0 \rightarrow \mathbb{Z} \xrightarrow{\times 12} \mathbb{Z} \rightarrow 0$ predict for torsion?

Solution. The single Smith invariant is 12, and the degree-zero homology is $\mathbb{Z}/12$. $\square$

Problem 16.59 (Euler rank check). For a finite exact complex of finite-rank free groups, what alternating rank should one expect?

Solution. Zero. Exactness lets ranks telescope across images and kernels; later the Euler–Poincaré formula formalizes this. $\square$

Problem 16.60 (Reduced degree zero). Why is reduced homology useful for connected spaces?

Solution. Ordinary H_0 always contains the universal $\mathbb{Z}$ component recording connectedness. The augmentation removes that component, making reduced H_0 vanish for connected spaces. $\square$

Chapter 17

Simplicial and Singular Homology

Chain complexes become topological invariants once their generators are chosen from a space. Simplicial homology uses oriented simplices in a triangulation. Singular homology replaces a triangulation by all continuous maps from standard simplices into the space. The two theories agree on triangulable spaces, but singular homology is defined for arbitrary topological spaces and is therefore the canonical general theory.

17.1 Simplicial chain groups

Let K be a simplicial complex.

Definition 17.1 (Simplicial chain group). The group $C_n(K)$ is the free abelian group generated by oriented n -simplices of K , with the relation that reversing orientation changes sign.

17.2 Simplicial boundary

For an oriented simplex,

$$[v_0, \dots, v_n],$$

define

$$\partial_n[v_0, \dots, v_n] = \sum_{i=0}^n (-1)^i [v_0, \dots, \widehat{v_i}, \dots, v_n].$$

Theorem 17.2 (Boundary squares to zero).

$$\partial_{n-1}\partial_n = 0.$$

Every codimension-two face appears twice with opposite signs.

17.3 Simplicial homology

Definition 17.3 (Simplicial homology).

$$H_n(K) = \ker \partial_n / \operatorname{im} \partial_{n+1}.$$

If $X \cong |K|$, one often writes these groups as homology groups of the triangulated space.

17.4 Degree zero

For a simplicial complex,

$$H_0(K)$$

records path components.

Theorem 17.4 (Zeroth homology). *If K has c path components, then*

$$H_0(K) \cong \mathbb{Z}^c.$$

17.5 The circle

Triangulate S^1 by a polygon. There is one independent one-cycle going around the loop and no two-simplex filling it.

Theorem 17.5 (Homology of the circle).

$$H_0(S^1) \cong \mathbb{Z}, \quad H_1(S^1) \cong \mathbb{Z}, \quad H_n(S^1) = 0 \quad (n \geq 2).$$

17.6 The sphere

For $n \geq 1$,

$$H_k(S^n) \cong \begin{cases} \mathbb{Z}, & k = 0, n, \\ 0, & \text{otherwise.} \end{cases}$$

The top-dimensional class records the fundamental cycle of the oriented sphere.

17.7 The disk

Because a disk is contractible,

$$H_0(D^n) \cong \mathbb{Z}, \quad H_k(D^n) = 0 \quad (k > 0).$$

A direct simplicial calculation sees every positive-dimensional cycle as a boundary after filling the triangulation.

17.8 Reduced homology

Definition 17.6 (Reduced homology). The reduced homology groups $\tilde{H}_n(X)$ are defined using the augmented chain complex. For nonempty X ,

$$H_0(X) \cong \tilde{H}_0(X) \oplus \mathbb{Z}.$$

If X is path connected,

$$\tilde{H}_0(X) = 0.$$

For a point,

$$\tilde{H}_n(*) = 0$$

for every n .

17.9 Singular simplices

Let

$$\Delta^n = \left\{ (t_0, \dots, t_n) \in \mathbb{R}^{n+1} : t_i \geq 0, \sum t_i = 1 \right\}.$$

Definition 17.7 (Singular simplex). A singular n -simplex in X is a continuous map

$$\sigma : \Delta^n \rightarrow X.$$

17.10 Singular chain groups

Definition 17.8 (Singular chains). The singular chain group

$$C_n^{\text{sing}}(X)$$

is the free abelian group generated by all singular n -simplices in X .

Unlike simplicial chains, there are generally infinitely many generators even for very simple spaces.

17.11 Face maps and singular boundary

Let

$$\delta_i : \Delta^{n-1} \rightarrow \Delta^n$$

include the i th face. Define

$$\partial\sigma = \sum_{i=0}^n (-1)^i \sigma \circ \delta_i.$$

The same cancellation proves

$$\partial^2 = 0.$$

17.12 Singular homology

Definition 17.9 (Singular homology).

$$H_n^{\text{sing}}(X) = H_n(C_*^{\text{sing}}(X)).$$

This construction requires no triangulation.

17.13 Functoriality

A continuous map

$$f : X \rightarrow Y$$

acts on a singular simplex by composition:

$$\sigma \mapsto f \circ \sigma.$$

Theorem 17.10 (Induced chain map). *The maps*

$$f_{\#} : C_n^{\text{sing}}(X) \rightarrow C_n^{\text{sing}}(Y)$$

defined by

$$f_{\#}(\sigma) = f \circ \sigma$$

form a chain map.

Hence every continuous map induces

$$f_* : H_n(X) \rightarrow H_n(Y).$$

17.14 Functoriality laws

$$(\mathrm{id}_X)_* = \mathrm{id}_{H_n(X)}, \quad (g \circ f)_* = g_* \circ f_*.$$

Thus homology is a functor from spaces and continuous maps to graded abelian groups.

17.15 Simplicial maps and homology

A simplicial map

$$f : K \rightarrow L$$

induces a simplicial chain map by sending an oriented simplex to its oriented image when the dimension is preserved, and to zero when vertices collapse and the image has smaller dimension.

This agrees with the induced map on singular homology after geometric realization.

17.16 Simplicial versus singular homology

Theorem 17.11 (Comparison theorem). *For a simplicial complex K ,*

$$H_n^{\mathrm{simp}}(K) \cong H_n^{\mathrm{sing}}(|K|).$$

The proof uses subdivision and approximation machinery. The theorem allows combinatorial calculations to compute the general topological invariant.

17.17 Homology of disjoint unions

For a disjoint union

$$X = \coprod_{\alpha} X_{\alpha},$$

singular simplices lie in individual components, so

$$H_n(X) \cong \bigoplus_{\alpha} H_n(X_{\alpha}).$$

17.18 Wedges in positive degree

For well-pointed spaces and positive degrees, reduced homology satisfies the wedge formula

$$\tilde{H}_n(X \vee Y) \cong \tilde{H}_n(X) \oplus \tilde{H}_n(Y).$$

This predicts, for example,

$$H_1(S^1 \vee S^1) \cong \mathbb{Z}^2.$$

17.19 Homology and holes

Homology does not record all homotopy information. Instead it linearizes cycles:

$$\text{cycles that do not bound} \quad \rightsquigarrow \quad \text{homology classes.}$$

The groups are abelian even when the fundamental group is not.

17.20 First homology and abelianization

Theorem 17.12 (First homology). *For a path-connected space,*

$$H_1(X; \mathbb{Z}) \cong \pi_1(X)_{\text{ab}},$$

the abelianization of the fundamental group.

This result provides a direct bridge between the covering/fundamental-group arc and homology.

17.21 Homotopy invariance preview

Homotopic maps induce the same maps in singular homology. The full chain-level proof uses prism operators and chain homotopies and is deferred to VIII/20, with chain-homotopy machinery developed further in VIII/22.

17.22 Bridge to cellular homology

Simplicial and singular chain groups can be very large. A CW complex offers a smaller chain complex with one generator per cell. VIII/18 derives that cellular chain complex from the skeletal filtration and computes its boundary maps from attaching maps.

17.23 Exercises

Exercise 17.13. Define $C_n(K)$.

Hint. Use oriented simplices as free generators. □

Solution. The free abelian group on oriented n -simplices modulo orientation reversal. □

Exercise 17.14. Write the simplicial boundary formula.

Hint. Delete one vertex with alternating signs. □

Solution. $\partial[v_0, \dots, v_n] = \sum_i (-1)^i [v_0, \dots, \widehat{v}_i, \dots, v_n]$. □

Exercise 17.15. What key identity makes simplicial chains a chain complex?

Hint. Apply boundary twice. □

Solution. $\partial^2 = 0$. □

Exercise 17.16. Define simplicial $H_n(K)$.

Hint. Cycles modulo boundaries. □

Solution. $\ker \partial_n / \text{im } \partial_{n+1}$. □

Exercise 17.17. What does H_0 count?

Hint. Think components. □

Solution. Free abelian rank equal to the number of path components. □

Exercise 17.18. Compute $H_1(S^1)$.

Hint. The fundamental loop does not bound. □

Solution. $\mathbb{Z}$. □

Exercise 17.19. Compute $H_k(S^n)$.

Hint. Only degrees zero and n . □

Solution. $\mathbb{Z}$ for $k = 0, n$, otherwise 0. □

Exercise 17.20. Compute positive-dimensional homology of D^n .

Hint. The disk is contractible. □

Solution. Zero. □

Exercise 17.21. Define reduced homology conceptually.

Hint. Use the augmented complex. □

Solution. It removes the universal degree-zero $\mathbb{Z}$ summand. □

Exercise 17.22. What is $\tilde{H}_0(X)$ for path-connected nonempty X ?

Hint. Only one component. □

Solution. Zero. □

Exercise 17.23. Define a singular n -simplex.

Hint. Map the standard simplex into X . □

Solution. A continuous $\sigma : \Delta^n \rightarrow X$. □

Exercise 17.24. Define $C_n^{\text{sing}}(X)$.

Hint. Free group on all singular simplices. □

Solution. The free abelian group generated by continuous maps $\Delta^n \rightarrow X$. □

Exercise 17.25. How is the singular boundary defined?

Hint. Restrict to faces with signs. □

Solution. $\partial\sigma = \sum_i (-1)^i \sigma \delta_i$. □

Exercise 17.26. Define singular homology.

Hint. Take homology of the singular chain complex. □

Solution. $H_n^{\text{sing}}(X) = H_n(C_*^{\text{sing}}(X))$. □

Exercise 17.27. How does $f : X \rightarrow Y$ act on a singular simplex?

Hint. Compose. □

Solution. $\sigma \mapsto f \circ \sigma$. □

Exercise 17.28. Why is $f_{\#}$ a chain map?

Hint. Composition commutes with face restriction. □

Solution. $\partial(f\sigma) = f_{\#}(\partial\sigma)$. □

Exercise 17.29. What map does f induce on homology?

Hint. Pass from the chain map. □

Solution. $f_* : H_n(X) \rightarrow H_n(Y)$. □

Exercise 17.30. State composition functoriality.

Hint. Induced maps compose. □

Solution. $(g \circ f)_* = g_* \circ f_*$. □

Exercise 17.31. Compare simplicial and singular homology on $|K|$.

Hint. Use the comparison theorem. □

Solution. They are naturally isomorphic. □

Exercise 17.32. State the disjoint-union formula.

Hint. Chains split by components. □

Solution. $H_n(\coprod X_{\alpha}) \cong \bigoplus H_n(X_{\alpha})$. □

Exercise 17.33. Compute $H_1(S^1 \vee S^1)$.

Hint. Use reduced wedge additivity. □

Solution. $\mathbb{Z}^2$. □

Exercise 17.34. How is H_1 related to π_1 ?

Hint. Abelianize. □

Solution. $H_1(X) \cong \pi_1(X)_{\text{ab}}$ for path-connected X . □

Exercise 17.35. Do homotopic maps induce the same homology map?

Hint. Preview VIII/20. □

Solution. Yes; VIII/20 proves this at chain level. □

Exercise 17.36. What is the bridge to VIII/18?

Hint. Replace huge chain groups by cells. □

Solution. Cellular homology uses one generator per CW cell. □

17.24 Solved problem dossiers

Problem 17.37 (Triangle boundary squared). Verify $\partial^2[v_0, v_1, v_2] = 0$.

Solution. First $\partial[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1]$. Applying $\partial[v_i, v_j] = [v_j] - [v_i]$, every vertex cancels. $\square$

Problem 17.38 (Polygonal circle). Compute simplicial H_1 of an m -gon with no 2-simplices.

Solution. The sum of all coherently oriented edges is a cycle. The kernel of ∂_1 has rank one and there are no 2-boundaries, so $H_1 \cong \mathbb{Z}$. $\square$

Problem 17.39 (Filled triangle). Compute H_1 of a filled 2-simplex.

Solution. The boundary cycle of the triangle is exactly the boundary of the 2-simplex, so it dies. The disk is contractible, hence $H_1 = 0$. $\square$

Problem 17.40 (Two components). Compute H_0 of two disjoint circles.

Solution. There are two path components, so $H_0 \cong \mathbb{Z}^2$. $\square$

Problem 17.41 (Reduced zero homology). Compute $\tilde{H}_0$ of a space with c nonempty path components.

Solution. Ordinary $H_0 \cong \mathbb{Z}^c$; removing the diagonal augmentation summand leaves $\mathbb{Z}^{c-1}$. $\square$

Problem 17.42 (Sphere top cycle). Why does an oriented triangulated sphere define a top-dimensional cycle?

Solution. Every codimension-one face belongs to two top simplices with opposite induced orientations, so those face contributions cancel in the total boundary. $\square$

Problem 17.43 (Disk boundary class). Why is the boundary sphere class zero in the homology of the disk?

Solution. The boundary fundamental cycle is the boundary of the sum of oriented top-dimensional simplices filling the disk. $\square$

Problem 17.44 (Singular 0-simplices). What are singular 0-simplices in X ?

Solution. Δ^0 is a point, so singular 0-simplices are exactly points of X . $\square$

Problem 17.45 (Singular 1-simplex). What familiar object is a singular 1-simplex?

Solution. A continuous map $I \cong \Delta^1 \rightarrow X$, i.e. a path with an orientation. $\square$

Problem 17.46 (Boundary of a singular path). For a singular 1-simplex σ , what is $\partial\sigma$?

Solution. It is the endpoint minus the starting point: $\sigma(1) - \sigma(0)$. $\square$

Problem 17.47 (Functorial boundary). Verify $f_{\#}\partial = \partial f_{\#}$ on a singular simplex.

Solution. Both expressions are $\sum_i (-1)^i f \circ \sigma \circ \delta_i$. $\square$

Problem 17.48 (Constant map on positive homology). What does a constant map induce on reduced positive-degree homology?

Solution. It factors through a point, whose reduced homology vanishes, so the induced map is zero. $\square$

Problem 17.49 (Identity map). What map does id_X induce on homology?

Solution. The identity on every $H_n(X)$. □

Problem 17.50 (Circle degree map). For $f : S^1 \rightarrow S^1$ of degree m , what is $f_* : H_1(S^1) \rightarrow H_1(S^1)$?

Solution. Under $H_1(S^1) \cong \mathbb{Z}$, it is multiplication by m . □

Problem 17.51 (Two-circle wedge). Interpret $H_1(S^1 \vee S^1)$ using abelianization.

Solution. $\pi_1 \cong F_2$, whose abelianization is $\mathbb{Z}^2$, so $H_1 \cong \mathbb{Z}^2$. □

Problem 17.52 (Torus first homology). Use $\pi_1(T^2) \cong \mathbb{Z}^2$ to compute $H_1(T^2)$.

Solution. The group is already abelian, so its abelianization is $\mathbb{Z}^2$. □

Problem 17.53 (Projective-plane first homology). Use $\pi_1(\mathbb{R}P^2) \cong \mathbb{Z}/2$ to compute H_1 .

Solution. The fundamental group is abelian already, so $H_1(\mathbb{R}P^2) \cong \mathbb{Z}/2$. □

Problem 17.54 (Simply connected space). What is $H_1(X)$ for path-connected simply connected X ?

Solution. The fundamental group is trivial, so its abelianization and therefore H_1 are zero. □

Problem 17.55 (Disjoint-union chains). Why do singular chain groups split over components?

Solution. The connected simplex Δ^n has image in a single component, so every singular generator belongs to one component and the chain group is the direct sum of component chain groups. □

Problem 17.56 (Comparison theorem use). Why is simplicial homology computationally useful if singular homology is the general invariant?

Solution. For a triangulated space, the comparison theorem identifies the small combinatorial simplicial chain complex with singular homology, avoiding infinitely many singular generators. □

Problem 17.57 (Simplicial collapse map). If a simplicial map sends an edge's two vertices to the same vertex, what happens to that oriented edge on chains?

Solution. Its image has dimension zero, so the induced simplicial 1-chain map sends the edge to zero. □

Problem 17.58 (Orientation sign). Why must simplicial chains impose sign under odd vertex permutations?

Solution. Without the sign convention, adjacent top simplices would not cancel correctly along shared faces and the alternating boundary formula would fail to encode orientation. □

Problem 17.59 (Homotopy-invariance boundary). Why is the proof that homotopic maps induce the same map postponed?

Solution. It requires a chain homotopy, concretely a prism operator on singular chains. VIII/20 develops this proof after the basic homology constructions are established. □

Problem 17.60 (Bridge to cellular chains). Why can cellular homology be much smaller than singular homology?

Solution. A CW complex with finitely many cells has one cellular generator per cell, while singular chains contain every continuous simplex map. The cellular boundary encodes attaching-map degrees. $\square$

Chapter 18

Cellular Homology

For a CW complex, singular chains are usually far larger than necessary. Cellular homology compresses the calculation to one generator for each cell. The cellular boundary map is determined by how each n -cell attaches to the $(n - 1)$ -skeleton. This makes the topology of CW constructions directly computable from degrees of attaching maps.

18.1 CW filtration and relative groups

Let

$$X^0 \subset X^1 \subset X^2 \subset \dots$$

be the skeletal filtration of a CW complex.

The cellular chain group in degree n is built from

$$H_n(X^n, X^{n-1}).$$

Theorem 18.1 (Relative cell decomposition). *If X has c_n n -cells, then*

$$H_k(X^n, X^{n-1}) \cong \begin{cases} \mathbb{Z}^{c_n}, & k = n, \\ 0, & k \neq n. \end{cases}$$

Thus every oriented n -cell contributes one free generator.

18.2 Cellular chain groups

Definition 18.2 (Cellular chains). Define

$$C_n^{\text{cell}}(X) = H_n(X^n, X^{n-1}).$$

After choosing orientations of the n -cells,

$$C_n^{\text{cell}}(X) \cong \mathbb{Z}^{c_n}.$$

18.3 Cellular boundary

The cellular boundary is the composite

$$d_n : H_n(X^n, X^{n-1}) \longrightarrow H_{n-1}(X^{n-1}, X^{n-2})$$

obtained from the connecting map in the long exact sequence of the triple

$$(X^n, X^{n-1}, X^{n-2}).$$

Theorem 18.3 (Cellular chain complex).

$$\cdots \rightarrow C_n^{\text{cell}}(X) \xrightarrow{d_n} C_{n-1}^{\text{cell}}(X) \rightarrow \cdots$$

is a chain complex, and its homology is naturally isomorphic to singular homology:

$$H_n(C_*^{\text{cell}}(X)) \cong H_n(X).$$

18.4 Boundary coefficients from attaching maps

Let e_α^n be an n -cell with attaching map

$$\varphi_\alpha : S^{n-1} \rightarrow X^{n-1}.$$

For an $(n-1)$ -cell e_β^{n-1} , collapse

$$X^{n-1} \setminus e_\beta^{n-1}$$

to a point. The resulting quotient is a sphere

$$X^{n-1} / (X^{n-1} \setminus e_\beta^{n-1}) \cong S^{n-1}.$$

The composite

$$S^{n-1} \xrightarrow{\varphi_\alpha} X^{n-1} \longrightarrow S^{n-1}$$

has an integer degree.

Theorem 18.4 (Cellular boundary formula). *The coefficient of e_β^{n-1} in*

$$d_n(e_\alpha^n)$$

is the degree of this composite map.

This is the computational heart of cellular homology.

18.5 A CW complex with no adjacent dimensions

If there are no $(n-1)$ -cells, then

$$C_{n-1}^{\text{cell}}(X) = 0$$

and therefore

$$d_n = 0.$$

This makes the minimal CW structure of a sphere immediate.

18.6 Spheres

For the CW structure on S^n with one 0-cell and one n -cell,

$$C_k = \begin{cases} \mathbb{Z}, & k = 0, n, \\ 0, & \text{otherwise.} \end{cases}$$

All differentials vanish, giving

$$H_k(S^n) \cong \begin{cases} \mathbb{Z}, & k = 0, n, \\ 0, & \text{otherwise.} \end{cases}$$

18.7 The torus

The standard torus CW structure has one 0-cell, two 1-cells, and one 2-cell:

$$C_2 \cong \mathbb{Z}, \quad C_1 \cong \mathbb{Z}^2, \quad C_0 \cong \mathbb{Z}.$$

The two-cell attaching word is

$$aba^{-1}b^{-1}.$$

Its total degree on each one-cell is zero, so

$$d_2 = 0.$$

Since the one-cells are loops at one vertex,

$$d_1 = 0.$$

Hence

$$H_2(T^2) \cong \mathbb{Z}, \quad H_1(T^2) \cong \mathbb{Z}^2, \quad H_0(T^2) \cong \mathbb{Z}.$$

18.8 Real projective spaces

The standard CW structure on

$$\mathbb{R}P^n$$

has one cell in each dimension $0, \dots, n$.

Theorem 18.5 (Projective cellular differential). *With suitable orientations,*

$$d_k = \begin{cases} 0, & k \text{ odd}, \\ \times 2, & k \text{ even}. \end{cases}$$

Thus the cellular chain complex alternates:

$$\dots \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z}.$$

18.9 Homology of real projective space

For $0 < k < n$,

$$H_k(\mathbb{R}P^n) \cong \begin{cases} \mathbb{Z}/2, & k \text{ odd}, \\ 0, & k \text{ even}. \end{cases}$$

At the top dimension,

$$H_n(\mathbb{R}P^n) \cong \begin{cases} \mathbb{Z}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

18.10 Degree- m two-cell attachments

Consider

$$X_m = S^1 \cup_m D^2.$$

Its cellular chain complex is

$$0 \rightarrow \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0.$$

For $m \neq 0$,

$$H_2(X_m) = 0, \quad H_1(X_m) \cong \mathbb{Z}/m, \quad H_0(X_m) \cong \mathbb{Z}.$$

For $m = 0$,

$$X_0 \simeq S^1 \vee S^2$$

and both H_1 and H_2 are $\mathbb{Z}$.

18.11 Moore spaces

For $n \geq 1$, attach an $(n+1)$ -cell to S^n by a degree- m map:

$$M(\mathbb{Z}/m, n) = S^n \cup_m D^{n+1}.$$

The relevant cellular boundary is

$$\mathbb{Z} \xrightarrow{\times m} \mathbb{Z}.$$

Hence, for $m \neq 0$,

$$\tilde{H}_k(M(\mathbb{Z}/m, n)) \cong \begin{cases} \mathbb{Z}/m, & k = n, \\ 0, & k \neq n. \end{cases}$$

18.12 Lens-space pattern

A lens space admits a CW structure with one cell in each dimension

$$0, 1, 2, 3.$$

The cellular boundary maps encode the finite cyclic fundamental-group data, and a standard three-dimensional lens space $L(p, q)$ has homology

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}/p, \quad H_2 = 0, \quad H_3 \cong \mathbb{Z}.$$

The parameter q affects finer topology but not these integral homology groups.

18.13 Closed orientable surfaces

A genus- g orientable surface has one 0-cell, $2g$ one-cells, and one 2-cell. The two-cell attaches by a product of commutators.

The total exponent of every one-cell in that attaching word is zero, so

$$d_2 = 0.$$

Also $d_1 = 0$.

Therefore

$$H_2(\Sigma_g) \cong \mathbb{Z}, \quad H_1(\Sigma_g) \cong \mathbb{Z}^{2g}, \quad H_0(\Sigma_g) \cong \mathbb{Z}.$$

18.14 Nonorientable surface pattern

For a closed nonorientable surface built from one vertex, g one-cells, and one two-cell, the two-cell boundary contains squared generators. Cellular homology therefore produces a nontrivial multiplication-by-2 component and detects 2-torsion in first homology.

18.15 Euler characteristic

For a finite CW complex,

$$\chi(X) = \sum_n (-1)^n c_n.$$

Since

$$C_n^{\text{cell}}(X) \cong \mathbb{Z}^{c_n},$$

the Euler–Poincaré principle later gives

$$\chi(X) = \sum_n (-1)^n \text{rank } H_n(X).$$

18.16 Cellular maps and induced maps

A cellular map

$$f : X \rightarrow Y$$

induces maps between the skeletal pairs and hence a chain map

$$C_*^{\text{cell}}(X) \rightarrow C_*^{\text{cell}}(Y).$$

The induced homology map agrees with the usual singular-homology map.

18.17 Why cellular homology is efficient

For a finite CW complex, the cellular chain groups have ranks equal to the finite cell counts. By contrast, singular chain groups are typically infinitely generated in every positive degree.

The price for this compression is that the cellular differential requires careful analysis of attaching maps.

18.18 Support-diagram provenance

The legacy corpus contains support figures for quotient, lens-space, and Moore space calculations mapped explicitly into this chapter. These belong with the cellular computations rather than as standalone chapters. Their mathematical content is represented here, while final one-to-one figure/dossier reconciliation is reserved for the post-VIII/35 corpus audit.

18.19 Bridge to relative homology

The definition

$$C_n^{\text{cell}}(X) = H_n(X^n, X^{n-1})$$

already uses relative homology. VIII/19 develops relative groups and the long exact sequence systematically, explaining the connecting maps that underlie the cellular boundary operator.

18.20 Exercises

Exercise 18.6. Define $C_n^{\text{cell}}(X)$.

Hint. Use adjacent skeleta. □

Solution. $H_n(X^n, X^{n-1})$. □

Exercise 18.7. If X has c_n n -cells, what is $C_n^{\text{cell}}(X)$?

Hint. Choose orientations. □

Solution. $\mathbb{Z}^{c_n}$. □

Exercise 18.8. What determines a cellular boundary coefficient?

Hint. Collapse all but one lower cell. □

Solution. The degree of the attaching-map composite $S^{n-1} \rightarrow S^{n-1}$. □

Exercise 18.9. What happens to d_n if there are no $(n-1)$ -cells?

Hint. The target chain group vanishes. □

Solution. $d_n = 0$. □

Exercise 18.10. Give the cellular groups for minimal S^n .

Hint. One 0-cell and one n -cell. □

Solution. $C_0 = C_n = \mathbb{Z}$, all others zero. □

Exercise 18.11. Compute $H_n(S^n)$ cellularly.

Hint. All differentials vanish. □

Solution. $\mathbb{Z}$. □

Exercise 18.12. Give the torus cellular ranks.

Hint. Use 1,2,1 cells. □

Solution. $C_2 = \mathbb{Z}$, $C_1 = \mathbb{Z}^2$, $C_0 = \mathbb{Z}$. □

Exercise 18.13. What is d_2 for the standard torus?

Hint. The commutator has zero exponent sum in each generator. □

Solution. Zero. □

Exercise 18.14. Compute $H_1(T^2)$.

Hint. Both differentials around C_1 vanish. □

Solution. $\mathbb{Z}^2$. □

Exercise 18.15. How many standard cells does $\mathbb{R}P^n$ have in each dimension?

Hint. One per dimension. □

Solution. Exactly one in each dimension $0, \dots, n$. □

Exercise 18.16. What are the projective cellular differentials?

Hint. Alternate by parity. □

Solution. $d_k = 0$ for odd k , and $d_k = \times 2$ for even k . □

Exercise 18.17. Compute $H_1(\mathbb{R}P^2)$.

Hint. Use $d_2 = \times 2$ and $d_1 = 0$.

□

Solution. $\mathbb{Z}/2$.

□

Exercise 18.18. Compute $H_2(\mathbb{R}P^2)$.

Hint. Take the kernel of multiplication by 2.

□

Solution. Zero.

□

Exercise 18.19. Compute $H_3(\mathbb{R}P^3)$.

Hint. The top differential is zero.

□

Solution. $\mathbb{Z}$.

□

Exercise 18.20. What chain complex computes $S^1 \cup_m D^2$?

Hint. One cell in dimensions 0,1,2.

□

Solution. $0 \rightarrow \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0$.

□

Exercise 18.21. For $m \neq 0$, compute $H_1(S^1 \cup_m D^2)$.

Hint. Take cokernel of $\times m$.

□

Solution. $\mathbb{Z}/m$.

□

Exercise 18.22. What is a Moore space $M(\mathbb{Z}/m, n)$?

Hint. Attach one higher cell by degree m .

□

Solution. $S^n \cup_m D^{n+1}$.

□

Exercise 18.23. What reduced homology does that Moore space realize?

Hint. Use the multiplication-by- m boundary.

□

Solution. $\tilde{H}_n \cong \mathbb{Z}/m$, all other reduced groups zero.

□

Exercise 18.24. State the homology of a 3D lens space $L(p, q)$.

Hint. Only H_1 sees p .

□

Solution. $H_0 = H_3 = \mathbb{Z}$, $H_1 = \mathbb{Z}/p$, $H_2 = 0$.

□

Exercise 18.25. Give H_1 of an orientable genus- g surface.

Hint. There are $2g$ one-cells and zero cellular differentials there.

□

Solution. $\mathbb{Z}^{2g}$.

□

Exercise 18.26. Give H_2 of an orientable closed surface.

Hint. The top cellular differential is zero.

□

Solution. $\mathbb{Z}$.

□

Exercise 18.27. How is Euler characteristic read from cells?

Hint. Take alternating cell counts.

□

Solution. $\chi(X) = \sum (-1)^n c_n$. □

Exercise 18.28. What does a cellular map induce?

Hint. Use skeletal pairs. □

Solution. A cellular chain map and hence maps on homology. □

Exercise 18.29. What is the bridge to VIII/19?

Hint. Cellular chains are relative groups. □

Solution. VIII/19 develops relative homology and long exact sequences underlying the cellular boundary. □

18.21 Solved problem dossiers

Problem 18.30 (Minimal sphere). Compute cellular homology of the two-cell CW structure on S^n .

Solution. The chain groups are $\mathbb{Z}$ in degrees n and 0 and zero elsewhere. Since there is no adjacent nonzero target, every differential is zero, giving $H_0 = H_n = \mathbb{Z}$. □

Problem 18.31 (Circle). Compute cellular homology of the one-vertex one-edge circle.

Solution. The chain complex is $0 \rightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0$, so $H_1 \cong H_0 \cong \mathbb{Z}$. □

Problem 18.32 (Two-circle wedge). Compute cellular homology of $S^1 \vee S^1$.

Solution. One vertex and two loop one-cells give $0 \rightarrow \mathbb{Z}^2 \xrightarrow{0} \mathbb{Z} \rightarrow 0$, so $H_1 \cong \mathbb{Z}^2$. □

Problem 18.33 (Torus). Compute all cellular homology groups of the standard torus.

Solution. The chain complex is $0 \rightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z}^2 \xrightarrow{0} \mathbb{Z} \rightarrow 0$, hence $H_2 = \mathbb{Z}$, $H_1 = \mathbb{Z}^2$, $H_0 = \mathbb{Z}$. □

Problem 18.34 (Genus two surface). Compute $H_*(\Sigma_2)$.

Solution. One 0-cell, four 1-cells, and one 2-cell with commutator-product attaching word give zero differentials, so $H_2 = \mathbb{Z}$, $H_1 = \mathbb{Z}^4$, $H_0 = \mathbb{Z}$. □

Problem 18.35 (Degree-three attachment). Compute $H_*(S^1 \cup_3 D^2)$.

Solution. The chain complex has $d_2 = \times 3$ and $d_1 = 0$. Thus $H_2 = 0$, $H_1 = \mathbb{Z}/3$, $H_0 = \mathbb{Z}$. □

Problem 18.36 (Degree-one attachment). Compute the homology of $S^1 \cup_1 D^2$.

Solution. Multiplication by one is an isomorphism, so positive homology vanishes and $H_0 = \mathbb{Z}$, agreeing with the disk model. □

Problem 18.37 (Degree-zero attachment). Compute the homology of $S^1 \cup_0 D^2$.

Solution. All differentials vanish, giving $H_2 = \mathbb{Z}$, $H_1 = \mathbb{Z}$, $H_0 = \mathbb{Z}$, as for $S^1 \vee S^2$. □

Problem 18.38 (Projective plane). Compute cellular homology of $\mathbb{R}P^2$.

Solution. The complex is $0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0$. Hence $H_2 = 0$, $H_1 = \mathbb{Z}/2$, $H_0 = \mathbb{Z}$. □

Problem 18.39 (Projective 3-space). Compute cellular homology of $\mathbb{R}P^3$.

Solution. The complex is $0 \rightarrow \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0$. Thus $H_3 = \mathbb{Z}$, $H_2 = 0$, $H_1 = \mathbb{Z}/2$, $H_0 = \mathbb{Z}$. $\square$

Problem 18.40 (Projective parity). Why is top homology of $\mathbb{R}P^n$ nonzero exactly for odd n ?

Solution. The top differential is zero for odd n , giving kernel $\mathbb{Z}$; for even n it is multiplication by 2, whose kernel is zero. $\square$

Problem 18.41 (Moore $M(\mathbb{Z}/5, n)$). Compute its reduced homology.

Solution. The only relevant differential is $\mathbb{Z} \xrightarrow{5} \mathbb{Z}$, so $\tilde{H}_n \cong \mathbb{Z}/5$ and all other reduced groups vanish. $\square$

Problem 18.42 (Moore $m = 1$). What homology does $S^n \cup_1 D^{n+1}$ have?

Solution. The multiplication-by-one differential is an isomorphism, so all reduced homology vanishes; the space is homologically contractible. $\square$

Problem 18.43 (Lens space). Explain why $H_1(L(p, q))$ is $\mathbb{Z}/p$ in the standard cell model.

Solution. The one-cell and two-cell portion contains a cellular differential whose essential coefficient is p , so the relevant cokernel is $\mathbb{Z}/p$. $\square$

Problem 18.44 (Lens parameter q). Why can cellular integral homology fail to distinguish $L(p, q)$ for different q ?

Solution. The standard cellular boundary ranks and the torsion coefficient p do not depend on q . The parameter affects finer invariants beyond these homology groups. $\square$

Problem 18.45 (Euler torus). Compute the torus Euler characteristic from its cells.

Solution. $\chi = 1 - 2 + 1 = 0$. $\square$

Problem 18.46 (Euler genus g). Compute $\chi(\Sigma_g)$ from the standard cells.

Solution. $\chi = 1 - 2g + 1 = 2 - 2g$. $\square$

Problem 18.47 (Euler projective plane). Compute $\chi(\mathbb{R}P^2)$.

Solution. There is one cell in dimensions 0,1,2, so $\chi = 1 - 1 + 1 = 1$. $\square$

Problem 18.48 (Boundary coefficient meaning). Explain how an attaching map contributes an integer to a cellular differential.

Solution. Project the attaching sphere onto the quotient sphere corresponding to one lower-dimensional cell; the degree of that sphere map is the incidence coefficient. $\square$

Problem 18.49 (Orientation change). What happens to a cellular boundary matrix if the orientation of one cell is reversed?

Solution. The corresponding basis generator changes sign, so one row or column changes sign. Homology is unchanged because this is a basis change by a unit. $\square$

Problem 18.50 (Cell subdivision). Why should subdividing a CW or simplicial model not change the final homology?

Solution. Subdivision changes the chain complex but not the underlying topological space; cellular/simplicial homology agrees with singular homology, which is an invariant of the space. $\square$

Problem 18.51 (Support-figure role). Where do the mapped quotient/lens/Moore support diagrams belong conceptually?

Solution. They illustrate cellular attaching maps and the resulting boundary matrices, so their canonical mathematical home is VIII/18 rather than an independent chapter. $\square$

Problem 18.52 (Cellular versus singular size). Compare chain sizes for the minimal CW sphere.

Solution. Cellular chains have only two generators total, while singular chains have infinitely many singular simplices in every relevant degree. Both yield the same homology. $\square$

Problem 18.53 (Bridge to relative exactness). Why does the cellular construction naturally lead to VIII/19?

Solution. Each cellular group is a relative group $H_n(X^n, X^{n-1})$, and the cellular differential is built from connecting homomorphisms in exact sequences of pairs and triples. $\square$

Chapter 19

Relative Homology and Exact Sequences

Relative homology measures what remains in a space after a chosen subspace is regarded as already filled or already understood. Algebraically, it is the homology of a quotient chain complex. Geometrically, it is the natural language for pairs, CW skeleta, excision, and connecting maps. This chapter develops the long exact sequence of a pair and the related exact-sequence machinery used throughout the rest of the volume.

19.1 Relative chains

Let $A \subset X$. The singular chain groups satisfy

$$C_n(A) \subset C_n(X),$$

since every singular simplex in A is also one in X .

Definition 19.1 (Relative chain group). The relative chain group is

$$C_n(X, A) = C_n(X)/C_n(A).$$

The singular boundary preserves $C_n(A)$, so it descends to a boundary map on the quotient.

Proposition 19.2 (Relative chain complex). *The groups $C_n(X, A)$ form a chain complex with*

$$\partial[\sigma] = [\partial\sigma].$$

19.2 Relative homology

Definition 19.3 (Relative homology).

$$H_n(X, A) = H_n(C_*(X, A)).$$

A relative cycle may have boundary inside A , because that boundary becomes zero in the quotient chain complex.

19.3 Basic examples

If $A = \emptyset$, then

$$H_n(X, \emptyset) \cong H_n(X).$$

If $A = X$, then

$$H_n(X, X) = 0.$$

If X is path connected and $A = \{x_0\}$, then positive-degree relative homology agrees with reduced homology:

$$H_n(X, \{x_0\}) \cong \tilde{H}_n(X) \quad (n > 0),$$

with the degree-zero relation controlled by the long exact sequence.

19.4 Short exact sequence of chain complexes

There is a natural short exact sequence

$$0 \longrightarrow C_*(A) \xrightarrow{i_\#} C_*(X) \xrightarrow{q_\#} C_*(X, A) \longrightarrow 0.$$

This is exact in every degree.

19.5 Connecting homomorphism

A short exact sequence of chain complexes

$$0 \rightarrow A_* \xrightarrow{i} B_* \xrightarrow{p} C_* \rightarrow 0$$

produces connecting homomorphisms

$$\delta_n : H_n(C) \rightarrow H_{n-1}(A).$$

To define δ_n , take a cycle $c \in C_n$, lift it to $b \in B_n$, observe that ∂b lies in the image of A_{n-1} , and take the corresponding homology class in $H_{n-1}(A)$.

19.6 Long exact sequence of a pair

Theorem 19.4 (Long exact sequence of a pair). *For every pair $A \subset X$, there is a natural long exact sequence*

$$\cdots \rightarrow H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{q_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \rightarrow \cdots$$

In low degrees,

$$H_1(X, A) \xrightarrow{\partial} H_0(A) \rightarrow H_0(X) \rightarrow H_0(X, A) \rightarrow 0.$$

19.7 Exactness meaning

Exactness at $H_n(X)$ means

$$\text{im } i_* = \ker q_*.$$

Thus an absolute homology class becomes zero relatively exactly when it comes from the subspace.

Exactness at $H_n(X, A)$ means

$$\text{im } q_* = \ker \partial.$$

Thus a relative class comes from an absolute cycle exactly when its connecting boundary class vanishes.

19.8 Naturality

A map of pairs

$$f : (X, A) \rightarrow (Y, B), \quad f(A) \subset B,$$

induces maps on absolute and relative homology.

Theorem 19.5 (Naturality of the pair sequence). *The induced maps fit into a commutative diagram of long exact sequences for (X, A) and (Y, B) .*

19.9 The pair (D^n, S^{n-1})

Because D^n is contractible and S^{n-1} has reduced homology concentrated in degree $n - 1$, the pair sequence gives

$$H_k(D^n, S^{n-1}) \cong \begin{cases} \mathbb{Z}, & k = n, \\ 0, & k \neq n. \end{cases}$$

This is the relative calculation underlying cellular chain groups.

19.10 Quotient-space interpretation

For a good pair, such as a CW pair (X, A) , collapsing A to a point relates relative homology to reduced homology of the quotient.

Theorem 19.6 (Relative quotient theorem). *For a CW pair (X, A) ,*

$$H_n(X, A) \cong \tilde{H}_n(X/A)$$

for all n .

This is one of the cleanest geometric interpretations of relative homology.

19.11 Excision

Theorem 19.7 (Excision). *Suppose $Z \subset A \subset X$ and*

$$\overline{Z} \subset \text{int } A.$$

Then inclusion induces an isomorphism

$$H_n(X \setminus Z, A \setminus Z) \cong H_n(X, A).$$

Excision says relative homology ignores material lying completely inside the interior of the distinguished subspace.

19.12 Why excision matters

Excision allows local decomposition. It is the technical engine behind:

- cellular homology,
- Mayer–Vietoris sequences,
- local homology,
- manifold orientation theory.

19.13 Triples

Let

$$B \subset A \subset X.$$

There is a short exact sequence of relative chain complexes

$$0 \rightarrow C_*(A, B) \rightarrow C_*(X, B) \rightarrow C_*(X, A) \rightarrow 0.$$

Theorem 19.8 (Long exact sequence of a triple). *There is a natural long exact sequence*

$$\cdots \rightarrow H_n(A, B) \rightarrow H_n(X, B) \rightarrow H_n(X, A) \rightarrow H_{n-1}(A, B) \rightarrow \cdots.$$

The cellular differential from VIII/18 can be interpreted through such triples of consecutive skeleta.

19.14 Cellular boundary revisited

For

$$X^{n-2} \subset X^{n-1} \subset X^n,$$

the triple sequence contains

$$H_n(X^n, X^{n-1}) \rightarrow H_{n-1}(X^{n-1}, X^{n-2}),$$

which is exactly the cellular boundary map.

Thus the exact-sequence framework explains the construction used in VIII/18.

19.15 Relative homology of attached cells

If

$$X = A \cup \bigcup_{\alpha} e_{\alpha}^n$$

is obtained from A by attaching only n -cells, then

$$H_k(X, A) \cong \begin{cases} \mathbb{Z}^{\#\{n\text{-cells}\}}, & k = n, \\ 0, & k \neq n. \end{cases}$$

This is the cell-by-cell relative calculation behind CW homology.

19.16 Relative H_0

The group $H_0(X, A)$ measures components of X not represented by A . If A meets every path component of X , then

$$H_0(X, A) = 0.$$

19.17 Reduced homology from a pair

For nonempty X with basepoint x_0 ,

$$\tilde{H}_n(X) \cong H_n(X, \{x_0\})$$

for $n > 0$, and the pair long exact sequence organizes the degree-zero adjustment.

19.18 Boundary map geometry

A relative class may be represented by a chain whose boundary lies in A . The connecting map sends that relative class to the homology class of its boundary inside A .

This is often the most intuitive way to understand ∂ .

19.19 Exactness as obstruction theory

The pair sequence can be read as a succession of obstruction statements:

- an A -class dies in X exactly when it comes from a relative class one degree higher;
- a relative class comes from X exactly when its boundary in A vanishes;
- an X -class dies relatively exactly when it comes from A .

19.20 Bridge to homotopy invariance

Relative homology is functorial, so homotopies of maps of pairs should induce the same relative maps. VIII/20 proves the stronger chain-level statement for singular chains using prism operators and chain homotopies.

19.21 Exercises

Exercise 19.9. Define $C_n(X, A)$.

Hint. Quotient absolute chains by chains in A . □

Solution. $C_n(X, A) = C_n(X)/C_n(A)$. □

Exercise 19.10. Why does the boundary descend to the quotient?

Hint. $C_*(A)$ is a subcomplex. □

Solution. $\partial C_n(A) \subset C_{n-1}(A)$, so $\partial[c] = [\partial c]$ is well defined. □

Exercise 19.11. Define $H_n(X, A)$.

Hint. Take homology of the relative chain complex. □

Solution. $H_n(X, A) = H_n(C_*(X, A))$. □

Exercise 19.12. What is $H_n(X, \emptyset)$?

Hint. The quotient changes nothing. □

Solution. $H_n(X)$. □

Exercise 19.13. What is $H_n(X, X)$?

Hint. The quotient chain complex is zero. □

Solution. Zero. □

Exercise 19.14. Write the chain short exact sequence for a pair.

Hint. Subspace chains, space chains, quotient chains. □

Solution. $0 \rightarrow C_*(A) \rightarrow C_*(X) \rightarrow C_*(X, A) \rightarrow 0$. □

Exercise 19.15. What does this short exact sequence produce?

Hint. Use homological algebra. □

Solution. The long exact sequence of the pair. □

Exercise 19.16. Write the central part of the pair LES.

Hint. Absolute, relative, then one lower subspace group. □

Solution. $H_n(A) \rightarrow H_n(X) \rightarrow H_n(X, A) \xrightarrow{\partial} H_{n-1}(A)$. □

Exercise 19.17. What does exactness at $H_n(X)$ say?

Hint. Image equals kernel. □

Solution. $\text{im } H_n(A) \rightarrow H_n(X) = \ker(H_n(X) \rightarrow H_n(X, A))$. □

Exercise 19.18. What does the connecting map do geometrically?

Hint. Take the boundary of a relative cycle in A . □

Solution. It sends a relative class to the homology class of its boundary inside A . □

Exercise 19.19. What is $H_k(D^n, S^{n-1})$?

Hint. Use the pair sequence. □

Solution. $\mathbb{Z}$ for $k = n$, zero otherwise. □

Exercise 19.20. For a CW pair, how is $H_n(X, A)$ related to X/A ?

Hint. Collapse A . □

Solution. $H_n(X, A) \cong \tilde{H}_n(X/A)$. □

Exercise 19.21. State excision.

Hint. Remove a subset whose closure lies in the interior of A . □

Solution. $H_n(X \setminus Z, A \setminus Z) \cong H_n(X, A)$. □

Exercise 19.22. What is the purpose of excision?

Hint. Ignore interior material of the subspace. □

Solution. It localizes relative calculations and underlies cellular and Mayer–Vietoris arguments. □

Exercise 19.23. Write the inclusions for a triple.

Hint. Three nested spaces. □

Solution. $B \subset A \subset X$. □

Exercise 19.24. What LES comes from a triple?

Hint. Use relative pairs. □

Solution. $\cdots \rightarrow H_n(A, B) \rightarrow H_n(X, B) \rightarrow H_n(X, A) \rightarrow H_{n-1}(A, B) \rightarrow \cdots$. □

Exercise 19.25. How does the cellular boundary arise from a triple?

Hint. Use consecutive skeleta. □

Solution. From $X^{n-2} \subset X^{n-1} \subset X^n$. □

Exercise 19.26. If X is obtained from A by r n -cells only, what is $H_n(X, A)$?

Hint. One generator per cell. □

Solution. $\mathbb{Z}^r$. □

Exercise 19.27. What is $H_k(X, A)$ in other degrees in that situation?

Hint. Only n -cells were added. □

Solution. Zero. □

Exercise 19.28. When is $H_0(X, A) = 0$?

Hint. Make A meet every component. □

Solution. When A meets every path component of X . □

Exercise 19.29. How is reduced homology related to a based pair?

Hint. Use a point subspace. □

Solution. For $n > 0$, $\tilde{H}_n(X) \cong H_n(X, \{x_0\})$. □

Exercise 19.30. What maps induce morphisms of pair LESs?

Hint. Maps preserving subspaces. □

Solution. Maps of pairs $f : (X, A) \rightarrow (Y, B)$. □

Exercise 19.31. Is the pair LES natural?

Hint. Yes; induced maps commute with all arrows. □

Solution. Yes. □

Exercise 19.32. What is the bridge to VIII/20?

Hint. Homotopies should induce equal homology maps. □

Solution. VIII/20 proves this by a chain homotopy on singular chains. □

19.22 Solved problem dossiers

Problem 19.33 (Pair of a disk and boundary). Compute $H_*(D^2, S^1)$.

Solution. The only nonzero relative group is $H_2(D^2, S^1) \cong \mathbb{Z}$. □

Problem 19.34 (Interval endpoints). Compute $H_1(I, \partial I)$.

Solution. Since $I/\partial I \cong S^1$, the quotient theorem gives $H_1(I, \partial I) \cong \tilde{H}_1(S^1) \cong \mathbb{Z}$. □

Problem 19.35 (Sphere from disk pair). Use the quotient theorem to rederive $H_n(D^n, S^{n-1})$.

Solution. Collapse S^{n-1} to a point: $D^n/S^{n-1} \cong S^n$, so relative homology equals $\tilde{H}_*(S^n)$. □

Problem 19.36 (Relative circle to point). Compute $H_1(S^1, \{*\})$.

Solution. By reduced homology, it is $\tilde{H}_1(S^1) \cong \mathbb{Z}$. □

Problem 19.37 (Relative H_0). Let X have three components and let A meet exactly two. Compute $H_0(X, A)$.

Solution. One component remains unrepresented by A , so $H_0(X, A) \cong \mathbb{Z}$. □

Problem 19.38 (Connecting map in disk pair). What does $\partial : H_n(D^n, S^{n-1}) \rightarrow H_{n-1}(S^{n-1})$ do?

Solution. Both groups are $\mathbb{Z}$, and the connecting map sends the relative fundamental class of the disk to the fundamental class of its boundary sphere; it is an isomorphism up to orientation sign. □

Problem 19.39 (Exactness at a sphere). Why does $H_n(D^n) = 0$ force the connecting map above to be injective?

Solution. In the pair LES, exactness gives $\ker \partial = \text{im}(H_n(D^n) \rightarrow H_n(D^n, S^{n-1})) = 0$. □

Problem 19.40 (Cell attachment pair). If $X = A \cup e^n$, compute $H_*(X, A)$.

Solution. The quotient X/A is S^n , so $H_n(X, A) \cong \mathbb{Z}$ and all other relative groups vanish. □

Problem 19.41 (Two attached cells). If two n -cells are attached to A , compute relative homology.

Solution. Collapsing A gives a wedge $S^n \vee S^n$ up to the relevant quotient cell structure, so $H_n(X, A) \cong \mathbb{Z}^2$. □

Problem 19.42 (Excision on a disk). Remove a small closed disk from the interior of D^2 while keeping an annular subspace inside A . Why can the relative homology remain unchanged?

Solution. If the removed set lies with closure inside the interior of the distinguished subspace, excision applies and the inclusion of complements induces an isomorphism. □

Problem 19.43 (Triple of skeleta). For a CW complex, identify the three spaces used to define d_n^{cell} .

Solution. Use $X^{n-2} \subset X^{n-1} \subset X^n$; the triple connecting map runs from $H_n(X^n, X^{n-1})$ to $H_{n-1}(X^{n-1}, X^{n-2})$. □

Problem 19.44 (Absolute class dies relatively). Suppose $[z] \in H_n(X)$ maps to zero in $H_n(X, A)$. What does exactness imply?

Solution. $[z]$ lies in the image of $H_n(A) \rightarrow H_n(X)$; it is represented, up to homology, by a class from the subspace. $\square$

Problem 19.45 (Relative class lifts absolutely). Suppose $\xi \in H_n(X, A)$ has $\partial\xi = 0$. What follows?

Solution. Exactness says ξ lies in the image of $H_n(X) \rightarrow H_n(X, A)$. $\square$

Problem 19.46 (Subspace class dies in space). Suppose $\eta \in H_{n-1}(A)$ maps to zero in $H_{n-1}(X)$. What follows?

Solution. Exactness says η lies in the image of the connecting map $H_n(X, A) \rightarrow H_{n-1}(A)$. $\square$

Problem 19.47 (Quotient pair). For a CW pair (X, A) , why is quotient homology reduced?

Solution. Collapsing A creates a distinguished basepoint. Relative chains ignore chains inside A , corresponding to reduced chains of the quotient rather than an extra degree-zero component. $\square$

Problem 19.48 (Relative homology of a point pair). Compute $H_*(X, X)$.

Solution. The quotient chain complex $C_*(X)/C_*(X)$ is zero, so every relative homology group is zero. $\square$

Problem 19.49 (Naturality square). What commutativity condition must hold for connecting maps under a map of pairs?

Solution. If $f : (X, A) \rightarrow (Y, B)$, then $f_*\partial = \partial f_*$ between $H_n(X, A) \rightarrow H_{n-1}(A)$ and $H_n(Y, B) \rightarrow H_{n-1}(B)$. $\square$

Problem 19.50 (Reduced H_0 of two components). Let X have two path components and basepoint in one. Compute $H_0(X, \{x_0\})$.

Solution. The pair kills the component containing the basepoint but leaves one independent component, so $H_0(X, \{x_0\}) \cong \mathbb{Z}$, matching $\tilde{H}_0(X)$. $\square$

Problem 19.51 (Relative torus skeleton). For the torus T^2 , what is $H_2(T^2, T^1)$ in the standard CW structure?

Solution. There is one 2-cell, so $H_2(T^2, T^1) \cong \mathbb{Z}$. $\square$

Problem 19.52 (Relative projective skeleton). For $\mathbb{R}P^n$, compute $H_n(\mathbb{R}P^n, \mathbb{R}P^{n-1})$.

Solution. There is one n -cell, so the relative group is $\mathbb{Z}$. $\square$

Problem 19.53 (Excision intuition). Why should relative homology ignore a region lying deep inside A ?

Solution. Relative chains already regard chains in A as zero. Removing material whose closure lies inside $\text{int } A$ does not affect how chains outside A meet its boundary. $\square$

Problem 19.54 (Boundary as obstruction). Interpret a nonzero connecting class $\partial\xi$.

Solution. It is the obstruction to representing the relative class ξ by an absolute cycle in X ; exactness says precisely that ξ lifts absolutely iff this boundary class vanishes. $\square$

Problem 19.55 (Long exact sequence source). What algebraic input produces the LES of a pair?

Solution. The short exact sequence of chain complexes $0 \rightarrow C_*(A) \rightarrow C_*(X) \rightarrow C_*(X, A) \rightarrow 0$. $\square$

Problem 19.56 (Bridge to homotopy). Why should a homotopy of maps of pairs preserve the entire long exact sequence map?

Solution. If the two maps induce chain-homotopic chain maps, they induce equal maps on absolute and relative homology, and naturality then preserves all exact-sequence arrows. $\square$

Chapter 20

Homotopy Invariance

Homology is useful only if it ignores deformations that preserve homotopy type. The key theorem is that homotopic maps induce the same homomorphism on homology. At chain level, the difference between the two induced chain maps is a boundary commutator. For singular chains this is implemented by the prism operator.

20.1 Chain homotopies

Let

$$f_{\bullet}, g_{\bullet} : C_{\bullet} \rightarrow D_{\bullet}$$

be chain maps.

Definition 20.1 (Chain homotopy). A chain homotopy from f to g is a family of homomorphisms

$$P_n : C_n \rightarrow D_{n+1}$$

such that

$$g_n - f_n = \partial_{n+1}P_n + P_{n-1}\partial_n.$$

The operator P has degree $+1$.

20.2 Chain-homotopic maps induce equal homology maps

Theorem 20.2 (Chain-homotopy invariance). *If chain maps $f, g : C_{\bullet} \rightarrow D_{\bullet}$ are chain homotopic, then*

$$H_n(f) = H_n(g)$$

for every n .

Proof. If z is a cycle, then

$$g(z) - f(z) = \partial P(z) + P(\partial z) = \partial P(z),$$

so $f(z)$ and $g(z)$ differ by a boundary and represent the same homology class. $\square$

20.3 Topological homotopies

Suppose

$$f, g : X \rightarrow Y$$

are homotopic through

$$H : X \times I \rightarrow Y.$$

The goal is to construct a chain homotopy between the induced singular chain maps

$$f_{\#}, g_{\#} : C_{*}^{\text{sing}}(X) \rightarrow C_{*}^{\text{sing}}(Y).$$

20.4 Prisms

For a singular simplex

$$\sigma : \Delta^n \rightarrow X,$$

the homotopy gives a map

$$\Delta^n \times I \rightarrow Y.$$

The prism $\Delta^n \times I$ can be subdivided into $n + 1$ $(n + 1)$ -simplices.

Definition 20.3 (Prism operator). The prism operator

$$P_n : C_n(X) \rightarrow C_{n+1}(Y)$$

is the signed sum of the singular $(n + 1)$ -simplices obtained by this standard prism subdivision followed by $H \circ (\sigma \times \text{id})$.

20.5 Prism identity

Theorem 20.4 (Prism identity). *The prism operator satisfies*

$$g_{\#} - f_{\#} = \partial P + P \partial.$$

Thus P is a chain homotopy between $f_{\#}$ and $g_{\#}$.

20.6 Homotopy invariance theorem

Theorem 20.5 (Homotopic maps induce the same map). *If*

$$f \simeq g : X \rightarrow Y,$$

then

$$f_* = g_* : H_n(X) \rightarrow H_n(Y)$$

for every n .

20.7 Homotopy equivalence

Corollary 20.6 (Homotopy equivalent spaces have isomorphic homology). *If*

$$X \simeq Y,$$

then

$$H_n(X) \cong H_n(Y)$$

for all n .

Indeed, if $f : X \rightarrow Y$ and $g : Y \rightarrow X$ are homotopy inverses, then

$$g_* f_* = (g \circ f)_* = (\text{id}_X)_*,$$

and similarly $f_* g_* = \text{id}$.

20.8 Contractible spaces

Corollary 20.7 (Homology of a contractible space). *If X is nonempty and contractible, then*

$$H_0(X) \cong \mathbb{Z}, \quad H_n(X) = 0 \quad (n > 0).$$

Equivalently,

$$\tilde{H}_n(X) = 0$$

for every n .

20.9 Deformation retracts

If $A \subset X$ is a deformation retract, then

$$A \simeq X.$$

Corollary 20.8 (Deformation retract). *The inclusion*

$$A \hookrightarrow X$$

induces homology isomorphisms in every degree.

20.10 Strong deformation retracts

A strong deformation retract gives the same conclusion, with the homotopy fixed on A . The additional relative control is important in pairs and CW constructions but does not alter the absolute homology conclusion.

20.11 Retracts and split maps

If

$$r : X \rightarrow A, \quad i : A \hookrightarrow X, \quad ri = \text{id}_A,$$

then

$$r_* i_* = \text{id}_{H_*(A)}.$$

Thus i_* is injective and r_* is surjective even without assuming a deformation retraction.

20.12 Null-homotopic maps

If

$$f : X \rightarrow Y$$

is null-homotopic, then on reduced homology

$$\tilde{f}_* = 0.$$

For positive degrees,

$$f_* : H_n(X) \rightarrow H_n(Y)$$

is zero.

20.13 Degree and homotopy

For maps

$$f, g : S^n \rightarrow S^n,$$

homotopy invariance of top homology gives

$$f \simeq g \implies \deg f = \deg g.$$

This recovers the degree obstruction from VIII/03 inside homology.

20.14 No-retraction theorem via homology

Suppose

$$r : D^{n+1} \rightarrow S^n$$

were a retraction. Then inclusion

$$i : S^n \hookrightarrow D^{n+1}$$

would satisfy

$$r_* i_* = \text{id}_{H_n(S^n)}.$$

But

$$H_n(D^{n+1}) = 0,$$

so $i_* = 0$, contradiction.

Theorem 20.9 (Homological no-retraction). *There is no retraction*

$$D^{n+1} \rightarrow S^n.$$

20.15 Homology detects noncontractibility

If a connected space has

$$\tilde{H}_n(X) \neq 0$$

for some n , then X is not contractible.

Thus:

$$S^n, \quad T^n, \quad \mathbb{R}P^{2k+1}$$

and many other spaces are immediately seen to be noncontractible.

20.16 Homology is not complete

The converse is false: spaces can have the same homology groups without being homotopy equivalent.

Homology is a powerful but coarse invariant. It is abelian and loses much noncommutative information visible to the fundamental group.

20.17 Acyclic does not mean contractible

A space is acyclic if its reduced homology vanishes. Acyclic spaces need not be contractible.

Thus

$$\tilde{H}_*(X) = 0$$

is necessary but not sufficient for contractibility.

20.18 Homotopy invariance for pairs

If maps of pairs

$$f, g : (X, A) \rightarrow (Y, B)$$

are homotopic through maps of pairs, then

$$f_* = g_* : H_n(X, A) \rightarrow H_n(Y, B).$$

The prism construction descends to relative quotient chain complexes.

20.19 Naturality with exact sequences

Homotopy invariance is compatible with the long exact sequence of a pair. Homotopic maps of pairs induce the same morphism of long exact sequences.

20.20 Cellular consequence

If two CW complexes are homotopy equivalent, cellular homology calculations for either must give isomorphic homology groups. This justifies simplifying cell structures through homotopy-preserving operations before computation.

20.21 Chain-homotopy viewpoint

The topological statement

$$f \simeq g$$

is converted into the algebraic statement

$$f_{\#} \simeq_{\text{ch}} g_{\#}.$$

This is the first major instance of a recurring principle:

$$\text{topological deformation} \quad \rightsquigarrow \quad \text{algebraic chain homotopy}.$$

VIII/22 develops chain homotopies as algebraic objects in their own right.

20.22 Bridge to Euler characteristic

For finite CW complexes, homology ranks determine Euler characteristic through the Euler–Poincare formula. Since homology is homotopy invariant, Euler characteristic becomes a homotopy invariant as well. VIII/21 develops this systematically.

20.23 Exercises

Exercise 20.10. Define a chain homotopy P between f and g .

Hint. Use a degree +1 operator. □

Solution. $g - f = \partial P + P\partial$. □

Exercise 20.11. What degree does a chain homotopy have?

Hint. It raises chain degree. ☐

Solution. $+1$. ☐

Exercise 20.12. Why do chain-homotopic maps agree on homology?

Hint. Evaluate on a cycle. ☐

Solution. For $\partial z = 0$, $g(z) - f(z) = \partial P(z)$, a boundary. ☐

Exercise 20.13. What topological object produces the singular-chain homotopy?

Hint. A homotopy $H : X \times I \rightarrow Y$. ☐

Solution. The prism operator. ☐

Exercise 20.14. What is the prism built from?

Hint. Use $\Delta^n \times I$. ☐

Solution. The product of a singular simplex with the interval, subdivided into $(n + 1)$ -simplices. ☐

Exercise 20.15. State the prism identity.

Hint. Compare endpoint chain maps. ☐

Solution. $g_{\#} - f_{\#} = \partial P + P\partial$. ☐

Exercise 20.16. What do homotopic maps induce on homology?

Hint. Apply the prism identity. ☐

Solution. The same homomorphism. ☐

Exercise 20.17. What does a homotopy equivalence induce?

Hint. Use homotopy inverses. ☐

Solution. Isomorphisms on all homology groups. ☐

Exercise 20.18. What is the reduced homology of a contractible space?

Hint. Compare to a point. ☐

Solution. Zero in every degree. ☐

Exercise 20.19. What is H_0 of a nonempty contractible space?

Hint. It is path connected. ☐

Solution. $\mathbb{Z}$. ☐

Exercise 20.20. What does a deformation retract induce?

Hint. It is a homotopy equivalence. ☐

Solution. Homology isomorphisms. ☐

Exercise 20.21. If $ri = \text{id}$, what follows for i_* ?

Hint. Apply homology. ☐

Solution. i_* is injective. ☐

Exercise 20.22. If $ri = \text{id}$, what follows for r_* ?

Hint. Same splitting equation. ☐

Solution. r_* is surjective. ☐

Exercise 20.23. What does a null-homotopic map induce on reduced homology?

Hint. Compare to a constant map. ☐

Solution. The zero map. ☐

Exercise 20.24. What does homotopy imply about degree of sphere maps?

Hint. Top homology maps agree. ☐

Solution. Homotopic sphere maps have equal degree. ☐

Exercise 20.25. Why is S^n not contractible?

Hint. Use nonzero reduced homology. ☐

Solution. $\tilde{H}_n(S^n) \cong \mathbb{Z} \neq 0$. ☐

Exercise 20.26. Can equal homology groups imply homotopy equivalence?

Hint. Homology is incomplete. ☐

Solution. No. ☐

Exercise 20.27. Does acyclic imply contractible?

Hint. Recall the warning. ☐

Solution. No. ☐

Exercise 20.28. State homotopy invariance for relative homology.

Hint. Use maps of pairs. ☐

Solution. Homotopic maps of pairs induce the same maps on $H_n(X, A)$. ☐

Exercise 20.29. How does the prism operator act on quotient chains?

Hint. It respects subspace chains under a homotopy of pairs. ☐

Solution. It descends to relative chain complexes. ☐

Exercise 20.30. Why does homotopy invariance help cellular computations?

Hint. Simplify by homotopy equivalence. ☐

Solution. Homotopy-equivalent CW models have isomorphic homology. ☐

Exercise 20.31. What algebraic relation corresponds to topological homotopy?

Hint. Pass to singular chains. ☐

Solution. Chain homotopy. □

Exercise 20.32. What chapter develops chain homotopy abstractly later?

Hint. Look ahead. □

Solution. VIII/22. □

Exercise 20.33. What is the bridge to VIII/21?

Hint. Euler characteristic is computed from homology ranks. □

Solution. Homotopy invariance of homology implies homotopy invariance of Euler characteristic for finite CW complexes. □

20.24 Solved problem dossiers

Problem 20.34 (Cycle proof). Prove chain-homotopic maps induce equal maps on homology.

Solution. If z is a cycle, $g(z) - f(z) = \partial P(z) + P\partial z = \partial P(z)$, so the two images differ by a boundary. □

Problem 20.35 (Homotopy inverse). Let $f : X \rightarrow Y$, $g : Y \rightarrow X$ be homotopy inverses. Prove f_* is an isomorphism.

Solution. Homotopy invariance gives $g_*f_* = (gf)_* = (\text{id}_X)_*$ and $f_*g_* = \text{id}_{H_*(Y)}$, so g_* is the inverse. □

Problem 20.36 (Contractible disk). Compute the homology of D^n from homotopy invariance.

Solution. $D^n \simeq *$, so $H_0(D^n) = \mathbb{Z}$ and all positive homology vanishes. □

Problem 20.37 (Deformation retract annulus). Compute the homology of an annulus by deformation retracting to its core circle.

Solution. The annulus is homotopy equivalent to S^1 , so $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}$, and higher groups vanish. □

Problem 20.38 (Punctured plane). Compute $H_*(\mathbb{R}^2 \setminus \{0\})$.

Solution. Radial deformation retract gives S^1 , hence $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}$, higher groups zero. □

Problem 20.39 (Punctured Euclidean space). Compute $H_*(\mathbb{R}^{n+1} \setminus \{0\})$.

Solution. It deformation retracts onto S^n , so homology is $\mathbb{Z}$ in degrees 0, n and zero otherwise. □

Problem 20.40 (No retraction). Give the homological proof that D^{n+1} does not retract onto S^n .

Solution. A retraction would give $r_*i_* = \text{id}$ on $H_n(S^n) = \mathbb{Z}$, but i_* factors through $H_n(D^{n+1}) = 0$, contradiction. □

Problem 20.41 (Null map). If $f : S^n \rightarrow Y$ is null-homotopic, what is f_* on H_n for $n > 0$?

Solution. Zero, because f is homotopic to a constant map and positive homology of a point vanishes. □

Problem 20.42 (Degree obstruction). Why can maps $S^n \rightarrow S^n$ of different degree not be homotopic?

Solution. The induced map on $H_n(S^n) \cong \mathbb{Z}$ is multiplication by degree. Homotopic maps induce the same homology map. $\square$

Problem 20.43 (Antipodal even sphere). Use homology to distinguish the antipodal map from the identity on even S^n .

Solution. The identity induces multiplication by 1 on H_n ; the antipodal map has degree -1 for even n , so the induced maps differ. $\square$

Problem 20.44 (Retract splitting). Suppose A is a retract of X . What algebraic information follows on homology?

Solution. The inclusion-induced map is split injective and the retraction-induced map is split surjective; $H_*(A)$ occurs as a direct summand in the appropriate algebraic sense. $\square$

Problem 20.45 (Noncontractible torus). Why is T^2 not contractible?

Solution. $H_1(T^2) \cong \mathbb{Z}^2 \neq 0$, while a contractible space has zero positive homology. $\square$

Problem 20.46 (Projective noncontractibility). Why is $\mathbb{R}P^2$ not contractible?

Solution. $H_1(\mathbb{R}P^2) \cong \mathbb{Z}/2 \neq 0$. $\square$

Problem 20.47 (Acyclic warning). Why can vanishing reduced homology not prove contractibility?

Solution. Homology forgets nonabelian and higher homotopy information; there exist noncontractible acyclic spaces. $\square$

Problem 20.48 (Relative deformation). If (X, A) deformation retracts as a pair onto (Y, B) , what happens to relative homology?

Solution. The pair maps are homotopy equivalences through maps of pairs, so they induce isomorphisms $H_n(X, A) \cong H_n(Y, B)$. $\square$

Problem 20.49 (Prism in dimension zero). What does the prism operator do to a singular 0-simplex x ?

Solution. The homotopy path $t \mapsto H(x, t)$ is a singular 1-simplex whose boundary is $g(x) - f(x)$, exactly the degree-zero prism identity. $\square$

Problem 20.50 (Prism boundary meaning). Geometrically, what are the endpoint terms in the boundary of a prism?

Solution. The top and bottom faces are the singular simplices obtained from $g \circ \sigma$ and $f \circ \sigma$; the side faces combine into $P(\partial\sigma)$. $\square$

Problem 20.51 (Identity homotopy). What chain homotopy is needed between $\text{id}_\#$ and itself?

Solution. The zero operator $P = 0$ satisfies $\text{id} - \text{id} = 0 = \partial P + P\partial$. $\square$

Problem 20.52 (Composition with homotopy). If $f_0 \simeq f_1 : X \rightarrow Y$ and $h : Y \rightarrow Z$, compare hf_0 and hf_1 on homology.

Solution. They are homotopic, so $(hf_0)_* = (hf_1)_*$; equivalently $h_*f_{0*} = h_*f_{1*}$. $\square$

Problem 20.53 (Precomposition). If $f_0 \simeq f_1 : X \rightarrow Y$ and $k : W \rightarrow X$, compare f_0k and f_1k .

Solution. They are homotopic via $H(k(w), t)$, hence induce the same homology map. $\square$

Problem 20.54 (Euler consequence). Why must finite CW complexes of different Euler characteristic fail to be homotopy equivalent?

Solution. Euler characteristic equals the alternating sum of homology ranks and is therefore preserved by homotopy equivalence. $\square$

Problem 20.55 (Bridge to algebraic chain homotopy). What additional structure will VIII/22 study beyond this chapter?

Solution. It treats chain homotopies abstractly, including chain contractions and algebraic equivalences, rather than only the prism operator arising from topological homotopies. $\square$

Chapter 21

Euler Characteristic

Euler characteristic compresses a finite cell decomposition into a single integer. Remarkably, the same integer can be computed from homology ranks, so it is independent of the chosen finite CW or simplicial structure and is preserved by homotopy equivalence. This chapter develops the combinatorial, homological, and structural forms of the invariant.

21.1 Cell-count definition

Let X be a finite CW complex with c_n cells of dimension n .

Definition 21.1 (Euler characteristic).

$$\chi(X) = \sum_{n \geq 0} (-1)^n c_n.$$

Only finitely many terms are nonzero.

21.2 Simplicial form

If a finite simplicial complex has f_n n -simplices, then

$$\chi(K) = \sum_n (-1)^n f_n.$$

For a triangulated space $X \cong |K|$, this agrees with the CW definition.

21.3 Basic examples

For a finite discrete set with m points,

$$\chi = m.$$

For an interval,

$$\chi = 2 - 1 = 1.$$

For a polygonal circle,

$$\chi = V - E = 0.$$

For the standard two-cell sphere structure,

$$\chi(S^n) = 1 + (-1)^n.$$

21.4 Graphs

For a finite graph,

$$\chi(X) = V - E.$$

If the graph is connected and has fundamental-group rank r , then

$$r = E - V + 1,$$

so

$$\chi(X) = 1 - r.$$

Thus a tree has

$$\chi = 1.$$

21.5 Closed orientable surfaces

The standard CW structure on a genus- g orientable surface has one 0-cell, $2g$ 1-cells, and one 2-cell. Hence

$$\chi(\Sigma_g) = 2 - 2g.$$

In particular,

$$\chi(S^2) = 2, \quad \chi(T^2) = 0.$$

21.6 Real projective spaces

The standard CW structure on $\mathbb{R}P^n$ has one cell in every dimension $0, \dots, n$, so

$$\chi(\mathbb{R}P^n) = \sum_{k=0}^n (-1)^k.$$

Therefore

$$\chi(\mathbb{R}P^n) = \begin{cases} 1, & n \text{ even,} \\ 0, & n \text{ odd.} \end{cases}$$

21.7 Euler–Poincare formula

Let X be a finite CW complex. Write

$$b_n = \text{rank } H_n(X; \mathbb{Z}).$$

Theorem 21.2 (Euler–Poincare).

$$\chi(X) = \sum_n (-1)^n b_n.$$

Torsion in homology does not contribute to the rank sum.

21.8 Algebraic proof

For a finite chain complex of finite-rank free abelian groups,

$$C_n,$$

write

$$Z_n = \ker \partial_n, \quad B_n = \operatorname{im} \partial_{n+1}.$$

From

$$0 \rightarrow Z_n \rightarrow C_n \rightarrow B_{n-1} \rightarrow 0$$

we obtain

$$\operatorname{rank} C_n = \operatorname{rank} Z_n + \operatorname{rank} B_{n-1}.$$

From

$$0 \rightarrow B_n \rightarrow Z_n \rightarrow H_n \rightarrow 0$$

we obtain

$$\operatorname{rank} Z_n = \operatorname{rank} B_n + \operatorname{rank} H_n.$$

Alternating sums telescope, leaving the homology-rank formula.

21.9 Independence of cell structure

Corollary 21.3 (Well-definedness). *The alternating cell count is independent of the chosen finite CW structure.*

Indeed both cell decompositions compute the same homology ranks.

21.10 Homotopy invariance

Corollary 21.4 (Homotopy invariance). *If finite CW complexes X and Y are homotopy equivalent, then*

$$\chi(X) = \chi(Y).$$

This follows immediately from VIII/20.

21.11 Contractible finite complexes

If X is a nonempty finite contractible CW complex, then

$$H_0(X) \cong \mathbb{Z}, \quad H_n(X) = 0 \ (n > 0),$$

so

$$\chi(X) = 1.$$

Therefore a finite CW complex with

$$\chi(X) \neq 1$$

cannot be contractible.

The converse is false.

21.12 Disjoint unions

For finite CW complexes,

$$\chi(X \sqcup Y) = \chi(X) + \chi(Y).$$

This is immediate from either cell counts or homology direct sums.

21.13 Wedges

For connected finite CW complexes with chosen basepoints,

$$\chi(X \vee Y) = \chi(X) + \chi(Y) - 1.$$

The subtraction corrects for identifying two zero-cells into one.

More generally,

$$\chi\left(\bigvee_{j=1}^r X_j\right) = \sum_j \chi(X_j) - (r - 1).$$

21.14 Product formula

Theorem 21.5 (Product formula). *For finite CW complexes,*

$$\chi(X \times Y) = \chi(X)\chi(Y).$$

One proof uses product cells: an i -cell times a j -cell contributes an $(i + j)$ -cell, and the alternating double sum factors.

21.15 Examples of products

Since

$$\chi(S^1) = 0,$$

every finite CW product

$$S^1 \times X$$

has

$$\chi = 0.$$

For example,

$$\chi(T^n) = 0$$

for every $n \geq 1$.

Also,

$$\chi(S^2 \times S^2) = 4.$$

21.16 Inclusion–exclusion

Suppose finite subcomplexes A, B cover a finite CW complex X :

$$X = A \cup B.$$

Theorem 21.6 (Euler inclusion–exclusion).

$$\chi(X) = \chi(A) + \chi(B) - \chi(A \cap B).$$

This follows from cell counts when the decompositions are compatible, and more generally from Mayer–Vietoris/homological arguments.

21.17 Attaching one cell

If Y is obtained from X by attaching one n -cell, then

$$\chi(Y) = \chi(X) + (-1)^n.$$

Thus:

- adding a 0-cell increases χ by 1;
- adding a 1-cell decreases χ by 1;
- adding a 2-cell increases χ by 1.

The attaching map does not affect this raw Euler change.

21.18 Relative Euler characteristic

For a finite CW pair (X, A) , define

$$\chi(X, A) = \chi(X) - \chi(A).$$

Equivalently, if the relative homology groups have finite rank,

$$\chi(X, A) = \sum_n (-1)^n \text{rank } H_n(X, A).$$

21.19 Exact sequences and Euler sums

For a finite exact sequence of finite-rank abelian groups

$$0 \rightarrow A_0 \rightarrow A_1 \rightarrow \cdots \rightarrow A_m \rightarrow 0,$$

the alternating sum of ranks is zero.

This elementary fact is the algebraic reason Euler characteristic behaves well with long exact sequences.

21.20 Euler characteristic and coverings

If

$$p : \tilde{X} \rightarrow X$$

is a finite d -sheeted covering of finite CW complexes, then

$$\chi(\tilde{X}) = d\chi(X).$$

Each cell downstairs lifts to d cells upstairs.

21.21 Covering examples

For the d -fold circle cover,

$$0 = \chi(S^1) = d \cdot 0.$$

For a finite-sheeted covering

$$\tilde{\Sigma} \rightarrow \Sigma_g$$

of closed orientable surfaces,

$$\chi(\tilde{\Sigma}) = d(2 - 2g).$$

This constrains the genus upstairs.

21.22 Odd-dimensional closed manifolds preview

For closed orientable odd-dimensional manifolds, Poincare duality later implies

$$\chi(M) = 0.$$

That theorem belongs to VIII/33; here it is only a preview of how deeper homological symmetry controls Euler characteristic.

21.23 What Euler characteristic cannot see

Euler characteristic is extremely coarse. Spaces with the same χ can have very different topology.

For example,

$$\chi(S^1) = 0 \quad \text{and} \quad \chi(T^2) = 0,$$

yet their homology and fundamental groups are different.

21.24 Bridge to homological machinery

The Euler–Poincare argument is a rank shadow of chain-level algebra. The next part develops that algebra more deeply through chain homotopies, contractions, mapping cones, coefficient changes, universal coefficient formulas, and products.

21.25 Exercises

Exercise 21.7. Define $\chi(X)$ from finite CW cells.

Hint. Take an alternating count.

□

Solution. $\chi(X) = \sum_n (-1)^n c_n$.

□

Exercise 21.8. What is the simplicial formula?

Hint. Replace cell counts by simplex counts.

□

Solution. $\chi(K) = \sum_n (-1)^n f_n$.

□

Exercise 21.9. Compute $\chi(*)$.

Hint. One 0-cell.

□

Solution. 1.

□

Exercise 21.10. Compute $\chi(I)$.

Hint. Two vertices and one edge.

□

Solution. $2 - 1 = 1$.

□

Exercise 21.11. Compute $\chi(S^1)$.

Hint. One 0-cell and one 1-cell.

□

Solution. 0.

□

Exercise 21.12. Compute $\chi(S^n)$.

Hint. One 0-cell and one n -cell.

□

Solution. $1 + (-1)^n$.

□

Exercise 21.13. For a finite graph, what is χ ?

Hint. Only dimensions 0 and 1.

□

Solution. $V - E$.

□

Exercise 21.14. For a connected graph of rank r , what is χ ?

Hint. Use $r = E - V + 1$.

□

Solution. $1 - r$.

□

Exercise 21.15. What is χ of a tree?

Hint. Rank zero.

□

Solution. 1.

□

Exercise 21.16. Compute $\chi(\Sigma_g)$.

Hint. Use 1, $2g$, 1 cells.

□

Solution. $2 - 2g$.

□

Exercise 21.17. Compute $\chi(\mathbb{R}P^n)$.

Hint. One cell per dimension.

□

Solution. 1 for even n , 0 for odd n .

□

Exercise 21.18. State Euler–Poincare.

Hint. Use homology ranks.

□

Solution. $\chi(X) = \sum_n (-1)^n \text{rank } H_n(X)$.

□

Exercise 21.19. Does torsion contribute to Euler–Poincare?

Hint. Ranks ignore finite torsion.

□

Solution. No.

□

Exercise 21.20. Why is Euler characteristic independent of finite CW structure?

Hint. Both compute the same homology rank sum.

□

Solution. Euler–Poincare makes the cell count equal a homological invariant.

□

Exercise 21.21. Is Euler characteristic a homotopy invariant for finite CW complexes?

Hint. Use homology invariance.

□

Solution. Yes.

□

Exercise 21.22. What is χ of a nonempty finite contractible CW complex?

Hint. Same homology as a point. □

Solution. 1. □

Exercise 21.23. State disjoint-union additivity.

Hint. Cell counts add. □

Solution. $\chi(X \sqcup Y) = \chi(X) + \chi(Y)$. □

Exercise 21.24. State wedge additivity.

Hint. One basepoint is identified. □

Solution. $\chi(X \vee Y) = \chi(X) + \chi(Y) - 1$. □

Exercise 21.25. State the product formula.

Hint. Product cell dimensions add. □

Solution. $\chi(X \times Y) = \chi(X)\chi(Y)$. □

Exercise 21.26. Compute $\chi(T^n)$ for $n \geq 1$.

Hint. It contains an S^1 factor. □

Solution. 0. □

Exercise 21.27. State inclusion–exclusion.

Hint. Subtract the overlap. □

Solution. $\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B)$. □

Exercise 21.28. What happens to χ when one n -cell is attached?

Hint. Add its alternating contribution. □

Solution. χ changes by $(-1)^n$. □

Exercise 21.29. How does a d -sheeted finite cover affect χ ?

Hint. Every cell lifts d times. □

Solution. $\chi(\tilde{X}) = d\chi(X)$. □

Exercise 21.30. What is the bridge beyond VIII/21?

Hint. Move from ranks to chain-level algebra. □

Solution. Part V develops chain homotopies, mapping cones, coefficients, UCT, and products. □

21.26 Solved problem dossiers

Problem 21.31 (Triangle disk). Compute Euler characteristic of a filled triangle.

Solution. There are 3 vertices, 3 edges, and 1 face, so $\chi = 3 - 3 + 1 = 1$. □

Problem 21.32 (Tetrahedral sphere). Compute χ of the boundary of a tetrahedron.

Solution. $V = 4, E = 6, F = 4$, so $\chi = 4 - 6 + 4 = 2$. □

Problem 21.33 (Square with diagonal). Compute χ of a square triangulated by one diagonal.

Solution. $V = 4, E = 5, F = 2$, so $\chi = 4 - 5 + 2 = 1$. □

Problem 21.34 (Graph rank). A connected finite graph has $V = 10, E = 13$. Compute its fundamental-group rank and Euler characteristic.

Solution. The rank is $E - V + 1 = 4$, and $\chi = V - E = -3 = 1 - 4$. □

Problem 21.35 (Rose). Compute χ of the wedge of r circles.

Solution. One vertex and r one-cells give $\chi = 1 - r$. □

Problem 21.36 (Genus three). Compute $\chi(\Sigma_3)$.

Solution. $2 - 2 \cdot 3 = -4$. □

Problem 21.37 (Projective plane). Compute $\chi(\mathbb{R}P^2)$ from cells and homology.

Solution. Cell count gives $1 - 1 + 1 = 1$. Homology ranks are $1, 0, 0$ since H_1 is torsion, so Euler–Poincaré also gives 1. □

Problem 21.38 (Projective 3-space). Compute $\chi(\mathbb{R}P^3)$.

Solution. Cell count $1 - 1 + 1 - 1 = 0$. Homology ranks $b_0 = b_3 = 1$ give $1 - 1 = 0$. □

Problem 21.39 (Torus). Compute $\chi(T^2)$ from homology.

Solution. The Betti numbers are $1, 2, 1$, so $\chi = 1 - 2 + 1 = 0$. □

Problem 21.40 (Sphere product). Compute $\chi(S^2 \times S^2)$.

Solution. $\chi(S^2) = 2$, so the product formula gives 4. □

Problem 21.41 (Circle product). Why does $S^1 \times X$ have zero Euler characteristic for finite CW X ?

Solution. $\chi(S^1) = 0$, so the product formula gives $0 \cdot \chi(X) = 0$. □

Problem 21.42 (Wedge of sphere and circle). Compute $\chi(S^2 \vee S^1)$.

Solution. $\chi(S^2) + \chi(S^1) - 1 = 2 + 0 - 1 = 1$. □

Problem 21.43 (Three-sphere wedge). Compute $\chi(S^2 \vee S^2 \vee S^2)$.

Solution. The sum is 6, and identifying three basepoints costs 2, so $\chi = 4$. □

Problem 21.44 (Attach a one-cell). If X has $\chi = 5$, what is the Euler characteristic after attaching one 1-cell?

Solution. It decreases by 1, giving 4. $\square$

Problem 21.45 (Attach a two-cell). Continue from the previous space by attaching one 2-cell.

Solution. It increases by 1, returning to 5. $\square$

Problem 21.46 (Contractibility obstruction). A finite CW complex has $\chi = 0$. What can you conclude?

Solution. It cannot be contractible, since every nonempty finite contractible CW complex has Euler characteristic 1. $\square$

Problem 21.47 (Same Euler warning). Why can $\chi = 0$ not distinguish S^1 from T^2 ?

Solution. Both have Euler characteristic zero, but their homology and fundamental groups differ. Euler characteristic is only one alternating integer. $\square$

Problem 21.48 (Covering surface genus). Let $\tilde{\Sigma} \rightarrow \Sigma_2$ be a 3-sheeted cover of closed orientable surfaces. Find the genus upstairs.

Solution. $\chi(\Sigma_2) = -2$, so upstairs $\chi = -6$. Solve $2 - 2\tilde{g} = -6$, giving $\tilde{g} = 4$. $\square$

Problem 21.49 (Cover of a torus). What Euler characteristic must every finite-sheeted cover of T^2 have?

Solution. Since $\chi(T^2) = 0$, the covering formula gives 0. $\square$

Problem 21.50 (Relative Euler disk pair). Compute $\chi(D^n, S^{n-1})$.

Solution. $\chi(D^n) - \chi(S^{n-1}) = 1 - [1 + (-1)^{n-1}] = (-1)^n$, matching the single relative n -cell. $\square$

Problem 21.51 (Exact sequence rank cancellation). Why does a finite exact sequence have zero alternating rank sum?

Solution. Each image rank appears once with each sign when the groups are decomposed into successive kernels/images, so the contributions telescope. $\square$

Problem 21.52 (Euler–Poincaré torsion). Why does $\mathbb{Z}/5$ in homology contribute nothing to Euler characteristic?

Solution. Euler–Poincaré uses free ranks (Betti numbers); a finite torsion group has rank zero. $\square$

Problem 21.53 (Subdivision). Why does barycentric subdivision preserve Euler characteristic?

Solution. It preserves the homeomorphism type, hence homology and Euler characteristic. Combinatorially the larger simplex counts have the same alternating sum. $\square$

Problem 21.54 (Bridge to duality). What later theorem forces $\chi = 0$ for closed orientable odd-dimensional manifolds?

Solution. Poincaré duality in VIII/33 pairs Betti numbers symmetrically across complementary dimensions, causing alternating cancellation. $\square$

Part V

Homological Machinery

Chapter 22

Chain Homotopies

Chain homotopy is the algebraic shadow of topological homotopy. It compares chain maps without requiring equality and is exactly the equivalence relation needed for homology: chain-homotopic maps induce the same maps on homology.

22.1 Definition

Definition 22.1 (Chain homotopy). Let $f, g : C_\bullet \rightarrow D_\bullet$ be chain maps. A chain homotopy $s : f \simeq g$ is a family

$$s_n : C_n \rightarrow D_{n+1}$$

such that

$$g - f = \partial s + s \partial.$$

22.2 Degree convention

The chain differential has degree -1 , while a chain homotopy has degree $+1$. Thus both terms

$$\partial s, \quad s \partial$$

have degree 0.

22.3 Homology consequence

Theorem 22.2 (Homology invariance). *If $f \simeq g$ as chain maps, then*

$$H_n(f) = H_n(g)$$

for every n .

Proof. For a cycle z ,

$$g(z) - f(z) = \partial s(z) + s(\partial z) = \partial s(z),$$

so the two images differ by a boundary. □

22.4 Reflexivity

The zero operator gives

$$f \simeq f.$$

22.5 Symmetry

If $s : f \simeq g$, then

$$-s : g \simeq f.$$

22.6 Transitivity

If $s : f \simeq g$ and $t : g \simeq h$, then

$$s + t : f \simeq h.$$

Hence chain homotopy is an equivalence relation on chain maps.

22.7 Compatibility with composition

If $f \simeq g : C \rightarrow D$ and $u : D \rightarrow E$ is a chain map, then

$$uf \simeq ug.$$

Similarly, if $v : B \rightarrow C$, then

$$fv \simeq gv.$$

22.8 Chain-homotopy equivalence

Definition 22.3 (Chain-homotopy equivalence). Chain complexes C and D are chain-homotopy equivalent if there are chain maps

$$f : C \rightarrow D, \quad g : D \rightarrow C$$

with

$$gf \simeq \text{id}_C, \quad fg \simeq \text{id}_D.$$

Theorem 22.4. *Chain-homotopy-equivalent complexes have naturally isomorphic homology groups.*

22.9 Null-homotopic chain maps

A chain map $f : C \rightarrow D$ is null-homotopic if

$$f \simeq 0.$$

Equivalently there are maps $s_n : C_n \rightarrow D_{n+1}$ satisfying

$$f = \partial s + s \partial.$$

Every null-homotopic chain map induces the zero map on homology.

22.10 The prism operator revisited

VIII/20 constructed, from a homotopy $F : X \times I \rightarrow Y$, a prism operator

$$P : C_n^{\text{sing}}(X) \rightarrow C_{n+1}^{\text{sing}}(Y)$$

with

$$g_{\#} - f_{\#} = \partial P + P \partial.$$

Thus topological homotopy becomes chain homotopy.

22.11 Homotopy of simplicial maps

After subdivision or through appropriate simplicial homotopy data, simplicial maps can likewise give chain-homotopic simplicial chain maps.

22.12 Chain homotopy and exact complexes

An exact complex need not automatically come equipped with a contraction. Chain homotopy provides the language for stating the stronger property that the identity itself is null-homotopic.

22.13 Hom-complex viewpoint

For graded homomorphisms $T : C \rightarrow D$, define

$$\delta(T) = \partial_D T - (-1)^{|T|} T \partial_C.$$

For $|s| = 1$,

$$\delta(s) = \partial s + s \partial.$$

Thus $g - f = \delta(s)$: chain-homotopic maps differ by a coboundary in the Hom-complex.

22.14 Matrices

For finite free complexes, a chain homotopy is represented by matrices S_n satisfying

$$G_n - F_n = D_{n+1}^D S_n + S_{n-1} D_n^C.$$

This converts chain-homotopy questions into linear Diophantine equations.

22.15 Direct sums

Chain homotopies add componentwise under direct sums.

22.16 Tensor-product preview

For tensor products, chain homotopies interact with Koszul signs. The full product machinery is deferred to VIII/27.

22.17 Relative chains

A homotopy of maps of pairs produces a chain homotopy on relative chain complexes because the prism operator preserves the distinguished subcomplex.

22.18 Augmented complexes

For augmented complexes, chain homotopies may be required to respect the augmentation. This is useful when reduced homology is the natural invariant.

22.19 Uniqueness up to chain homotopy

Many chain-level constructions are not canonical on the nose but are canonical up to chain homotopy. Since homology kills this ambiguity, homology maps remain canonical.

22.20 Why this notion matters

Chain homotopy is the correct algebraic equivalence because it is:

- weaker than equality of chain maps;
- strong enough to force equality on homology;
- stable under composition;
- produced naturally by topological homotopies.

22.21 Bridge to contractions

A chain contraction is precisely a chain homotopy

$$\mathrm{id}_C \simeq 0.$$

VIII/23 studies this special but powerful case and its relation to acyclic complexes.

22.22 Exercises

Exercise 22.5. Exercise 1

Hint. Use the defining identity for chain homotopy. □

Solution. If $s : f \simeq g$, write the identity $g - f = \partial s + s\partial$. □

Exercise 22.6. Exercise 2

Hint. Use the defining identity for chain homotopy. □

Solution. The homotopy operator has degree +1. □

Exercise 22.7. Exercise 3

Hint. Use the defining identity for chain homotopy. □

Solution. The zero homotopy proves reflexivity. □

Exercise 22.8. Exercise 4

Hint. Use the defining identity for chain homotopy. □

Solution. If $s : f \simeq g$, then $-s : g \simeq f$. □

Exercise 22.9. Exercise 5

Hint. Use the defining identity for chain homotopy. □

Solution. If $s : f \simeq g$ and $t : g \simeq h$, then $s + t : f \simeq h$. □

Exercise 22.10. Exercise 6

Hint. Use the defining identity for chain homotopy. ☐

Solution. On cycles, $g(z) - f(z) = \partial s(z)$. ☐

Exercise 22.11. Exercise 7

Hint. Use the defining identity for chain homotopy. ☐

Solution. Therefore chain-homotopic maps induce the same homology map. ☐

Exercise 22.12. Exercise 8

Hint. Use the defining identity for chain homotopy. ☐

Solution. A null-homotopic map satisfies $f = \partial s + s\partial$. ☐

Exercise 22.13. Exercise 9

Hint. Use the defining identity for chain homotopy. ☐

Solution. A null-homotopic chain map induces zero on homology. ☐

Exercise 22.14. Exercise 10

Hint. Use the defining identity for chain homotopy. ☐

Solution. Postcomposition by a chain map preserves chain homotopy. ☐

Exercise 22.15. Exercise 11

Hint. Use the defining identity for chain homotopy. ☐

Solution. Precomposition by a chain map preserves chain homotopy. ☐

Exercise 22.16. Exercise 12

Hint. Use the defining identity for chain homotopy. ☐

Solution. Chain-homotopy equivalence requires homotopy inverses, not strict inverses. ☐

Exercise 22.17. Exercise 13

Hint. Use the defining identity for chain homotopy. ☐

Solution. Chain-homotopy-equivalent complexes have isomorphic homology. ☐

Exercise 22.18. Exercise 14

Hint. Use the defining identity for chain homotopy. ☐

Solution. The prism operator is a topological source of chain homotopies. ☐

Exercise 22.19. Exercise 15

Hint. Use the defining identity for chain homotopy. ☐

Solution. For finite free complexes the identity becomes a matrix equation. ☐

Exercise 22.20. Exercise 16

Hint. Use the defining identity for chain homotopy. □

Solution. Direct sums of chain homotopies are defined componentwise. □

Exercise 22.21. Exercise 17

Hint. Use the defining identity for chain homotopy. □

Solution. In the Hom-complex, $g - f$ is a coboundary. □

Exercise 22.22. Exercise 18

Hint. Use the defining identity for chain homotopy. □

Solution. For a degree-one operator s , $\delta(s) = \partial s + s\partial$. □

Exercise 22.23. Exercise 19

Hint. Use the defining identity for chain homotopy. □

Solution. A homotopy of maps of pairs descends to relative chains. □

Exercise 22.24. Exercise 20

Hint. Use the defining identity for chain homotopy. □

Solution. Augmented chain homotopies can be required to preserve augmentation. □

Exercise 22.25. Exercise 21

Hint. Use the defining identity for chain homotopy. □

Solution. Homology ignores chain-level choices differing by chain homotopy. □

Exercise 22.26. Exercise 22

Hint. Use the defining identity for chain homotopy. □

Solution. Equality of chain maps implies chain homotopy via $s = 0$. □

Exercise 22.27. Exercise 23

Hint. Use the defining identity for chain homotopy. □

Solution. Chain homotopy is weaker than chain isomorphism. □

Exercise 22.28. Exercise 24

Hint. Use the defining identity for chain homotopy. □

Solution. A contraction is a homotopy from the identity map to zero. □

22.23 Solved problem dossiers

Problem 22.29 (Cycle argument). Prove directly that chain-homotopic maps induce equal homology maps.

Solution. For a cycle z , $g(z) - f(z) = \partial s(z)$, so their classes agree. □

Problem 22.30 (Equivalence relation). Verify reflexivity, symmetry and transitivity.

Solution. Use 0 , $-s$, and $s + t$, respectively. □

Problem 22.31 (Postcomposition). If $f \simeq g$ and u is a chain map, find a homotopy from uf to ug .

Solution. Use us ; the chain-map identity moves u past the differential. □

Problem 22.32 (Precomposition). If $f \simeq g$ and v is a chain map, find a homotopy from fv to gv .

Solution. Use sv . □

Problem 22.33 (Null map). Show a null-homotopic map induces zero in homology.

Solution. It is chain-homotopic to the zero chain map, so the induced maps agree. □

Problem 22.34 (Homotopy equivalence). Show chain-homotopy equivalence implies homology isomorphism.

Solution. Apply homology to $gf \simeq \text{id}$ and $fg \simeq \text{id}$. □

Problem 22.35 (Strict isomorphism). Why is every chain isomorphism a chain-homotopy equivalence?

Solution. Use its strict inverse; the composites equal the identities, hence are homotopic via zero. □

Problem 22.36 (Prism). Interpret the prism identity algebraically.

Solution. It says the singular chain maps induced by homotopic topological maps differ by $\partial P + P\partial$. □

Problem 22.37 (Relative prism). Why does the prism operator descend to relative chains for a homotopy of pairs?

Solution. Prisms on chains lying in the subspace remain in the target subspace, so the operator respects the quotient. □

Problem 22.38 (Matrix criterion). State the matrix equation for a chain homotopy.

Solution. $G_n - F_n = D_{n+1}^D S_n + S_{n-1} D_n^C$. □

Problem 22.39 (Zero differential). If both complexes have zero differentials, when are two chain maps chain homotopic?

Solution. The homotopy equation becomes $g - f = 0$, so exactly when the maps are equal. □

Problem 22.40 (Concentrated complex). Let C be concentrated in one degree. What does chain homotopy between endomorphisms reduce to?

Solution. Again all differentials vanish, so homotopy is equality. $\square$

Problem 22.41 (Two-term example). Explain why nontrivial chain homotopies can occur only when adjacent target degrees exist.

Solution. The operator $s_n : C_n \rightarrow D_{n+1}$ must have somewhere to land; if all $D_{n+1} = 0$, that component vanishes. $\square$

Problem 22.42 (Hom-complex). Explain $g - f = \delta(s)$.

Solution. The differential in the Hom-complex is $\delta(s) = \partial_D s + s \partial_C$ for degree one. $\square$

Problem 22.43 (Canonical up to homotopy). Why can a chain-level construction still define a canonical homology map if choices are chain homotopic?

Solution. All choices induce the same map on homology. $\square$

Problem 22.44 (Direct sums). Combine homotopies $s^\alpha : f^\alpha \simeq g^\alpha$.

Solution. The direct sum $\bigoplus s^\alpha$ satisfies the homotopy identity componentwise. $\square$

Problem 22.45 (Topological composition). Relate postcomposition of a homotopy F to chain homotopy.

Solution. Postcomposing F by a continuous map corresponds to postcomposing its prism chain homotopy by the induced chain map. $\square$

Problem 22.46 (Augmentation). What extra condition is natural for augmented chain homotopies?

Solution. They should be compatible with the augmentation so the homotopy descends correctly to reduced complexes. $\square$

Problem 22.47 (Homology cannot detect). Can homology distinguish chain maps that are chain homotopic?

Solution. No; by definition of chain-homotopy invariance, their induced maps are identical. $\square$

Problem 22.48 (Bridge). Why is a chain contraction a special case?

Solution. It is a chain homotopy from id_C to 0, i.e. $\text{id}_C = \partial s + s \partial$. $\square$

Problem 22.49 (Two streams). Why is this chapter broader than VIII/20's prism proof?

Solution. It treats chain homotopy abstractly for arbitrary chain complexes, not only singular chains arising from topological homotopies. $\square$

Problem 22.50 (Coefficient stability). Why can chain-homotopy identities be transported after coefficient change?

Solution. Applying an additive coefficient functor preserves sums and compositions in the defining identity, subject to the usual tensor exactness caveats for later homology calculations. $\square$

Problem 22.51 (Naturality). Explain why chain homotopy is compatible with functorial constructions.

Solution. Because the identity $g - f = \partial s + s \partial$ is preserved by additive functors that commute with the relevant differentials. $\square$

Problem 22.52 (Final principle). State the main invariant content.

Solution. Chain-homotopy classes of chain maps, not raw chain maps, are what homology sees. $\square$

Chapter 23

Chain Contractions

A chain contraction is a chain homotopy from the identity to zero. It gives an explicit algebraic certificate that every cycle bounds. Thus contractible chain complexes are acyclic, while the converse requires additional splitting hypotheses.

23.1 Definition

Definition 23.1 (Chain contraction). A chain contraction on $C_\bullet$ is a family

$$s_n : C_n \rightarrow C_{n+1}$$

such that

$$\text{id}_C = \partial s + s \partial.$$

A complex admitting such an s is contractible as a chain complex.

23.2 Contractible implies acyclic

Theorem 23.2. *If $C_\bullet$ admits a chain contraction, then*

$$H_n(C) = 0$$

for all n .

Proof. For a cycle z ,

$$z = \partial s(z) + s(\partial z) = \partial s(z),$$

so every cycle is a boundary. □

23.3 Acyclic versus contractible

Acyclic means only $H_*(C) = 0$. Contractible means the stronger identity map is null-homotopic. In general:

$$\text{contractible} \implies \text{acyclic},$$

but not conversely without hypotheses.

23.4 Split exact complexes

Theorem 23.3 (Split exact complexes contract). *An exact chain complex of modules whose short exact sequences*

$$0 \rightarrow Z_n \rightarrow C_n \rightarrow B_{n-1} \rightarrow 0$$

split admits a chain contraction.

23.5 Free complexes

Over a field, every short exact sequence of vector spaces splits. Hence every acyclic chain complex of vector spaces is contractible.

For free abelian complexes, splitting must be checked more carefully.

23.6 Elementary contractible pair

Consider

$$0 \rightarrow A \xrightarrow{\text{id}} A \rightarrow 0$$

in adjacent degrees. It is contractible.

23.7 Disk-like complexes

The augmented simplicial chain complex of a simplex is contractible. An explicit contraction can be obtained by coning every simplex to a fixed vertex.

23.8 Cone-to-a-vertex operator

Fix a vertex v . On a simplex $[v_0, \dots, v_n]$, define informally

$$s([v_0, \dots, v_n]) = [v, v_0, \dots, v_n]$$

when this simplex is valid, with the standard conventions when v is already present. Then

$$\partial s + s \partial = \text{id}$$

on the reduced chain complex of a simplex.

23.9 Reduced versus unreduced contraction

A nonempty contractible space has ordinary $H_0 \cong \mathbb{Z}$, so its ordinary chain complex is not acyclic. The reduced or augmented chain complex is the correct object to contract.

23.10 Strong deformation retract analogy

Topologically, a contraction deforms a space to a point. Algebraically, a chain contraction deforms the identity chain map to zero. For reduced chains of a contractible space, these ideas align.

23.11 Chain contraction and null maps

If C is contractible, every chain map

$$f : C \rightarrow D$$

is null-homotopic:

$$f = f \operatorname{id}_C = f(\partial s + s\partial) = \partial(fs) + (fs)\partial.$$

Similarly, every chain map $D \rightarrow C$ is null-homotopic.

23.12 Direct sums

A direct sum of contractible complexes is contractible by taking the direct sum of their contractions.

23.13 Tensor preview

Tensoring a contraction with the identity often produces a contraction on a tensor product, with the usual grading signs. This becomes useful in VIII/27.

23.14 Filtration-compatible contractions

If a contraction respects a filtration, it can simplify spectral or filtered computations by collapsing contractible pieces without altering homology.

23.15 Cancellation of a basis pair

In a finite free chain complex, if a boundary matrix contains a unit entry that pairs one basis generator with another, an elementary chain contraction can often cancel that pair.

This is the algebraic basis of many chain-complex reduction algorithms.

23.16 Gaussian elimination over a field

Over a field, any nonzero pivot is a unit. Acyclic finite-dimensional chain complexes can therefore be reduced to direct sums of elementary contractible two-term complexes.

23.17 Smith normal form over $\mathbb{Z}$

Over $\mathbb{Z}$, diagonal entries equal to ± 1 correspond to contractible summands, while entries of absolute value greater than 1 produce torsion homology and therefore cannot be contracted away.

23.18 Acyclic finite free example

The complex

$$0 \rightarrow \mathbb{Z} \xrightarrow{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} \mathbb{Z}^2 \xrightarrow{\begin{pmatrix} 0 & 1 \end{pmatrix}} \mathbb{Z} \rightarrow 0$$

is exact and splits, hence contractible.

23.19 Noncontractible torsion example

The two-term complex

$$0 \rightarrow \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \rightarrow 0$$

with $|m| > 1$ has

$$H_0 \cong \mathbb{Z}/m,$$

so it is not contractible.

23.20 Acyclicity certificate

A chain contraction is stronger than merely knowing homology vanishes: it gives a formula that explicitly fills each cycle.

23.21 Homological perturbation preview

If a differential is modified slightly, one may try to correct an existing contraction rather than recompute homology from scratch. This idea belongs to more advanced homological perturbation theory.

23.22 Bridge to mapping cones

Mapping cones detect chain-homotopy equivalences. In particular, the cone of a chain map is acyclic when the map is a quasi-isomorphism, and is contractible when the map is a chain-homotopy equivalence under the standard algebraic hypotheses. VIII/24 develops this construction.

23.23 Exercises

Exercise 23.4. Exercise 1

Hint. Use the contraction identity. □

Solution. A chain contraction satisfies $\text{id} = \partial s + s\partial$. □

Exercise 23.5. Exercise 2

Hint. Use the contraction identity. □

Solution. A contractible chain complex has zero homology. □

Exercise 23.6. Exercise 3

Hint. Use the contraction identity. □

Solution. For a cycle z , the formula gives $z = \partial s(z)$. □

Exercise 23.7. Exercise 4

Hint. Use the contraction identity. □

Solution. Acyclicity alone does not always provide a contraction. □

Exercise 23.8. Exercise 5

Hint. Use the contraction identity. ☐

Solution. Split exact complexes admit contractions. ☐

Exercise 23.9. Exercise 6

Hint. Use the contraction identity. ☐

Solution. Every acyclic complex of vector spaces is contractible. ☐

Exercise 23.10. Exercise 7

Hint. Use the contraction identity. ☐

Solution. The elementary complex $0 \rightarrow A \xrightarrow{\text{id}} A \rightarrow 0$ is contractible. ☐

Exercise 23.11. Exercise 8

Hint. Use the contraction identity. ☐

Solution. The reduced chain complex of a simplex is contractible. ☐

Exercise 23.12. Exercise 9

Hint. Use the contraction identity. ☐

Solution. Ordinary chains of a point retain H_0 , so reduced chains are used. ☐

Exercise 23.13. Exercise 10

Hint. Use the contraction identity. ☐

Solution. If C is contractible, every map $C \rightarrow D$ is null-homotopic. ☐

Exercise 23.14. Exercise 11

Hint. Use the contraction identity. ☐

Solution. If C is contractible, every map $D \rightarrow C$ is null-homotopic. ☐

Exercise 23.15. Exercise 12

Hint. Use the contraction identity. ☐

Solution. Direct sums of contractions remain contractions. ☐

Exercise 23.16. Exercise 13

Hint. Use the contraction identity. ☐

Solution. A unit pivot in a boundary matrix often determines a cancellable pair. ☐

Exercise 23.17. Exercise 14

Hint. Use the contraction identity. ☐

Solution. Over a field every nonzero pivot is a unit. ☐

Exercise 23.18. Exercise 15

Hint. Use the contraction identity. ☐

Solution. Smith entries ± 1 correspond to contractible summands. ☐

Exercise 23.19. Exercise 16

Hint. Use the contraction identity. ☐

Solution. Smith entries $|m| > 1$ can produce torsion homology. ☐

Exercise 23.20. Exercise 17

Hint. Use the contraction identity. ☐

Solution. A contraction is an explicit certificate that cycles bound. ☐

Exercise 23.21. Exercise 18

Hint. Use the contraction identity. ☐

Solution. A contraction has degree $+1$. ☐

Exercise 23.22. Exercise 19

Hint. Use the contraction identity. ☐

Solution. Contractibility is preserved by chain isomorphism. ☐

Exercise 23.23. Exercise 20

Hint. Use the contraction identity. ☐

Solution. A chain-homotopy-equivalent complex to zero is contractible. ☐

Exercise 23.24. Exercise 21

Hint. Use the contraction identity. ☐

Solution. Reduced chains align chain contraction with topological contraction. ☐

Exercise 23.25. Exercise 22

Hint. Use the contraction identity. ☐

Solution. Filtration-compatible contractions can simplify filtered complexes. ☐

Exercise 23.26. Exercise 23

Hint. Use the contraction identity. ☐

Solution. Tensor contractions require grading signs. ☐

Exercise 23.27. Exercise 24

Hint. Use the contraction identity. ☐

Solution. Mapping cones provide the next major test for chain equivalences. ☐

23.24 Solved problem dossiers

Problem 23.28 (Cycle filling). Show a contraction fills every cycle.

Solution. If $\partial z = 0$, then $z = \partial s(z) + s\partial z = \partial s(z)$. □

Problem 23.29 (Elementary pair). Construct a contraction on $0 \rightarrow A \xrightarrow{\text{id}} A \rightarrow 0$.

Solution. Take s from the lower copy of A back to the upper copy as the identity and all other components zero. □

Problem 23.30 (Split exact). Explain how splittings yield a contraction.

Solution. Decompose each C_n into a boundary part and a chosen complement mapping isomorphically onto B_{n-1} ; invert that isomorphism to define s . □

Problem 23.31 (Vector spaces). Why does acyclic imply contractible over a field?

Solution. All short exact sequences of vector spaces split, so the split-exact theorem applies. □

Problem 23.32 (Torsion obstruction). Why is $0 \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \rightarrow 0$ not contractible?

Solution. Its H_0 is $\mathbb{Z}/2$, but contractible complexes are acyclic. □

Problem 23.33 (Simplex contraction). Describe the vertex-coning contraction on reduced simplicial chains.

Solution. Choose a vertex v and prepend it to each oriented simplex, with the usual sign/convention; the boundary identity gives $\partial s + s\partial = \text{id}$. □

Problem 23.34 (Reduced chains). Why can the unreduced chain complex of a point not be contracted?

Solution. It has $H_0 \cong \mathbb{Z} \neq 0$, contradicting contractible-implies-acyclic. □

Problem 23.35 (Map out). If C contracts, prove every chain map $f : C \rightarrow D$ is null-homotopic.

Solution. Compose the contraction with f : $f = f\partial s + fs\partial = \partial(fs) + (fs)\partial$. □

Problem 23.36 (Map in). If C contracts, prove every chain map $g : D \rightarrow C$ is null-homotopic.

Solution. Use sg : $g = \partial(sg) + (sg)\partial$. □

Problem 23.37 (Direct sum). Construct a contraction on $C \oplus D$ from contractions on each.

Solution. Use $s_C \oplus s_D$. □

Problem 23.38 (Unit cancellation). Why does a unit incidence coefficient permit cancellation?

Solution. The corresponding differential component is an isomorphism on rank-one summands, producing an elementary contractible two-term summand. □

Problem 23.39 (Field elimination). Explain finite-dimensional acyclic complexes over a field as sums of elementary pairs.

Solution. Choose bases adapted to images and complements; exactness pairs each complement with the next boundary space by an isomorphism. $\square$

Problem 23.40 (Smith form). What does a Smith diagonal entry 1 mean homologically?

Solution. It yields an isomorphism between rank-one summands, hence no homology and a contractible pair. $\square$

Problem 23.41 (Smith torsion). What does a Smith diagonal entry $m > 1$ mean?

Solution. It contributes a cokernel $\mathbb{Z}/m$, so the summand is not contractible. $\square$

Problem 23.42 (Explicit exact example). Verify the displayed $\mathbb{Z} \rightarrow \mathbb{Z}^2 \rightarrow \mathbb{Z}$ complex is exact.

Solution. The first image is $\mathbb{Z}(1, 0)$, exactly the kernel of $(0 \ 1)$, and the second map is surjective. $\square$

Problem 23.43 (Contraction strength). Why is a contraction stronger data than vanishing homology?

Solution. It gives an explicit degree-one operator solving $\text{id} = \partial s + s \partial$, not just an existence statement about boundaries. $\square$

Problem 23.44 (Homotopy to zero). Restate chain contractibility in VIII/22 language.

Solution. The identity chain map is chain homotopic to the zero chain map. $\square$

Problem 23.45 (Invariance). If $C \cong D$ as chain complexes and C contracts, why does D contract?

Solution. Transport the contraction through the chain isomorphism and its inverse. $\square$

Problem 23.46 (Topological analogy). What topological object suggests the chain contraction of a simplex?

Solution. A simplex is contractible to a vertex; the coning operator is the chain-level analogue. $\square$

Problem 23.47 (Filtration). Why might a filtration-compatible contraction be useful?

Solution. It removes contractible pieces without mixing filtration levels, preserving the structure needed for later filtered calculations. $\square$

Problem 23.48 (Bridge to cones). What should happen to the mapping cone of a chain-homotopy equivalence?

Solution. It should be contractible; VIII/24 proves the cone packages failure of invertibility up to chain homotopy. $\square$

Problem 23.49 (Acyclic cone preview). What should happen to the cone of a quasi-isomorphism?

Solution. Its homology vanishes, so the cone is acyclic. $\square$

Problem 23.50 (Contraction versus quasi-isomorphism). Why is contractible stronger than quasi-isomorphic to zero?

Solution. Quasi-isomorphic to zero only says homology vanishes; contractible supplies a homotopy-level inverse to zero. $\square$

Problem 23.51 (Final principle). State the practical use of a contraction.

Solution. It gives explicit algebraic cancellation and a formula for filling cycles. $\square$

Chapter 24

Mapping Cones of Chain Maps

The mapping cone of a chain map packages the map into a new chain complex. Its homology measures the failure of the map to be a quasi-isomorphism, while its contractibility detects chain-homotopy equivalences.

24.1 Definition

Let $f : C_\bullet \rightarrow D_\bullet$ be a chain map.

Definition 24.1 (Mapping cone). Define

$$\text{Cone}(f)_n = D_n \oplus C_{n-1}$$

with differential

$$d(d, c) = (\partial_D d + f(c), -\partial_C c).$$

24.2 Chain condition

Proposition 24.2. *The cone differential satisfies*

$$d^2 = 0.$$

The verification uses exactly the chain-map identity

$$\partial_D f = f \partial_C.$$

24.3 Shifted complex

Define the suspension/shift $C[1]$ by

$$C[1]_n = C_{n-1}, \quad \partial_{C[1]} = -\partial_C.$$

Then there is a short exact sequence

$$0 \rightarrow D \rightarrow \text{Cone}(f) \rightarrow C[1] \rightarrow 0.$$

24.4 Cone exact sequence

The short exact sequence yields

$$\cdots \rightarrow H_n(D) \rightarrow H_n(\text{Cone}(f)) \rightarrow H_{n-1}(C) \xrightarrow{H_{n-1}(f)} H_{n-1}(D) \rightarrow \cdots$$

Equivalently one can arrange it as

$$\cdots \rightarrow H_n(C) \xrightarrow{H_n(f)} H_n(D) \rightarrow H_n(\text{Cone}(f)) \rightarrow H_{n-1}(C) \rightarrow \cdots$$

24.5 Cone of the identity

Theorem 24.3 (Identity cone contracts).

$$\text{Cone}(\text{id}_C)$$

is contractible.

24.6 Cone of the zero map

If $f = 0$, then

$$\text{Cone}(0) \cong D \oplus C[1]$$

as chain complexes.

Thus

$$H_n(\text{Cone}(0)) \cong H_n(D) \oplus H_{n-1}(C).$$

24.7 Quasi-isomorphisms

Definition 24.4 (Quasi-isomorphism). A chain map $f : C \rightarrow D$ is a quasi-isomorphism if

$$H_n(f) : H_n(C) \rightarrow H_n(D)$$

is an isomorphism for every n .

Theorem 24.5 (Cone criterion). *A chain map f is a quasi-isomorphism if and only if*

$$H_n(\text{Cone}(f)) = 0$$

for all n .

24.8 Chain-homotopy equivalences

Theorem 24.6 (Homotopy-equivalence cone). *If f is a chain-homotopy equivalence, then*

$$\text{Cone}(f)$$

is contractible.

Thus chain-homotopy equivalence is stronger than quasi-isomorphism, matching contractible versus merely acyclic cones.

24.9 Cone of an inclusion

If $i : A \hookrightarrow B$ is an inclusion of chain complexes, the cone provides a chain model closely related to the quotient B/A .

Under suitable splitting/cofibration-style algebraic hypotheses, the cone and quotient are quasi-isomorphic.

24.10 Relative chains

For a subspace $A \subset X$, the inclusion

$$C_*(A) \rightarrow C_*(X)$$

has a cone whose homology is naturally related to relative homology

$$H_*(X, A).$$

This mirrors the topological mapping-cone picture from VIII/07.

24.11 Topological versus algebraic cones

A map of spaces

$$f : A \rightarrow X$$

has a topological mapping cone C_f . Applying singular chains gives a chain complex naturally compared with

$$\text{Cone}(C_*(f)).$$

The two constructions encode the same cofiber geometry up to the expected chain equivalences.

24.12 Functoriality

A commutative square of chain maps induces a chain map between mapping cones.

24.13 Homotopy-commutative squares

If a square commutes only up to chain homotopy, a corrected cone map can still be built by including the homotopy term. This is one reason mapping cones are central in derived constructions.

24.14 Connecting morphism

The cone exact sequence realizes the map

$$H_{n-1}(C) \rightarrow H_{n-1}(D)$$

as the homology map induced by f . Thus the cone is a universal algebraic device for measuring this map.

24.15 Two-term example

For

$$f : \mathbb{Z} \rightarrow \mathbb{Z}, \quad f = \times m$$

with both complexes concentrated in degree 0, the cone is the two-term complex

$$0 \rightarrow \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \rightarrow 0.$$

Hence for $m \neq 0$,

$$H_0(\text{Cone}(f)) \cong \mathbb{Z}/m.$$

24.16 Moore-space connection

The multiplication-by- m cone is the algebraic analogue of the degree- m cell attachment producing a Moore space. Both produce $\mathbb{Z}/m$ torsion.

24.17 Cone of an isomorphism

If f is a strict chain isomorphism, its cone is contractible. This is the strict special case of the homotopy-equivalence theorem.

24.18 Cone and exactness

If

$$0 \rightarrow C \xrightarrow{f} D \rightarrow Q \rightarrow 0$$

is a short exact sequence, the cone of f is frequently quasi-isomorphic to the quotient complex Q .

24.19 Distinguished-triangle preview

The pattern

$$C \xrightarrow{f} D \rightarrow \text{Cone}(f) \rightarrow C[1]$$

is the prototype of a distinguished triangle. Derived categories formalize this pattern, but no derived-category machinery is needed in this volume.

24.20 Why signs matter

The minus sign in the shifted differential is essential. Without it, the cone differential would generally fail to square to zero.

24.21 Bridge to coefficients

The next chapter changes coefficient groups. Mapping-cone calculations remain valuable there because tensoring a cone produces the expected cone of the tensored map, subject to standard exactness considerations.

24.22 Exercises

Exercise 24.7. Exercise 1

Hint. Use the cone differential and exact sequence. □

Solution. $\text{Cone}(f)_n = D_n \oplus C_{n-1}$. □

Exercise 24.8. Exercise 2

Hint. Use the cone differential and exact sequence. □

Solution. The cone differential is $d(d, c) = (\partial_D d + f(c), -\partial_C c)$. □

Exercise 24.9. Exercise 3

Hint. Use the cone differential and exact sequence. □

Solution. The minus sign belongs to the shifted C -summand. □

Exercise 24.10. Exercise 4

Hint. Use the cone differential and exact sequence. □

Solution. The chain-map identity is needed to prove $d^2 = 0$. □

Exercise 24.11. Exercise 5

Hint. Use the cone differential and exact sequence. □

Solution. The shifted complex satisfies $C[1]_n = C_{n-1}$. □

Exercise 24.12. Exercise 6

Hint. Use the cone differential and exact sequence. □

Solution. Its differential is $-\partial_C$. □

Exercise 24.13. Exercise 7

Hint. Use the cone differential and exact sequence. □

Solution. There is a short exact sequence $0 \rightarrow D \rightarrow \text{Cone}(f) \rightarrow C[1] \rightarrow 0$. □

Exercise 24.14. Exercise 8

Hint. Use the cone differential and exact sequence. □

Solution. The cone therefore has a long exact homology sequence. □

Exercise 24.15. Exercise 9

Hint. Use the cone differential and exact sequence. □

Solution. The cone of the identity is contractible. □

Exercise 24.16. Exercise 10

Hint. Use the cone differential and exact sequence. □

Solution. The cone of zero is $D \oplus C[1]$. □

Exercise 24.17. Exercise 11

Hint. Use the cone differential and exact sequence. ☐

Solution. A quasi-isomorphism induces homology isomorphisms. ☐

Exercise 24.18. Exercise 12

Hint. Use the cone differential and exact sequence. ☐

Solution. A map is a quasi-isomorphism iff its cone is acyclic. ☐

Exercise 24.19. Exercise 13

Hint. Use the cone differential and exact sequence. ☐

Solution. A chain-homotopy equivalence has contractible cone. ☐

Exercise 24.20. Exercise 14

Hint. Use the cone differential and exact sequence. ☐

Solution. Contractible cone is stronger than acyclic cone. ☐

Exercise 24.21. Exercise 15

Hint. Use the cone differential and exact sequence. ☐

Solution. The cone of a strict isomorphism is contractible. ☐

Exercise 24.22. Exercise 16

Hint. Use the cone differential and exact sequence. ☐

Solution. The cone of a subcomplex inclusion models relative information. ☐

Exercise 24.23. Exercise 17

Hint. Use the cone differential and exact sequence. ☐

Solution. Topological and algebraic mapping cones are parallel constructions. ☐

Exercise 24.24. Exercise 18

Hint. Use the cone differential and exact sequence. ☐

Solution. A commutative square induces a map of cones. ☐

Exercise 24.25. Exercise 19

Hint. Use the cone differential and exact sequence. ☐

Solution. Homotopy-commutative squares require a correction term. ☐

Exercise 24.26. Exercise 20

Hint. Use the cone differential and exact sequence. ☐

Solution. For $f = \times m$ in degree zero, the cone is a two-term complex. ☐

Exercise 24.27. Exercise 21

Hint. Use the cone differential and exact sequence. □

Solution. Its degree-zero homology is $\mathbb{Z}/m$ for $m \neq 0$. □

Exercise 24.28. Exercise 22

Hint. Use the cone differential and exact sequence. □

Solution. Multiplication-by- m cones parallel Moore-space torsion. □

Exercise 24.29. Exercise 23

Hint. Use the cone differential and exact sequence. □

Solution. The pattern $C \rightarrow D \rightarrow \text{Cone}(f) \rightarrow C[1]$ is a triangle. □

Exercise 24.30. Exercise 24

Hint. Use the cone differential and exact sequence. □

Solution. VIII/25 will revisit cones after changing coefficients. □

24.23 Solved problem dossiers

Problem 24.31 (Square-zero check). Verify the cone differential squares to zero.

Solution. Compute $d^2(d, c)$: the first component is $\partial_D^2 d + \partial_D f(c) - f \partial_C c = 0$ by the chain-map identity; the second is $\partial_C^2 c = 0$. □

Problem 24.32 (Identity cone). Explain why $\text{Cone}(\text{id})$ contracts.

Solution. An explicit degree-one operator swaps the shifted and unshifted components appropriately; equivalently the cone models an algebraic cylinder pair and is chain-homotopy equivalent to zero. □

Problem 24.33 (Zero cone). Compute the homology of $\text{Cone}(0)$.

Solution. The differential is block diagonal, so the cone is $D \oplus C[1]$ and $H_n \cong H_n(D) \oplus H_{n-1}(C)$. □

Problem 24.34 (Quasi-isomorphism criterion). Prove the cone criterion using the long exact sequence.

Solution. Exactness shows all cone homology vanishes exactly when every $H_n(C) \rightarrow H_n(D)$ is both injective and surjective. □

Problem 24.35 (Homotopy equivalence). Why does chain-homotopy equivalence imply contractible cone?

Solution. A homotopy inverse and the two chain homotopies can be assembled into an explicit cone contraction. □

Problem 24.36 (Strict isomorphism). Deduce the cone of a chain isomorphism contracts.

Solution. A strict isomorphism is a chain-homotopy equivalence with zero homotopies. □

Problem 24.37 (Multiplication by m). Compute the cone of $\times m : \mathbb{Z} \rightarrow \mathbb{Z}$ concentrated in degree zero.

Solution. It is $0 \rightarrow \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \rightarrow 0$, with $H_0 = \mathbb{Z}/m$ and $H_1 = 0$ for $m \neq 0$. □

Problem 24.38 (Case $m = 0$). Compute the same cone for $m = 0$.

Solution. The differential is zero, so $H_1 \cong \mathbb{Z}$ and $H_0 \cong \mathbb{Z}$. □

Problem 24.39 (Case $m = 1$). What happens for $m = 1$?

Solution. The differential is an isomorphism, so the cone is acyclic and in fact contractible. □

Problem 24.40 (Relative model). Relate the cone of $C_*(A) \rightarrow C_*(X)$ to $H_*(X, A)$.

Solution. The cone is a standard chain model quasi-isomorphic to the quotient relative complex, up to the conventional degree arrangement. □

Problem 24.41 (Topological cone). Why should singular chains on a topological mapping cone compare with the algebraic cone?

Solution. Both encode the same cofiber construction: one geometrically before applying chains, the other algebraically after applying the chain functor. □

Problem 24.42 (Map of cones). Given a strictly commutative square, define the obvious map on cones.

Solution. Apply the target map on the D -summand and the source map on the shifted C -summand; commutativity makes it a chain map. □

Problem 24.43 (Homotopy square). Why is an extra term needed if the square commutes only up to chain homotopy?

Solution. The failure of strict commutation appears in the cone differential; the chosen homotopy supplies exactly the correction needed to cancel it. □

Problem 24.44 (Quotient comparison). When does the cone of an inclusion resemble the quotient complex?

Solution. For a suitably split inclusion, the cone contains a contractible copy of the subcomplex and reduces up to chain homotopy/quasi-isomorphism to the quotient. □

Problem 24.45 (Acyclic versus contractible). Why can an acyclic cone fail to be contractible?

Solution. Acyclicity detects only quasi-isomorphism of f ; contractibility of the cone requires the stronger chain-homotopy-equivalence property. □

Problem 24.46 (Moore analogy). Explain the algebraic/topological parallel for multiplication by m .

Solution. The chain cone has $\mathbb{Z}/m$ homology, matching the cellular chain complex of a Moore space built by degree- m attachment. □

Problem 24.47 (Triangle). Why is the cone sequence naturally written $C \rightarrow D \rightarrow \text{Cone}(f) \rightarrow C[1]$?

Solution. The inclusion of D and projection to the shifted C -summand form the short exact cone sequence and its associated cofiber pattern. □

Problem 24.48 (Sign). Show what fails if the shifted differential had no minus sign.

Solution. The cross terms in d^2 would add instead of canceling, leaving generally $2f\partial_C$ or equivalent nonzero terms depending on convention. $\square$

Problem 24.49 (Homology map failure). What does nonzero $H_n(\text{Cone}(f))$ tell you?

Solution. It witnesses failure of $H_*(f)$ to be an isomorphism in adjacent degrees, as organized by the cone long exact sequence. $\square$

Problem 24.50 (Coefficient bridge). Why are cones useful with changed coefficients?

Solution. They convert the question whether a coefficient-changed map remains a quasi-isomorphism into acyclicity of the corresponding coefficient-changed cone. $\square$

Problem 24.51 (Derived preview). What later formalism abstracts mapping-cone triangles?

Solution. Derived and triangulated categories, though this volume only uses the concrete chain-complex construction. $\square$

Problem 24.52 (Final principle). State the main diagnostic role of a cone.

Solution. The cone measures the failure of a chain map to be an equivalence: acyclic cone means quasi-isomorphism; contractible cone means chain-homotopy equivalence. $\square$

Chapter 25

Homology with Coefficients

Integral homology is only one member of a family of invariants. Replacing the coefficient group $\mathbb{Z}$ by an abelian group G can reveal torsion, simplify calculations, and produce operations such as Bockstein connecting homomorphisms.

25.1 Chains with coefficients

For a chain complex $C_\bullet$ of free abelian groups and an abelian group G , define

$$C_n(C; G) = C_n \otimes G$$

with differential

$$\partial \otimes \text{id}_G.$$

Definition 25.1 (Homology with coefficients).

$$H_n(C; G) = H_n(C_\bullet \otimes G).$$

For a space X ,

$$H_n(X; G) = H_n(C_*^{\text{sing}}(X) \otimes G).$$

25.2 Integer coefficients

Taking $G = \mathbb{Z}$ recovers ordinary integral homology.

25.3 Field coefficients

If $G = \mathbb{F}_p = \mathbb{Z}/p$, every chain group becomes a vector space over $\mathbb{F}_p$. Boundary matrices may be reduced modulo p , and homology can be computed by ordinary linear algebra.

25.4 Cellular chains with coefficients

For a CW complex,

$$C_n^{\text{cell}}(X; G) \cong C_n^{\text{cell}}(X; \mathbb{Z}) \otimes G.$$

Thus the same cellular boundary matrices are interpreted in G .

25.5 Degree- m attachment

For

$$X_m = S^1 \cup_m D^2,$$

the coefficient chain complex is

$$0 \rightarrow G \xrightarrow{\times m} G \rightarrow G \rightarrow 0.$$

Hence the result depends strongly on whether multiplication by m is injective or zero in G .

25.6 Projective plane mod 2

The integral cellular differential of $\mathbb{R}P^2$ is multiplication by 2:

$$0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0.$$

With $\mathbb{F}_2$ -coefficients, this differential becomes zero.

Therefore

$$H_k(\mathbb{R}P^2; \mathbb{F}_2) \cong \mathbb{F}_2 \quad (k = 0, 1, 2).$$

25.7 Projective spaces mod 2

More generally, every cellular differential in $\mathbb{R}P^n$ becomes zero modulo 2, so

$$H_k(\mathbb{R}P^n; \mathbb{F}_2) \cong \mathbb{F}_2 \quad 0 \leq k \leq n.$$

25.8 Moore spaces

For

$$M(\mathbb{Z}/m, n) = S^n \cup_m D^{n+1},$$

the coefficient chain complex contains

$$G \xrightarrow{\times m} G.$$

Thus

$$H_{n+1}(M; G) = \ker(\times m : G \rightarrow G),$$

and

$$\tilde{H}_n(M; G) = G/mG.$$

25.9 Example with mod p

For $G = \mathbb{F}_p$:

- if $p \nmid m$, multiplication by m is an isomorphism;
- if $p \mid m$, multiplication by m is zero.

So mod- p homology detects exactly the p -primary part of this Moore attachment.

25.10 Tensor products of abelian groups

Basic identities include

$$\mathbb{Z} \otimes G \cong G, \quad \mathbb{Z}/m \otimes G \cong G/mG,$$

and

$$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m, n).$$

These formulas foreshadow the universal coefficient theorem.

25.11 Tor preview

Tensoring is right exact but not exact in general. The failure of left exactness is measured by

$$\mathrm{Tor}_1^{\mathbb{Z}}(-, G).$$

For cyclic groups,

$$\mathrm{Tor}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n).$$

VIII/26 makes this correction precise.

25.12 Short exact coefficient sequences

A short exact sequence of coefficient groups

$$0 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 0$$

produces a short exact sequence of singular chain complexes because singular chain groups are free abelian.

Therefore it yields a long exact sequence in homology:

$$\cdots \rightarrow H_n(X; G') \rightarrow H_n(X; G) \rightarrow H_n(X; G'') \xrightarrow{\beta} H_{n-1}(X; G') \rightarrow \cdots .$$

25.13 Bockstein homomorphism

The connecting homomorphism

$$\beta : H_n(X; G'') \rightarrow H_{n-1}(X; G')$$

is called a Bockstein homomorphism for the chosen coefficient sequence.

A central example is

$$0 \rightarrow \mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \rightarrow \mathbb{Z}/p \rightarrow 0.$$

25.14 Bockstein and torsion

The Bockstein can detect when a mod- p class comes from p -torsion in integral homology.

For projective spaces this relates the rich mod-2 homology to integral 2-torsion.

25.15 Naturality

Continuous maps induce homomorphisms on homology with arbitrary coefficients, and coefficient homomorphisms

$$G \rightarrow H$$

also induce maps

$$H_n(X; G) \rightarrow H_n(X; H).$$

Thus homology is functorial in both the space and the coefficient group.

25.16 Reduced homology

Reduced homology with coefficients is defined using the augmented coefficient chain complex. For a connected space,

$$\tilde{H}_0(X; G) = 0.$$

25.17 Relative coefficients

For a pair (X, A) ,

$$H_n(X, A; G) = H_n(C_*(X, A) \otimes G).$$

Long exact sequences of pairs remain valid with coefficients.

25.18 Orientation with coefficients

A manifold that is not orientable over $\mathbb{Z}$ may become orientable over $\mathbb{F}_2$, because sign reversal is invisible modulo 2.

This phenomenon is developed systematically in the manifold chapters.

25.19 Characteristic dependence

Different coefficient fields can produce different Betti numbers. One should therefore always record the coefficient ring when reporting homology:

$$H_n(X; \mathbb{Z}), \quad H_n(X; \mathbb{Q}), \quad H_n(X; \mathbb{F}_p)$$

need not contain equivalent information.

25.20 Rational homology

Tensoring with $\mathbb{Q}$ kills finite torsion. Thus rational homology records the free part of integral homology.

25.21 Support-diagram provenance

The legacy support corpus contains explicit Moore-space, Bockstein, and projective-space coefficient diagrams mapped to this chapter. Their mathematical roles are integrated here; exact visual-instance reconciliation is reserved for the post-VIII/35 audit.

25.22 Bridge to UCT

Computing directly after tensoring chains works, but the universal coefficient theorem gives a far more efficient answer in terms of integral homology: a tensor term plus a Tor correction. VIII/26 develops that theorem.

25.23 Exercises

Exercise 25.2. Exercise 1

Hint. Use coefficient chains and basic tensor identities. □

Solution. $H_n(X; G)$ is the homology of $C_*(X) \otimes G$. □

Exercise 25.3. Exercise 2

Hint. Use coefficient chains and basic tensor identities. □

Solution. Taking $G = \mathbb{Z}$ recovers integral homology. □

Exercise 25.4. Exercise 3

Hint. Use coefficient chains and basic tensor identities. □

Solution. With $G = \mathbb{F}_p$, chain groups are vector spaces. □

Exercise 25.5. Exercise 4

Hint. Use coefficient chains and basic tensor identities. □

Solution. Cellular boundary matrices may be reduced modulo p . □

Exercise 25.6. Exercise 5

Hint. Use coefficient chains and basic tensor identities. □

Solution. For $S^1 \cup_m D^2$, the key coefficient differential is multiplication by m on G . □

Exercise 25.7. Exercise 6

Hint. Use coefficient chains and basic tensor identities. □

Solution. For $\mathbb{R}P^2 \bmod 2$, the multiplication-by-2 boundary becomes zero. □

Exercise 25.8. Exercise 7

Hint. Use coefficient chains and basic tensor identities. □

Solution. $H_k(\mathbb{R}P^2; \mathbb{F}_2) \cong \mathbb{F}_2$ for $k = 0, 1, 2$. □

Exercise 25.9. Exercise 8

Hint. Use coefficient chains and basic tensor identities. □

Solution. For a Moore space, $\tilde{H}_n(-; G) \cong G/mG$. □

Exercise 25.10. Exercise 9

Hint. Use coefficient chains and basic tensor identities. □

Solution. For the same Moore space, $H_{n+1}(-; G) = \ker(\times m)$. □

Exercise 25.11. Exercise 10

Hint. Use coefficient chains and basic tensor identities. □

Solution. If $p \nmid m$, multiplication by m is invertible on $\mathbb{F}_p$. □

Exercise 25.12. Exercise 11

Hint. Use coefficient chains and basic tensor identities. □

Solution. If $p \mid m$, multiplication by m is zero on $\mathbb{F}_p$. □

Exercise 25.13. Exercise 12

Hint. Use coefficient chains and basic tensor identities. □

Solution. $\mathbb{Z} \otimes G \cong G$. □

Exercise 25.14. Exercise 13

Hint. Use coefficient chains and basic tensor identities. □

Solution. $\mathbb{Z}/m \otimes G \cong G/mG$. □

Exercise 25.15. Exercise 14

Hint. Use coefficient chains and basic tensor identities. □

Solution. $\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m, n)$. □

Exercise 25.16. Exercise 15

Hint. Use coefficient chains and basic tensor identities. □

Solution. Tor measures the failure of tensoring to be left exact. □

Exercise 25.17. Exercise 16

Hint. Use coefficient chains and basic tensor identities. □

Solution. A short exact coefficient sequence yields a long exact homology sequence. □

Exercise 25.18. Exercise 17

Hint. Use coefficient chains and basic tensor identities. □

Solution. The connecting map in a coefficient LES is a Bockstein. □

Exercise 25.19. Exercise 18

Hint. Use coefficient chains and basic tensor identities. □

Solution. The sequence $0 \rightarrow \mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \rightarrow \mathbb{Z}/p \rightarrow 0$ is the standard Bockstein source. □

Exercise 25.20. Exercise 19

Hint. Use coefficient chains and basic tensor identities. □

Solution. Homology is functorial in coefficient homomorphisms. □

Exercise 25.21. Exercise 20

Hint. Use coefficient chains and basic tensor identities. □

Solution. Relative homology is defined with coefficients by tensoring the relative chain complex. □

Exercise 25.22. Exercise 21

Hint. Use coefficient chains and basic tensor identities. □

Solution. Reduced H_0 of a connected space is zero over any coefficient group. □

Exercise 25.23. Exercise 22

Hint. Use coefficient chains and basic tensor identities. □

Solution. Mod-2 coefficients erase orientation signs. □

Exercise 25.24. Exercise 23

Hint. Use coefficient chains and basic tensor identities. □

Solution. Rational coefficients kill finite torsion. □

Exercise 25.25. Exercise 24

Hint. Use coefficient chains and basic tensor identities. □

Solution. VIII/26 expresses coefficient homology through tensor and Tor terms. □

25.24 Solved problem dossiers

Problem 25.26 (RP2 mod 2). Compute $H_*(\mathbb{R}P^2; \mathbb{F}_2)$ cellularly.

Solution. The integral differential $d_2 = 2$ becomes zero modulo 2, as does d_1 ; hence one copy of $\mathbb{F}_2$ survives in degrees 0,1,2. □

Problem 25.27 (RP2 mod 3). Compute positive homology of $\mathbb{R}P^2; \mathbb{F}_3$.

Solution. Multiplication by 2 is invertible mod 3, so $H_2 = H_1 = 0$. □

Problem 25.28 (Degree m mod p). For $S^1 \cup_m D^2$, compute $H_1(-; \mathbb{F}_p)$.

Solution. It is the cokernel of multiplication by m on $\mathbb{F}_p$: zero if $p \nmid m$, and $\mathbb{F}_p$ if $p \mid m$. □

Problem 25.29 (Moore mod p). Compute reduced homology of $M(\mathbb{Z}/m, n)$ over $\mathbb{F}_p$.

Solution. If $p \nmid m$, it vanishes; if $p \mid m$, copies of $\mathbb{F}_p$ occur in degrees n and $n+1$. □

Problem 25.30 (Tensor cyclic). Compute $\mathbb{Z}/12 \otimes \mathbb{Z}/18$.

Solution. It is $\mathbb{Z}/\gcd(12, 18) \cong \mathbb{Z}/6$. □

Problem 25.31 (Tensor with $\mathbb{Q}$). Compute $\mathbb{Z}/m \otimes \mathbb{Q}$.

Solution. Zero, since multiplication by m is invertible on $\mathbb{Q}$. □

Problem 25.32 (Rational projective). Compute rational homology of $\mathbb{R}P^2$.

Solution. Integral $H_1 = \mathbb{Z}/2$ is killed by tensoring with $\mathbb{Q}$; only $H_0 \cong \mathbb{Q}$ remains. $\square$

Problem 25.33 (Rational torus). Compute $H_1(T^2; \mathbb{Q})$.

Solution. $\mathbb{Z}^2 \otimes \mathbb{Q} \cong \mathbb{Q}^2$. $\square$

Problem 25.34 (Coefficient map). What map does $\mathbb{Z} \rightarrow \mathbb{Z}/p$ induce?

Solution. It gives reduction mod p : $H_n(X; \mathbb{Z}) \rightarrow H_n(X; \mathbb{Z}/p)$, though not every mod- p class need arise this way because of Tor/Bockstein effects. $\square$

Problem 25.35 (Bockstein source). Explain the Bockstein from $0 \rightarrow \mathbb{Z} \xrightarrow{p} \mathbb{Z} \rightarrow \mathbb{Z}/p \rightarrow 0$.

Solution. It is the connecting map $H_n(X; \mathbb{Z}/p) \rightarrow H_{n-1}(X; \mathbb{Z})$. $\square$

Problem 25.36 (Bockstein meaning). What can a nonzero Bockstein indicate?

Solution. It indicates an obstruction to lifting a mod- p class to integral coefficients and often detects p -torsion one degree lower. $\square$

Problem 25.37 (Field computation). Why are mod- p cellular calculations often easier?

Solution. Boundary matrices become matrices over a field, so Gaussian elimination computes kernels and images directly. $\square$

Problem 25.38 (Orientation mod 2). Why can every manifold be locally oriented mod 2?

Solution. Changing an orientation sign multiplies by -1 , which equals 1 in $\mathbb{F}_2$. $\square$

Problem 25.39 (Relative coefficients). State the coefficient LES of a pair.

Solution. $\cdots \rightarrow H_n(A; G) \rightarrow H_n(X; G) \rightarrow H_n(X, A; G) \rightarrow H_{n-1}(A; G) \rightarrow \cdots$. $\square$

Problem 25.40 (Disjoint union). How does coefficient homology behave on disjoint unions?

Solution. Chains split by components, so $H_n(\coprod X_\alpha; G) \cong \bigoplus H_n(X_\alpha; G)$. $\square$

Problem 25.41 (Sphere coefficients). Compute $H_*(S^n; G)$.

Solution. For nonempty G , $H_0 \cong G$, $H_n \cong G$, and other groups vanish. $\square$

Problem 25.42 (Circle coefficients). Compute $H_1(S^1; G)$.

Solution. It is G , since the cellular differential is zero. $\square$

Problem 25.43 (Moore G/mG). Explain the quotient G/mG .

Solution. It is the cokernel of the degree- m cellular boundary $G \xrightarrow{\times m} G$. $\square$

Problem 25.44 (Moore kernel). Explain the extra H_{n+1} coefficient group.

Solution. It is the kernel of multiplication by m on G , which may be nonzero even though integral top homology vanishes. $\square$

Problem 25.45 (Mod 2 projective spaces). Compute $H_k(\mathbb{R}P^n; \mathbb{F}_2)$.

Solution. Every cellular differential becomes zero, so it is $\mathbb{F}_2$ in every degree $0 \leq k \leq n$. $\square$

Problem 25.46 (Coefficient dependence). Give a space whose Betti numbers depend on the field.

Solution. $\mathbb{R}P^2$: over $\mathbb{Q}$, positive homology vanishes; over $\mathbb{F}_2$, both H_1 and H_2 are nonzero. $\square$

Problem 25.47 (Tor preview). Compute $\text{Tor}(\mathbb{Z}/12, \mathbb{Z}/18)$.

Solution. It is also $\mathbb{Z}/6$, illustrating that tensor and Tor have the same cyclic gcd size but occur in different UCT degrees. $\square$

Problem 25.48 (Support figures). Where do Bockstein/RP2 and Moore coefficient support diagrams belong?

Solution. Their canonical mathematical destination is VIII/25 because they visualize coefficient change and torsion detection. $\square$

Problem 25.49 (Bridge). Why is direct chain tensoring not the end of the story?

Solution. Homology does not generally commute with tensor products exactly; VIII/26 adds the Tor correction through the universal coefficient theorem. $\square$

Chapter 26

The Universal Coefficient Theorem

The universal coefficient theorem converts integral homology into homology with arbitrary coefficients. The price for tensoring is a Tor correction term.

26.1 Statement for homology

Theorem 26.1 (Universal coefficient theorem for homology). *For a space X and abelian group G , there is a natural short exact sequence*

$$0 \rightarrow H_n(X; \mathbb{Z}) \otimes G \longrightarrow H_n(X; G) \longrightarrow \operatorname{Tor}(H_{n-1}(X; \mathbb{Z}), G) \rightarrow 0.$$

The sequence splits, but not naturally in general.

26.2 Meaning

The first term is the expected coefficient change from degree n . The Tor term imports torsion information from degree $n - 1$.

26.3 Free homology

If $H_{n-1}(X; \mathbb{Z})$ is free abelian, then

$$\operatorname{Tor}(H_{n-1}(X), G) = 0,$$

and therefore

$$H_n(X; G) \cong H_n(X; \mathbb{Z}) \otimes G.$$

26.4 Field coefficients

For $G = \mathbb{F}_p$,

$$A \otimes \mathbb{F}_p \cong A/pA,$$

while

$$\operatorname{Tor}(A, \mathbb{F}_p)$$

detects p -torsion in A .

26.5 Cyclic formulas

For positive integers m, n ,

$$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m, n),$$

and

$$\mathrm{Tor}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n).$$

26.6 Free summands

$$\mathrm{Tor}(\mathbb{Z}, G) = 0, \quad \mathbb{Z} \otimes G \cong G.$$

26.7 Finite direct sums

Tensor and Tor distribute over finite direct sums, so finitely generated integral homology can be handled one cyclic summand at a time.

26.8 Projective plane

For $\mathbb{R}P^2$,

$$H_0 = \mathbb{Z}, \quad H_1 = \mathbb{Z}/2, \quad H_2 = 0.$$

With $G = \mathbb{F}_2$, UCT gives

$$H_1(\mathbb{R}P^2; \mathbb{F}_2) \cong \mathbb{F}_2$$

from the tensor term and

$$H_2(\mathbb{R}P^2; \mathbb{F}_2) \cong \mathbb{F}_2$$

from the Tor term.

26.9 Why a top mod-2 class appears

The integral top group $H_2(\mathbb{R}P^2; \mathbb{Z})$ vanishes, yet mod-2 H_2 is nonzero. UCT explains this as

$$\mathrm{Tor}(H_1(\mathbb{R}P^2), \mathbb{F}_2) \cong \mathbb{F}_2.$$

26.10 Moore spaces

For a Moore space with

$$\tilde{H}_n(M; \mathbb{Z}) = \mathbb{Z}/m,$$

UCT gives coefficient homology in degrees n and $n+1$:

$$\tilde{H}_n(M; G) \cong (\mathbb{Z}/m) \otimes G,$$

$$\tilde{H}_{n+1}(M; G) \cong \mathrm{Tor}(\mathbb{Z}/m, G).$$

26.11 Rational coefficients

Since $\mathbb{Q}$ is torsion-free and divisible, for finitely generated homology the Tor contribution vanishes and finite torsion tensors to zero. Thus

$$H_n(X; \mathbb{Q}) \cong H_n(X; \mathbb{Z})_{\mathrm{free}} \otimes \mathbb{Q}.$$

26.12 Proof idea

Singular chain groups are free abelian. One analyzes

$$0 \rightarrow Z_n \rightarrow C_n \rightarrow B_{n-1} \rightarrow 0$$

and the quotient

$$0 \rightarrow B_n \rightarrow Z_n \rightarrow H_n \rightarrow 0,$$

then tensors with G . The possible failure of exactness is precisely encoded by Tor .

26.13 Free resolutions

For an abelian group A , choose a free resolution

$$0 \rightarrow F_1 \rightarrow F_0 \rightarrow A \rightarrow 0.$$

Then

$$\text{Tor}(A, G) = \ker(F_1 \otimes G \rightarrow F_0 \otimes G).$$

26.14 Cyclic resolution

For $A = \mathbb{Z}/m$, use

$$0 \rightarrow \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \rightarrow \mathbb{Z}/m \rightarrow 0.$$

Tensoring with G shows

$$\text{Tor}(\mathbb{Z}/m, G) = \ker(\times m : G \rightarrow G).$$

26.15 Tensor cyclic quotient

The same resolution gives

$$(\mathbb{Z}/m) \otimes G \cong G/mG.$$

Thus the UCT terms match the direct Moore-space calculation from VIII/25.

26.16 Naturality

The UCT short exact sequence is natural in both X and G .

A map $X \rightarrow Y$ induces a commutative diagram of UCT sequences, as does a homomorphism of coefficient groups.

26.17 Nonnatural splitting

Although the UCT short exact sequence splits as a sequence of abelian groups, there is generally no preferred splitting compatible with all maps.

Therefore one should treat

$$H_n(X; G)$$

as an extension unless a canonical identification is available.

26.18 Dimension over a field

For finite-type homology and $G = \mathbb{F}_p$,

$$\dim_{\mathbb{F}_p} H_n(X; \mathbb{F}_p)$$

receives contributions from the free rank of $H_n(X; \mathbb{Z})$, from p -torsion in H_n , and from p -torsion in H_{n-1} .

26.19 Betti numbers

Over $\mathbb{Q}$, the dimension is exactly the integral Betti number:

$$b_n = \dim_{\mathbb{Q}} H_n(X; \mathbb{Q}).$$

26.20 Relative UCT

The same theorem applies to relative chain complexes:

$$0 \rightarrow H_n(X, A; \mathbb{Z}) \otimes G \rightarrow H_n(X, A; G) \rightarrow \operatorname{Tor}(H_{n-1}(X, A; \mathbb{Z}), G) \rightarrow 0.$$

26.21 Reduced UCT

Reduced homology satisfies the same formula in the expected degrees.

26.22 Computational workflow

For finitely generated integral homology:

1. decompose each group into free and cyclic torsion summands;
2. tensor the degree- n group with G ;
3. compute Tor from the degree- $(n-1)$ group;
4. assemble the resulting short exact sequence.

26.23 Bridge to Künneth

UCT concerns one chain complex with changed coefficients. The Künneth theorem concerns tensor products of two chain complexes. Again, a tensor term gives the main answer and Tor provides the correction.

26.24 Exercises

Exercise 26.2. Exercise 1

Hint. Apply the UCT short exact sequence. □

Solution. UCT has a tensor term from $H_n(X; \mathbb{Z})$. □

Exercise 26.3. Exercise 2

Hint. Apply the UCT short exact sequence. ☐

Solution. Its correction is $\text{Tor}(H_{n-1}(X; \mathbb{Z}), G)$. ☐

Exercise 26.4. Exercise 3

Hint. Apply the UCT short exact sequence. ☐

Solution. The UCT sequence splits but generally not naturally. ☐

Exercise 26.5. Exercise 4

Hint. Apply the UCT short exact sequence. ☐

Solution. If H_{n-1} is free, the Tor term vanishes. ☐

Exercise 26.6. Exercise 5

Hint. Apply the UCT short exact sequence. ☐

Solution. $\mathbb{Z} \otimes G \cong G$. ☐

Exercise 26.7. Exercise 6

Hint. Apply the UCT short exact sequence. ☐

Solution. $\text{Tor}(\mathbb{Z}, G) = 0$. ☐

Exercise 26.8. Exercise 7

Hint. Apply the UCT short exact sequence. ☐

Solution. $\mathbb{Z}/m \otimes G \cong G/mG$. ☐

Exercise 26.9. Exercise 8

Hint. Apply the UCT short exact sequence. ☐

Solution. $\text{Tor}(\mathbb{Z}/m, G) = \ker(\times m : G \rightarrow G)$. ☐

Exercise 26.10. Exercise 9

Hint. Apply the UCT short exact sequence. ☐

Solution. $\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\text{gcd}(m, n)$. ☐

Exercise 26.11. Exercise 10

Hint. Apply the UCT short exact sequence. ☐

Solution. $\text{Tor}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\text{gcd}(m, n)$. ☐

Exercise 26.12. Exercise 11

Hint. Apply the UCT short exact sequence. ☐

Solution. UCT explains the top mod-2 class of $\mathbb{R}P^2$. ☐

Exercise 26.13. Exercise 12

Hint. Apply the UCT short exact sequence. ☐

Solution. For $\mathbb{R}P^2$, mod-2 H_2 comes from Tor of integral H_1 . □

Exercise 26.14. Exercise 13

Hint. Apply the UCT short exact sequence. □

Solution. Moore-space coefficient homology has tensor and Tor in adjacent degrees. □

Exercise 26.15. Exercise 14

Hint. Apply the UCT short exact sequence. □

Solution. Rational coefficients kill finite torsion. □

Exercise 26.16. Exercise 15

Hint. Apply the UCT short exact sequence. □

Solution. Over $\mathbb{Q}$, homology dimensions are Betti numbers. □

Exercise 26.17. Exercise 16

Hint. Apply the UCT short exact sequence. □

Solution. UCT is natural in space maps. □

Exercise 26.18. Exercise 17

Hint. Apply the UCT short exact sequence. □

Solution. UCT is natural in coefficient homomorphisms. □

Exercise 26.19. Exercise 18

Hint. Apply the UCT short exact sequence. □

Solution. A cyclic free resolution computes both tensor quotient and Tor kernel. □

Exercise 26.20. Exercise 19

Hint. Apply the UCT short exact sequence. □

Solution. Relative homology satisfies UCT. □

Exercise 26.21. Exercise 20

Hint. Apply the UCT short exact sequence. □

Solution. Reduced homology satisfies UCT. □

Exercise 26.22. Exercise 21

Hint. Apply the UCT short exact sequence. □

Solution. Finite direct sums can be handled summandwise. □

Exercise 26.23. Exercise 22

Hint. Apply the UCT short exact sequence. □

Solution. Mod- p dimensions detect p -torsion in adjacent integral degrees. □

Exercise 26.24. Exercise 23

Hint. Apply the UCT short exact sequence. □

Solution. The splitting is algebraic, not canonically functorial. □

Exercise 26.25. Exercise 24

Hint. Apply the UCT short exact sequence. □

Solution. VIII/27 applies the same tensor-plus-Tor philosophy to products. □

26.25 Solved problem dossiers

Problem 26.26 (RP2 H2 mod2). Use UCT to compute $H_2(\mathbb{R}P^2; \mathbb{F}_2)$.

Solution. The tensor term is zero because integral $H_2 = 0$. The Tor term is $\text{Tor}(\mathbb{Z}/2, \mathbb{F}_2) \cong \mathbb{F}_2$. □

Problem 26.27 (RP2 H1 mod2). Compute $H_1(\mathbb{R}P^2; \mathbb{F}_2)$.

Solution. $(\mathbb{Z}/2) \otimes \mathbb{F}_2 \cong \mathbb{F}_2$, while Tor from $H_0 = \mathbb{Z}$ vanishes. □

Problem 26.28 (RP2 mod3). Compute positive homology with $\mathbb{F}_3$.

Solution. Both tensor and Tor of $\mathbb{Z}/2$ with $\mathbb{F}_3$ vanish, so positive homology is zero. □

Problem 26.29 (Moore mod p). Apply UCT to $M(\mathbb{Z}/m, n)$ with $\mathbb{F}_p$.

Solution. If $p \mid m$, tensor gives $\mathbb{F}_p$ in degree n and Tor gives $\mathbb{F}_p$ in degree $n + 1$; otherwise both vanish. □

Problem 26.30 (Cyclic G). Compute $(\mathbb{Z}/12) \otimes \mathbb{Z}/18$ and Tor.

Solution. Both are $\mathbb{Z}/6$. □

Problem 26.31 (Free plus torsion). Let $H_n = \mathbb{Z}^2 \oplus \mathbb{Z}/6$. Tensor with $\mathbb{F}_2$.

Solution. The free part gives $\mathbb{F}_2^2$ and the torsion summand gives $\mathbb{F}_2$, so the tensor term is $\mathbb{F}_2^3$. □

Problem 26.32 (Adjacent Tor). If $H_{n-1} = \mathbb{Z}/4$, what Tor contribution appears with $\mathbb{F}_2$?

Solution. $\text{Tor}(\mathbb{Z}/4, \mathbb{F}_2) \cong \mathbb{F}_2$. □

Problem 26.33 (Rationalization). Let $H_n = \mathbb{Z}^3 \oplus \mathbb{Z}/12$. Compute $H_n(-; \mathbb{Q})$ assuming adjacent Tor vanishes.

Solution. $\mathbb{Q}^3$. □

Problem 26.34 (Cyclic resolution). Use the standard resolution to compute $\text{Tor}(\mathbb{Z}/m, G)$.

Solution. Tensor $0 \rightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \rightarrow \mathbb{Z}/m \rightarrow 0$; Tor is the kernel of multiplication by m on G . □

Problem 26.35 (Tensor quotient). Use the same resolution to compute $\mathbb{Z}/m \otimes G$.

Solution. It is the cokernel of multiplication by m , namely G/mG . □

Problem 26.36 (Natural map). What happens to UCT under $f : X \rightarrow Y$?

Solution. Tensor, coefficient homology, and Tor maps form a commutative diagram of short exact UCT sequences. $\square$

Problem 26.37 (Nonnatural split). Why should one not write a canonical direct sum in UCT?

Solution. A splitting exists as groups, but no choice is generally compatible with all maps; only the short exact sequence is canonical. $\square$

Problem 26.38 (Relative example). Apply UCT to $H_n(D^n, S^{n-1}; G)$.

Solution. Integral relative homology is $\mathbb{Z}$ in degree n , so coefficient relative homology is G there and zero elsewhere. $\square$

Problem 26.39 (Sphere). Use UCT to compute $H_n(S^n; G)$.

Solution. Integral top homology is free $\mathbb{Z}$ and adjacent lower homology is zero for $n > 1$, so top homology is G . $\square$

Problem 26.40 (Circle). Use UCT to compute $H_1(S^1; G)$.

Solution. $\mathbb{Z} \otimes G \cong G$, and Tor from $H_0 = \mathbb{Z}$ vanishes. $\square$

Problem 26.41 (Betti). Why does rational homology recover Betti numbers?

Solution. Finite torsion disappears and Tor vanishes, leaving one $\mathbb{Q}$ for each free integral generator. $\square$

Problem 26.42 (Mod p dimension). Why can $\dim H_n(X; \mathbb{F}_p)$ exceed b_n ?

Solution. Besides free rank, p -torsion in H_n contributes through tensor and p -torsion in H_{n-1} contributes through Tor. $\square$

Problem 26.43 (UCT workflow). Given finitely generated integral homology, what is the first computational step?

Solution. Decompose each group into free and cyclic torsion summands, then apply tensor and Tor summandwise. $\square$

Problem 26.44 (Extension issue). What information can remain after tensor and Tor are known?

Solution. One must still know how the two terms fit into the short exact sequence; the theorem says it splits noncanonically for homology over $\mathbb{Z}$. $\square$

Problem 26.45 (Coefficient consistency). How does UCT confirm direct cellular mod- p calculations?

Solution. Both methods produce the same coefficient homology; UCT explains which classes come from tensor and which are new Tor classes. $\square$

Problem 26.46 (Bridge). How does Künneth resemble UCT?

Solution. Both have a primary tensor expression and a Tor correction measuring failure of exactness. $\square$

Problem 26.47 (Final principle). What does UCT save computationally?

Solution. It avoids rebuilding the entire coefficient chain complex once integral homology is known. $\square$

Chapter 27

Products and the Künneth Theorem

Products of spaces lead to tensor products of chains. The Künneth theorem expresses the homology of $X \times Y$ from the homology of the factors, with Tor terms measuring torsion interactions.

27.1 Product cells

If $e^p \subset X$ and $e^q \subset Y$ are cells, then

$$e^p \times e^q$$

is a cell of dimension $p + q$ in the product CW structure.

This immediately gives

$$\chi(X \times Y) = \chi(X)\chi(Y)$$

for finite CW complexes.

27.2 Tensor product of chain complexes

For chain complexes C, D , define

$$(C \otimes D)_n = \bigoplus_{p+q=n} C_p \otimes D_q.$$

The differential on homogeneous tensors is

$$\partial(c \otimes d) = \partial c \otimes d + (-1)^p c \otimes \partial d, \quad c \in C_p.$$

27.3 Koszul sign

The sign $(-1)^p$ is necessary for

$$\partial^2 = 0.$$

27.4 Eilenberg–Zilber principle

For spaces X, Y , singular chains on the product are chain-homotopy equivalent to the tensor product:

$$C_*^{\text{sing}}(X \times Y) \simeq C_*^{\text{sing}}(X) \otimes C_*^{\text{sing}}(Y).$$

This is the chain-level bridge from topology to algebra.

27.5 Cross product

The Eilenberg–Zilber construction yields the homology cross product

$$\times : H_p(X) \otimes H_q(Y) \rightarrow H_{p+q}(X \times Y).$$

27.6 Künneth theorem over $\mathbb{Z}$

Theorem 27.1 (Künneth theorem). *For suitable spaces such as CW complexes, there is a natural short exact sequence*

$$0 \rightarrow \bigoplus_{p+q=n} H_p(X) \otimes H_q(Y) \longrightarrow H_n(X \times Y) \longrightarrow \bigoplus_{p+q=n-1} \text{Tor}(H_p(X), H_q(Y)) \rightarrow 0.$$

The sequence splits noncanonically.

27.7 Field coefficients

Over a field k , Tor vanishes and Künneth becomes

$$H_n(X \times Y; k) \cong \bigoplus_{p+q=n} H_p(X; k) \otimes_k H_q(Y; k).$$

27.8 Product of circles

Since

$$H_0(S^1) = H_1(S^1) = \mathbb{Z},$$

Künneth gives

$$H_0(T^2) = \mathbb{Z}, \quad H_1(T^2) = \mathbb{Z}^2, \quad H_2(T^2) = \mathbb{Z}.$$

27.9 Higher tori

For

$$T^n = (S^1)^n,$$

$$H_k(T^n; \mathbb{Z}) \cong \mathbb{Z}^{\binom{n}{k}}.$$

The total homology ranks are the binomial coefficients.

27.10 Products of spheres

For $S^m \times S^n$, homology is free and concentrated in degrees

$$0, \quad m, \quad n, \quad m+n,$$

with the middle degrees combining if $m = n$.

27.11 $S^2 \times S^2$

$$H_0 = \mathbb{Z}, \quad H_2 = \mathbb{Z}^2, \quad H_4 = \mathbb{Z},$$

and all other groups vanish.

27.12 Cylinder

$$S^1 \times I \simeq S^1.$$

Künneth agrees: I has only $H_0 = \mathbb{Z}$, so tensoring introduces no new positive-dimensional classes.

27.13 Product with a contractible space

If Y is nonempty and contractible,

$$X \times Y \simeq X,$$

and Künneth reduces correspondingly.

27.14 Torsion interaction

If both factors have torsion, Tor can create homology one degree higher than the naive tensor terms predict.

27.15 Example with projective spaces

Since

$$H_1(\mathbb{R}P^2) = \mathbb{Z}/2,$$

the product $\mathbb{R}P^2 \times \mathbb{R}P^2$ has tensor torsion and a Tor contribution. In particular,

$$\mathrm{Tor}(\mathbb{Z}/2, \mathbb{Z}/2) \cong \mathbb{Z}/2$$

appears in degree 3.

27.16 Moore-space products

For Moore spaces, the tensor and Tor terms can be read directly from the cyclic torsion groups:

$$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathrm{Tor}(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\mathrm{gcd}(m, n).$$

27.17 Naturality

The Künneth short exact sequence is natural in maps of both factors.

27.18 Nonnatural splitting

As with UCT, the splitting is generally not natural. The short exact sequence, rather than a chosen direct-sum decomposition, is the canonical object.

27.19 Reduced Künneth

For pointed connected spaces, reduced homology often gives a cleaner product formula by separating the H_0 terms.

27.20 Relative Künneth preview

Pairs and products of pairs admit relative forms of Künneth under suitable hypotheses. These are useful in manifold and bundle calculations.

27.21 Euler characteristic from Künneth

Over a field, alternating dimensions factor:

$$\chi(X \times Y) = \chi(X)\chi(Y).$$

This recovers the product formula from homology.

27.22 Betti-number convolution

Over a field,

$$b_n(X \times Y) = \sum_{p+q=n} b_p(X)b_q(Y).$$

Thus Poincaré polynomials multiply:

$$P_{X \times Y}(t) = P_X(t)P_Y(t).$$

27.23 Poincaré polynomial

Define

$$P_X(t) = \sum_n b_n(X)t^n.$$

For the torus,

$$P_{T^n}(t) = (1+t)^n.$$

27.24 Chain-homotopy input

The Eilenberg–Zilber and Alexander–Whitney maps are inverse up to chain homotopy, not literally inverse. The machinery of VIII/22 is exactly what allows them to induce inverse homology isomorphisms.

27.25 Why the theorem matters

Künneth replaces a potentially large product-space chain calculation with:

- tensor products of known homology groups;
- Tor corrections for torsion.

27.26 Bridge to cohomology

Products become even richer in cohomology because the diagonal map and cross product combine to form the cup product. VIII/28 begins cohomology, and VIII/29 develops its multiplicative structure.

27.27 Exercises

Exercise 27.2. Exercise 1

Hint. Apply the product chain complex or Künneth formula. □

Solution. $(C \otimes D)_n = \bigoplus_{p+q=n} C_p \otimes D_q$. □

Exercise 27.3. Exercise 2

Hint. Apply the product chain complex or Künneth formula. □

Solution. The product differential uses the sign $(-1)^p$. □

Exercise 27.4. Exercise 3

Hint. Apply the product chain complex or Künneth formula. □

Solution. The sign is needed for $\partial^2 = 0$. □

Exercise 27.5. Exercise 4

Hint. Apply the product chain complex or Künneth formula. □

Solution. Eilenberg–Zilber compares product chains with tensor-product chains. □

Exercise 27.6. Exercise 5

Hint. Apply the product chain complex or Künneth formula. □

Solution. The cross product sends $H_p(X) \otimes H_q(Y)$ to $H_{p+q}(X \times Y)$. □

Exercise 27.7. Exercise 6

Hint. Apply the product chain complex or Künneth formula. □

Solution. Künneth has a tensor term in total degree n . □

Exercise 27.8. Exercise 7

Hint. Apply the product chain complex or Künneth formula. □

Solution. Its Tor term uses total degree $n - 1$. □

Exercise 27.9. Exercise 8

Hint. Apply the product chain complex or Künneth formula. □

Solution. Over a field, Tor vanishes. □

Exercise 27.10. Exercise 9

Hint. Apply the product chain complex or Künneth formula. □

Solution. $H_1(T^2) \cong \mathbb{Z}^2$. □

Exercise 27.11. Exercise 10

Hint. Apply the product chain complex or Künneth formula. □

Solution. $H_2(T^2) \cong \mathbb{Z}$. □

Exercise 27.12. Exercise 11

Hint. Apply the product chain complex or Künneth formula. ☐

Solution. $H_k(T^n) \cong \mathbb{Z}^{\binom{n}{k}}$. ☐

Exercise 27.13. Exercise 12

Hint. Apply the product chain complex or Künneth formula. ☐

Solution. $H_2(S^2 \times S^2) \cong \mathbb{Z}^2$. ☐

Exercise 27.14. Exercise 13

Hint. Apply the product chain complex or Künneth formula. ☐

Solution. Product with a contractible factor preserves homology. ☐

Exercise 27.15. Exercise 14

Hint. Apply the product chain complex or Künneth formula. ☐

Solution. Torsion in both factors can create Tor classes. ☐

Exercise 27.16. Exercise 15

Hint. Apply the product chain complex or Künneth formula. ☐

Solution. $\text{Tor}(\mathbb{Z}/2, \mathbb{Z}/2) \cong \mathbb{Z}/2$. ☐

Exercise 27.17. Exercise 16

Hint. Apply the product chain complex or Künneth formula. ☐

Solution. Moore-space products produce gcd torsion in tensor and Tor. ☐

Exercise 27.18. Exercise 17

Hint. Apply the product chain complex or Künneth formula. ☐

Solution. Künneth is natural in both factors. ☐

Exercise 27.19. Exercise 18

Hint. Apply the product chain complex or Künneth formula. ☐

Solution. Its splitting is generally nonnatural. ☐

Exercise 27.20. Exercise 19

Hint. Apply the product chain complex or Künneth formula. ☐

Solution. Betti numbers convolve under products over a field. ☐

Exercise 27.21. Exercise 20

Hint. Apply the product chain complex or Künneth formula. ☐

Solution. Poincaré polynomials multiply. ☐

Exercise 27.22. Exercise 21

Hint. Apply the product chain complex or Künneth formula. □

Solution. $P_{T^n}(t) = (1+t)^n$. □

Exercise 27.23. Exercise 22

Hint. Apply the product chain complex or Künneth formula. □

Solution. Euler characteristic is multiplicative. □

Exercise 27.24. Exercise 23

Hint. Apply the product chain complex or Künneth formula. □

Solution. Eilenberg–Zilber inverse maps are only inverse up to chain homotopy. □

Exercise 27.25. Exercise 24

Hint. Apply the product chain complex or Künneth formula. □

Solution. VIII/29 will turn products into the cup-product ring structure. □

27.28 Solved problem dossiers

Problem 27.26 (Torus). Compute $H_*(S^1 \times S^1)$ with Künneth.

Solution. All groups are free, so no Tor occurs: tensor terms give $\mathbb{Z}$ in degrees 0 and 2 and $\mathbb{Z}^2$ in degree 1. □

Problem 27.27 (Three-torus). Compute the ranks of $H_k(T^3)$.

Solution. They are $\binom{3}{0}, \binom{3}{1}, \binom{3}{2}, \binom{3}{3} = 1, 3, 3, 1$. □

Problem 27.28 (Four-torus). Compute $b_2(T^4)$.

Solution. $\binom{4}{2} = 6$. □

Problem 27.29 (Sphere product). Compute $H_*(S^2 \times S^3)$.

Solution. Free groups $\mathbb{Z}$ occur in degrees 0,2,3,5 and vanish elsewhere. □

Problem 27.30 (Equal spheres). Compute $H_2(S^2 \times S^2)$.

Solution. There are two tensor contributions $H_2(S^2) \otimes H_0(S^2)$ and $H_0 \otimes H_2$, giving $\mathbb{Z}^2$. □

Problem 27.31 (Top class). Where does the top class of $S^m \times S^n$ come from?

Solution. From $H_m(S^m) \otimes H_n(S^n) \cong \mathbb{Z}$ in total degree $m+n$. □

Problem 27.32 (Contractible factor). Use Künneth for $X \times I$.

Solution. Only $H_0(I) = \mathbb{Z}$ is nonzero, so $H_n(X \times I) \cong H_n(X)$. □

Problem 27.33 (RP2 product Tor). Locate a Tor contribution in $\mathbb{R}P^2 \times \mathbb{R}P^2$.

Solution. Since both factors have $H_1 = \mathbb{Z}/2$, $\text{Tor}(H_1, H_1) = \mathbb{Z}/2$ contributes in total degree $1+1+1=3$. □

Problem 27.34 (RP2 product H2). Give tensor contributions to $H_2(\mathbb{R}P^2 \times \mathbb{R}P^2)$.

Solution. The $H_1 \otimes H_1$ term gives $\mathbb{Z}/2$, while integral H_2 of each factor vanishes. $\square$

Problem 27.35 (Moore gcd). For Moore torsion groups $\mathbb{Z}/12$ and $\mathbb{Z}/18$, compute tensor and Tor.

Solution. Both are $\mathbb{Z}/6$. $\square$

Problem 27.36 (Field Künneth). Why is Künneth especially simple over $\mathbb{F}_p$?

Solution. Vector spaces are flat and Tor vanishes, leaving a direct sum of tensor products. $\square$

Problem 27.37 (Betti convolution). If $P_X = 1 + 2t + t^2$ and $P_Y = 1 + t$, compute $P_{X \times Y}$.

Solution. Multiply to get $1 + 3t + 3t^2 + t^3$. $\square$

Problem 27.38 (Euler product). Use Poincaré polynomials to recover $\chi(X \times Y)$.

Solution. Euler characteristic is $P(-1)$; since product polynomials multiply, $\chi(X \times Y) = \chi(X)\chi(Y)$. $\square$

Problem 27.39 (Cross product). What homology class does $a \times b$ have degree?

Solution. If $a \in H_p(X)$ and $b \in H_q(Y)$, then $a \times b \in H_{p+q}(X \times Y)$. $\square$

Problem 27.40 (Naturality). How does $f \times g$ act on cross products?

Solution. $(f \times g)_*(a \times b) = f_*(a) \times g_*(b)$. $\square$

Problem 27.41 (Nonnatural split). Why is the Künneth exact sequence preferable to a chosen direct sum?

Solution. The exact sequence is natural; a splitting usually depends on arbitrary choices. $\square$

Problem 27.42 (Reduced product). Why can reduced homology simplify product formulas?

Solution. It separates the ubiquitous degree-zero $\mathbb{Z}$ terms, making the genuinely positive-dimensional cross terms more transparent. $\square$

Problem 27.43 (Eilenberg–Zilber). What is the role of chain homotopy in the theorem?

Solution. The Alexander–Whitney and shuffle/Eilenberg–Zilber maps compose to maps chain-homotopic to identities, so they induce actual inverse homology isomorphisms. $\square$

Problem 27.44 (Koszul sign). Verify the cross terms in $\partial^2(c \otimes d)$ cancel.

Solution. The second application of the differential produces cross terms with opposite signs because the degree of ∂c is $p - 1$. $\square$

Problem 27.45 (Product cells). How does a p -cell times a q -cell contribute?

Solution. It gives a $(p + q)$ -cell, explaining both the tensor grading and Euler multiplicativity. $\square$

Problem 27.46 (Two streams). Why is VIII/27 sourced from both topology-13 and topology-17?

Solution. One stream carries the main product/Künneth theorem while the other carries complementary product calculations and homological machinery; both have the same canonical destination. $\square$

Problem 27.47 (Bridge). How do products lead toward cup products?

Solution. The cohomology cross product composed with the diagonal $X \rightarrow X \times X$ produces the cup product. $\square$

Problem 27.48 (Final principle). What does Künneth compute?

Solution. It reconstructs homology of a product from factor homologies via tensor products plus Tor corrections. $\square$

Problem 27.49 (Comparison with UCT). State the analogy with VIII/26.

Solution. UCT changes coefficients for one complex; Künneth combines two complexes. Both are controlled by tensor products corrected by Tor. $\square$

Part VI

Cohomology, Bundles and Manifolds

Chapter 28

Cohomology

Homology is built from chains and boundaries. Cohomology reverses the direction: a cochain assigns algebraic data to chains, and the coboundary operator is obtained by precomposing with the chain boundary. The resulting groups are contravariant in spaces and support additional structures that have no direct homological analogue.

28.1 Cochains

Let $C_\bullet$ be a chain complex and G an abelian group.

Definition 28.1 (Cochain group). The group of n -cochains with coefficients in G is

$$C^n(C; G) = \text{Hom}(C_n, G).$$

For a space X ,

$$C^n(X; G) = \text{Hom}(C_n^{\text{sing}}(X), G).$$

28.2 Coboundary

Definition 28.2 (Coboundary operator). For $\varphi \in C^n(C; G)$, define

$$\delta\varphi = \varphi \circ \partial_{n+1}.$$

Thus

$$\delta : C^n \rightarrow C^{n+1}.$$

Since $\partial^2 = 0$,

$$\delta^2 = 0.$$

28.3 Cocycles and coboundaries

Definition 28.3.

$$Z^n(C; G) = \ker \delta, \quad B^n(C; G) = \text{im } \delta.$$

Elements of Z^n are cocycles and elements of B^n are coboundaries.

28.4 Cohomology groups

Definition 28.4 (Cohomology).

$$H^n(C; G) = Z^n(C; G)/B^n(C; G).$$

For a space X ,

$$H^n(X; G) = H^n(C^*(X; G)).$$

28.5 Contravariance

A continuous map

$$f : X \rightarrow Y$$

induces

$$f_{\#} : C_*(X) \rightarrow C_*(Y).$$

Precomposition defines

$$f^{\#} : C^n(Y; G) \rightarrow C^n(X; G), \quad f^{\#}(\varphi) = \varphi \circ f_{\#}.$$

Theorem 28.5 (Functoriality). *The cochain maps $f^{\#}$ induce*

$$f^* : H^n(Y; G) \rightarrow H^n(X; G),$$

with

$$(g \circ f)^* = f^* \circ g^*.$$

28.6 Degree zero

For a path-connected space and coefficient group G ,

$$H^0(X; G) \cong G.$$

More generally, H^0 is the group of locally constant G -valued functions on path components.

28.7 Spheres

With integer coefficients,

$$H^k(S^n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, n, \\ 0, & \text{otherwise.} \end{cases}$$

28.8 Cohomology of a point

$$H^0(*; G) \cong G, \quad H^n(*; G) = 0 \quad (n > 0).$$

28.9 Simplicial cohomology

For a simplicial complex, cochains are homomorphisms from simplicial chain groups to G . With chosen oriented simplices, a cochain can be regarded as an assignment of coefficients to oriented simplices, compatible with sign.

Simplicial and singular cohomology agree on simplicial complexes.

28.10 Cellular cohomology

For a CW complex, dualizing the cellular chain complex gives the cellular cochain complex

$$C_{\text{cell}}^n(X; G) = \text{Hom}(C_n^{\text{cell}}(X), G).$$

For finite CW complexes, this often gives the fastest computation.

28.11 Evaluation pairing

A cochain evaluates on a chain. Passing to homology and cohomology gives the Kronecker pairing

$$\langle \ , \ \rangle : H^n(X; G) \times H_n(X; \mathbb{Z}) \rightarrow G.$$

If $[\varphi] \in H^n$ and $[z] \in H_n$, then

$$\langle [\varphi], [z] \rangle = \varphi(z).$$

This is well defined because cocycles vanish on boundaries and coboundaries vanish on cycles.

28.12 Relative cohomology

For a pair $A \subset X$,

$$C^n(X, A; G) = \text{Hom}(C_n(X, A), G).$$

Definition 28.6 (Relative cohomology).

$$H^n(X, A; G) = H^n(C^*(X, A; G)).$$

28.13 Long exact sequence of a pair

A pair has a natural long exact cohomology sequence

$$\cdots \rightarrow H^n(X, A; G) \rightarrow H^n(X; G) \rightarrow H^n(A; G) \xrightarrow{\delta} H^{n+1}(X, A; G) \rightarrow \cdots.$$

Notice the connecting map raises degree.

28.14 Connecting homomorphism

If a cocycle on A extends to a cochain on X , its coboundary vanishes on A and therefore determines a relative cocycle one degree higher. This is the cohomological connecting map.

28.15 Annulus example

Let A be one boundary circle of an annulus $X = S^1 \times I$. Since the annulus deformation retracts onto A , the relative cohomology

$$H^*(X, A; G)$$

vanishes. The pair exact sequence reflects that the restriction

$$H^1(X; G) \rightarrow H^1(A; G)$$

is an isomorphism.

This is the canonical home of the mapped legacy annulus/connecting visual.

28.16 Universal coefficient theorem for cohomology

For integral homology and coefficient group G , one has a natural short exact sequence

$$0 \rightarrow \operatorname{Ext}(H_{n-1}(X; \mathbb{Z}), G) \rightarrow H^n(X; G) \rightarrow \operatorname{Hom}(H_n(X; \mathbb{Z}), G) \rightarrow 0.$$

The sequence splits noncanonically in general.

28.17 Free homology case

If $H_{n-1}(X; \mathbb{Z})$ is free, then the Ext term vanishes and

$$H^n(X; G) \cong \operatorname{Hom}(H_n(X; \mathbb{Z}), G).$$

28.18 Cyclic Ext formula

For positive m ,

$$\operatorname{Ext}(\mathbb{Z}/m, G) \cong G/mG.$$

For $G = \mathbb{Z}$,

$$\operatorname{Ext}(\mathbb{Z}/m, \mathbb{Z}) \cong \mathbb{Z}/m.$$

Thus torsion in homology may appear one degree higher in integral cohomology.

28.19 Projective-plane example

Integral homology of $\mathbb{R}P^2$ has

$$H_1 \cong \mathbb{Z}/2, \quad H_2 = 0.$$

The cohomology UCT gives

$$H^2(\mathbb{R}P^2; \mathbb{Z}) \cong \mathbb{Z}/2.$$

28.20 Homotopy invariance

Homotopic maps induce the same cohomology homomorphisms because their chain maps are chain homotopic and precomposition converts that chain homotopy into a cochain homotopy.

Thus homotopy-equivalent spaces have isomorphic cohomology groups.

28.21 Reduced cohomology

Reduced cohomology removes the degree-zero contribution of a basepoint, in parallel with reduced homology. For $n > 0$,

$$\tilde{H}^n(X; G) \cong H^n(X; G).$$

28.22 Why cohomology is richer

As graded groups, cohomology often resembles homology. But cohomology admits a natural multiplication

$$H^p(X; R) \times H^q(X; R) \rightarrow H^{p+q}(X; R),$$

the cup product. VIII/29 develops this ring structure.

28.23 SVG visual laboratory

This chapter ships with editable SVG assets for:

- the chain-to-cochain reversal and evaluation pairing;
- the annulus pair exact-sequence/connecting-map geometry.

The SVG sources are tracked as production assets and do not require an external SVG renderer for the LaTeX build.

28.24 Exercises

Exercise 28.7. Exercise 1

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. $C^n(C; G) = \text{Hom}(C_n, G)$. □

Exercise 28.8. Exercise 2

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. The coboundary is $\delta\varphi = \varphi\partial$. □

Exercise 28.9. Exercise 3

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. $\delta^2 = 0$ follows from $\partial^2 = 0$. □

Exercise 28.10. Exercise 4

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. Cocycles are $\ker \delta$. □

Exercise 28.11. Exercise 5

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. Coboundaries are $\text{im } \delta$. □

Exercise 28.12. Exercise 6

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. $H^n = Z^n/B^n$. □

Exercise 28.13. Exercise 7

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. A map $f : X \rightarrow Y$ induces $f^* : H^n(Y) \rightarrow H^n(X)$. □

Exercise 28.14. Exercise 8

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. Cohomology is contravariant. □

Exercise 28.15. Exercise 9

Hint. Use the definitions of cochains, coboundaries, and functoriality. ☐

Solution. For path-connected X , $H^0(X; G) \cong G$. ☐

Exercise 28.16. Exercise 10

Hint. Use the definitions of cochains, coboundaries, and functoriality. ☐

Solution. $H^n(*; G) = 0$ for $n > 0$. ☐

Exercise 28.17. Exercise 11

Hint. Use the definitions of cochains, coboundaries, and functoriality. ☐

Solution. Simplicial cochains dualize simplicial chains. ☐

Exercise 28.18. Exercise 12

Hint. Use the definitions of cochains, coboundaries, and functoriality. ☐

Solution. Cellular cochains dualize cellular chains. ☐

Exercise 28.19. Exercise 13

Hint. Use the definitions of cochains, coboundaries, and functoriality. ☐

Solution. The Kronecker pairing evaluates cohomology classes on homology classes. ☐

Exercise 28.20. Exercise 14

Hint. Use the definitions of cochains, coboundaries, and functoriality. ☐

Solution. A cocycle vanishes on boundaries. ☐

Exercise 28.21. Exercise 15

Hint. Use the definitions of cochains, coboundaries, and functoriality. ☐

Solution. A coboundary vanishes on cycles. ☐

Exercise 28.22. Exercise 16

Hint. Use the definitions of cochains, coboundaries, and functoriality. ☐

Solution. Relative cochains dualize relative chains. ☐

Exercise 28.23. Exercise 17

Hint. Use the definitions of cochains, coboundaries, and functoriality. ☐

Solution. The cohomology connecting map raises degree. ☐

Exercise 28.24. Exercise 18

Hint. Use the definitions of cochains, coboundaries, and functoriality. ☐

Solution. The annulus relative to one boundary circle has zero relative cohomology. ☐

Exercise 28.25. Exercise 19

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. The cohomology UCT has Hom and Ext terms. □

Exercise 28.26. Exercise 20

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. If H_{n-1} is free, the Ext term vanishes. □

Exercise 28.27. Exercise 21

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. $\text{Ext}(\mathbb{Z}/m, \mathbb{Z}) \cong \mathbb{Z}/m$. □

Exercise 28.28. Exercise 22

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. $H^2(\mathbb{R}P^2; \mathbb{Z}) \cong \mathbb{Z}/2$. □

Exercise 28.29. Exercise 23

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. Cohomology is homotopy invariant. □

Exercise 28.30. Exercise 24

Hint. Use the definitions of cochains, coboundaries, and functoriality. □

Solution. VIII/29 equips cohomology with the cup product. □

28.25 Solved problem dossiers

Problem 28.31 (Coboundary squared). Prove $\delta^2 = 0$.

Solution. For a cochain φ , $\delta^2\varphi = \varphi\partial\partial = 0$. □

Problem 28.32 (Contravariance). Verify $(g \circ f)^* = f^*g^*$.

Solution. Precomposition gives $\varphi \mapsto \varphi g_{\#} f_{\#}$, which is exactly $f^*(g^{\#}\varphi)$. □

Problem 28.33 (H0). Why is $H^0(X; G) \cong G^c$ for c path components?

Solution. A zero-cocycle must take the same value on endpoints of every path, hence is constant on each component; there are no negative-degree coboundaries. □

Problem 28.34 (Sphere). Compute integral cohomology of S^n .

Solution. The homology is free in degrees 0 and n , so the cohomology UCT gives $\mathbb{Z}$ in those degrees and zero elsewhere. □

Problem 28.35 (Kronecker). Why is the evaluation pairing well defined?

Solution. Changing the cycle by a boundary does not change a cocycle's value, and changing the cocycle by a coboundary does not change its value on a cycle. □

Problem 28.36 (Circle generator). Interpret a generator of $H^1(S^1; \mathbb{Z})$.

Solution. It is a cohomology class evaluating to 1 on the fundamental one-cycle. □

Problem 28.37 (Torus H1). Compute $H^1(T^2; \mathbb{Z})$.

Solution. Since $H_1(T^2) = \mathbb{Z}^2$ is free, $H^1 \cong \text{Hom}(\mathbb{Z}^2, \mathbb{Z}) \cong \mathbb{Z}^2$. □

Problem 28.38 (RP2 H2). Use cohomology UCT to compute $H^2(\mathbb{R}P^2; \mathbb{Z})$.

Solution. The Hom term from $H_2 = 0$ vanishes; the Ext term from $H_1 = \mathbb{Z}/2$ is $\mathbb{Z}/2$. □

Problem 28.39 (RP2 H1). Compute $H^1(\mathbb{R}P^2; \mathbb{Z})$.

Solution. $\text{Hom}(\mathbb{Z}/2, \mathbb{Z}) = 0$ and Ext from $H_0 = \mathbb{Z}$ vanishes, so $H^1 = 0$. □

Problem 28.40 (Relative disk). Compute $H^n(D^n, S^{n-1}; G)$.

Solution. The pair has one relative n -class; cohomology is G in degree n and zero otherwise. □

Problem 28.41 (Annulus pair). Why does $H^*(S^1 \times I, S^1 \times \{0\}; G) = 0$?

Solution. The chosen boundary circle is a deformation retract of the annulus, so the pair is homotopy equivalent to (A, A) . □

Problem 28.42 (Connecting map). Describe the cohomology connecting homomorphism geometrically.

Solution. Extend a cocycle on A to a cochain on X ; its coboundary vanishes on A and defines a relative cocycle one degree higher. □

Problem 28.43 (Cellular cochains). For a finite CW complex with c_n n -cells, what is $C_{\text{cell}}^n(X; G)$?

Solution. It is $\text{Hom}(\mathbb{Z}^{c_n}, G) \cong G^{c_n}$. □

Problem 28.44 (Finite graph). Compute H^1 of a connected graph of rank r over $\mathbb{Z}$.

Solution. Its H_1 is $\mathbb{Z}^r$, so $H^1 \cong \mathbb{Z}^r$. □

Problem 28.45 (Moore cohomology). If $H_n(M) = \mathbb{Z}/m$ and other reduced homology vanishes, where does integral cohomology torsion appear?

Solution. $H^{n+1}(M; \mathbb{Z}) \cong \text{Ext}(\mathbb{Z}/m, \mathbb{Z}) \cong \mathbb{Z}/m$. □

Problem 28.46 (Free case). Why does free integral homology make cohomology a dual group?

Solution. Ext vanishes, leaving $H^n(X; G) \cong \text{Hom}(H_n(X), G)$. □

Problem 28.47 (Homotopy). Why do homotopic maps induce equal cohomology maps?

Solution. The induced chain maps are chain homotopic; dualizing yields cochain-homotopic maps, which agree on cohomology. □

Problem 28.48 (Reduced). For $n > 0$, compare H^n and reduced H^n .

Solution. They agree; only degree zero is altered. □

Problem 28.49 (Coefficient field). What simplification occurs over a field for finite-dimensional chain groups?

Solution. Cochain groups are ordinary vector-space duals and Ext phenomena over the field disappear. $\square$

Problem 28.50 (SVG annulus). What concept should the annulus SVG emphasize?

Solution. The deformation-retract geometry together with the restriction/connecting portion of the cohomology long exact sequence. $\square$

Chapter 29

Cup Products

Cohomology becomes substantially richer when its groups are multiplied. The cup product turns $H^*(X; R)$ into a graded ring and detects topology that the additive groups alone can miss.

29.1 Cochain-level cup product

Let R be a commutative coefficient ring. For singular cochains

$$\alpha \in C^p(X; R), \quad \beta \in C^q(X; R),$$

define their cup product using the Alexander–Whitney front/back face formula.

Definition 29.1 (Cup product). For a singular $(p + q)$ -simplex

$$\sigma : [v_0, \dots, v_{p+q}] \rightarrow X,$$

set

$$(\alpha \smile \beta)(\sigma) = \alpha(\sigma| [v_0, \dots, v_p]) \beta(\sigma| [v_p, \dots, v_{p+q}]).$$

29.2 Leibniz rule

Theorem 29.2 (Coboundary derivation rule).

$$\delta(\alpha \smile \beta) = \delta\alpha \smile \beta + (-1)^p \alpha \smile \delta\beta.$$

Therefore cocycles multiply to cocycles, and changing a factor by a coboundary changes the product by a coboundary.

29.3 Product on cohomology

The cochain product descends to

$$\smile : H^p(X; R) \times H^q(X; R) \rightarrow H^{p+q}(X; R).$$

29.4 Cohomology ring

Definition 29.3 (Cohomology ring). The graded group

$$H^*(X; R) = \bigoplus_{n \geq 0} H^n(X; R)$$

with cup product is the cohomology ring of X .

The unit is the class

$$1 \in H^0(X; R)$$

represented by the constant 1-valued zero-cochain on a connected space.

29.5 Associativity

Cup product is associative on cohomology:

$$(a \smile b) \smile c = a \smile (b \smile c).$$

29.6 Naturality

Theorem 29.4 (Naturality). *For $f : X \rightarrow Y$,*

$$f^*(a \smile b) = f^*a \smile f^*b.$$

Thus $f^* : H^*(Y; R) \rightarrow H^*(X; R)$ is a graded-ring homomorphism.

29.7 Graded commutativity

Theorem 29.5 (Graded commutativity). *For*

$$a \in H^p(X; R), \quad b \in H^q(X; R),$$

one has

$$a \smile b = (-1)^{pq} b \smile a.$$

This equality generally holds only after passing to cohomology, not strictly on the standard cochain-level cup product.

29.8 Odd-degree squares

If a has odd degree and 2 is invertible in R , then

$$a \smile a = 0.$$

Over characteristic 2, the sign disappears, so odd-degree squares need not vanish for sign reasons.

29.9 Sphere ring

For $n > 0$,

$$H^*(S^n; \mathbb{Z}) \cong \mathbb{Z}[u]/(u^2), \quad |u| = n,$$

where $u^2 = 0$ for dimensional reasons.

29.10 Circle ring

$$H^*(S^1; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}(a), \quad |a| = 1.$$

Thus

$$a^2 = 0.$$

29.11 Torus ring

Let $a, b \in H^1(T^2; \mathbb{Z})$ be dual to the two fundamental one-cycles. Then

$$H^*(T^2; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}(a, b).$$

The top class is

$$a \smile b \in H^2(T^2; \mathbb{Z}) \cong \mathbb{Z},$$

with

$$b \smile a = -a \smile b.$$

29.12 Higher tori

For

$$T^n = (S^1)^n,$$

$$H^*(T^n; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}(a_1, \dots, a_n), \quad |a_i| = 1.$$

This multiplicative structure refines the binomial-rank computation from Künneth.

29.13 Real projective space mod 2

Let

$$x \in H^1(\mathbb{R}P^n; \mathbb{F}_2)$$

be the degree-one generator.

A fundamental result is

$$H^*(\mathbb{R}P^n; \mathbb{F}_2) \cong \mathbb{F}_2[x]/(x^{n+1}), \quad |x| = 1.$$

Thus

$$x^k \neq 0 \quad (0 \leq k \leq n).$$

29.14 Why the ring is stronger

As graded vector spaces, many spaces can have the same Betti numbers. The cup product records which classes multiply nontrivially and can distinguish such spaces.

29.15 Example: wedge versus product

Both

$$S^1 \vee S^1 \vee S^2$$

and

$$T^2$$

have additive homology ranks

$$b_0 = 1, \quad b_1 = 2, \quad b_2 = 1.$$

But on the wedge, products of positive-degree classes from different wedge summands vanish, whereas on the torus

$$a \smile b$$

generates H^2 .

Hence the spaces are not homotopy equivalent.

29.16 Cross product and diagonal

The cup product may be described conceptually as

$$a \smile b = \Delta^*(a \times b),$$

where

$$\Delta : X \rightarrow X \times X, \quad \Delta(x) = (x, x),$$

is the diagonal and $\times$ is the cohomology cross product.

29.17 Kronecker pairing compatibility

Cup products interact with homology cross products through evaluation: cohomology products detect product cycles and, on manifolds, later encode intersection pairings.

29.18 Relative cup products

There are relative forms

$$H^p(X, A; R) \times H^q(X, B; R) \rightarrow H^{p+q}(X, A \cup B; R)$$

under suitable hypotheses.

These become important for Thom classes and duality.

29.19 Cup length

A basic ring invariant is cup length: the largest number of positive-degree classes whose product is nonzero.

Cup length provides lower bounds for several topological complexity invariants.

29.20 Maps and ring obstructions

If $f : X \rightarrow Y$ is a homotopy equivalence, then

$$f^* : H^*(Y; R) \rightarrow H^*(X; R)$$

is a graded-ring isomorphism.

Therefore a mismatch of cohomology rings obstructs homotopy equivalence.

29.21 SVG visual laboratory

This chapter ships with editable SVG assets for:

- the diagonal/cross-product construction of the cup product;
- the contrast between the torus ring and a wedge with the same Betti numbers.

29.22 Bridge to bundles

Cohomology rings are the natural receptacle for characteristic classes of bundles. VIII/30 first develops vector bundles and clutching; Thom and Euler classes appear in VIII/31–VIII/32.

29.23 Exercises

Exercise 29.6. Exercise 1

Hint. Use the cup-product laws and standard cohomology rings. ☐

Solution. The cup product raises degree additively. ☐

Exercise 29.7. Exercise 2

Hint. Use the cup-product laws and standard cohomology rings. ☐

Solution. The Leibniz rule is $\delta(\alpha \smile \beta) = \delta\alpha \smile \beta + (-1)^p \alpha \smile \delta\beta$. ☐

Exercise 29.8. Exercise 3

Hint. Use the cup-product laws and standard cohomology rings. ☐

Solution. Cocycles multiply to cocycles. ☐

Exercise 29.9. Exercise 4

Hint. Use the cup-product laws and standard cohomology rings. ☐

Solution. The product descends to cohomology. ☐

Exercise 29.10. Exercise 5

Hint. Use the cup-product laws and standard cohomology rings. ☐

Solution. The class $1 \in H^0$ is the multiplicative unit. ☐

Exercise 29.11. Exercise 6

Hint. Use the cup-product laws and standard cohomology rings. ☐

Solution. Cup product is associative on cohomology. ☐

Exercise 29.12. Exercise 7

Hint. Use the cup-product laws and standard cohomology rings. ☐

Solution. Pullback preserves cup products. ☐

Exercise 29.13. Exercise 8

Hint. Use the cup-product laws and standard cohomology rings. ☐

Solution. Cup product is graded commutative. ☐

Exercise 29.14. Exercise 9

Hint. Use the cup-product laws and standard cohomology rings. ☐

Solution. Odd-degree squares vanish when 2 is invertible. ☐

Exercise 29.15. Exercise 10

Hint. Use the cup-product laws and standard cohomology rings. ☐

Solution. In characteristic two, the sign disappears. ☐

Exercise 29.16. Exercise 11

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. $H^*(S^n; \mathbb{Z}) \cong \mathbb{Z}[u]/(u^2)$ for $n > 0$. □

Exercise 29.17. Exercise 12

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. $H^*(S^1; \mathbb{Z})$ is an exterior algebra on one degree-one generator. □

Exercise 29.18. Exercise 13

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. $H^*(T^2; \mathbb{Z})$ is exterior on two degree-one generators. □

Exercise 29.19. Exercise 14

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. The torus top class is $a \smile b$. □

Exercise 29.20. Exercise 15

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. $b \smile a = -a \smile b$ integrally. □

Exercise 29.21. Exercise 16

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. $H^*(T^n; \mathbb{Z})$ is exterior on n generators. □

Exercise 29.22. Exercise 17

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. $H^*(\mathbb{R}P^n; \mathbb{F}_2) \cong \mathbb{F}_2[x]/(x^{n+1})$. □

Exercise 29.23. Exercise 18

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. The projective generator x has degree one. □

Exercise 29.24. Exercise 19

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. On $\mathbb{R}P^n$, $x^n \neq 0$. □

Exercise 29.25. Exercise 20

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. The cup product can be written $\Delta^*(a \times b)$. □

Exercise 29.26. Exercise 21

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. Cohomology rings can distinguish spaces with equal Betti numbers. □

Exercise 29.27. Exercise 22

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. Relative cup products land in the union pair. □

Exercise 29.28. Exercise 23

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. Homotopy equivalences induce cohomology-ring isomorphisms. □

Exercise 29.29. Exercise 24

Hint. Use the cup-product laws and standard cohomology rings. □

Solution. Characteristic classes will live in these cohomology rings. □

29.24 Solved problem dossiers

Problem 29.30 (Leibniz). Why does the cup product descend to cohomology?

Solution. The Leibniz rule shows cocycles multiply to cocycles; if one factor changes by a coboundary, the product changes by a coboundary. □

Problem 29.31 (Naturality). Prove $f^*(a \smile b) = f^*a \smile f^*b$.

Solution. The Alexander–Whitney face formula commutes with precomposition by f , so the equality already holds on cochains. □

Problem 29.32 (Odd square). Let $a \in H^1(X; \mathbb{Q})$. Show $a^2 = 0$.

Solution. Graded commutativity gives $a^2 = -a^2$, so $2a^2 = 0$; over $\mathbb{Q}$, 2 is invertible. □

Problem 29.33 (Circle). Compute the integral cohomology ring of S^1 .

Solution. It is $\Lambda(a)$ with $|a| = 1$; dimensional reasons and graded commutativity give $a^2 = 0$. □

Problem 29.34 (Torus). Describe the ring $H^*(T^2; \mathbb{Z})$.

Solution. It is the exterior algebra on $a, b \in H^1$, with top generator $a \smile b$ and $b \smile a = -a \smile b$. □

Problem 29.35 (Three-torus). How many degree-two ring generators arise as products in T^3 ?

Solution. The products a_1a_2, a_1a_3, a_2a_3 give three independent degree-two classes. □

Problem 29.36 (Top torus class). What generates $H^n(T^n; \mathbb{Z})$?

Solution. The product $a_1 \smile \cdots \smile a_n$. □

Problem 29.37 ($\mathbb{R}P^2 \bmod 2$). Describe $H^*(\mathbb{R}P^2; \mathbb{F}_2)$.

Solution. $\mathbb{F}_2[x]/(x^3)$ with $|x| = 1$; thus x^2 is the nonzero top class. $\square$

Problem 29.38 (RP3 mod2). Describe the nonzero powers of the degree-one generator.

Solution. $1, x, x^2, x^3$ are nonzero and $x^4 = 0$. $\square$

Problem 29.39 (Wedge versus torus). Explain why $S^1 \vee S^1 \vee S^2$ is not homotopy equivalent to T^2 despite equal Betti numbers.

Solution. The wedge has trivial products between positive-degree classes from different summands, while the torus has $a \smile b \neq 0$. $\square$

Problem 29.40 (Diagonal). Explain $a \smile b = \Delta^*(a \times b)$.

Solution. Take the cohomology cross product on $X \times X$ and pull it back along the diagonal; the Alexander–Whitney formula realizes this construction. $\square$

Problem 29.41 (Ring map obstruction). If $f : X \rightarrow Y$ is a homotopy equivalence, what must happen to cup products?

Solution. f^* is an isomorphism preserving multiplication, so the graded cohomology rings must be isomorphic. $\square$

Problem 29.42 (Cup length). What does a nonzero product $a_1 \cdots a_r$ prove?

Solution. The cup length is at least r , providing a multiplicative lower bound on topological complexity. $\square$

Problem 29.43 (Characteristic two). Why need an odd-degree mod-2 class not square to zero?

Solution. Graded commutativity contributes the sign -1 , but $-1 = 1$ over $\mathbb{F}_2$, so no sign-forced cancellation occurs. $\square$

Problem 29.44 (Sphere). Why is the square of the top sphere class zero?

Solution. It would lie in degree $2n$, where $H^{2n}(S^n) = 0$ for $n > 0$. $\square$

Problem 29.45 (Relative product). State a useful relative cup product.

Solution. $H^p(X, A) \times H^q(X, B) \rightarrow H^{p+q}(X, A \cup B)$. $\square$

Problem 29.46 (Pairing). How can a cup product be tested against homology?

Solution. Evaluate $a \smile b$ on an appropriate $(p+q)$ -cycle using the Kronecker pairing. $\square$

Problem 29.47 (Product vs additive). What extra information does the ring contain beyond cohomology groups?

Solution. It records which additive classes multiply to nonzero higher-degree classes and the relations among these products. $\square$

Problem 29.48 (SVG torus). What should the torus-vs-wedge SVG show?

Solution. Identical Betti-number rows but different multiplication: a nonzero $a \smile b$ cell for the torus and zero mixed products for the wedge. $\square$

Problem 29.49 (Bridge). Why are cup products relevant to bundles?

Solution. Characteristic classes lie in cohomology and multiply; their ring relations encode bundle and manifold structure. $\square$

Problem 29.50 (Override provenance). Why is cup-product material assigned here rather than to Poincaré duality?

Solution. The migration ledger contains a raw-source-verified targeted override for cup-product/cohomology-ring sections into VIII/29. $\square$

Problem 29.51 (Final principle). State the main improvement from cohomology groups to the cohomology ring.

Solution. The ring retains multiplicative interactions between classes, making it strictly more informative than the graded groups alone. $\square$

Chapter 30

Vector Bundles and Clutching

A vector bundle is a continuously varying family of vector spaces over a base space. Local triviality makes bundles look like products in small neighborhoods, while transition functions record the global twisting. Clutching constructs bundles over spheres by gluing trivial bundles across the equator.

30.1 Vector bundles

Definition 30.1 (Real vector bundle). A rank- k real vector bundle over X is a map

$$\pi : E \rightarrow X$$

such that each fiber

$$E_x = \pi^{-1}(x)$$

is a k -dimensional real vector space and every $x \in X$ has a neighborhood U with a homeomorphism

$$\Phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$$

that is linear on fibers.

Complex vector bundles are defined similarly with $\mathbb{C}^k$.

30.2 Local trivializations

A local trivialization identifies nearby fibers with one fixed model vector space. The bundle can be globally nontrivial even though every small piece is a product.

30.3 Transition functions

If Φ_i, Φ_j are trivializations over U_i, U_j , then on the overlap

$$U_i \cap U_j$$

their change of coordinates has the form

$$(x, v) \mapsto (x, g_{ij}(x)v)$$

with

$$g_{ij} : U_i \cap U_j \rightarrow GL_k(\mathbb{R}).$$

They satisfy

$$g_{ii} = I, \quad g_{ij}g_{jk} = g_{ik}.$$

30.4 Trivial bundles

The product

$$X \times \mathbb{R}^k \rightarrow X$$

is the trivial rank- k real bundle.

A rank- k bundle is trivial if it is isomorphic to this product bundle.

30.5 Sections

Definition 30.2 (Section). A section is a map

$$s : X \rightarrow E$$

with

$$\pi \circ s = \text{id}_X.$$

The zero vector in every fiber defines the zero section.

A nowhere-zero section is one satisfying

$$s(x) \neq 0$$

for every x .

30.6 Bundle maps

A bundle map over $f : X \rightarrow Y$ is a map of total spaces carrying fibers linearly into fibers and covering f .

An isomorphism of bundles over the same base is fiberwise linear and a homeomorphism on total spaces.

30.7 Pullback bundles

Given

$$f : Y \rightarrow X$$

and a vector bundle $E \rightarrow X$, define

$$f^*E = \{(y, e) \in Y \times E : f(y) = \pi(e)\}.$$

Theorem 30.3 (Pullback). *The projection*

$$f^*E \rightarrow Y$$

is a vector bundle of the same rank.

30.8 Whitney sum

For bundles $E, F \rightarrow X$, define

$$(E \oplus F)_x = E_x \oplus F_x.$$

This is the Whitney sum bundle

$$E \oplus F \rightarrow X.$$

30.9 Subbundles and quotient bundles

A vector subbundle $F \subset E$ has fiberwise vector subspaces of constant rank with local triviality. Under suitable hypotheses one can form the quotient bundle

$$E/F.$$

30.10 Line bundles

Rank-one bundles are line bundles.

Real line bundles have transition functions in

$$GL_1(\mathbb{R}) = \mathbb{R}^\times.$$

After choosing a metric, the structure group reduces to

$$O(1) = \{\pm 1\}.$$

30.11 The Möbius line bundle

The Möbius strip is the total space of a nontrivial real line bundle over S^1 . It can be constructed from

$$[0, 1] \times \mathbb{R}$$

by the identification

$$(0, v) \sim (1, -v).$$

The sign change is the twisting transition function.

30.12 No nowhere-zero section criterion for line bundles

A real line bundle is trivial if and only if it admits a nowhere-zero section.

Hence the Möbius line bundle has no nowhere-zero global section.

30.13 Tangent bundles

For a smooth n -manifold M ,

$$TM = \coprod_{x \in M} T_x M$$

is a rank- n vector bundle.

The tangent bundle of S^1 is trivial.

The tangent bundle of S^2 is nontrivial; the hairy-ball theorem is a manifestation of the absence of a nowhere-zero section.

30.14 Bundle metrics

Every vector bundle over a paracompact base admits a fiberwise inner product. This allows the structure group of a real rank- k bundle to be reduced from $GL_k(\mathbb{R})$ to $O(k)$.

For complex bundles one similarly reduces to $U(k)$.

30.15 Clutching over a sphere

Write

$$S^n = D_+^n \cup D_-^n, \quad D_+^n \cap D_-^n \simeq S^{n-1}.$$

A bundle is trivial over each hemisphere because each hemisphere is contractible. The global bundle is therefore determined by how the two trivial bundles are glued along the equator.

Definition 30.4 (Clutching map). A clutching map for a rank- k real bundle over S^n is a map

$$g : S^{n-1} \rightarrow GL_k(\mathbb{R})$$

used to identify

$$(x, v)_+ \sim (x, g(x)v)_-$$

on the equator.

30.16 Homotopic clutching maps

Theorem 30.5 (Clutching homotopy invariance). *Homotopic clutching maps determine isomorphic vector bundles over S^n .*

After reducing structure group, one can work with

$$g : S^{n-1} \rightarrow O(k)$$

for real bundles or

$$g : S^{n-1} \rightarrow U(k)$$

for complex bundles.

30.17 Classification by clutching

Under the standard rank/base hypotheses, isomorphism classes of rank- k bundles over S^n are classified by homotopy classes

$$[S^{n-1}, GL_k(\mathbb{R})]$$

or equivalently, after metric reduction,

$$[S^{n-1}, O(k)].$$

For complex bundles use $U(k)$.

30.18 Real line bundles over the circle

A rank-one real bundle over S^1 is clutched along

$$S^0 = \{-1, +1\}.$$

Up to isomorphism there are two possibilities:

- the trivial cylinder bundle;
- the Möbius bundle.

30.19 Complex line bundles over S^2

A complex line bundle over S^2 is clutched by

$$g : S^1 \rightarrow U(1) \cong S^1.$$

Its homotopy class is classified by an integer winding number.

This integer will later be identified with the first Chern class after the appropriate characteristic-class machinery is introduced.

30.20 Tangent bundle clutching

The tangent bundle of a sphere can be described by a clutching map on the equator. The nontriviality of this clutching map encodes the global twisting of the tangent bundle.

30.21 Stable viewpoint

Adding a trivial bundle can simplify classification:

$$E \sim_{\text{st}} F \iff E \oplus \varepsilon^r \cong F \oplus \varepsilon^s$$

for suitable trivial summands.

This is the entry point to K -theory, beyond the scope of this volume.

30.22 SVG visual laboratory

This chapter ships with editable SVG assets for:

- local trivializations and transition functions, including the Möbius twist;
- hemisphere clutching over S^n , with the equatorial map $S^{n-1} \rightarrow O(k)$.

30.23 Bridge to Thom classes

A vector bundle has a total space, a zero section, and a disk/sphere bundle. Cohomology of these pairs produces the Thom class. VIII/31 develops that construction and the Thom isomorphism.

30.24 Exercises

Exercise 30.6. Exercise 1

Hint. Use local triviality, transition functions, and clutching. □

Solution. A rank- k bundle has k -dimensional vector-space fibers. □

Exercise 30.7. Exercise 2

Hint. Use local triviality, transition functions, and clutching. □

Solution. Local trivializations identify $\pi^{-1}(U)$ with $U \times \mathbb{R}^k$. □

Exercise 30.8. Exercise 3

Hint. Use local triviality, transition functions, and clutching. ☐

Solution. Transition functions take values in $GL_k(\mathbb{R})$. ☐

Exercise 30.9. Exercise 4

Hint. Use local triviality, transition functions, and clutching. ☐

Solution. They satisfy the cocycle law $g_{ij}g_{jk} = g_{ik}$. ☐

Exercise 30.10. Exercise 5

Hint. Use local triviality, transition functions, and clutching. ☐

Solution. The product bundle is trivial. ☐

Exercise 30.11. Exercise 6

Hint. Use local triviality, transition functions, and clutching. ☐

Solution. A section satisfies $\pi s = \text{id}$. ☐

Exercise 30.12. Exercise 7

Hint. Use local triviality, transition functions, and clutching. ☐

Solution. The zero section exists for every vector bundle. ☐

Exercise 30.13. Exercise 8

Hint. Use local triviality, transition functions, and clutching. ☐

Solution. Pullback preserves bundle rank. ☐

Exercise 30.14. Exercise 9

Hint. Use local triviality, transition functions, and clutching. ☐

Solution. Whitney sum adds fiber dimensions. ☐

Exercise 30.15. Exercise 10

Hint. Use local triviality, transition functions, and clutching. ☐

Solution. A real line bundle has rank one. ☐

Exercise 30.16. Exercise 11

Hint. Use local triviality, transition functions, and clutching. ☐

Solution. With a metric, a real line bundle has structure group $O(1) = \{\pm 1\}$. ☐

Exercise 30.17. Exercise 12

Hint. Use local triviality, transition functions, and clutching. ☐

Solution. The Möbius bundle uses the gluing $(0, v) \sim (1, -v)$. ☐

Exercise 30.18. Exercise 13

Hint. Use local triviality, transition functions, and clutching. ☐

Solution. A real line bundle is trivial iff it has a nowhere-zero section. □

Exercise 30.19. Exercise 14

Hint. Use local triviality, transition functions, and clutching. □

Solution. TS^1 is trivial. □

Exercise 30.20. Exercise 15

Hint. Use local triviality, transition functions, and clutching. □

Solution. TS^2 is nontrivial. □

Exercise 30.21. Exercise 16

Hint. Use local triviality, transition functions, and clutching. □

Solution. A bundle metric reduces $GL_k(\mathbb{R})$ to $O(k)$. □

Exercise 30.22. Exercise 17

Hint. Use local triviality, transition functions, and clutching. □

Solution. Each hemisphere of S^n trivializes a bundle. □

Exercise 30.23. Exercise 18

Hint. Use local triviality, transition functions, and clutching. □

Solution. A clutching map lives on the equator S^{n-1} . □

Exercise 30.24. Exercise 19

Hint. Use local triviality, transition functions, and clutching. □

Solution. Real rank- k clutching maps may be taken into $O(k)$. □

Exercise 30.25. Exercise 20

Hint. Use local triviality, transition functions, and clutching. □

Solution. Complex rank- k clutching maps may be taken into $U(k)$. □

Exercise 30.26. Exercise 21

Hint. Use local triviality, transition functions, and clutching. □

Solution. Homotopic clutching maps yield isomorphic bundles. □

Exercise 30.27. Exercise 22

Hint. Use local triviality, transition functions, and clutching. □

Solution. Real line bundles over S^1 give the cylinder and Möbius cases. □

Exercise 30.28. Exercise 23

Hint. Use local triviality, transition functions, and clutching. □

Solution. Complex line bundles over S^2 are classified by winding number. □

Exercise 30.29. Exercise 24

Hint. Use local triviality, transition functions, and clutching. □

Solution. VIII/31 uses the disk/sphere bundle pair to define Thom classes. □

30.25 Solved problem dossiers

Problem 30.30 (Transition cocycle). Derive $g_{ij}g_{jk} = g_{ik}$.

Solution. Changing coordinates from chart k to j and then j to i must equal the direct change from k to i . $\square$

Problem 30.31 (Pullback). Describe the fiber of $f^*E \rightarrow Y$ over y .

Solution. It is naturally the fiber $E_{f(y)}$, so the pullback has the same rank. $\square$

Problem 30.32 (Whitney rank). If E has rank r and F rank s , what is $\text{rank}(E \oplus F)$?

Solution. $r + s$. $\square$

Problem 30.33 (Möbius gluing). Why is $(0, v) \sim (1, -v)$ a nontrivial line-bundle gluing?

Solution. Going once around the base reverses the fiber coordinate, so no consistent nonzero orientation/vector can return to itself globally. $\square$

Problem 30.34 (Möbius section). Why has the Möbius line bundle no nowhere-zero section?

Solution. A nowhere-zero section would trivialize a real line bundle, contradicting the non-trivial sign monodromy. $\square$

Problem 30.35 (Cylinder). What clutching produces the trivial real line bundle over S^1 ?

Solution. The constant $+1$ gluing gives the cylinder. $\square$

Problem 30.36 (Two line bundles). Why are there exactly two real line bundles over S^1 up to isomorphism?

Solution. After metric reduction, the clutching data take values in $O(1) = \{\pm 1\}$, yielding trivial and sign-twisted classes. $\square$

Problem 30.37 (Hemisphere triviality). Why is a bundle over each hemisphere of S^n trivial?

Solution. Each hemisphere is contractible; vector bundles over a contractible paracompact base are trivial. $\square$

Problem 30.38 (Clutching homotopy). Why do homotopic clutching maps give isomorphic bundles?

Solution. A homotopy of transition data extends across a collar of the equator and gives equivalent gluing data. $\square$

Problem 30.39 (Complex line S^2). Classify complex line bundles over S^2 by clutching.

Solution. Clutching maps are $S^1 \rightarrow U(1) \cong S^1$, classified by degree/winding number $\mathbb{Z}$. $\square$

Problem 30.40 (Winding zero). What complex line bundle corresponds to clutching degree zero?

Solution. The clutching map is null-homotopic, so the bundle is trivial. $\square$

Problem 30.41 (Bundle metric). Why does a metric reduce the structure group?

Solution. Choose orthonormal frames in each trivialization; changes between orthonormal frames lie in $O(k)$ or $U(k)$. $\square$

Problem 30.42 (Tangent circle). Construct a nowhere-zero section of TS^1 .

Solution. The unit tangent vector field $s(e^{it}) = ie^{it}$ is continuous and nowhere zero, trivializing the line bundle. $\square$

Problem 30.43 (Hairy sphere). What bundle fact is expressed by the hairy-ball theorem on S^2 ?

Solution. The tangent bundle TS^2 has no nowhere-zero section and is therefore nontrivial. $\square$

Problem 30.44 (Bundle isomorphism). What must a bundle isomorphism preserve?

Solution. It covers the identity on the base and is a linear isomorphism on every fiber, with a homeomorphic total-space map. $\square$

Problem 30.45 (Stable equivalence). What does adding a trivial summand do conceptually?

Solution. It can erase unstable distinctions while retaining stable bundle information, leading toward K -theory. $\square$

Problem 30.46 (Clutching tangent sphere). Why can the tangent bundle of S^n be encoded on S^{n-1} ?

Solution. Trivialize over northern and southern hemispheres; their mismatch along the equator is the clutching map. $\square$

Problem 30.47 (SVG transition). What should the transition-function SVG show?

Solution. Two overlapping base charts, product-like fibers above each, and a matrix $g_{ij}(x)$ converting fiber coordinates on the overlap. $\square$

Problem 30.48 (SVG clutching). What should the clutching SVG show?

Solution. Two trivial hemisphere bundles glued along the equator by $g : S^{n-1} \rightarrow O(k)$, making the global twist visually explicit. $\square$

Problem 30.49 (Bridge). Why does the zero section matter for Thom theory?

Solution. The Thom class lives in relative cohomology of the bundle total space relative to the complement of the zero section, or equivalently the disk/sphere bundle pair. $\square$

Problem 30.50 (Corpus ownership). Why does this chapter own both vector-bundle and clutching legacy streams?

Solution. The migration map sends both explicit sections and all inherited theorem/problem descendants in topology-14 to VIII/30. $\square$

Problem 30.51 (Final principle). What does clutching convert a global bundle problem into?

Solution. A homotopy-class problem for a map from the equator into the structure group. $\square$

Chapter 31

Thom Classes

Thom classes turn the local orientation data of a vector bundle into a global cohomology class. They are the bridge from vector bundles to Euler classes, Gysin maps, Poincaré duality, and intersection theory.

31.1 Disk and sphere bundles

Let $\pi : E \rightarrow B$ be a rank- r real vector bundle equipped with a bundle metric. Define

$$D(E) = \{v \in E : \|v\| \leq 1\}, \quad S(E) = \{v \in E : \|v\| = 1\}.$$

The pair $(D(E), S(E))$ is the natural relative space in which the Thom class lives.

31.2 Fiber orientation

An orientation of E is a continuous choice of orientation on every fiber. For each $b \in B$,

$$(D(E_b), S(E_b)) \cong (D^r, S^{r-1}), \quad H^r(D^r, S^{r-1}; R) \cong R.$$

Definition 31.1 (Thom class). For an oriented rank- r vector bundle $E \rightarrow B$, a Thom class is

$$u_E \in H^r(D(E), S(E); R)$$

whose restriction to every oriented fiber pair is the chosen generator.

31.3 The Thom isomorphism

Theorem 31.2 (Thom isomorphism). *For an oriented rank- r vector bundle over a suitable paracompact base,*

$$\Phi_E : H^k(B; R) \longrightarrow H^{k+r}(D(E), S(E); R), \quad \alpha \longmapsto \pi^* \alpha \smile u_E$$

is an isomorphism.

31.4 Why relative cohomology appears

The sphere bundle is the fiberwise boundary of the disk bundle. Relative cohomology records the transverse orientation class while forgetting boundary data.

31.5 The zero section

The zero section $s : B \rightarrow E$ is a homotopy inverse to the projection of the disk bundle, but the relative pair remembers the fiber orientation.

31.6 Naturality

For $f : Y \rightarrow B$, the pulled-back bundle satisfies

$$\tilde{f}^*(u_E) = u_{f^*E}.$$

Theorem 31.3 (Naturality). *The square formed by pullback on base cohomology and pullback on relative cohomology commutes with the two Thom isomorphisms.*

31.7 Compactly supported formulation

Under the standard local-compactness hypotheses,

$$H^{k+r}(D(E), S(E); R) \cong H_c^{k+r}(E; R),$$

hence

$$H^k(B; R) \cong H_c^{k+r}(E; R).$$

31.8 Trivial bundles and suspension

For $E = B \times \mathbb{R}^r$,

$$\mathrm{Th}(E) = D(E)/S(E) \simeq B_+ \wedge S^r,$$

so the Thom shift becomes an r -fold suspension phenomenon.

31.9 Orientation reversal

Reversing the orientation negates the integral Thom class. Over $\mathbb{Z}/2$, the sign disappears.

31.10 Normal bundles and tubular neighborhoods

If $i : N \hookrightarrow M$ is a closed embedded submanifold with oriented normal bundle ν , a tubular neighborhood identifies a neighborhood of N with $D(\nu)$. The normal Thom class is therefore a cohomological representative concentrated transversely to N .

31.11 The Thom space and collapse

Definition 31.4 (Thom space).

$$\mathrm{Th}(E) = D(E)/S(E).$$

Collapsing the complement of a tubular neighborhood gives the Thom collapse

$$M \longrightarrow \mathrm{Th}(\nu).$$

31.12 Gysin viewpoint

Cup product with a Thom class, followed by excision and extension, produces wrong-way maps in cohomology. This is the local mechanism behind geometric duality.

31.13 From Thom to Euler

Pulling the Thom class back by the zero section gives

$$e(E) = s^*(u_E) \in H^r(B; R).$$

This is the central bridge to VIII/32.

31.14 From normal Thom classes to dual cycles

For an oriented embedded submanifold, the normal Thom class extends to ambient cohomology and becomes the class Poincaré dual to that submanifold. This is the bridge to VIII/33 and VIII/34.

31.15 Solved dossiers

Problem 31.5. Compute the relative cohomology of a fiber pair (D^r, S^{r-1}) .

Solution. It is $\mathbb{Z}$ in degree r and zero otherwise, with the generator fixed by orientation. $\square$

Problem 31.6. Why is the Thom class naturally relative?

Solution. The sphere bundle is the fiberwise boundary; relative cohomology retains the orientation class of the disk fiber while killing boundary data. $\square$

Problem 31.7. Find the Thom class for a trivial oriented line bundle.

Solution. It is the pullback of the generator of $H^1(D^1, S^0)$ along the fiber factor. $\square$

Problem 31.8. State the rank-five Thom shift.

Solution. $H^k(B) \cong H^{k+5}(D(E), S(E))$. $\square$

Problem 31.9. What happens under orientation reversal?

Solution. The integral Thom class changes sign. $\square$

Problem 31.10. Why do $\mathbb{Z}/2$ coefficients remove the orientability requirement?

Solution. The two signs agree modulo two. $\square$

Problem 31.11. Describe the Thom space of a trivial rank- r bundle.

Solution. It is naturally $B_+ \wedge S^r$. $\square$

Problem 31.12. What role does the zero section play?

Solution. It identifies the base in the total space and pulls the Thom class back to the Euler class. $\square$

Problem 31.13. What does a tubular neighborhood provide?

Solution. It identifies a neighborhood of an embedded submanifold with the disk bundle of its normal bundle. $\square$

Problem 31.14. Define the Thom collapse.

Solution. Collapse the complement of a tubular neighborhood to the basepoint of the Thom space. $\square$

Problem 31.15. State naturality under $f : Y \rightarrow B$.

Solution. For the pulled-back orientation, $\tilde{f}^*u_E = u_{f^*E}$. $\square$

Problem 31.16. How is compact support related to the Thom pair?

Solution. Under standard hypotheses, $H^*(D(E), S(E)) \cong H_c^*(E)$. $\square$

Problem 31.17. Where does the bundle rank appear?

Solution. As the cohomological degree shift. $\square$

Problem 31.18. How does the Thom class encode normal orientation?

Solution. Its restriction to each normal disk fiber is the chosen orientation generator. $\square$

Problem 31.19. What is the prototype of a cohomological wrong-way map?

Solution. Cup with the Thom class, followed by excision and extension from a tubular neighborhood. $\square$

Problem 31.20. Relate the Thom space to reduced cohomology.

Solution. $\tilde{H}^{k+r}(\text{Th}(E)) \cong H^k(B)$. $\square$

Problem 31.21. Why is a bundle metric useful?

Solution. It defines disk and sphere subbundles without changing the underlying homotopy-theoretic data. $\square$

Problem 31.22. What characteristic class comes next?

Solution. $e(E) = s^*u_E$. $\square$

Problem 31.23. How does a submanifold acquire a dual cohomology class?

Solution. Transfer the Thom class of its oriented normal bundle into the ambient manifold. $\square$

Problem 31.24. Why is the Thom theorem a bridge theorem?

Solution. It connects bundle orientation, relative cohomology, embeddings, Gysin maps, Euler classes, and duality. $\square$

31.16 Exercises, Hints, and Solutions

Exercise 31.25. Compute $H^*(D^r, S^{r-1}; \mathbb{Z})$.

Hint. Use $D^r/S^{r-1} \cong S^r$. □

Solution. Since $D^r/S^{r-1} \cong S^r$, relative cohomology is reduced cohomology of S^r :

$$H^q(D^r, S^{r-1}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & q = r, \\ 0, & q \neq r. \end{cases}$$

The generator is the orientation class of the disk fiber. □

Exercise 31.26. Write the Thom isomorphism for a rank-three bundle over S^2 .

Hint. Shift degree by the rank. □

Solution. For a rank-three oriented bundle $E \rightarrow S^2$,

$$\Phi_E : H^k(S^2) \xrightarrow{\cong} H^{k+3}(D(E), S(E)), \quad \alpha \mapsto \pi^* \alpha \smile u_E.$$

Thus $1 \in H^0(S^2)$ maps to $u_E \in H^3$, while the degree-two generator maps to a degree-five relative class. □

Exercise 31.27. Check the fiber normalization for a trivial oriented line bundle.

Hint. Restrict to each fiber pair. □

Solution. For $E = B \times \mathbb{R}$, the fiber pair over b is (D^1, S^0) . Choose u_E so that

$$u_E|_{(D^1, S^0)} = 1 \in H^1(D^1, S^0; \mathbb{Z}).$$

This fiberwise generator condition is exactly the Thom normalization. □

Exercise 31.28. Show that reversing orientation changes the integral Thom class by a sign.

Hint. Track the chosen generator. □

Solution. Reversing the orientation reverses the chosen generator of every fiber group $H^r(D^r, S^{r-1}; \mathbb{Z})$. By uniqueness of the normalized class, the new Thom class is $-u_E$. □

Exercise 31.29. Explain why mod-2 Thom theory does not require orientability.

Hint. Reduce signs modulo two. □

Solution. The only ambiguity in an integral fiber generator is its sign. In $\mathbb{Z}/2$, $1 = -1$, so local generators glue without an orientation choice. Hence every real vector bundle has a mod-two Thom class. □

Exercise 31.30. Identify the Thom space of a trivial rank- r bundle.

Hint. Use $B_+ \wedge S^r$. □

Solution. For $E = B \times \mathbb{R}^r$,

$$\text{Th}(E) = D(E)/S(E) \cong B_+ \wedge (D^r/S^{r-1}) \cong B_+ \wedge S^r.$$

□

Exercise 31.31. Compute its reduced cohomology when the base is S^1 .

Hint. Use suspension. □

Solution. With $B = S^1$,

$$\mathrm{Th}(E) \simeq S_+^1 \wedge S^r \simeq S^r \vee S^{r+1}.$$

Therefore its reduced integral cohomology is $\mathbb{Z}$ in degrees r and $r+1$, and zero otherwise. □

Exercise 31.32. Describe the pullback Thom class under a map $f : Y \rightarrow B$.

Hint. Compute the pulled-back bundle first. □

Solution. If $f : Y \rightarrow B$, the bundle map $\tilde{f} : D(f^*E) \rightarrow D(E)$ covering f pulls the Thom class back:

$$u_{f^*E} = \tilde{f}^* u_E.$$

Its restriction to every pulled-back fiber is the same oriented generator. □

Exercise 31.33. Verify naturality of the Thom isomorphism.

Hint. Apply naturality of cup products. □

Solution. For $\alpha \in H^k(B)$,

$$\tilde{f}^*(\pi^*\alpha \smile u_E) = (\pi')^*(f^*\alpha) \smile u_{f^*E}.$$

Thus pullback commutes with the two Thom isomorphisms. □

Exercise 31.34. Describe the normal bundle of an equatorial $S^{n-1} \subset S^n$.

Hint. The normal direction is one-dimensional. □

Solution. The equator $S^{n-1} \subset S^n$ has a one-dimensional normal bundle. A continuous unit normal can be chosen pointing toward one hemisphere, so this normal line bundle is trivial:

$$\nu \cong S^{n-1} \times \mathbb{R}.$$

□

Exercise 31.35. Construct a tubular neighborhood in that example.

Hint. Use a product collar or exponential map. □

Solution. Using the normal coordinate, a tubular neighborhood is

$$S^{n-1} \times (-\varepsilon, \varepsilon) \longrightarrow S^n,$$

with $S^{n-1} \times \{0\}$ identified with the equator. Radial/geodesic normal displacement gives the embedding. □

Exercise 31.36. Describe the corresponding Thom collapse map.

Hint. Collapse the complement. □

Solution. Collapse the complement of the chosen tubular neighborhood to one point, and identify the neighborhood with $D(\nu)$. This gives

$$S^n \longrightarrow D(\nu)/S(\nu) = \mathrm{Th}(\nu).$$

□

Exercise 31.37. Relate relative cohomology of $(D(E), S(E))$ to the Thom quotient.

Hint. Use reduced cohomology of a quotient. □

Solution. For any good pair,

$$H^q(D(E), S(E)) \cong \tilde{H}^q(D(E)/S(E)).$$

Since $D(E)/S(E) = \text{Th}(E)$, relative Thom cohomology is exactly reduced cohomology of the Thom quotient. □

Exercise 31.38. Explain the compact-support formulation for $E = \mathbb{R}^r$.

Hint. Compactify the fiber direction. □

Solution. For the bundle $\mathbb{R}^r \rightarrow \{*\}$, one-point compactification gives S^r . Hence

$$H_c^q(\mathbb{R}^r; \mathbb{Z}) \cong \tilde{H}^q(S^r; \mathbb{Z}),$$

which is $\mathbb{Z}$ in degree r and zero otherwise. □

Exercise 31.39. Track the degree shift in a rank-five example.

Hint. Write the rank explicitly. □

Solution. For a rank-five oriented bundle,

$$H^k(B) \xrightarrow{\cong} H^{k+5}(D(E), S(E)).$$

For example a class in $H^2(B)$ corresponds to a relative class in degree 7. □

Exercise 31.40. Compare Thom classes obtained from two bundle metrics.

Hint. Metrics are homotopic through positive-definite metrics. □

Solution. Two bundle metrics are joined by a continuous path of positive-definite metrics. Fiberwise radial rescaling identifies their disk/sphere pairs through an isotopy, and the normalized fiber generator is preserved. Thus the induced Thom classes correspond. □

Exercise 31.41. Show that an orientation-preserving bundle isomorphism preserves Thom classes.

Hint. Use the fiber orientation generator. □

Solution. An orientation-preserving bundle isomorphism sends each oriented fiber generator to the oriented fiber generator. Pullback of the target Thom class therefore satisfies the defining normalization, so by uniqueness it equals the source Thom class. □

Exercise 31.42. Analyze an orientation-reversing bundle isomorphism.

Hint. Orientation reversal changes its sign. □

Solution. An orientation-reversing bundle isomorphism sends the fiber orientation generator to its negative. Consequently the pulled-back integral Thom class is $-u_E$; modulo two there is no sign change. □

Exercise 31.43. Describe the first step in the Gysin construction for an embedding.

Hint. Start inside a tubular neighborhood. □

Solution. For an oriented embedding $i : N \hookrightarrow M$, first choose a tubular neighborhood $D(\nu) \subset M$ of the normal bundle. Cup with u_ν on that neighborhood, then use excision and extension to ambient cohomology. This is the geometric core of the Gysin map. $\square$

Exercise 31.44. Explain how the zero section meets every fiber disk.

Hint. The zero section selects the zero vector. $\square$

Solution. The zero section is $s(b) = 0_b$. Since 0_b lies in the interior of every fiber disk $D(E_b)$, the image $s(B)$ meets every disk fiber at its center and never lies in the sphere boundary. $\square$

Exercise 31.45. Compute s^*u_E for a trivial positive-rank bundle.

Hint. A trivial positive-rank bundle admits a nonzero constant section. $\square$

Solution. A trivial positive-rank bundle has a nowhere-zero constant section, so its Euler class vanishes. Since $e(E) = s^*u_E$,

$$s^*u_E = 0.$$

$\square$

Exercise 31.46. Preview why a nowhere-zero section constrains s^*u_E .

Hint. Compare zero and nonzero sections. $\square$

Solution. A nowhere-zero section can be normalized to a section of $S(E)$. The Euler class is the obstruction represented by the zero section's pullback of the Thom class; a section avoiding zero makes that obstruction vanish. Hence a nowhere-zero section forces $s^*u_E = e(E) = 0$. $\square$

Exercise 31.47. Relate a normal Thom class to the cohomology class of a submanifold.

Hint. Extend the normal Thom class into ambient cohomology. $\square$

Solution. For an oriented embedded submanifold $N \subset M$, take the Thom class of its oriented normal bundle, identify the disk bundle with a tubular neighborhood, and extend by excision into M . The resulting ambient cohomology class is $\text{PD}[N]$. $\square$

Exercise 31.48. Summarize the chain Thom class $\rightarrow$ Euler class $\rightarrow$ Poincaré duality.

Hint. Track the degree change at every arrow. $\square$

Solution. The chain is

$$u_E \in H^r(D(E), S(E)) \xrightarrow{s^*} e(E) \in H^r(B) \xrightarrow{\frown[B]} H_{\dim B - r}(B).$$

Thus the Thom class produces the Euler class, and cap product with the fundamental class converts it into its geometric dual cycle. $\square$

Chapter 32

Sphere Bundles and Euler Classes

The unit sphere bundle converts a nowhere-zero vector field into a genuine section problem. The Euler class measures the primary obstruction to solving that problem globally.

32.1 Unit sphere bundles

For a rank- r real vector bundle $E \rightarrow B$ with metric,

$$S(E) = \{v \in E : \|v\| = 1\} \longrightarrow B$$

has fiber S^{r-1} .

32.2 Nowhere-zero sections

A nowhere-zero section of E normalizes to a section of $S(E)$, and every section of $S(E)$ is a nowhere-zero section of E .

32.3 Euler class

Definition 32.1 (Euler class). For an oriented rank- r bundle with Thom class u_E and zero section $s : B \rightarrow E$,

$$e(E) = s^*u_E \in H^r(B; R).$$

32.4 The obstruction theorem

Theorem 32.2 (Nowhere-zero section obstruction). *If an oriented vector bundle admits a nowhere-zero section, then*

$$e(E) = 0.$$

The converse is an obstruction-theoretic statement that requires appropriate hypotheses; the Euler class is the primary obstruction, not in every setting a complete classifier of sections.

32.5 Naturality

For $f : Y \rightarrow B$,

$$e(f^*E) = f^*e(E).$$

32.6 Whitney sums

For oriented bundles,

$$e(E \oplus F) = e(E) \smile e(F).$$

In particular, adding a positive-rank trivial summand forces the Euler class to vanish.

32.7 Sphere bundles and the Gysin sequence

For an oriented rank- r bundle, the sphere bundle $S(E) \rightarrow B$ has a long exact Gysin sequence. The characteristic map in the base is cup product with $e(E)$:

$$\cdots \rightarrow H^{k-r}(B) \xrightarrow{\smile e(E)} H^k(B) \rightarrow H^k(S(E)) \rightarrow H^{k-r+1}(B) \rightarrow \cdots.$$

32.8 Tangent bundles

For a closed oriented n -manifold,

$$e(TM) \in H^n(M).$$

Its evaluation on the fundamental class is the Euler number.

Theorem 32.3 (Euler class and Euler characteristic). *For a closed oriented manifold M ,*

$$\langle e(TM), [M] \rangle = \chi(M).$$

32.9 The hairy-ball theorem

Since $\chi(S^{2m}) = 2$, the tangent Euler class of an even-dimensional sphere is nonzero. Thus S^{2m} admits no nowhere-zero tangent vector field.

32.10 Odd-dimensional manifolds

A closed oriented odd-dimensional manifold has Euler characteristic zero. In particular, the tangent Euler number vanishes.

32.11 Zeros of transverse sections

For an oriented rank- r bundle over an oriented r -manifold, a generic section has isolated transverse zeros. Each zero has a local sign, and the signed sum equals the Euler number.

32.12 Poincaré–Hopf viewpoint

For $E = TM$, the preceding statement becomes the Poincaré–Hopf theorem: the sum of indices of isolated zeros of a vector field equals $\chi(M)$.

32.13 Zero loci as dual cycles

More generally, a transverse section of an oriented rank- r bundle has a codimension- r zero set whose fundamental class is Poincaré dual to $e(E)$. This is the direct bridge to VIII/33.

32.14 Solved dossiers

Problem 32.4. Show that a nowhere-zero section of E is equivalent to a section of $S(E)$ after choosing a metric.

Solution. Normalize the vector field fiberwise; conversely include the unit sphere bundle in E . □

Problem 32.5. Define the Euler class from the Thom class.

Solution. For an oriented rank- r bundle, $e(E) = s^*u_E \in H^r(B)$. □

Problem 32.6. Why does a nowhere-zero section force $e(E) = 0$?

Solution. Move the zero section through a homotopy to a disjoint section; the pulled-back Thom class vanishes. □

Problem 32.7. Compute the Euler class of a trivial positive-rank oriented bundle.

Solution. It is zero because a constant nowhere-zero section exists. □

Problem 32.8. What is the Euler class of an oriented line bundle with a nowhere-zero section?

Solution. Zero. □

Problem 32.9. State multiplicativity for Whitney sums.

Solution. $e(E \oplus F) = e(E) \smile e(F)$ for oriented bundles. □

Problem 32.10. Relate $e(TM)$ to the Euler characteristic.

Solution. For a closed oriented manifold, $\langle e(TM), [M] \rangle = \chi(M)$. □

Problem 32.11. Evaluate the Euler number of TS^2 .

Solution. It equals $\chi(S^2) = 2$. □

Problem 32.12. What happens for a closed oriented odd-dimensional manifold?

Solution. Its Euler characteristic, hence tangent Euler number, is zero. □

Problem 32.13. Why does S^{2m} have no nowhere-zero tangent vector field?

Solution. Its Euler class pairs to $2 \neq 0$. □

Problem 32.14. Why is an Euler class an obstruction rather than always a complete classifier?

Solution. Vanishing is necessary for a nonzero section, but further obstructions can exist outside the standard obstruction-theoretic range. □

Problem 32.15. Where does the Euler class appear in the Gysin sequence?

Solution. It is the class whose cup-product map links adjacent base-cohomology terms. □

Problem 32.16. Describe the sphere bundle of a rank- r bundle.

Solution. It has fiber S^{r-1} and records nonzero directions in each vector fiber. □

Problem 32.17. What is the geometric meaning of a zero of a transverse section?

Solution. It contributes a local index; the signed sum represents the Euler number. □

Problem 32.18. State the Poincaré–Hopf connection.

Solution. For an oriented closed manifold, the sum of vector-field indices equals $\chi(M)$. □

Problem 32.19. What coefficient system avoids orientability issues?

Solution. Mod-2 coefficients. □

Problem 32.20. How does pullback affect Euler classes?

Solution. $e(f^*E) = f^*e(E)$. □

Problem 32.21. What is the Euler class of $E \oplus \varepsilon^1$?

Solution. Zero, since the trivial summand gives a nowhere-zero section; equivalently multiplicativity includes $e(\varepsilon^1) = 0$. □

Problem 32.22. How does VIII/32 lead to duality?

Solution. The zero locus of a transverse section represents the Poincaré dual of the Euler class. □

Problem 32.23. Why are sphere bundles visually natural here?

Solution. A global section is literally a continuous choice of one point on every sphere fiber. □

32.15 Exercises, Hints, and Solutions

Exercise 32.24. Construct the unit sphere bundle of a trivial rank-two bundle over S^1 .

Hint. Normalize fiberwise. □

Solution. For $E = S^1 \times \mathbb{R}^2$, the unit vectors in each fiber form S^1 , so

$$S(E) \cong S^1 \times S^1 \rightarrow S^1.$$

It is the trivial circle bundle. □

Exercise 32.25. Show that a nonzero section normalizes to a sphere-bundle section.

Hint. Use $e = s^*u$. □

Solution. If $s(b) \neq 0$ for all b , define $\hat{s}(b) = s(b)/\|s(b)\|$. Continuity of the norm gives a continuous section $\hat{s}: B \rightarrow S(E)$. Conversely, including $S(E) \subset E$ turns every sphere-bundle section into a nowhere-zero section. □

Exercise 32.26. Prove naturality $e(f^*E) = f^*e(E)$.

Hint. Apply Thom naturality. □

Solution. Thom naturality gives $u_{f^*E} = \tilde{f}^*u_E$. Pulling back along the zero sections yields

$$e(f^*E) = s_Y^*u_{f^*E} = s_Y^*\tilde{f}^*u_E = f^*s_B^*u_E = f^*e(E).$$

□

Exercise 32.27. Compute the Euler class of a trivial rank-three bundle.

Hint. Use a constant vector. □

Solution. A trivial rank-three bundle admits a constant nowhere-zero section. Therefore its Euler class is zero:

$$e(B \times \mathbb{R}^3) = 0 \in H^3(B).$$

□

Exercise 32.28. Use multiplicativity to compute $e(E \oplus \varepsilon^1)$.

Hint. Use multiplicativity. □

Solution. By multiplicativity,

$$e(E \oplus \varepsilon^1) = e(E) \smile e(\varepsilon^1).$$

The trivial line has a nowhere-zero section, so $e(\varepsilon^1) = 0$, and the product vanishes. □

Exercise 32.29. Show that $e(TS^2)$ pairs to 2.

Hint. Pair with the fundamental class. □

Solution. For the oriented two-sphere,

$$\langle e(TS^2), [S^2] \rangle = \chi(S^2) = 2.$$

Thus the Euler class is twice the preferred generator of $H^2(S^2; \mathbb{Z})$. □

Exercise 32.30. Show that TS^2 has no nowhere-zero section.

Hint. Use nonvanishing of the pairing. □

Solution. If TS^2 had a nowhere-zero section, the Euler class would vanish. But its evaluation on $[S^2]$ is 2, so $e(TS^2) \neq 0$. Hence no nowhere-zero tangent vector field exists. □

Exercise 32.31. Compute the Euler number of TS^{2m} .

Hint. Use $\chi(S^{2m}) = 2$. □

Solution. For every $m \geq 1$,

$$\langle e(TS^{2m}), [S^{2m}] \rangle = \chi(S^{2m}) = 2.$$

So the tangent Euler number of every even-dimensional sphere is 2. □

Exercise 32.32. Explain why the Euler number of TS^{2m+1} vanishes.

Hint. Use $\chi(S^{2m+1}) = 0$. □

Solution. For S^{2m+1} ,

$$\langle e(TS^{2m+1}), [S^{2m+1}] \rangle = \chi(S^{2m+1}) = 0.$$

Indeed odd spheres admit explicit nowhere-zero tangent vector fields. □

Exercise 32.33. Relate the hairy-ball theorem to the Euler class.

Hint. Translate vector fields into sections. □

Solution. A tangent vector field is a section of TM , and after normalization a nowhere-zero vector field is a section of $S(TM)$. Since $e(TS^{2m}) \neq 0$, the Euler obstruction rules out such a section. This is the hairy-ball theorem. $\square$

Exercise 32.34. Write the relevant fragment of the Gysin sequence for an oriented circle bundle.

Hint. Take fiber S^1 . $\square$

Solution. For an oriented rank-two bundle $E \rightarrow B$, the sphere fiber is S^1 , and the Gysin sequence contains

$$\cdots \rightarrow H^{k-2}(B) \xrightarrow{\smile e(E)} H^k(B) \rightarrow H^k(S(E)) \rightarrow H^{k-1}(B) \xrightarrow{\smile e(E)} H^{k+1}(B) \rightarrow \cdots$$

$\square$

Exercise 32.35. Identify the cup-with-Euler-class map in that sequence.

Hint. Look for cup product by $e(E)$. $\square$

Solution. In the Gysin sequence the characteristic map from base cohomology to base cohomology is precisely

$$\alpha \mapsto \alpha \smile e(E).$$

Its degree shift equals the rank of the oriented vector bundle. $\square$

Exercise 32.36. Describe zeros of a generic section of an oriented rank- r bundle over an r -manifold.

Hint. Use transversality. $\square$

Solution. If $\text{rank } E = r = \dim B$, transversality to the zero section makes the zero set of a generic section a submanifold of dimension $r - r = 0$. On a compact base it is therefore a finite set of isolated zeros. $\square$

Exercise 32.37. Assign signs to isolated transverse zeros.

Hint. Compare orientations of base and fiber. $\square$

Solution. At an isolated transverse zero x , trivialize E and orient both base and fiber. The derivative of the local section is an isomorphism $T_x B \rightarrow E_x$. Assign sign $+1$ if it preserves orientation and -1 if it reverses orientation. $\square$

Exercise 32.38. Relate their signed sum to the Euler number.

Hint. Sum local degrees. $\square$

Solution. The local signs are local degrees. Their sum equals the Euler number:

$$\sum_{x:s(x)=0} \text{ind}_x(s) = \langle e(E), [B] \rangle.$$

For $E = TB$ this is Poincaré–Hopf. $\square$

Exercise 32.39. Explain the mod-2 version without orientation.

Hint. Ignore signs modulo two. $\square$

Solution. With $\mathbb{Z}/2$ coefficients, the signs $+1$ and -1 coincide. Thus the zero locus is counted modulo two, and the mod-two Thom/Euler obstruction is defined without orienting the bundle. $\square$

Exercise 32.40. Compare Euler classes before and after pullback to a point.

Hint. Pullback to the fiber over the chosen point. $\square$

Solution. Pulling a positive-rank bundle back to a point gives a vector space over a point. Since $H^r(\{*\}) = 0$ for $r > 0$,

$$e(i^*E) = i^*e(E) = 0.$$

This is consistent with every vector space admitting nonzero vectors. $\square$

Exercise 32.41. Analyze a product bundle over $S^2 \times S^2$.

Hint. Use Künneth and naturality. $\square$

Solution. For example, on $S^2 \times S^2$ let

$$E = p_1^*TS^2 \oplus p_2^*TS^2.$$

Then

$$e(E) = p_1^*e(TS^2) \smile p_2^*e(TS^2),$$

and its evaluation on $[S^2 \times S^2]$ is $2 \cdot 2 = 4$. $\square$

Exercise 32.42. Use Whitney-sum multiplicativity in a tangent-bundle example.

Hint. Split the tangent bundle if possible. $\square$

Solution. Since $T(M \times N) \cong p_M^*TM \oplus p_N^*TN$,

$$e(T(M \times N)) = p_M^*e(TM) \smile p_N^*e(TN).$$

Evaluating gives $\chi(M \times N) = \chi(M)\chi(N)$, illustrating Whitney-sum multiplicativity. $\square$

Exercise 32.43. Explain why vanishing Euler class need not classify a bundle.

Hint. Remember that characteristic classes are generally not complete invariants. $\square$

Solution. The Euler class is only one characteristic class. In higher rank or on complicated bases, two nonisomorphic bundles can share the same Euler class but differ in other characteristic data such as Stiefel–Whitney or Pontryagin classes. Therefore $e(E) = 0$ alone is not a complete classification. $\square$

Exercise 32.44. Describe the first obstruction to a section of $S(E)$.

Hint. Use obstruction theory on the sphere fiber. $\square$

Solution. For an oriented rank- r bundle, the fiber of $S(E)$ is S^{r-1} . The primary obstruction to extending a section lies in

$$H^r(B; \pi_{r-1}(S^{r-1})) \cong H^r(B; \mathbb{Z})$$

(with the appropriate orientation local system), and this obstruction is the Euler class. $\square$

Exercise 32.45. Relate a transverse zero set to a Thom class.

Hint. Pull back the Thom class by the section. □

Solution. A transverse section s meets the zero section cleanly. Pulling the normal Thom class through this intersection identifies the cohomology class detecting the zero set with $s^*u_E = e(E)$. Thus the Thom class localizes the zero locus. □

Exercise 32.46. Explain how the zero set becomes a dual homology class.

Hint. Apply Poincaré duality in the base. □

Solution. If $Z = s^{-1}(0)$ is transverse, then Z has codimension r and its normal bundle is naturally $E|_Z$. The Thom construction therefore yields

$$\text{PD}[Z] = e(E).$$

□

Exercise 32.47. Summarize sphere bundle $\rightarrow$ section obstruction $\rightarrow$ Euler class $\rightarrow$ duality.

Hint. Track geometry and cohomology simultaneously. □

Solution. The geometry and algebra line up as

$$\text{section of } S(E) \iff \text{nowhere-zero section of } E,$$

while failure of such a section is detected first by $e(E)$. For a transverse section, its zero set represents $\text{PD}(e(E))$, leading directly to Poincaré duality. □

Chapter 33

Poincaré Duality

Poincaré duality is the organizing theorem that turns cohomology classes into geometric homology classes on an oriented manifold. The Thom-class viewpoint from VIII/31 supplies the local normal model; the Euler-class viewpoint from VIII/32 supplies a first family of dual cycles.

33.1 Orientation and local homology

For an n -manifold M and $x \in M$,

$$H_n(M, M \setminus \{x\}; \mathbb{Z}) \cong \mathbb{Z}.$$

An orientation chooses compatible generators of these local groups.

33.2 The fundamental class

Definition 33.1 (Fundamental class). For a connected closed oriented n -manifold, the fundamental class

$$[M] \in H_n(M; \mathbb{Z})$$

is the unique class restricting to the chosen local orientation generator at every point.

33.3 Cap product

Cap product combines cohomology and homology:

$$\frown: H^k(M; R) \times H_n(M; R) \longrightarrow H_{n-k}(M; R).$$

33.4 Poincaré duality

Theorem 33.2 (Poincaré duality). *If M is a closed oriented n -manifold, then*

$$D_M: H^k(M; R) \longrightarrow H_{n-k}(M; R), \quad D_M(\alpha) = \alpha \frown [M]$$

is an isomorphism.

33.5 Betti-number symmetry

Over a field,

$$b_k(M) = b_{n-k}(M).$$

This gives immediate global restrictions, including vanishing Euler characteristic in odd dimension.

33.6 The complementary-degree pairing

Duality can be written as the nondegenerate pairing

$$H^k(M; R) \times H^{n-k}(M; R) \longrightarrow R, \quad (\alpha, \beta) \longmapsto \langle \alpha \smile \beta, [M] \rangle.$$

33.7 Spheres and surfaces

For S^n , degree zero and degree n are dual. For a closed oriented surface of genus g , the $2g$ degree-one generators pair through the symplectic intersection form.

33.8 The torus

Choose $a, b \in H^1(T^2; \mathbb{Z})$ so that

$$a \smile b = [T^2]^*, \quad b \smile a = -[T^2]^*.$$

The algebra records the signed intersection of the two standard generating circles.

33.9 Complex projective space

For $\mathbb{CP}^n$,

$$H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[h]/(h^{n+1}), \quad |h| = 2,$$

and the top class h^n evaluates to 1 after choosing the complex orientation.

33.10 Geometric Poincaré duals

If $N^{n-r} \hookrightarrow M^n$ is a closed oriented submanifold with oriented normal bundle, its normal Thom class extends to

$$\text{PD}[N] \in H^r(M).$$

33.11 Transverse intersection and cup product

If dual submanifolds meet transversely, then cup product corresponds to intersection:

$$\text{PD}[A] \smile \text{PD}[B] = \text{PD}[A \pitchfork B]$$

with the usual orientation conventions.

33.12 Intersection numbers

For complementary dimensions, the signed number of transverse intersection points is

$$A \cdot B = \langle \text{PD}[A] \smile \text{PD}[B], [M] \rangle.$$

33.13 Euler classes revisited

If a transverse section of an oriented rank- r bundle has zero set Z , then

$$\text{PD}[Z] = e(E).$$

Thus Euler classes are geometric duals of zero loci.

33.14 Manifolds with boundary

Theorem 33.3 (Poincaré–Lefschetz duality). *For a compact oriented n -manifold with boundary,*

$$H^k(M; R) \cong H_{n-k}(M, \partial M; R), \quad H^k(M, \partial M; R) \cong H_{n-k}(M; R).$$

33.15 Nonorientable manifolds

Integral duality requires the orientation local system. With $\mathbb{Z}/2$ coefficients the sign obstruction disappears and a mod-2 duality statement holds without orientability.

33.16 Bridge to intersection forms

In dimension $4m$, the middle-degree pairing

$$H^{2m}(M) \times H^{2m}(M) \rightarrow R$$

is a bilinear form obtained by cup product and evaluation on $[M]$. VIII/34 will study this intersection form as an invariant in its own right.

33.17 Solved dossiers

Problem 33.4. State Poincaré duality for a closed oriented n -manifold.

Solution. Cap product with $[M]$ gives $H^k(M; R) \cong H_{n-k}(M; R)$. □

Problem 33.5. Define the duality map.

Solution. $D_M(\alpha) = \alpha \frown [M]$. □

Problem 33.6. What is the fundamental class?

Solution. The homology class whose local image is the chosen orientation generator at every point. □

Problem 33.7. What symmetry follows for Betti numbers over a field?

Solution. $b_k = b_{n-k}$. □

Problem 33.8. Write the degree-pairing form.

Solution. $(\alpha, \beta) \mapsto \langle \alpha \smile \beta, [M] \rangle$. □

Problem 33.9. Why is the pairing nondegenerate?

Solution. Poincaré duality identifies one factor with the dual of complementary homology/cohomology. □

Problem 33.10. Compute the dual of the point class on a connected oriented surface.

Solution. It corresponds to the top-degree cohomology generator. $\square$

Problem 33.11. Describe duality on S^n .

Solution. H^0 pairs with H^n and vice versa; all intermediate groups vanish. $\square$

Problem 33.12. Describe the first-cohomology pairing on T^2 .

Solution. Two degree-one generators have nonzero cup product equal to the orientation class. $\square$

Problem 33.13. How does a submanifold define a dual class?

Solution. Use the Thom class of its oriented normal bundle and extend it to ambient cohomology. $\square$

Problem 33.14. What does cup product represent geometrically for transverse dual submanifolds?

Solution. Their intersection. $\square$

Problem 33.15. What is the intersection number of complementary oriented transverse submanifolds?

Solution. The evaluation of the cup product of their dual classes on $[M]$. $\square$

Problem 33.16. How does boundary change the theorem?

Solution. Poincaré–Lefschetz duality relates absolute cohomology to relative homology and relative cohomology to absolute homology. $\square$

Problem 33.17. State one Poincaré–Lefschetz isomorphism.

Solution. $H^k(M) \cong H_{n-k}(M, \partial M)$ for compact oriented M . $\square$

Problem 33.18. What changes for nonorientable manifolds?

Solution. Use the orientation local system, or mod-2 coefficients when appropriate. $\square$

Problem 33.19. How is the Euler characteristic constrained in odd dimension?

Solution. Betti numbers pair in complementary degrees, giving zero Euler characteristic for closed oriented odd-dimensional manifolds. $\square$

Problem 33.20. What is the dual of a codimension- r transverse zero locus?

Solution. The Euler class of the rank- r bundle whose section cuts it out. $\square$

Problem 33.21. How is the Thom class used in the proof philosophy?

Solution. Local duality in normal disks is transported globally through tubular neighborhoods and excision. $\square$

Problem 33.22. Why does VIII/33 lead directly to intersection forms?

Solution. Middle-degree cup products evaluated on $[M]$ become bilinear forms. $\square$

Problem 33.23. What is the key synthesis of VIII/31–VIII/33?

Solution. Thom classes create dual cohomology classes, Euler classes arise from zero sections, and cap product with the fundamental class identifies cohomology with geometric homology. $\square$

33.18 Exercises, Hints, and Solutions

Exercise 33.24. Compute Poincaré duality for S^n .

Hint. Only degrees 0 and n survive. □

Solution. For a closed oriented n -sphere,

$$H^0(S^n) \cong H_n(S^n) \cong \mathbb{Z}, \quad H^n(S^n) \cong H_0(S^n) \cong \mathbb{Z},$$

and all other groups vanish. Cap product with $[S^n]$ gives these two isomorphisms. □

Exercise 33.25. Compute the Betti-number symmetry for a genus- g surface.

Hint. Use $b_0 = b_2 = 1$ and $b_1 = 2g$. □

Solution. For a genus- g closed oriented surface,

$$(b_0, b_1, b_2) = (1, 2g, 1).$$

Poincaré duality gives the symmetry $b_0 = b_2$, while the middle group is self-dual. □

Exercise 33.26. Write the cup-product pairing matrix for T^2 in a standard basis.

Hint. Choose generators a, b with $a \smile b = [T^2]^*$. □

Solution. Choose $a, b \in H^1(T^2; \mathbb{Z})$ with $\langle a \smile b, [T^2] \rangle = 1$. Since degree-one cup product is skew-commutative,

$$(\langle x_i \smile x_j, [T^2] \rangle) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

□

Exercise 33.27. Identify the Poincaré dual of an embedded oriented circle in T^2 .

Hint. Use a transverse complementary circle. □

Solution. Let $A = S^1 \times \{*\} \subset T^2$. Its Poincaré dual is the degree-one class that evaluates by intersection with A ; in the standard basis it is the cohomology class dual to the transverse factor circle. □

Exercise 33.28. Intersect two standard generating circles on the torus.

Hint. There is one signed intersection. □

Solution. The two standard circles $S^1 \times \{*\}$ and $\{*\} \times S^1$ meet in exactly one point. With the product orientation and compatible orientations of the circles, the local intersection sign is $+1$. □

Exercise 33.29. Relate their signed intersection to cup product.

Hint. Evaluate the cup product on $[T^2]$. □

Solution. If α, β are the dual cohomology classes of the two standard circles, then

$$A \cdot B = \langle \alpha \smile \beta, [T^2] \rangle = 1.$$

Thus the signed geometric intersection is exactly the cup-product pairing. □

Exercise 33.30. Compute the duality groups of $\mathbb{CP}^2$.

Hint. Use one generator in degrees 0, 2, 4. □

Solution. For $\mathbb{CP}^2$,

$$H_k(\mathbb{CP}^2; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & k = 0, 2, 4, \\ 0, & \text{otherwise,} \end{cases}$$

and Poincaré duality pairs degrees $0 \leftrightarrow 4$ and $2 \leftrightarrow 2$. □

Exercise 33.31. Evaluate the square of the degree-two generator on $[\mathbb{CP}^2]$.

Hint. The square evaluates to 1 in the complex orientation. □

Solution. If $h \in H^2(\mathbb{CP}^2; \mathbb{Z})$ is the positive generator, then h^2 is the orientation class and

$$\langle h^2, [\mathbb{CP}^2] \rangle = 1.$$

□

Exercise 33.32. Show that closed oriented odd-dimensional manifolds have Euler characteristic zero.

Hint. Pair complementary Betti numbers. □

Solution. For a closed oriented $n = 2m + 1$ manifold, $b_k = b_{n-k}$. The paired terms in

$$\chi(M) = \sum_k (-1)^k b_k$$

have opposite signs because n is odd, so they cancel and $\chi(M) = 0$. □

Exercise 33.33. Describe the local orientation classes defining $[M]$.

Hint. Use $H_n(M, M \setminus \{x\})$. □

Solution. An orientation is a compatible family of generators of

$$H_n(M, M \setminus \{x\}; \mathbb{Z}) \cong \mathbb{Z}.$$

Compactness permits these compatible local generators to assemble into the unique global class $[M] \in H_n(M; \mathbb{Z})$ restricting to each chosen generator. □

Exercise 33.34. Explain the cap-product degree shift.

Hint. Cap lowers degree by k . □

Solution. If $\alpha \in H^k(M)$ and $c \in H_n(M)$, then

$$\alpha \frown c \in H_{n-k}(M).$$

Thus cap product lowers homological degree by exactly the cohomological degree. □

Exercise 33.35. Verify naturality of evaluation under an orientation-preserving homeomorphism.

Hint. Track the fundamental class. □

Solution. If $f : M \rightarrow N$ is orientation preserving, then $f_*[M] = [N]$. Naturality of cup/cap evaluation gives

$$\langle f^*\alpha, [M] \rangle = \langle \alpha, f_*[M] \rangle = \langle \alpha, [N] \rangle.$$

□

Exercise 33.36. Use a tubular neighborhood to build the dual class of a submanifold.

Hint. Use the normal Thom class. □

Solution. For $i : N^{n-r} \hookrightarrow M^n$, choose an oriented tubular neighborhood $D(\nu)$. The Thom class $u_\nu \in H^r(D(\nu), S(\nu))$ extends by excision to a class in $H^r(M, M \setminus N)$ and then to $H^r(M)$. This is $\text{PD}[N]$. □

Exercise 33.37. Relate that construction to the Thom collapse.

Hint. Collapse the complement of the tubular neighborhood. □

Solution. The Thom collapse

$$M \rightarrow \text{Th}(\nu)$$

pulls the Thom class back to the ambient dual class of N . Thus the collapse-map construction and the tubular-neighborhood construction give the same Poincaré dual. □

Exercise 33.38. Show that a transverse zero set represents the dual of an Euler class.

Hint. Use the section's zero locus. □

Solution. For a transverse section of an oriented rank- r bundle, the zero set Z has normal bundle $E|_Z$. Its normal Thom class is the restriction of the bundle Thom class, so

$$\text{PD}[Z] = e(E).$$

□

Exercise 33.39. State Poincaré–Lefschetz duality for a compact manifold with boundary.

Hint. Replace absolute homology by relative homology. □

Solution. For a compact oriented n -manifold with boundary,

$$H^k(M; R) \cong H_{n-k}(M, \partial M; R), \quad H^k(M, \partial M; R) \cong H_{n-k}(M; R).$$

The fundamental class is relative: $[M, \partial M] \in H_n(M, \partial M)$. □

Exercise 33.40. Compute it for an interval.

Hint. Check endpoints as boundary. □

Solution. For $I = [0, 1]$,

$$H^0(I) \cong \mathbb{Z} \cong H_1(I, \partial I),$$

while $H^1(I) = 0 \cong H_0(I, \partial I)$. The relative fundamental class is the oriented interval. □

Exercise 33.41. Compute it for a disk.

Hint. Use (D^n, S^{n-1}) . □

Solution. For D^n ,

$$H^0(D^n) \cong \mathbb{Z} \cong H_n(D^n, S^{n-1}),$$

and all other corresponding groups vanish. This is the basic local model for Poincaré–Lefschetz duality. $\square$

Exercise 33.42. Explain why ordinary integral duality fails without orientation.

Hint. The local signs cannot be chosen coherently. $\square$

Solution. On a nonorientable manifold the local orientation generators cannot be chosen coherently, so there is no global integral fundamental class in ordinary coefficients. Consequently the ordinary integral cap-product isomorphism fails. $\square$

Exercise 33.43. Repair the statement using mod-2 coefficients.

Hint. Signs disappear modulo two. $\square$

Solution. With $\mathbb{Z}/2$ coefficients, the sign ambiguity disappears. Every manifold has a mod-two fundamental class, and

$$H^k(M; \mathbb{Z}/2) \cong H_{n-k}(M; \mathbb{Z}/2)$$

for closed manifolds. $\square$

Exercise 33.44. Describe the complementary-degree pairing in dimension four.

Hint. Use degree two against degree two. $\square$

Solution. For a closed oriented four-manifold, middle-degree duality is

$$H^2(M; R) \times H^2(M; R) \rightarrow R, \quad (\alpha, \beta) \mapsto \langle \alpha \smile \beta, [M] \rangle.$$

Because degree two is even, this pairing is symmetric. $\square$

Exercise 33.45. Identify the middle-dimensional bilinear form that will appear in VIII/34.

Hint. Evaluate cup products on $[M]$. $\square$

Solution. The bilinear form in the previous exercise is the intersection form

$$Q_M(\alpha, \beta) = \langle \alpha \smile \beta, [M] \rangle$$

on middle cohomology (or, integrally, on its free lattice). This is the invariant studied in VIII/34. $\square$

Exercise 33.46. Explain how transverse intersections produce its coefficients.

Hint. Represent classes by transverse surfaces. $\square$

Solution. Represent $\text{PD}^{-1}(\alpha)$ and $\text{PD}^{-1}(\beta)$ by transverse oriented surfaces. Each intersection point contributes a sign, and the signed total equals

$$\langle \alpha \smile \beta, [M] \rangle.$$

$\square$

Exercise 33.47. Summarize fundamental class $\rightarrow$ cap product $\rightarrow$ duality $\rightarrow$ intersection form.

Hint. Track the middle degree. □

Solution. The fundamental class turns cohomology into complementary-dimensional homology by cap product. Geometric representatives of those dual homology classes intersect according to cup product. In the middle dimension this complementary-degree pairing becomes the intersection form. □

Chapter 34

Intersection Forms

Poincaré duality becomes especially rigid in the middle dimension. On a closed oriented manifold, cup product followed by evaluation on the fundamental class produces a bilinear pairing whose algebra records geometric intersection. In dimension four this pairing is one of the basic topological invariants.

34.1 The middle-dimensional pairing

Let M be a closed oriented $2n$ -manifold. Define

$$Q_M(\alpha, \beta) = \langle \alpha \smile \beta, [M] \rangle, \quad \alpha, \beta \in H^n(M; \mathbb{Z})/\text{tors}.$$

Definition 34.1 (Intersection form). The integral middle-dimensional intersection form is the bilinear form

$$Q_M : (H^n(M; \mathbb{Z})/\text{tors})^2 \longrightarrow \mathbb{Z}$$

given by cup product and evaluation on the fundamental class.

34.2 Symmetry and graded symmetry

Graded commutativity gives

$$Q_M(\beta, \alpha) = (-1)^n Q_M(\alpha, \beta).$$

Thus the middle pairing is symmetric when n is even and skew-symmetric when n is odd.

34.3 Unimodularity

Theorem 34.2 (Unimodularity). *For a closed oriented manifold, the integral intersection form on the free middle cohomology is unimodular.*

This is the lattice form of integral Poincaré duality.

34.4 Geometric intersections

If α and β are Poincaré dual to complementary transverse cycles, then

$$Q_M(\alpha, \beta)$$

is their signed intersection number.

34.5 Self-intersection

A middle-dimensional submanifold can be pushed slightly in a normal direction. Its signed intersection with the push-off is its self-intersection number. For an oriented normal bundle, this number is the Euler number of the normal bundle.

34.6 Four-manifolds

For a closed oriented 4-manifold,

$$Q_M : H^2(M; \mathbb{Z})_{\text{free}} \times H^2(M; \mathbb{Z})_{\text{free}} \rightarrow \mathbb{Z}$$

is symmetric.

34.7 Positive and negative directions

After tensoring with $\mathbb{R}$, a symmetric form decomposes into positive and negative directions. Write

$$b_2^+(M), \quad b_2^-(M)$$

for their dimensions.

Definition 34.3 (Signature).

$$\sigma(M) = b_2^+(M) - b_2^-(M).$$

34.8 Definite and indefinite forms

The form is positive definite if every nonzero vector has positive square, negative definite if every nonzero vector has negative square, and indefinite if both signs occur.

34.9 Even and odd forms

An integral symmetric form is even when

$$Q(x, x) \in 2\mathbb{Z}$$

for every integral vector x ; otherwise it is odd.

34.10 The four-sphere

Since

$$H^2(S^4; \mathbb{Z}) = 0,$$

the intersection form of S^4 has rank zero.

34.11 Complex projective plane

Let $h \in H^2(\mathbb{CP}^2; \mathbb{Z})$ be the standard generator with

$$\langle h^2, [\mathbb{CP}^2] \rangle = 1.$$

Then

$$Q_{\mathbb{CP}^2} = [1].$$

It is positive definite, odd, unimodular, and has signature 1.

34.12 Orientation reversal

Changing the orientation changes the fundamental class by a sign:

$$Q_{-M} = -Q_M.$$

Thus

$$Q_{-\mathbb{CP}^2} = [-1].$$

34.13 The product $S^2 \times S^2$

Let $a, b \in H^2(S^2 \times S^2; \mathbb{Z})$ be pulled back from the two factors. Then

$$a^2 = b^2 = 0, \quad \langle a \smile b, [S^2 \times S^2] \rangle = 1.$$

Hence

$$Q_{S^2 \times S^2} = H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The hyperbolic form H is even, indefinite, unimodular, and has signature zero.

34.14 Connected sums

For oriented closed 4-manifolds,

$$Q_{M \# N} \cong Q_M \oplus Q_N.$$

Consequently

$$Q_{\mathbb{CP}^2 \# (-\mathbb{CP}^2)} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

34.15 Integral change of basis

If A represents Q_M and $P \in GL_r(\mathbb{Z})$ changes the integral basis, then

$$A \mapsto P^T A P.$$

Thus rank, unimodularity, parity, definiteness, and signature are invariants of the integral congruence class.

34.16 Intersection matrices from geometry

Choosing geometric representatives for a basis of middle homology turns the form into an intersection matrix. Diagonal entries are self-intersections; off-diagonal entries count pairwise signed intersections.

34.17 Bridge to Lefschetz theory

For a self-map $f : M \rightarrow M$, fixed points are intersections of the graph $\Gamma_f \subset M \times M$ with the diagonal $\Delta \subset M \times M$. Lefschetz theory converts that global intersection number into a trace on homology.

34.18 Solved dossiers

Problem 34.4. Define the middle-dimensional intersection form of a closed oriented $2n$ -manifold.

Solution. On the free part, $Q_M(\alpha, \beta) = \langle \alpha \smile \beta, [M] \rangle$ for $\alpha, \beta \in H^n(M; \mathbb{Z})/\text{tors}$. □

Problem 34.5. How does graded commutativity control the symmetry of Q_M ?

Solution. $Q_M(\beta, \alpha) = (-1)^n Q_M(\alpha, \beta)$; it is symmetric for even n and skew-symmetric for odd n . □

Problem 34.6. Why is the integral form unimodular on the free part?

Solution. Integral Poincaré duality identifies the free middle cohomology lattice with its integral dual, so the representing matrix has determinant ± 1 . □

Problem 34.7. What is the intersection form of S^4 ?

Solution. $H^2(S^4) = 0$, so the form has rank zero. □

Problem 34.8. Compute the form of $\mathbb{CP}^2$ with its complex orientation.

Solution. If $h \in H^2(\mathbb{CP}^2)$ and $\langle h^2, [\mathbb{CP}^2] \rangle = 1$, then the matrix is $[1]$. □

Problem 34.9. What does orientation reversal do to the form?

Solution. $[-M] = -[M]$, hence $Q_{-M} = -Q_M$. □

Problem 34.10. Compute the form of $S^2 \times S^2$.

Solution. With a, b dual to the two factors, $a^2 = b^2 = 0$ and ab evaluates to 1, so the matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. □

Problem 34.11. What is the signature of the hyperbolic form?

Solution. Its eigenvalues are 1 and -1 , so the signature is 0. □

Problem 34.12. Define positive and negative indices in dimension four.

Solution. After extending Q_M to $\mathbb{R}$, b_2^+ and b_2^- are the dimensions of maximal positive- and negative-definite subspaces. □

Problem 34.13. Define the signature.

Solution. $\sigma(M) = b_2^+ - b_2^-$. □

Problem 34.14. How does connected sum affect the form for oriented closed 4-manifolds?

Solution. On the middle cohomology free parts, $Q_{M\#N} \cong Q_M \oplus Q_N$. □

Problem 34.15. Compute $Q_{\mathbb{CP}^2 \# (-\mathbb{CP}^2)}$.

Solution. It is $\text{diag}(1, -1)$, hence indefinite of signature zero. □

Problem 34.16. What does even mean for an integral symmetric form?

Solution. $Q(x, x)$ is even for every integral lattice vector x . □

Problem 34.17. Is the $\mathbb{CP}^2$ form even or odd?

Solution. Odd, because the generator has self-intersection 1. □

Problem 34.18. Is the hyperbolic form even?

Solution. Yes: for (x, y) its square is $2xy$, always even. □

Problem 34.19. Give the geometric meaning of $Q_M(\alpha, \beta)$ in dimension four.

Solution. Represent the dual classes by oriented embedded or immersed surfaces in general position; the form is their signed intersection number. □

Problem 34.20. What is self-intersection geometrically?

Solution. It is the intersection of a surface with a transverse push-off, equivalently the Euler number of its oriented normal bundle. □

Problem 34.21. Why does torsion not enter the ordinary integral matrix directly?

Solution. The cup-evaluation pairing to $\mathbb{Z}$ vanishes on torsion against the free fundamental class, so the lattice form is defined on the free quotient. □

Problem 34.22. How does a change of integral basis affect the matrix?

Solution. A changes by integral congruence $A \mapsto P^T A P$ with $P \in GL_r(\mathbb{Z})$. □

Problem 34.23. Why is the chapter a direct continuation of Poincaré duality?

Solution. The intersection form is precisely the middle-degree Poincaré pairing promoted from a duality statement to a structured bilinear invariant. □

34.19 Exercises, Hints, and Solutions

Exercise 34.24. Compute the intersection form of S^4 .

Hint. Use $H^2(S^4) = 0$. □

Solution. Since $H^2(S^4; \mathbb{Z}) = 0$, the free middle cohomology lattice is zero. Hence Q_{S^4} is the unique rank-zero bilinear form. □

Exercise 34.25. Compute the form of $\mathbb{CP}^2$.

Hint. Use the generator h with $h^2[M] = 1$. □

Solution. With h the positive generator of $H^2(\mathbb{CP}^2; \mathbb{Z})$,

$$\langle h^2, [\mathbb{CP}^2] \rangle = 1.$$

Thus the matrix is $[1]$. □

Exercise 34.26. Compute the form of $-\mathbb{CP}^2$.

Hint. Reverse the fundamental class. □

Solution. Orientation reversal replaces the fundamental class by its negative. Therefore

$$Q_{-\mathbb{CP}^2} = -Q_{\mathbb{CP}^2} = [-1].$$

□

Exercise 34.27. Compute the form of $S^2 \times S^2$.

Hint. Use the two factor classes. □

Solution. For the factor classes a, b ,

$$a^2 = b^2 = 0, \quad \langle ab, [S^2 \times S^2] \rangle = 1.$$

Hence

$$Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = H.$$

□

Exercise 34.28. Diagonalize the hyperbolic form over $\mathbb{R}$.

Hint. Find eigenvectors $(1, 1)$ and $(1, -1)$. □

Solution. Over $\mathbb{R}$, the vectors $(1, 1)$ and $(1, -1)$ are eigenvectors of H with eigenvalues 1 and -1 , respectively. So H is congruent over $\mathbb{R}$ to $\text{diag}(1, -1)$. □

Exercise 34.29. Compute its signature.

Hint. Count positive minus negative eigenvalues. □

Solution. The hyperbolic form has one positive and one negative eigenvalue, so

$$b_2^+ = 1, \quad b_2^- = 1, \quad \sigma(H) = 0.$$

□

Exercise 34.30. Show that the hyperbolic form is even.

Hint. Evaluate $Q((x, y), (x, y))$. □

Solution. For $v = (x, y) \in \mathbb{Z}^2$,

$$Q(v, v) = 2xy,$$

which is always even. Thus H is an even integral form. □

Exercise 34.31. Show that $[1]$ is odd.

Hint. Test the generator. □

Solution. For the generator $1 \in \mathbb{Z}$, the form $[1]$ has $Q(1, 1) = 1$, which is odd. Hence it is an odd form. □

Exercise 34.32. Compute the form of $\mathbb{CP}^2 \# \mathbb{CP}^2$.

Hint. Use orthogonal direct sum under connected sum. □

Solution. Connected sum gives orthogonal direct sum:

$$Q_{\mathbb{CP}^2 \# \mathbb{CP}^2} \cong [1] \oplus [1] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

It is positive definite with signature 2. □

Exercise 34.33. Compute the form of $\mathbb{CP}^2 \# (-\mathbb{CP}^2)$.

Hint. Combine $[1]$ and $[-1]$. □

Solution. Similarly,

$$Q_{\mathbb{CP}^2 \# (-\mathbb{CP}^2)} \cong [1] \oplus [-1] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

It is indefinite and has signature 0. □

Exercise 34.34. Determine b_2^+ and b_2^- in the preceding two examples.

Hint. Read signs from the diagonal. □

Solution. For $\mathbb{CP}^2 \# \mathbb{CP}^2$, $(b_2^+, b_2^-) = (2, 0)$. For $\mathbb{CP}^2 \# (-\mathbb{CP}^2)$, $(b_2^+, b_2^-) = (1, 1)$. □

Exercise 34.35. Verify that orientation reversal negates the matrix.

Hint. Replace $[M]$ by $-[M]$. □

Solution. Since $Q_M(\alpha, \beta) = \langle \alpha \smile \beta, [M] \rangle$ and $[-M] = -[M]$,

$$Q_{-M}(\alpha, \beta) = -Q_M(\alpha, \beta).$$

Thus every representing matrix is negated. □

Exercise 34.36. Show that integral change of basis acts by congruence.

Hint. Write the new basis matrix P . □

Solution. If the old coordinate vector is $x = Py$, then

$$Q(x, x') = y^T (P^T A P) y'.$$

Hence an integral basis change $P \in GL_r(\mathbb{Z})$ sends A to $P^T A P$. □

Exercise 34.37. Show that unimodularity is basis independent.

Hint. Use $\det(P) = \pm 1$. □

Solution. If A is unimodular, $\det A = \pm 1$. Under $A \mapsto P^T A P$,

$$\det(P^T A P) = \det(P)^2 \det(A) = \det(A)$$

because $\det P = \pm 1$. So unimodularity is basis independent. □

Exercise 34.38. Derive symmetry in dimension four from graded commutativity.

Hint. Degree two has even parity. □

Solution. In dimension four the middle degree is 2. Graded commutativity gives

$$\alpha \smile \beta = (-1)^{2 \cdot 2} \beta \smile \alpha = \beta \smile \alpha,$$

so Q_M is symmetric. □

Exercise 34.39. Derive skew-symmetry in middle degree on a closed oriented surface.

Hint. Degree one has odd parity. □

Solution. For a closed oriented surface the middle degree is 1, so

$$\alpha \smile \beta = -\beta \smile \alpha.$$

Therefore the degree-one intersection pairing is skew-symmetric. $\square$

Exercise 34.40. Represent the standard generators of $H_2(S^2 \times S^2)$ geometrically.

Hint. Use the two factor spheres. $\square$

Solution. Take

$$A = S^2 \times \{q\}, \quad B = \{p\} \times S^2.$$

Their homology classes generate $H_2(S^2 \times S^2) \cong \mathbb{Z}^2$. $\square$

Exercise 34.41. Compute their mutual intersections.

Hint. Perturb to transverse representatives. $\square$

Solution. The factor spheres A and B meet transversely at the single point (p, q) . With product orientations the sign is $+1$, so $A \cdot B = 1$. $\square$

Exercise 34.42. Compute their self-intersections using push-offs.

Hint. Push a factor sphere in its normal direction. $\square$

Solution. A small push-off of $A = S^2 \times \{q\}$ is $S^2 \times \{q'\}$, disjoint from A ; hence $A^2 = 0$. The same argument gives $B^2 = 0$. $\square$

Exercise 34.43. Relate self-intersection to the normal Euler number.

Hint. Use the normal-bundle Euler class. $\square$

Solution. For an oriented middle-dimensional submanifold A , a transverse push-off is determined by a section of its normal bundle. Its zeros count the self-intersection, so

$$A \cdot A = \langle e(\nu_A), [A] \rangle.$$

$\square$

Exercise 34.44. Explain why torsion is removed from the lattice form.

Hint. Pair torsion with an integer-valued functional. $\square$

Solution. If x is torsion, say $mx = 0$, then for any y ,

$$mQ(x, y) = Q(mx, y) = 0$$

in $\mathbb{Z}$, hence $Q(x, y) = 0$. Therefore the integer-valued lattice form factors through the free quotient. $\square$

Exercise 34.45. Compare rank, parity, definiteness and signature as invariants.

Hint. Keep the invariants logically distinct. $\square$

Solution. Rank records lattice dimension; parity records whether all squares are even; definiteness records the sign pattern over $\mathbb{R}$; signature is $b^+ - b^-$. These are preserved under integral congruence but capture different information. $\square$

Exercise 34.46. Write the intersection matrix for an orthogonal sum $H \oplus [1]$.

Hint. Write a block diagonal matrix. □

Solution. For

$$H \oplus [1],$$

a natural basis gives

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It is unimodular and indefinite, with signature 1. □

Exercise 34.47. Explain how VIII/34 prepares the graph–diagonal intersection picture in Lefschetz theory.

Hint. Think of fixed points as intersections of graph and diagonal. □

Solution. For $f : M \rightarrow M$, fixed points satisfy $(x, f(x)) = (x, x)$, so they are intersections of the graph Γ_f with the diagonal Δ in $M \times M$. VIII/35 turns their signed intersection number into the Lefschetz trace. □

Chapter 35

Lefschetz Theory

Lefschetz theory is a synthesis theorem. A self-map acts linearly on homology, fixed points appear geometrically as intersections of a graph with a diagonal, and Poincaré duality proves that the two counts agree.

35.1 The Lefschetz number

Let X be a finite CW complex and $f : X \rightarrow X$ a continuous map.

Definition 35.1 (Lefschetz number). With coefficients in a field such as $\mathbb{Q}$,

$$L(f) = \sum_{k \geq 0} (-1)^k \operatorname{tr}(f_* : H_k(X; \mathbb{Q}) \rightarrow H_k(X; \mathbb{Q})).$$

35.2 Homotopy invariance

If $f \simeq g$, then $f_* = g_*$ on homology, hence

$$L(f) = L(g).$$

35.3 The fixed-point theorem

Theorem 35.2 (Lefschetz fixed-point theorem). *For a self-map of a suitable compact triangulable space, if*

$$L(f) \neq 0,$$

then f has a fixed point.

The converse is false: fixed points can occur with total index zero.

35.4 The identity map

For the identity,

$$L(\operatorname{id}_X) = \sum_k (-1)^k \dim H_k(X; \mathbb{Q}) = \chi(X).$$

Thus Euler characteristic is the Lefschetz number of the identity.

35.5 Contractible spaces and Brouwer

If X is connected and contractible, only H_0 contributes:

$$L(f) = 1.$$

Applied to D^n , this gives Brouwer's fixed-point theorem.

35.6 Sphere maps

If $f : S^n \rightarrow S^n$ has degree d , then

$$L(f) = 1 + (-1)^n d.$$

Whenever this number is nonzero, f must have a fixed point.

35.7 Constant maps

A constant self-map of a connected finite CW complex induces the identity on H_0 and zero on positive-dimensional reduced homology, so

$$L(f) = 1.$$

35.8 The graph and diagonal

For a self-map $f : M \rightarrow M$ of a closed manifold, define

$$\Gamma_f = \{(x, f(x)) : x \in M\}, \quad \Delta = \{(x, x) : x \in M\}.$$

Then

$$x \in \text{Fix}(f) \iff (x, f(x)) \in \Gamma_f \cap \Delta.$$

35.9 Intersection-theoretic meaning

Under the usual transversality and orientation hypotheses,

$$L(f) = \Gamma_f \cdot \Delta.$$

This is the geometric bridge from VIII/34 to the homological trace formula.

35.10 Local fixed-point index

If a smooth fixed point x is nondegenerate, so

$$\det(I - Df_x) \neq 0,$$

its local index is

$$\text{ind}_x(f) = \text{sign } \det(I - Df_x).$$

Theorem 35.3 (Lefschetz–Hopf). *When the fixed points are isolated and the standard hypotheses hold,*

$$L(f) = \sum_{x \in \text{Fix}(f)} \text{ind}_x(f).$$

35.11 Cancellation

The theorem only says that a nonzero Lefschetz number forces a fixed point. If positive and negative local indices cancel, fixed points can exist while

$$L(f) = 0.$$

35.12 Torus endomorphisms

A linear torus map induced by

$$A \in M_2(\mathbb{Z})$$

acts by 1 on H_0 , by A on H_1 , and by $\det A$ on H_2 . Therefore

$$L(f) = 1 - \operatorname{tr}(A) + \det(A) = \det(I - A).$$

35.13 A fixed-point criterion on the torus

If

$$\det(I - A) \neq 0,$$

then the induced torus map has nonzero Lefschetz number and therefore a fixed point.

35.14 Poincaré-duality proof philosophy

The diagonal class in $M \times M$ is expressed using dual bases in homology and cohomology. Pulling it against the graph of f converts geometric intersection into the alternating trace of f_* .

35.15 Supertrace viewpoint

The Lefschetz number is the supertrace of the graded homology endomorphism:

$$L(f) = \operatorname{str}(f_*).$$

This compactly packages the alternating signs.

35.16 Endpoint of the canonical arc

The chapter closes the Volume VIII progression

homology $\rightarrow$ cohomology $\rightarrow$ cup/cap products $\rightarrow$ Thom and Euler classes $\rightarrow$ Poincaré duality $\rightarrow$ intersection

35.17 Solved dossiers

Problem 35.4. Define the Lefschetz number of a self-map of a finite CW complex.

Solution. $L(f) = \sum_k (-1)^k \operatorname{tr}(f_* : H_k(X; \mathbb{Q}) \rightarrow H_k(X; \mathbb{Q}))$. □

Problem 35.5. Why may rational homology be used in the basic trace formula?

Solution. It removes torsion from the finite-dimensional trace calculation while preserving the classical Lefschetz number. $\square$

Problem 35.6. State the Lefschetz fixed-point theorem.

Solution. If $L(f) \neq 0$ for a self-map of a suitable compact triangulable space, then f has a fixed point. $\square$

Problem 35.7. What is $L(\text{id}_X)$?

Solution. $\chi(X)$. $\square$

Problem 35.8. How does Lefschetz imply Brouwer for a map $D^n \rightarrow D^n$?

Solution. D^n is contractible, so only H_0 contributes and $L(f) = 1 \neq 0$. $\square$

Problem 35.9. Compute $L(f)$ for $f : S^n \rightarrow S^n$ of degree d .

Solution. $L(f) = 1 + (-1)^n d$. $\square$

Problem 35.10. What condition on the degree of a sphere map guarantees a fixed point?

Solution. $1 + (-1)^n d \neq 0$. $\square$

Problem 35.11. What is the Lefschetz number of a constant self-map of a connected finite complex?

Solution. 1. $\square$

Problem 35.12. Why is the converse of the Lefschetz theorem false?

Solution. A map can have fixed points whose local indices cancel, so $L(f) = 0$ does not imply fixed-point-free. $\square$

Problem 35.13. Why is $L(f)$ homotopy invariant?

Solution. Homotopic maps induce the same homomorphisms on homology and therefore the same traces. $\square$

Problem 35.14. Describe fixed points using graph and diagonal.

Solution. x is fixed exactly when $(x, f(x))$ lies in both Γ_f and Δ . $\square$

Problem 35.15. What does the global Lefschetz number measure geometrically?

Solution. Under transversality/nondegeneracy hypotheses it is the signed intersection number $\Gamma_f \cdot \Delta$. $\square$

Problem 35.16. What is a local fixed-point index in the nondegenerate smooth case?

Solution. It is the local intersection sign; equivalently the sign of $\det(I - Df_x)$ when that determinant is nonzero. $\square$

Problem 35.17. State the Lefschetz–Hopf refinement.

Solution. For isolated fixed points satisfying the usual hypotheses, $L(f)$ is the sum of their local fixed-point indices. $\square$

Problem 35.18. Compute the Lefschetz number of a torus map induced by $A \in M_2(\mathbb{Z})$.

Solution. $L(f) = 1 - \text{tr}(A) + \det(A) = \det(I - A)$. □

Problem 35.19. What does $\det(I - A) \neq 0$ imply for the induced torus map?

Solution. Its Lefschetz number is nonzero, so it has a fixed point. □

Problem 35.20. What happens for a fixed-point-free map?

Solution. Necessarily $L(f) = 0$. □

Problem 35.21. How does the identity map connect Lefschetz theory to Euler characteristic?

Solution. Its local/global fixed-point count is encoded by $L(\text{id}) = \chi(X)$. □

Problem 35.22. How does Lefschetz theory use Poincaré duality?

Solution. On manifolds, duality converts the graph–diagonal intersection class into the alternating trace on homology. □

Problem 35.23. Why is VIII/35 a natural endpoint for Volume VIII?

Solution. It synthesizes homology, cohomology, cup/cap products, Poincaré duality and intersection theory into a computable fixed-point invariant. □

35.18 Exercises, Hints, and Solutions

Exercise 35.24. Compute $L(\text{id}_{S^n})$.

Hint. Use $L(\text{id}) = \chi$. □

Solution. For the identity, $L(\text{id}_X) = \chi(X)$. Hence

$$L(\text{id}_{S^n}) = \chi(S^n) = 1 + (-1)^n,$$

equal to 2 for even n and 0 for odd n . □

Exercise 35.25. Compute $L(f)$ for a degree- d self-map of S^n .

Hint. Only H_0 and H_n contribute. □

Solution. Only $H_0(S^n)$ and $H_n(S^n)$ are nonzero. The induced maps have traces 1 and d , so

$$L(f) = 1 + (-1)^n d.$$

□

Exercise 35.26. Determine for which d the Lefschetz theorem forces a fixed point on S^2 .

Hint. Set $1 + d \neq 0$. □

Solution. For S^2 ,

$$L(f) = 1 + d.$$

Lefschetz forces a fixed point whenever $d \neq -1$. The theorem is silent when $d = -1$. □

Exercise 35.27. Do the same on S^1 .

Hint. Set $1 - d \neq 0$. □

Solution. For S^1 ,

$$L(f) = 1 - d.$$

Thus a fixed point is forced whenever $d \neq 1$. □

Exercise 35.28. Compute the Lefschetz number of a constant map on a connected finite CW complex.

Hint. Only H_0 contributes nontrivially. □

Solution. A constant map on a connected space induces the identity on H_0 and zero on positive reduced homology. Therefore

$$L(f) = 1.$$

□

Exercise 35.29. Deduce Brouwer's fixed-point theorem for D^n .

Hint. Use contractibility. □

Solution. Every self-map $f : D^n \rightarrow D^n$ has $L(f) = 1$, since the disk is contractible and connected. The Lefschetz fixed-point theorem therefore guarantees a fixed point. This is Brouwer's theorem. □

Exercise 35.30. Show that homotopic self-maps have the same Lefschetz number.

Hint. Use homotopy invariance of induced homology maps. □

Solution. Homotopic maps induce identical maps on every homology group. Their traces agree degree by degree, hence their alternating trace sums agree:

$$f \simeq g \implies L(f) = L(g).$$

□

Exercise 35.31. Show that a fixed-point-free map must have Lefschetz number zero.

Hint. Use the contrapositive of Lefschetz. □

Solution. The Lefschetz theorem says $L(f) \neq 0 \implies \text{Fix}(f) \neq \emptyset$. Taking the contrapositive gives

$$\text{Fix}(f) = \emptyset \implies L(f) = 0.$$

□

Exercise 35.32. Give an example where $L(f) = 0$ but fixed points may still occur.

Hint. Cancellation of local indices is possible. □

Solution. The identity on S^1 has every point fixed but

$$L(\text{id}_{S^1}) = \chi(S^1) = 0.$$

Thus $L(f) = 0$ does not imply that the fixed-point set is empty. □

Exercise 35.33. Write the graph Γ_f and diagonal Δ for a self-map of a manifold.

Hint. Use $x \mapsto (x, f(x))$. □

Solution. The graph and diagonal are

$$\Gamma_f = \{(x, f(x)) : x \in M\}, \quad \Delta = \{(x, x) : x \in M\} \subset M \times M.$$

□

Exercise 35.34. Explain why their intersections are fixed points.

Hint. Solve $(x, f(x)) = (x, x)$.

□

Solution. A point lies in both submanifolds exactly when

$$(x, f(x)) = (x, x),$$

which is equivalent to $f(x) = x$. Hence $\Gamma_f \cap \Delta$ is naturally the fixed-point set.

□

Exercise 35.35. Relate a transverse intersection sign to local index.

Hint. Orient the graph and diagonal.

□

Solution. At a transverse intersection of Γ_f and Δ , the orientations of their tangent spaces determine a local intersection sign. For a nondegenerate smooth fixed point this sign is

$$\text{sign det}(I - Df_x),$$

the local fixed-point index.

□

Exercise 35.36. Compute $L(f)$ for the torus map induced by $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$.

Hint. Use $1 - \text{tr } A + \det A$.

□

Solution. For $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$,

$$L(f) = 1 - \text{tr } A + \det A = 1 - 4 + 4 = 1.$$

Equivalently $\det(I - A) = \det(-I) = 1$, so a fixed point is forced.

□

Exercise 35.37. Compute it for $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Hint. Compute trace and determinant.

□

Solution. For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$,

$$L(f) = 1 - 2 + 1 = 0.$$

Indeed $I - A$ is singular, so Lefschetz gives no forced fixed point.

□

Exercise 35.38. Verify $L(f) = \det(I - A)$ for a general 2×2 integer matrix.

Hint. Expand $\det(I - A)$.

□

Solution. For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$\det(I - A) = (1 - a)(1 - d) - bc = 1 - (a + d) + (ad - bc) = 1 - \text{tr } A + \det A.$$

This is exactly the torus Lefschetz number.

□

Exercise 35.39. Determine when the torus trace formula guarantees a fixed point.

Hint. Require $\det(I - A) \neq 0$. □

Solution. The trace formula gives $L(f) = \det(I - A)$. Therefore

$$\det(I - A) \neq 0$$

is sufficient for the induced torus map to have a fixed point. □

Exercise 35.40. Compute L for an orientation-reversing degree -1 map of S^2 .

Hint. Use $1 + d$ in even dimension. □

Solution. An orientation-reversing map $S^2 \rightarrow S^2$ of degree -1 has

$$L(f) = 1 + (-1) = 0.$$

Thus Lefschetz alone does not force a fixed point. □

Exercise 35.41. Compute L for the antipodal map on S^n .

Hint. The antipodal degree is $(-1)^{n+1}$. □

Solution. The antipodal map on S^n has degree $(-1)^{n+1}$. Hence

$$L(a) = 1 + (-1)^n(-1)^{n+1} = 1 - 1 = 0.$$

□

Exercise 35.42. Compare the conclusion with the actual fixed-point set.

Hint. Check that antipodal maps have no fixed points. □

Solution. The antipodal map has no fixed points because $x = -x$ on a sphere would imply $x = 0$, which is impossible. Its Lefschetz number is correspondingly zero, consistent with the theorem. □

Exercise 35.43. Use the identity map to recover Euler characteristic.

Hint. Sum alternating Betti numbers. □

Solution. For the identity,

$$L(\text{id}_X) = \sum_k (-1)^k \text{tr}(\text{id}|H_k) = \sum_k (-1)^k b_k = \chi(X).$$

□

Exercise 35.44. Express the Lefschetz number as an alternating supertrace.

Hint. Regard the total homology as a graded vector space. □

Solution. If $H_*(X; \mathbb{Q}) = H_{\text{even}} \oplus H_{\text{odd}}$ as a graded vector space, then

$$L(f) = \text{tr}(f_*|H_{\text{even}}) - \text{tr}(f_*|H_{\text{odd}}),$$

which is the supertrace. □

Exercise 35.45. Explain the role of finite-dimensional homology in defining the trace.

Hint. Use a field such as $\mathbb{Q}$. □

Solution. The trace is defined for finite-dimensional vector spaces. A finite CW complex has finite-dimensional $H_k(X; \mathbb{Q})$, and only finitely many degrees are nonzero, so the alternating trace sum is finite and well-defined. □

Exercise 35.46. Describe how Poincaré duality enters the graph–diagonal proof.

Hint. Dualize the diagonal class. □

Solution. Poincaré duality identifies the homology class of the diagonal with a sum built from dual homology/cohomology bases. Pairing the graph class with this diagonal class converts the graph–diagonal intersection number into the alternating traces of f_* . □

Exercise 35.47. Summarize homology trace $\leftrightarrow$ intersection number $\leftrightarrow$ fixed point.

Hint. Track the three descriptions of the same number. □

Solution. The same integer has three forms:

$$L(f) = \sum_k (-1)^k \operatorname{tr}(f_* | H_k) = \Gamma_f \cdot \Delta = \sum_{x \in \operatorname{Fix}(f)} \operatorname{ind}_x(f)$$

when the local-index hypotheses hold. This is the algebraic, geometric, and local fixed-point synthesis of Lefschetz theory. □